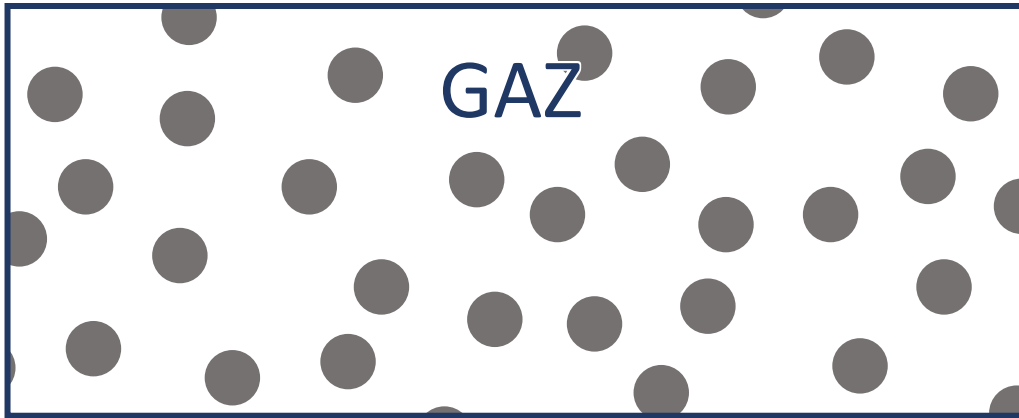
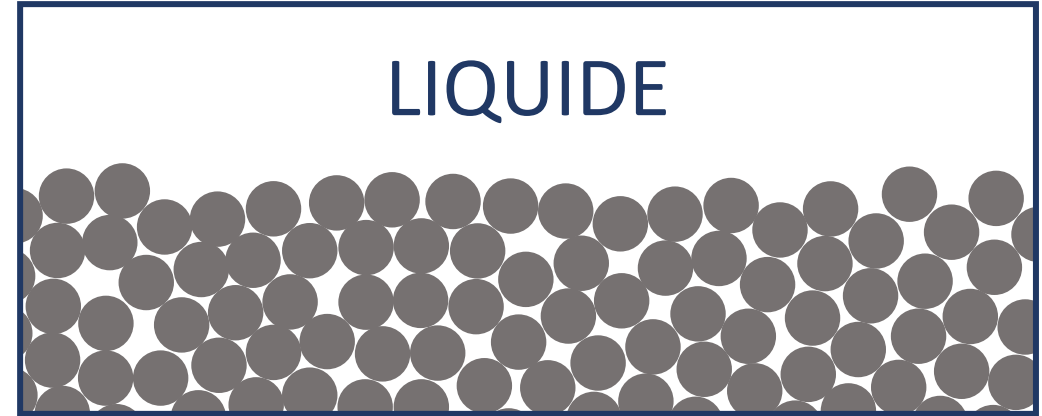
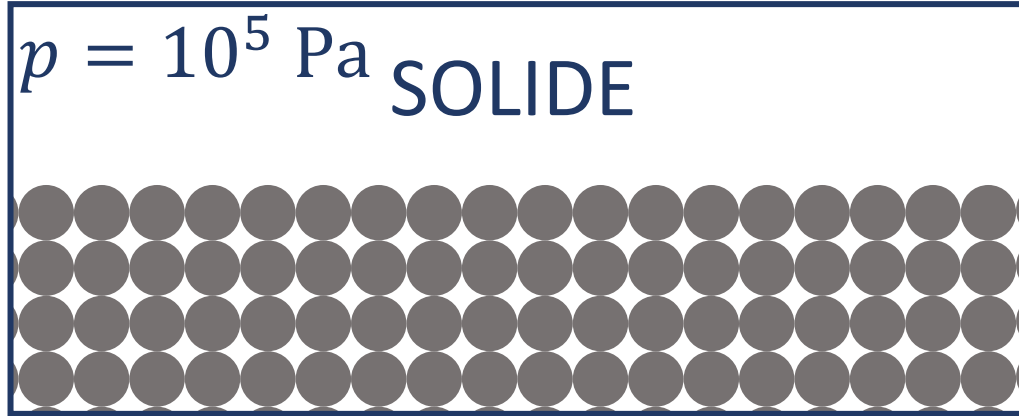

Contrôle et stabilité des colonnes de plasma magnétisées : interactions plasma-cathode et opération d'un plasma hélicon



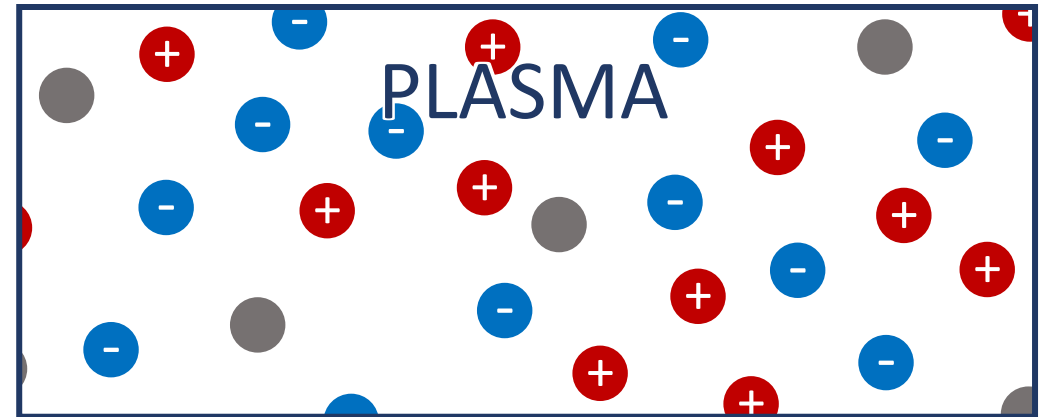
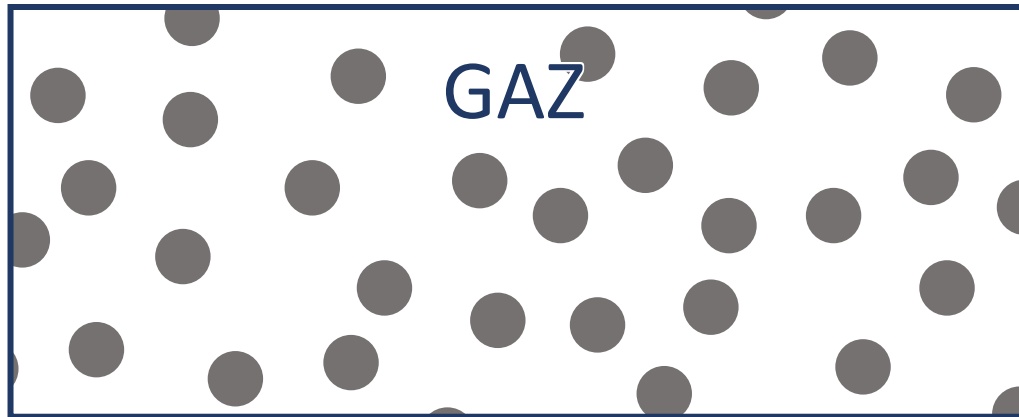
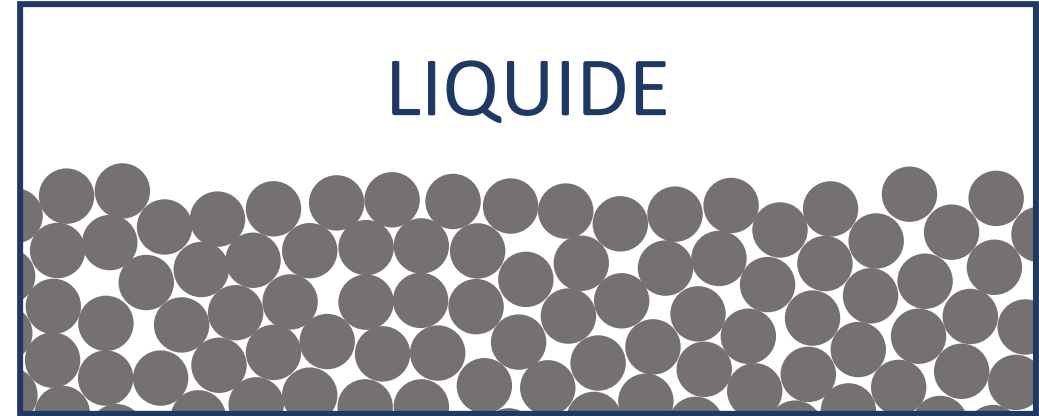
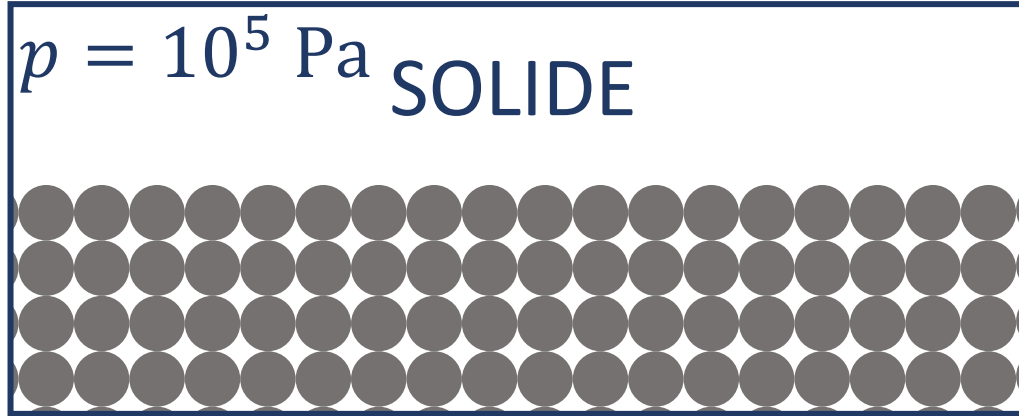
Qu'est-ce que le plasma ?

Contrôle et stabilité des colonnes de **plasma** magnétisées :
interactions plasma-cathode et opération d'un plasma hélicon

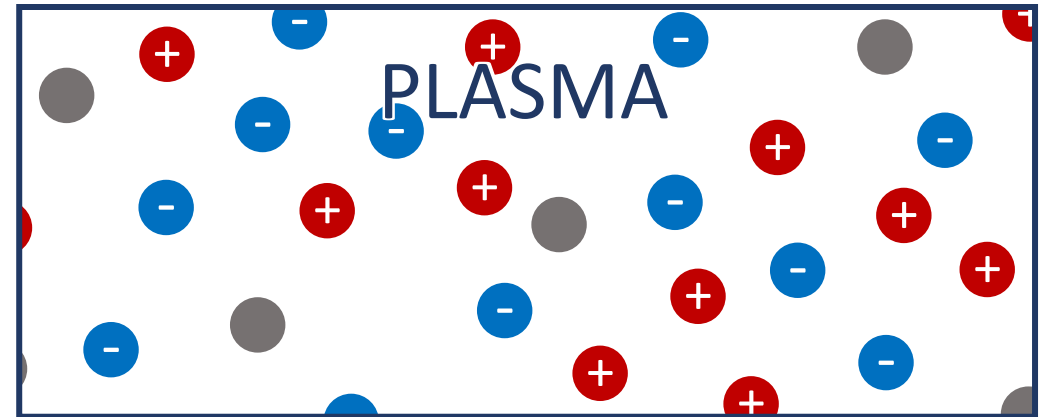
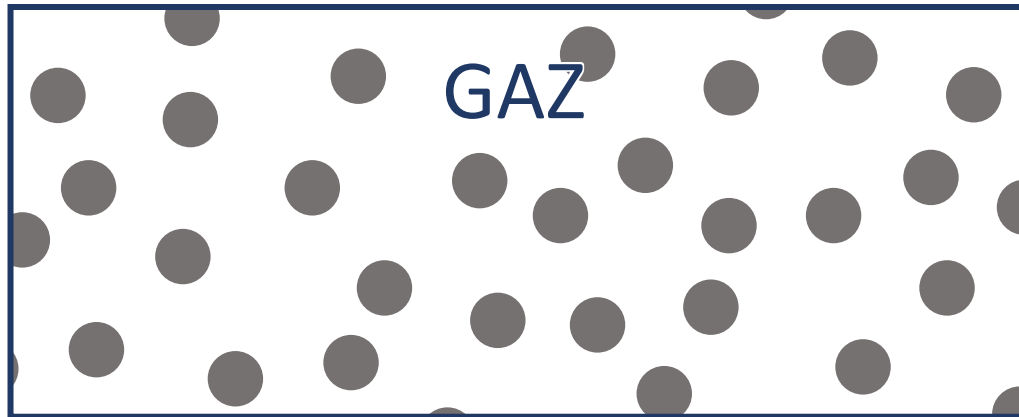
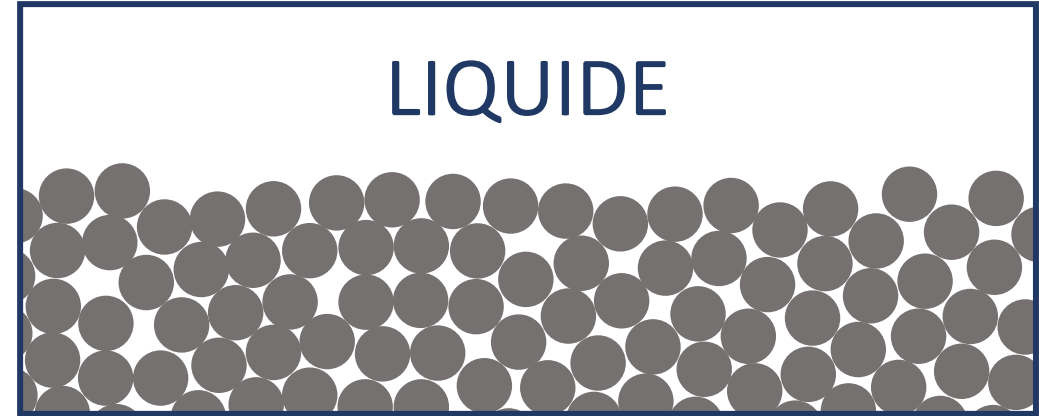
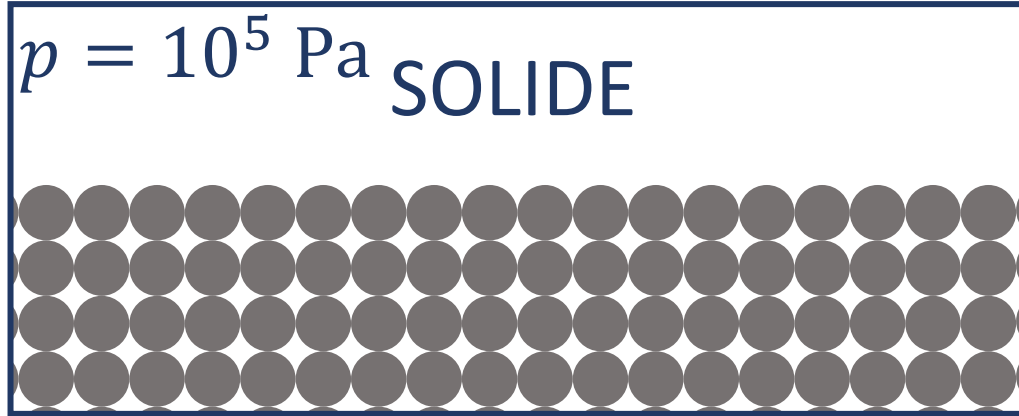
Le quatrième état de la matière



Le quatrième état de la matière



Le quatrième état de la matière



- Transition d'état continue : **ionisation**
- Multi-espèces : neutres, ions, électrons

Le plasma dans l'univers



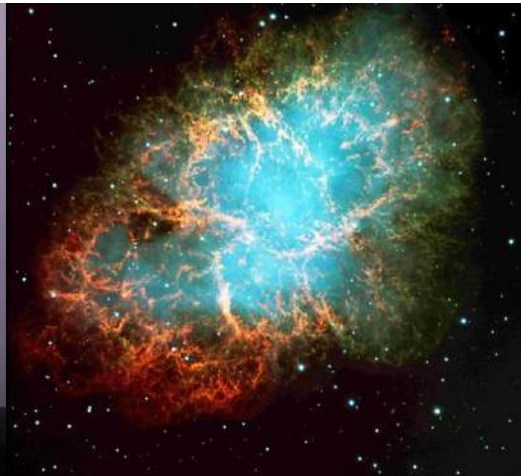
Flamme



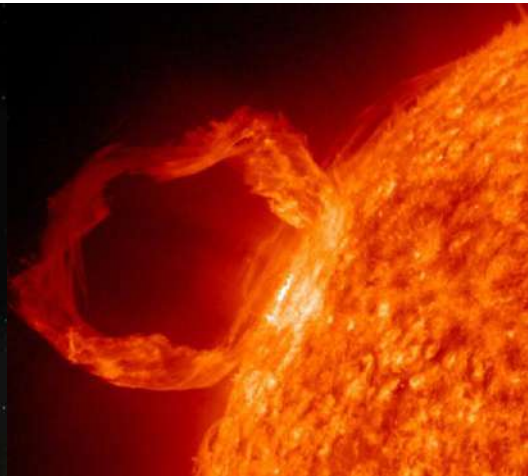
**Aurore
polaire**



Foudre

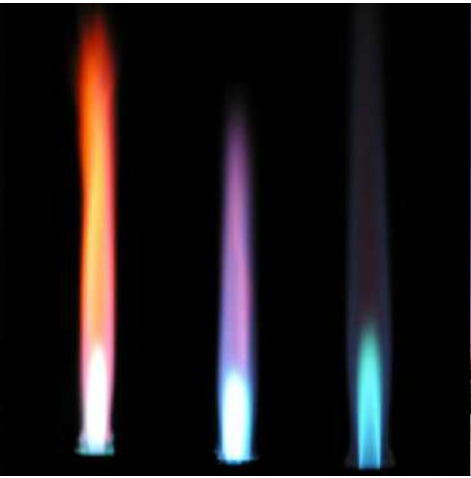


Nébuleuse



Etoile

Le plasma dans l'univers



Flamme

$$n = 10^{16} \text{ m}^{-3}$$

$$T = 2\,500 \text{ K}$$



**Aurore
polaire**

$$n = 10^{11} \text{ m}^{-3}$$

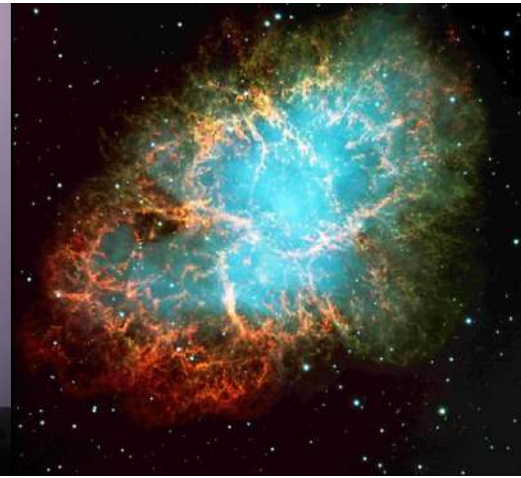
$$T = 5\,000 \text{ K}$$



Foudre

$$n = 10^{19} \text{ m}^{-3}$$

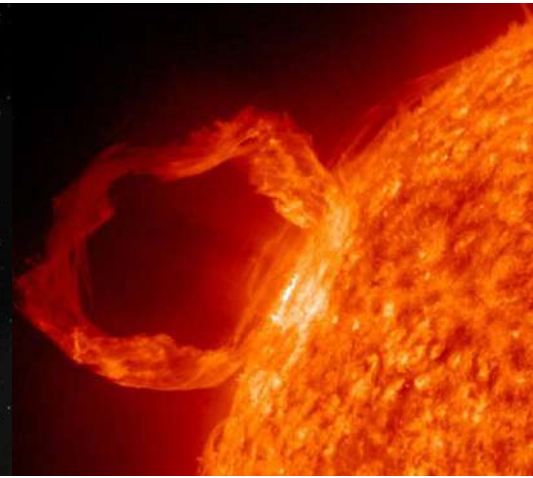
$$T = 30\,000 \text{ K}$$



Nébuleuse

$$n = 10^8 \text{ m}^{-3}$$

$$T = 15\,000 \text{ K}$$

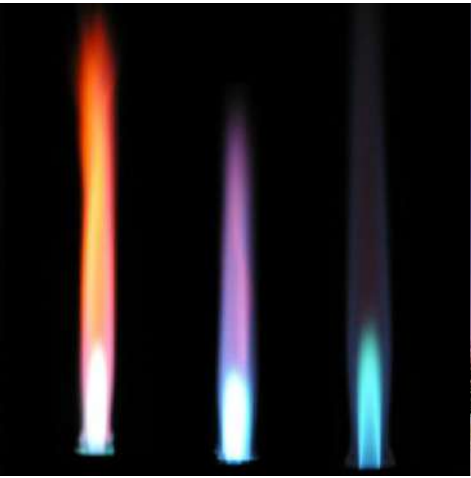


Etoile

$$n = 10^{31} \text{ m}^{-3}$$

$$T = 1\,000\,000 \text{ K}$$

Le plasma dans l'univers



Flamme

$$n = 10^{16} \text{ m}^{-3}$$

$$T = 2\,500 \text{ K}$$

$$\gamma = \frac{n}{n_{\text{tot}}} = 10^{-9}$$



**Aurore
polaire**

$$n = 10^{11} \text{ m}^{-3}$$

$$T = 5\,000 \text{ K}$$

$$\gamma = 10^{-5}$$

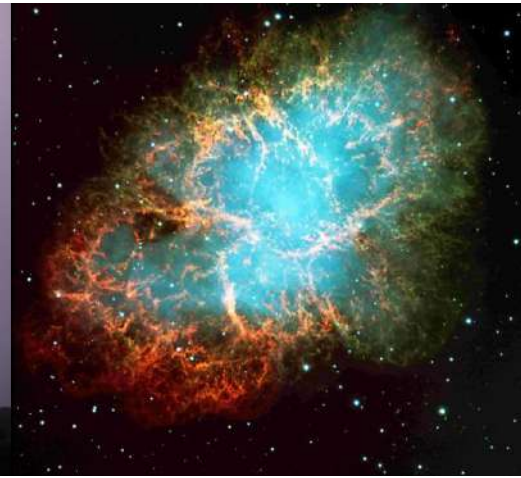


Foudre

$$n = 10^{19} \text{ m}^{-3}$$

$$T = 30\,000 \text{ K}$$

$$\gamma = 10^{-4}$$

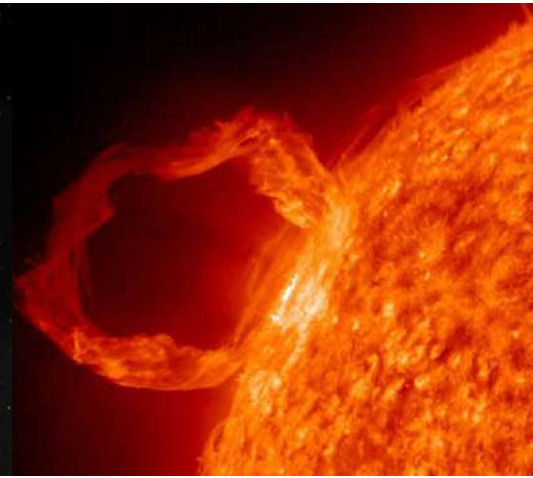


Nébuleuse

$$n = 10^8 \text{ m}^{-3}$$

$$T = 15\,000 \text{ K}$$

$$\gamma = 1$$



Etoile

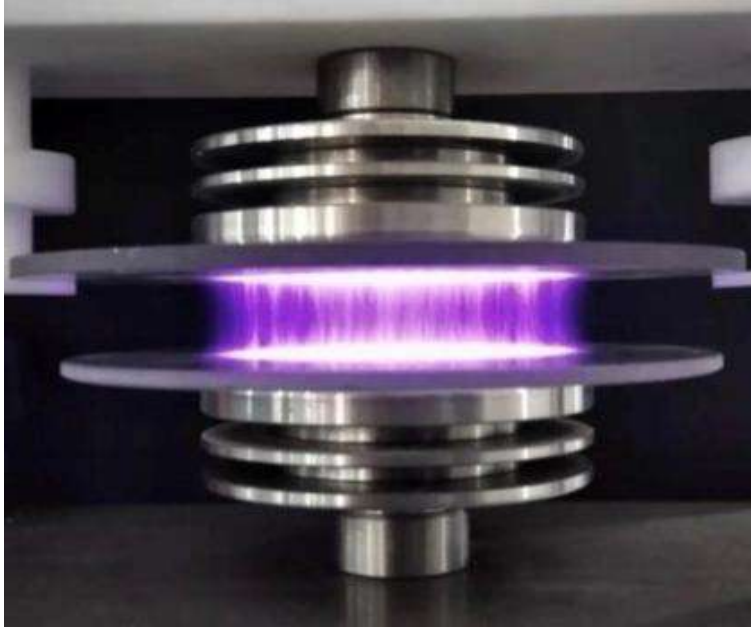
$$n = 10^{31} \text{ m}^{-3}$$

$$T = 1\,000\,000 \text{ K}$$

$$\gamma = 1$$

➤ Fraction de ionisation : impacte la dynamique du plasma

Le plasma et ses applications



Traitement de surface

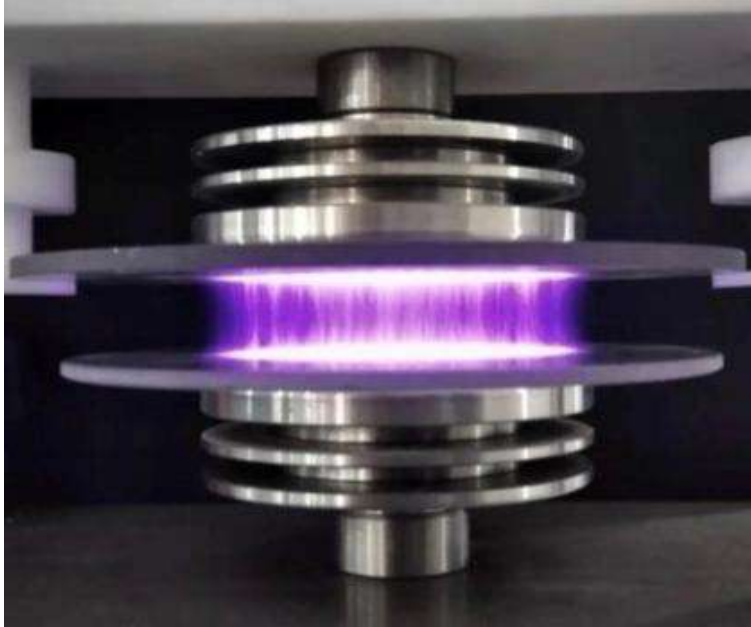


Aérospatial



**Fusion par confinement
magnétique**

Le plasma et ses applications



Traitement de surface

*Gravure, circuits électroniques
Couches minces*

Contrôler l'homogénéité



Aérospatial

Propulseurs embarqués

Contrôler la poussée



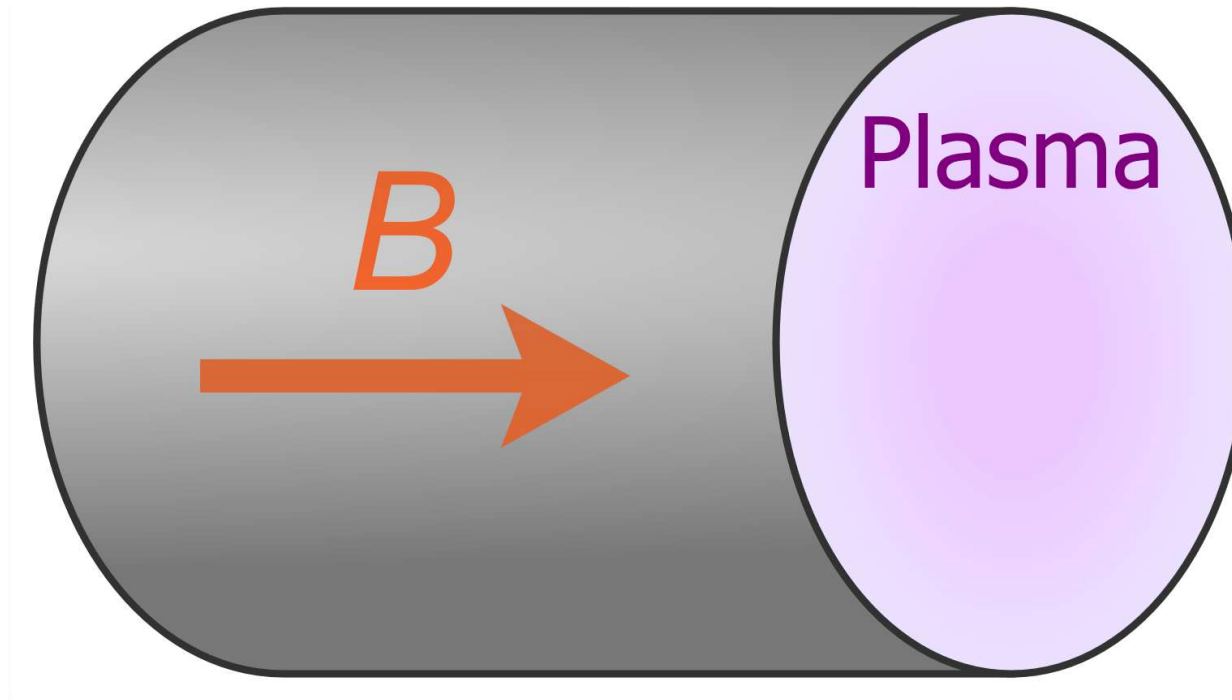
Fusion par confinement magnétique

Production d'énergie décarbonée

Contrôler l'énergie

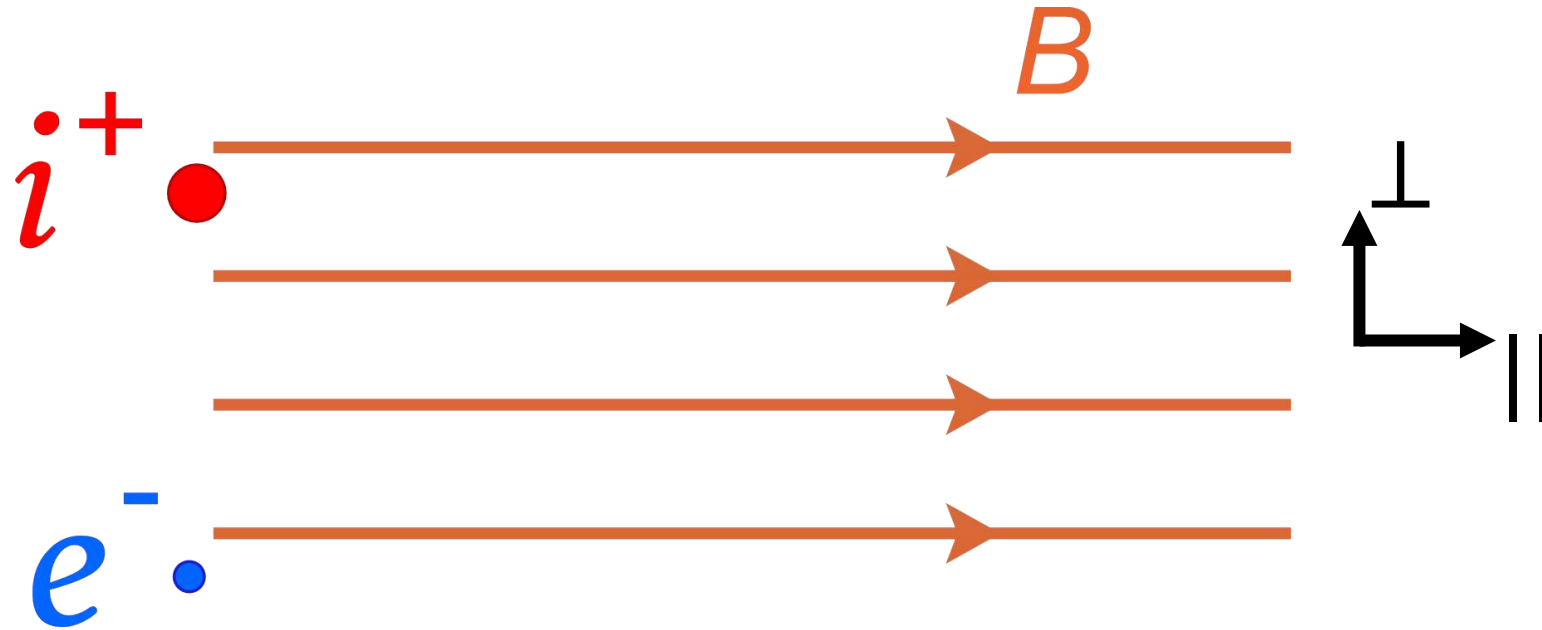
Colonne de plasma magnétisée

Plasma en conformation cylindrique



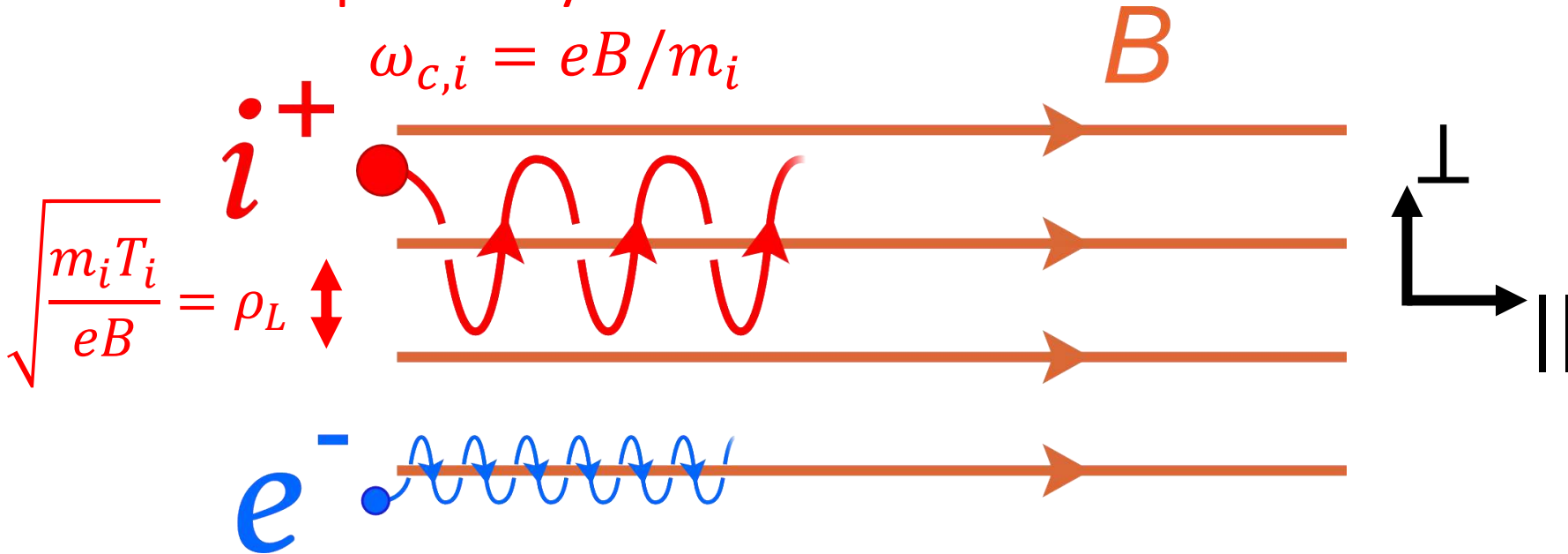
- Quelle stabilité du plasma ?
- Quels outils de contrôle ?

Confinement magnétique



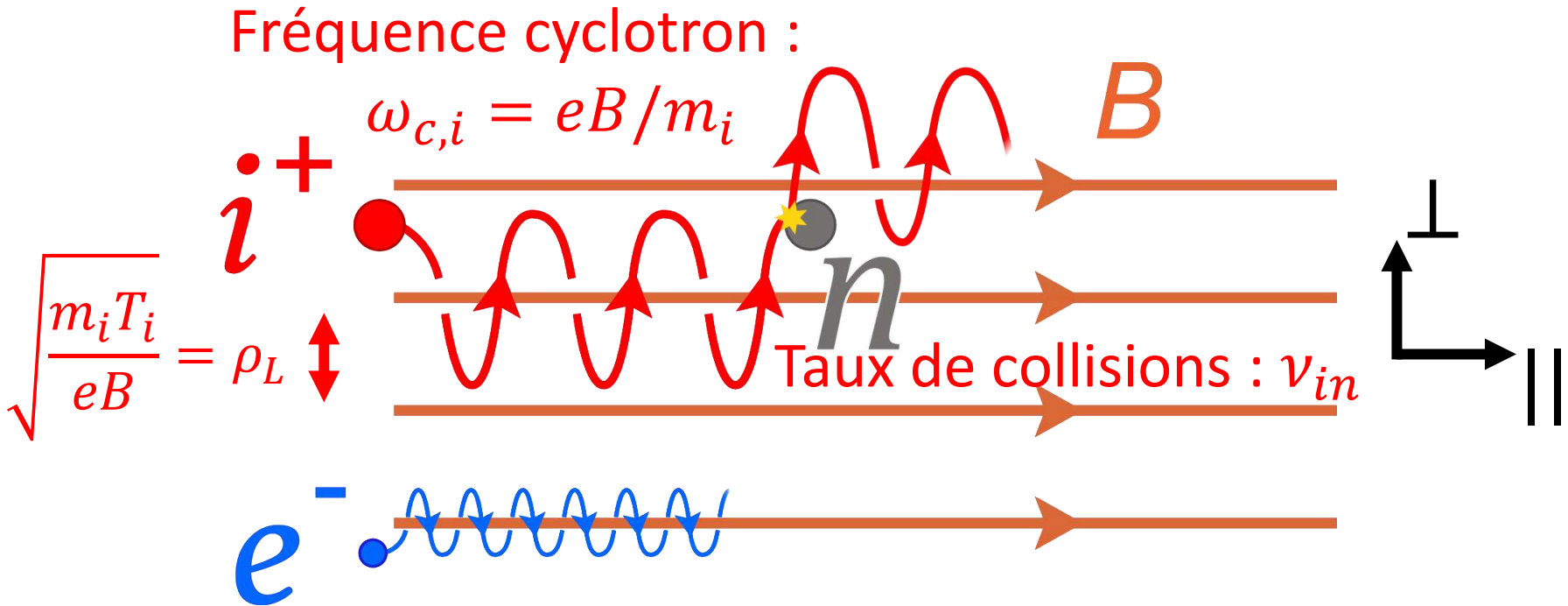
Confinement magnétique

Fréquence cyclotron :



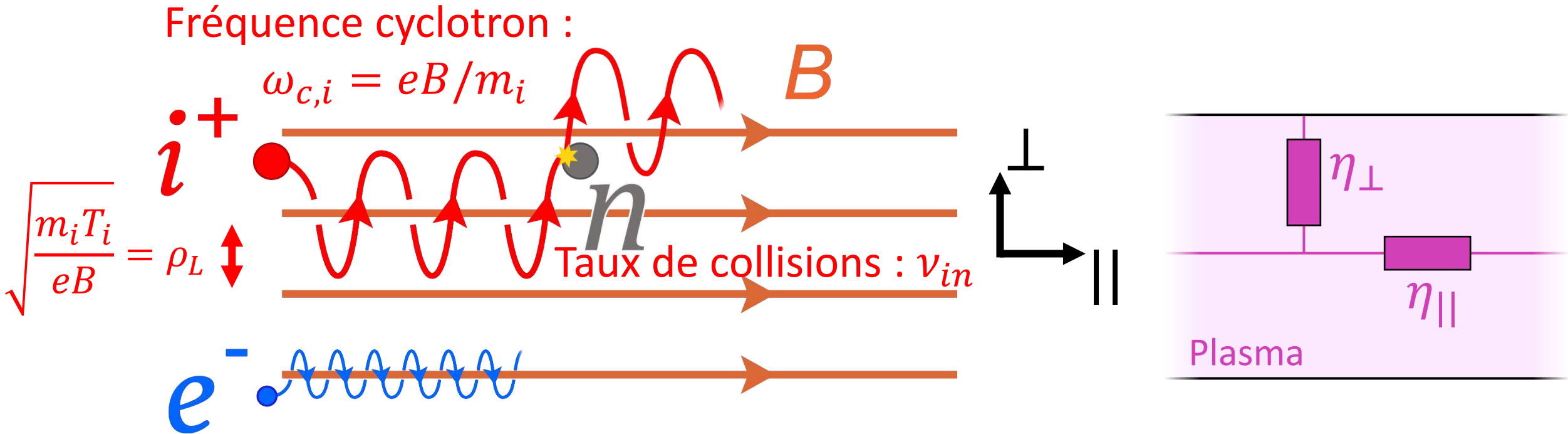
- Confinement magnétique : plasma anisotrope
 - Diffusion collisionnelle : transport radial
 - $\omega_{c,i} / \nu_{in} = 0.1 - 2$; $\omega_{c,e} / \nu_{en} \gg 1$
 - $\eta_{\perp} \gg \eta_{||}$

Confinement magnétique



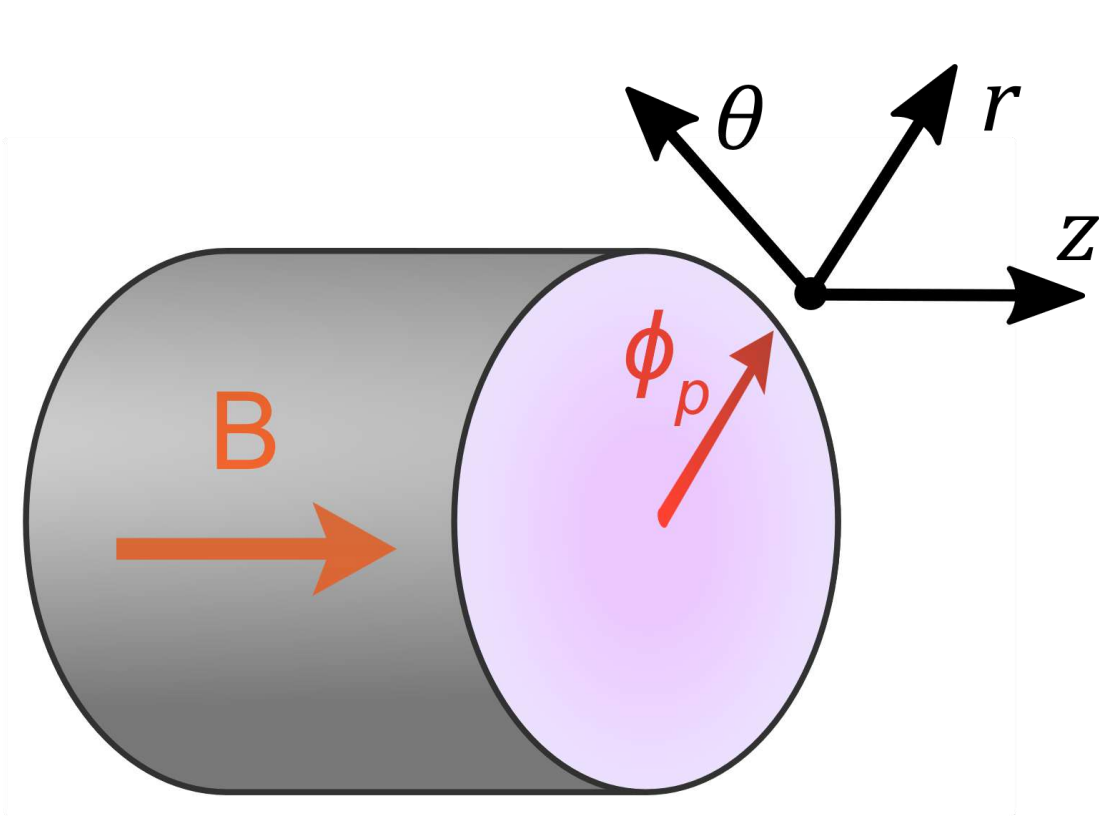
- Confinement magnétique : plasma anisotrope
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 - $\eta_{\perp} \gg \eta_{\parallel}$

Confinement magnétique

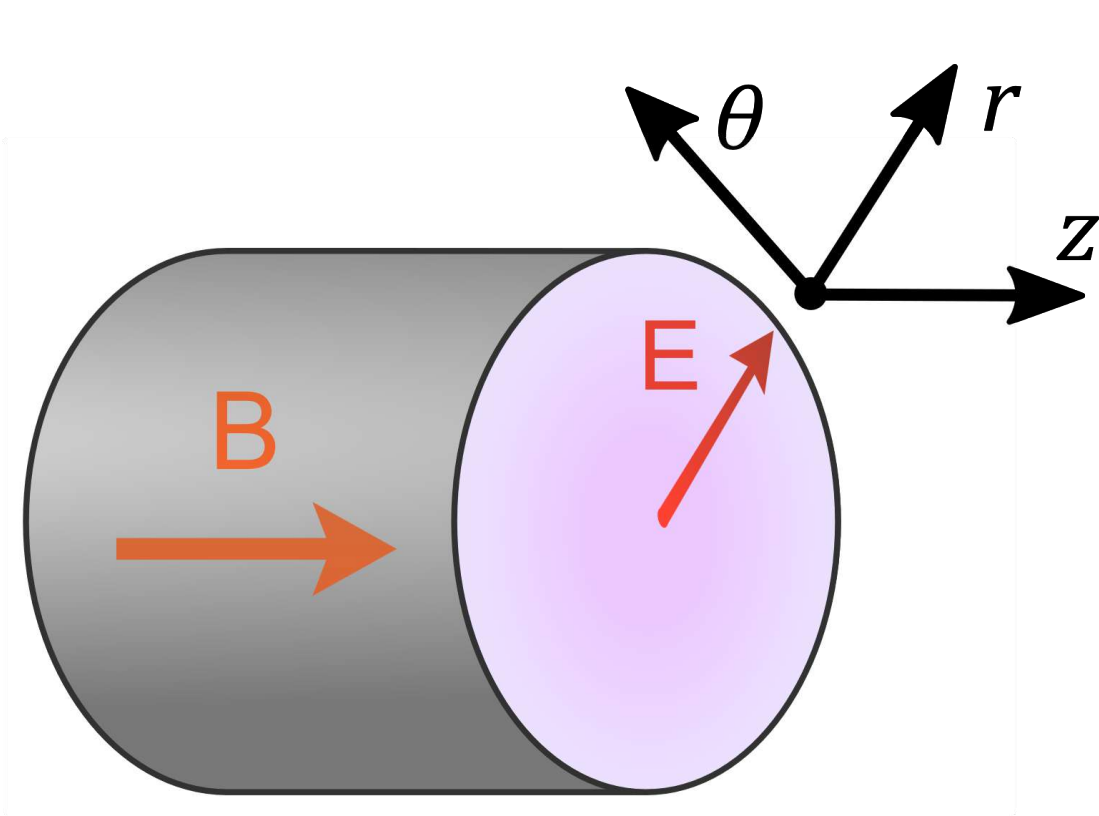


- Confinement magnétique : plasma anisotrope
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 - $\omega_{c,i}/\nu_{in} = 0.1 - 2$; $\omega_{c,e}/\nu_{en} \gg 1$
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Rotation du plasma : dérive $E \times B$

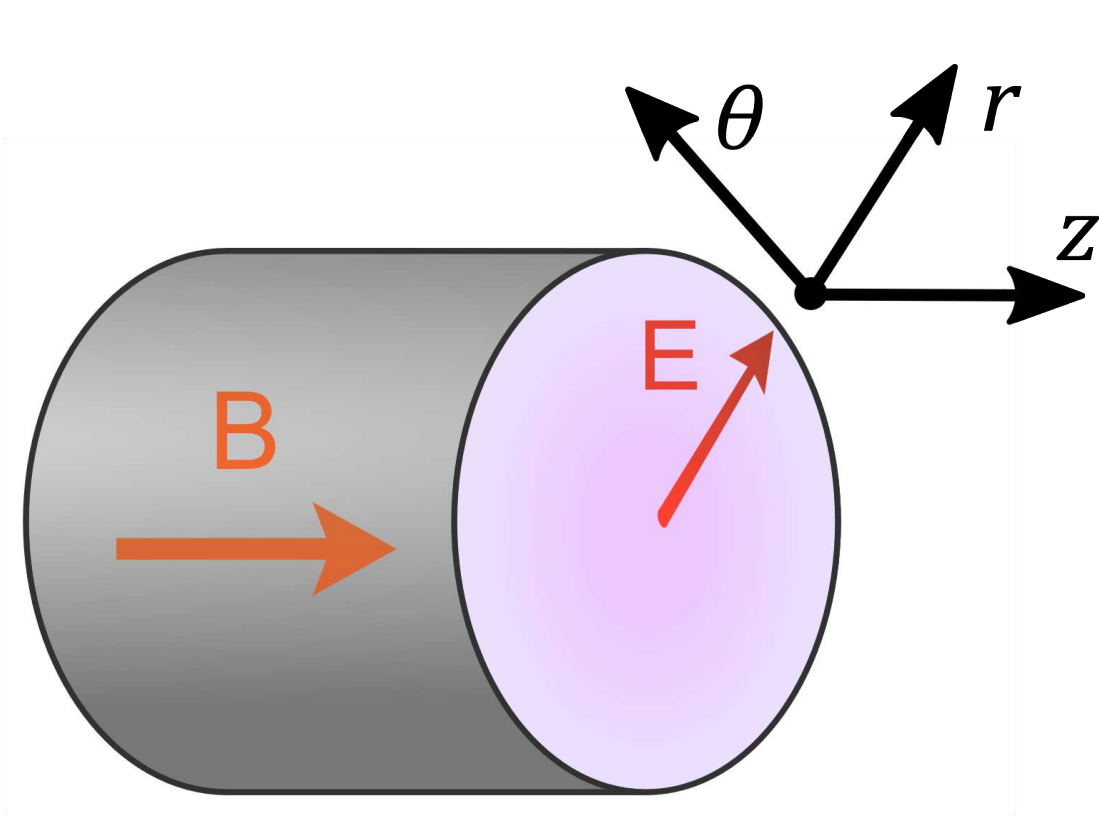


Rotation du plasma : dérive $E \times B$

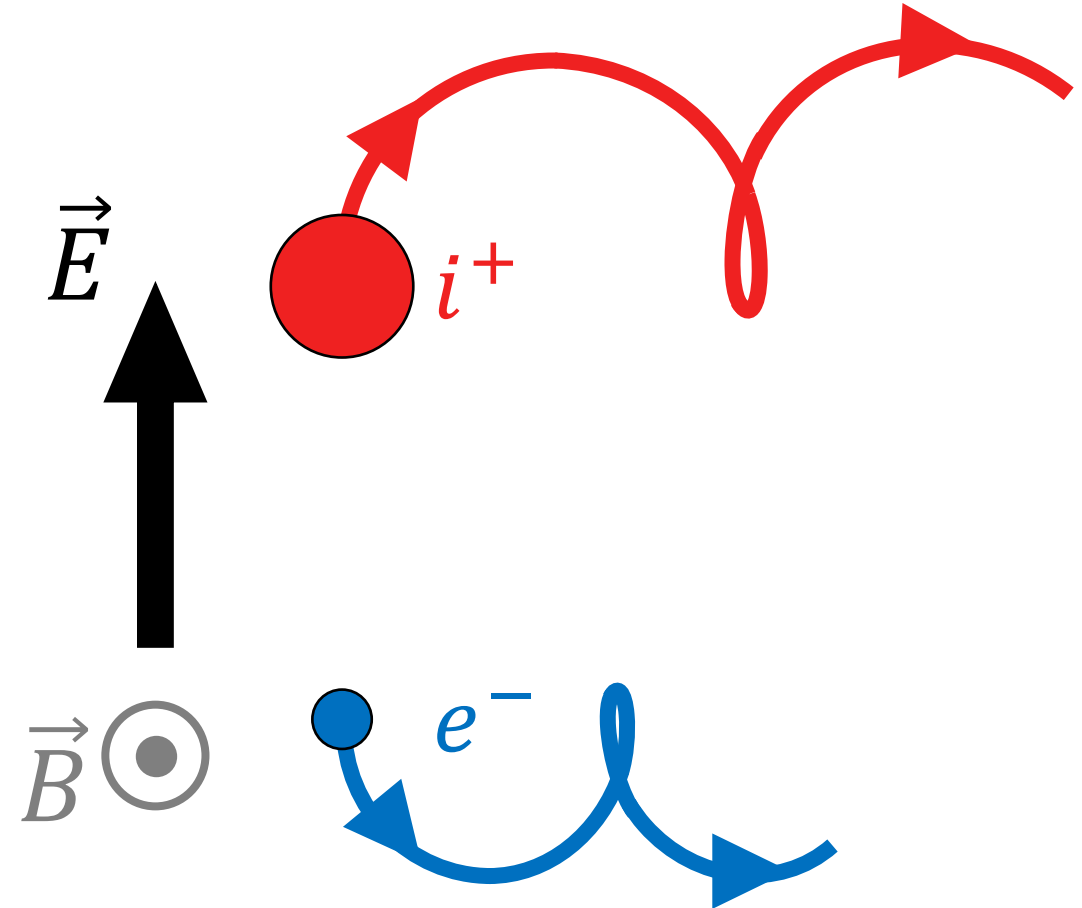


➤ $E(r) = -\nabla\phi_p(r)$

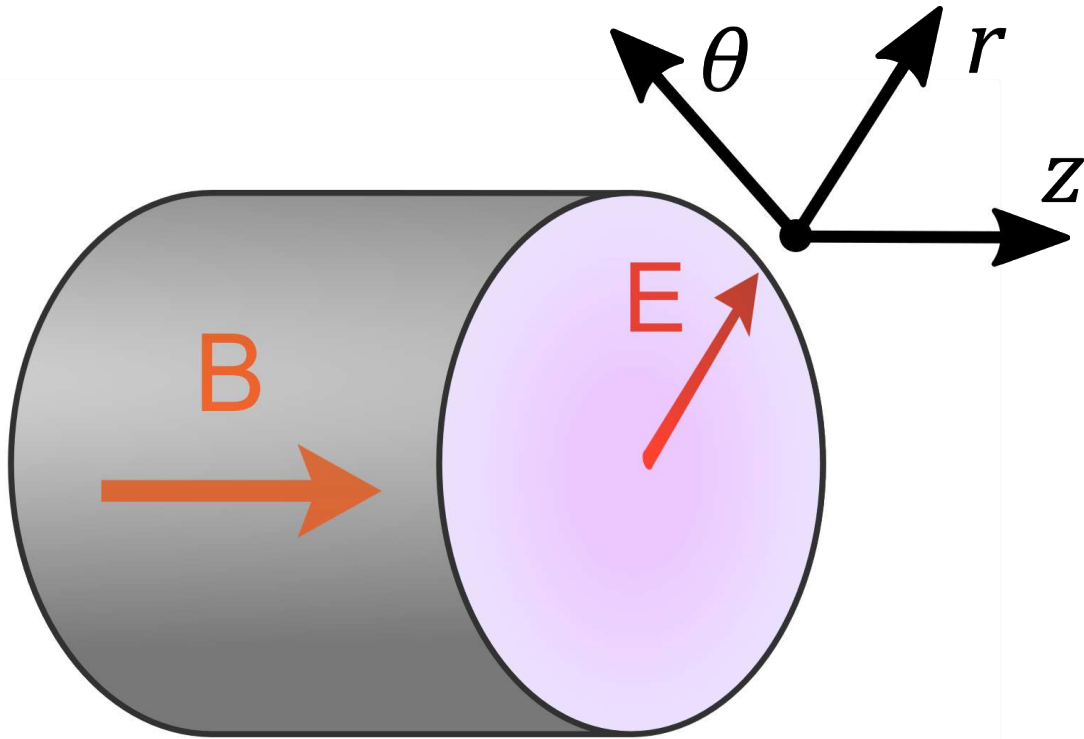
Rotation du plasma : dérive $E \times B$



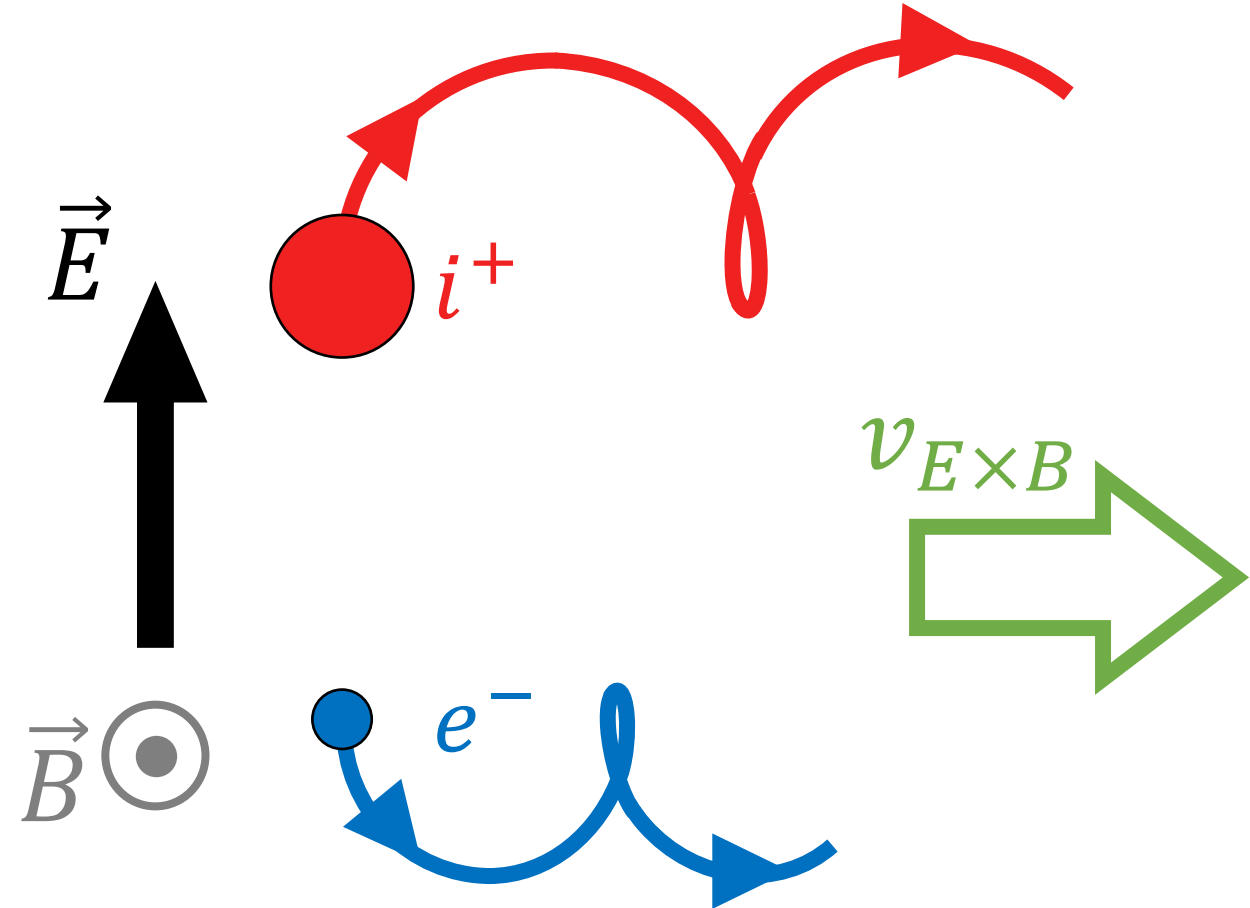
➤ $E(r) = -\nabla\phi_p(r)$



Rotation du plasma : dérive $E \times B$

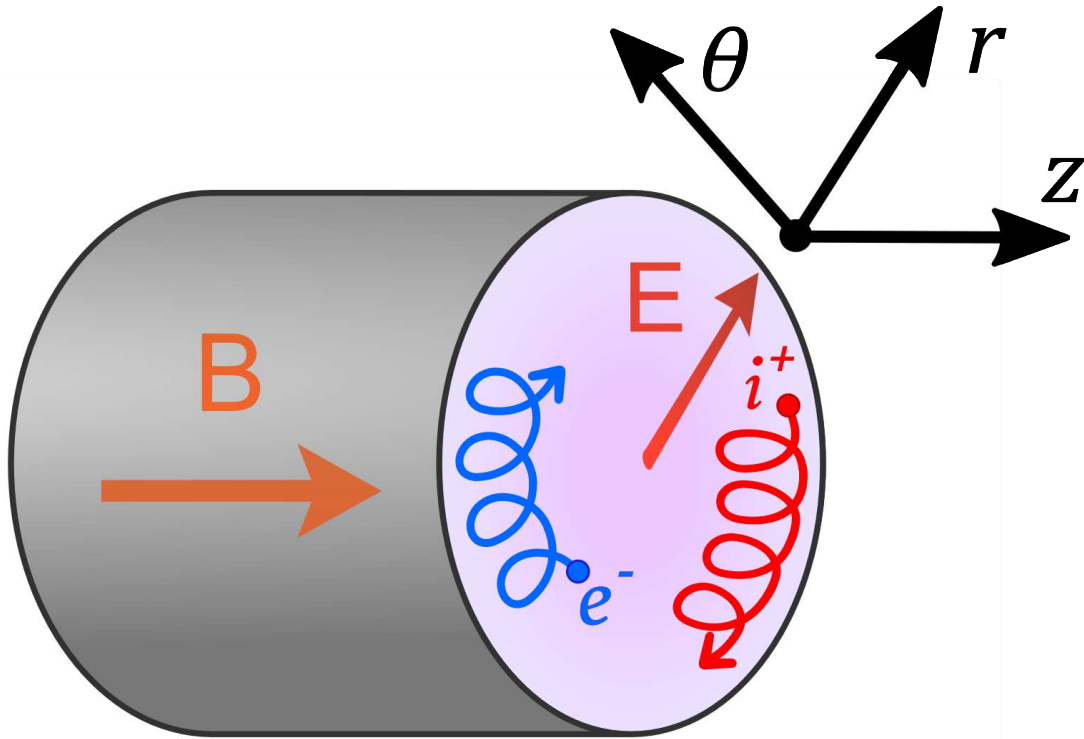


➤ $E(r) = -\nabla\phi_p(r)$

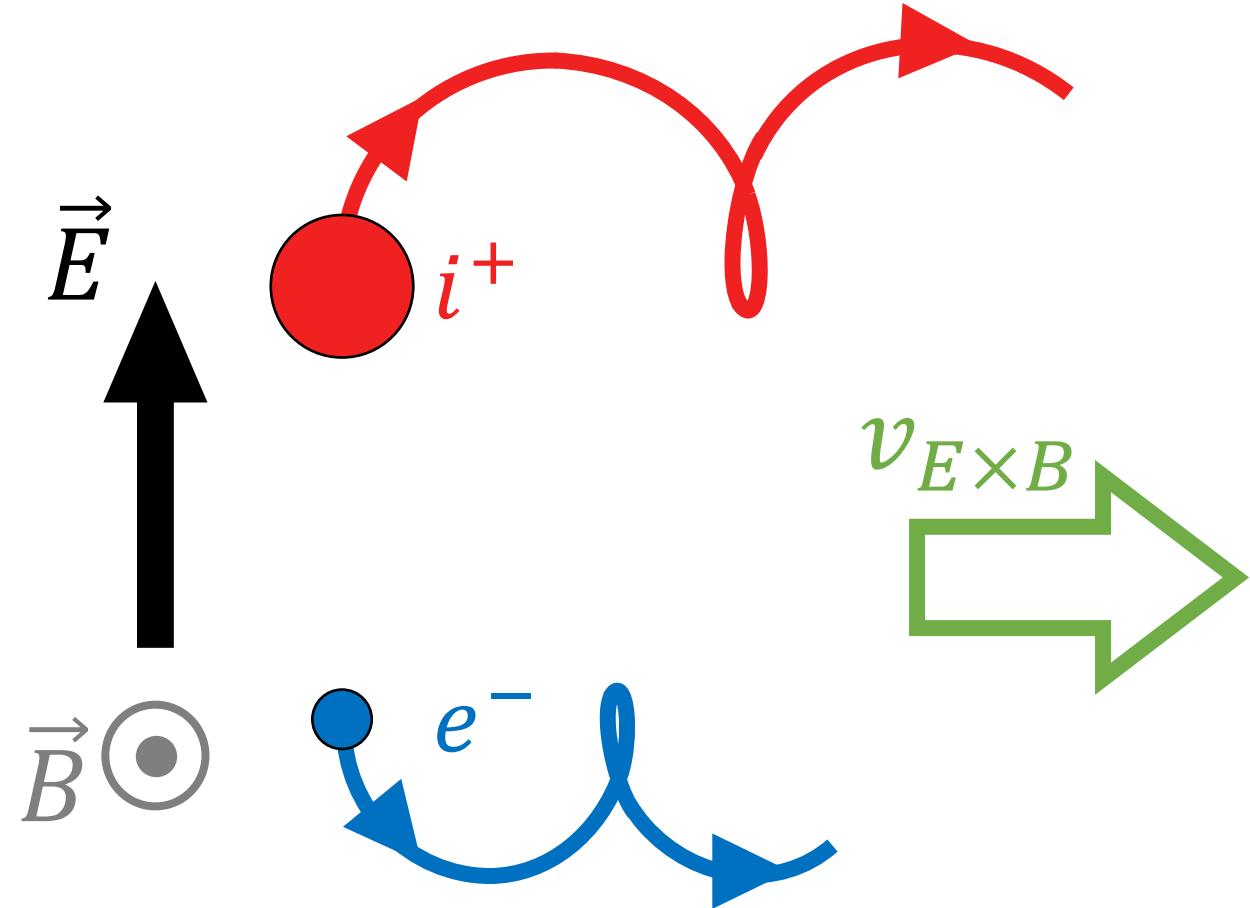


➤ $v = \frac{\mathbf{E} \times \mathbf{B}}{B^2} \sim 1 \text{ km/s}$

Rotation du plasma : dérive $E \times B$

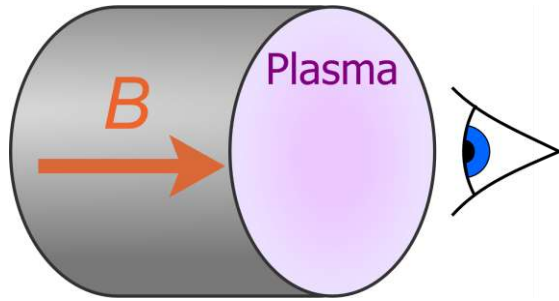


➤ $E(r) = -\nabla\phi_p(r)$



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Plasma multi-instabilités



- Cisaillement
Instabilité de Kelvin-Helmholtz

- Force centrifuge
Instabilité de Rayleigh-Taylor

- Gradient de densité
Ondes de dérive

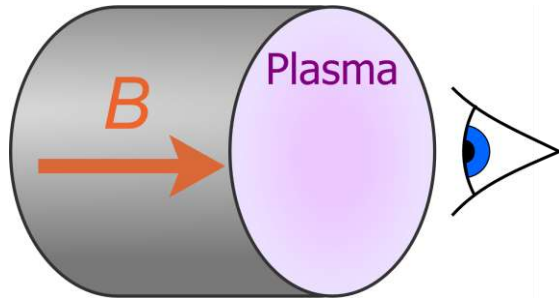
Thakur *et al.*, Plasma Sources Sci. Technol. (2014)

Brochard *et al.*, Phys. Plasmas (2005)

Gulke *et al.*, AIP Conf. Proc. (2008)

Vincent *et al.*, Phys. Plasmas (2023)

Plasma multi-instabilités



- Cisaillement

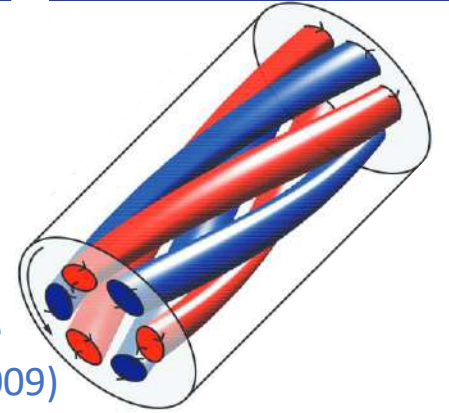
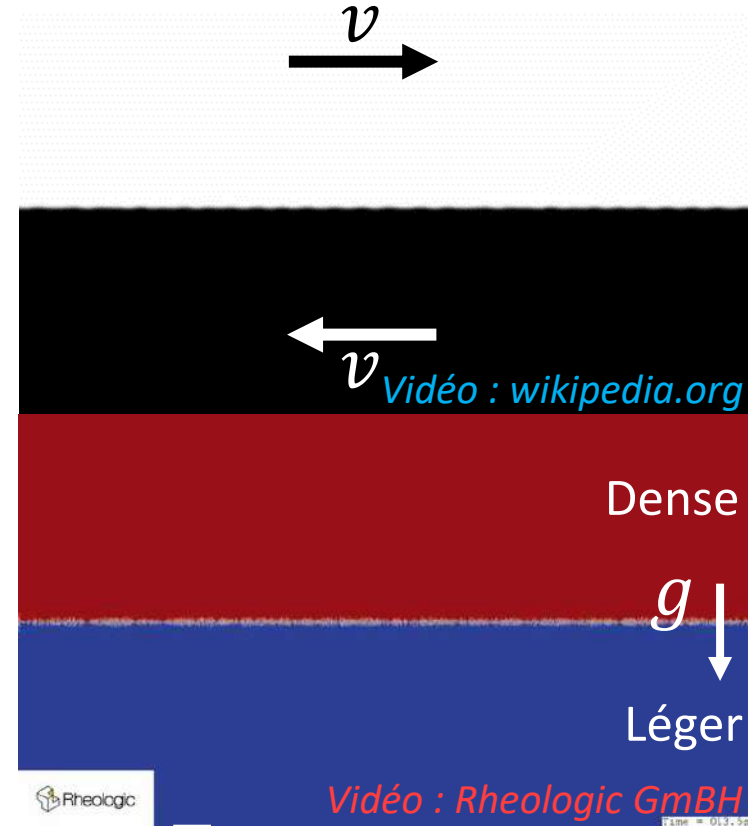
Instabilité de Kelvin-Helmholtz →

- Force centrifuge

Instabilité de Rayleigh-Taylor →

- Gradient de densité

Ondes de dérive →



Thakur *et al.*, Plasma Sources Sci. Technol. (2014)

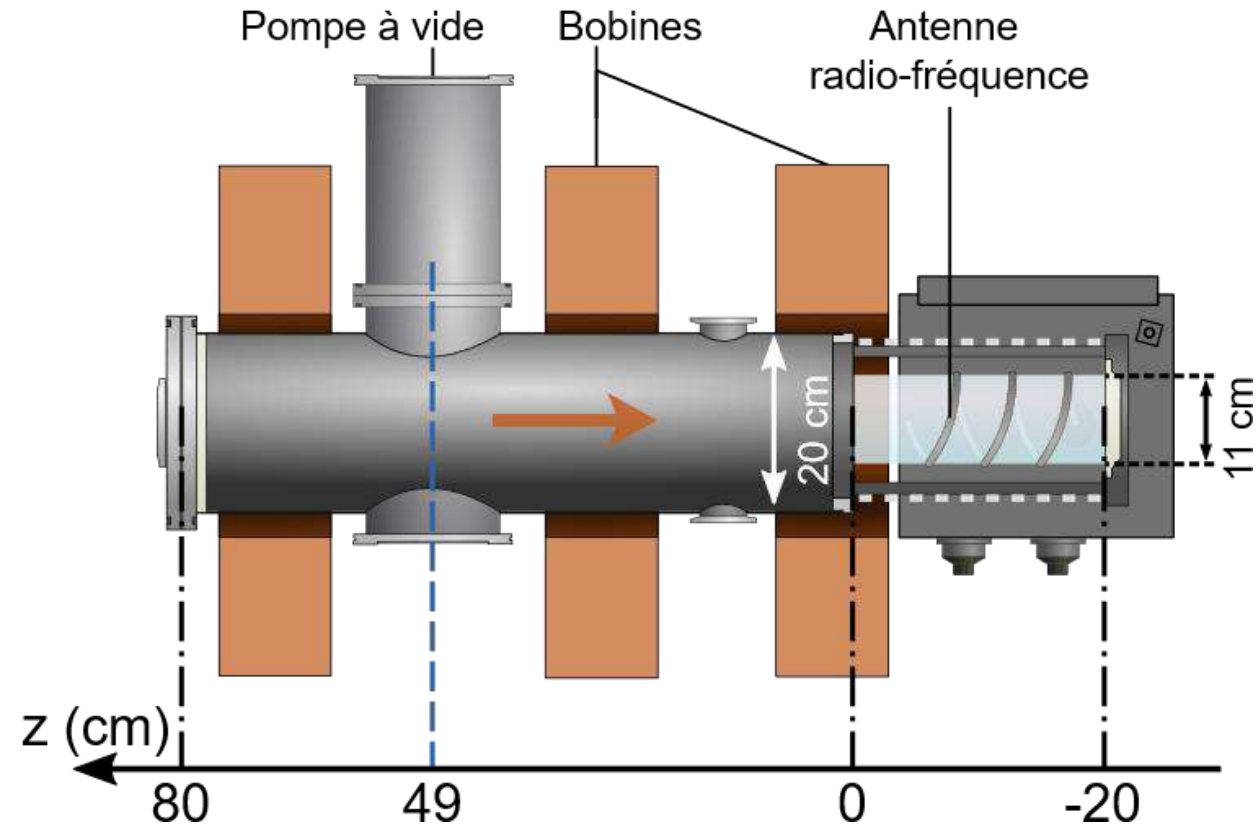
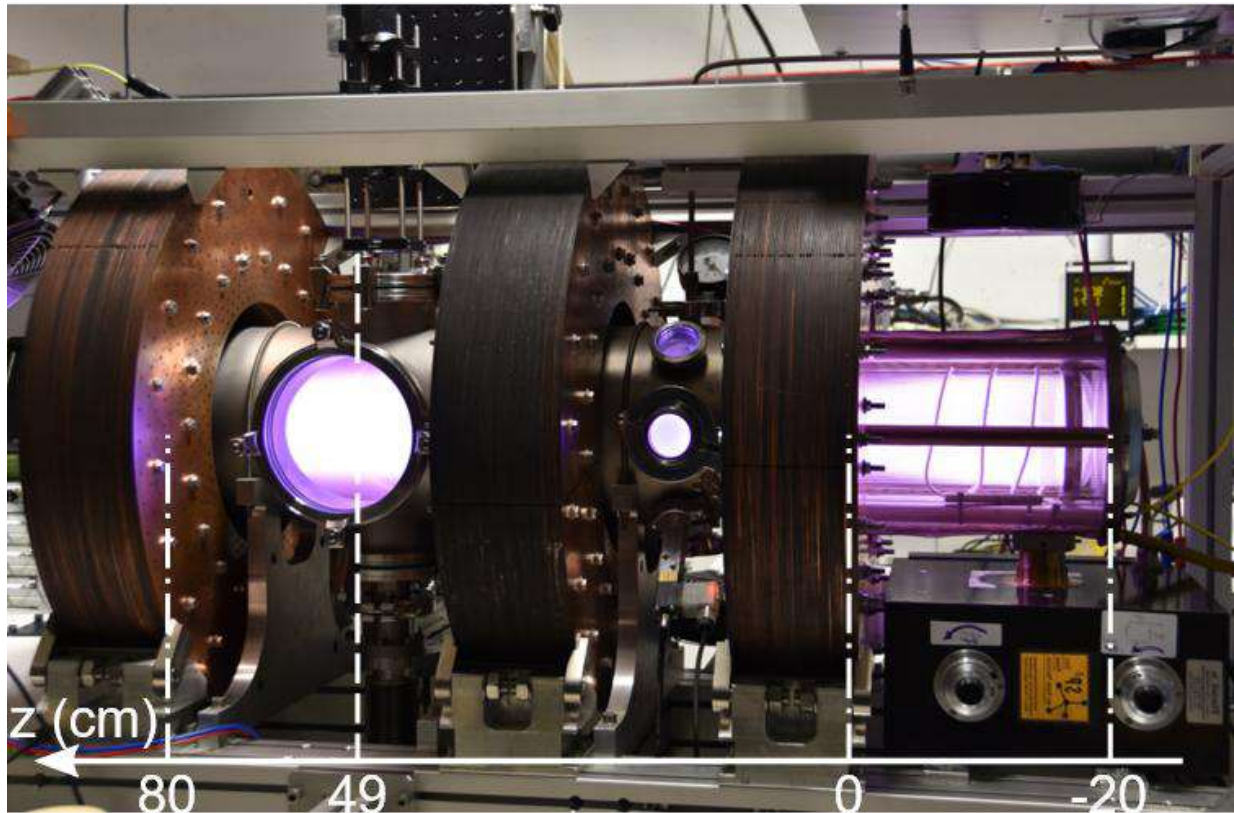
Brochard *et al.*, Phys. Plasmas (2005)

Grulke *et al.*, AIP Conf. Proc. (2008)

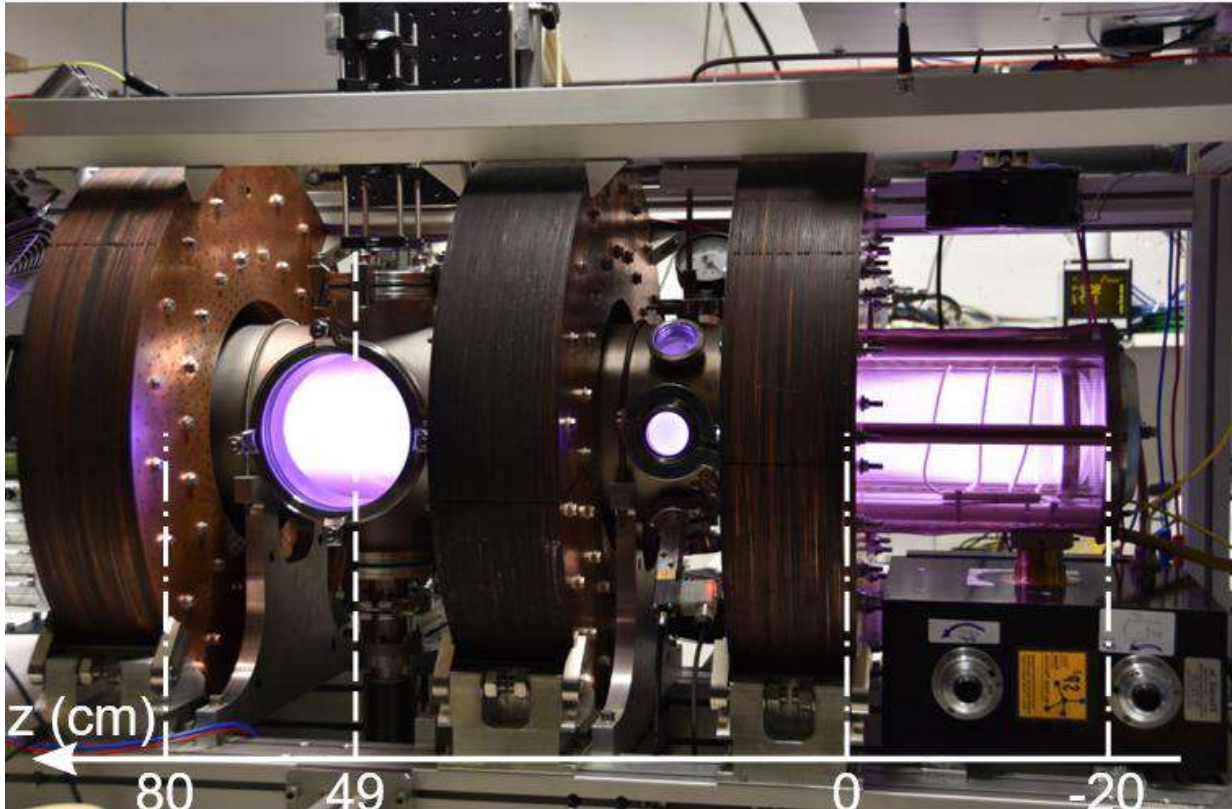
Vincent *et al.*, Phys. Plasmas (2023)

C. Brandt, thèse
de doctorat (2009)

Le dispositif Von-Kármán Plasma (VKP)



Le dispositif Von-Kármán Plasma (VKP)

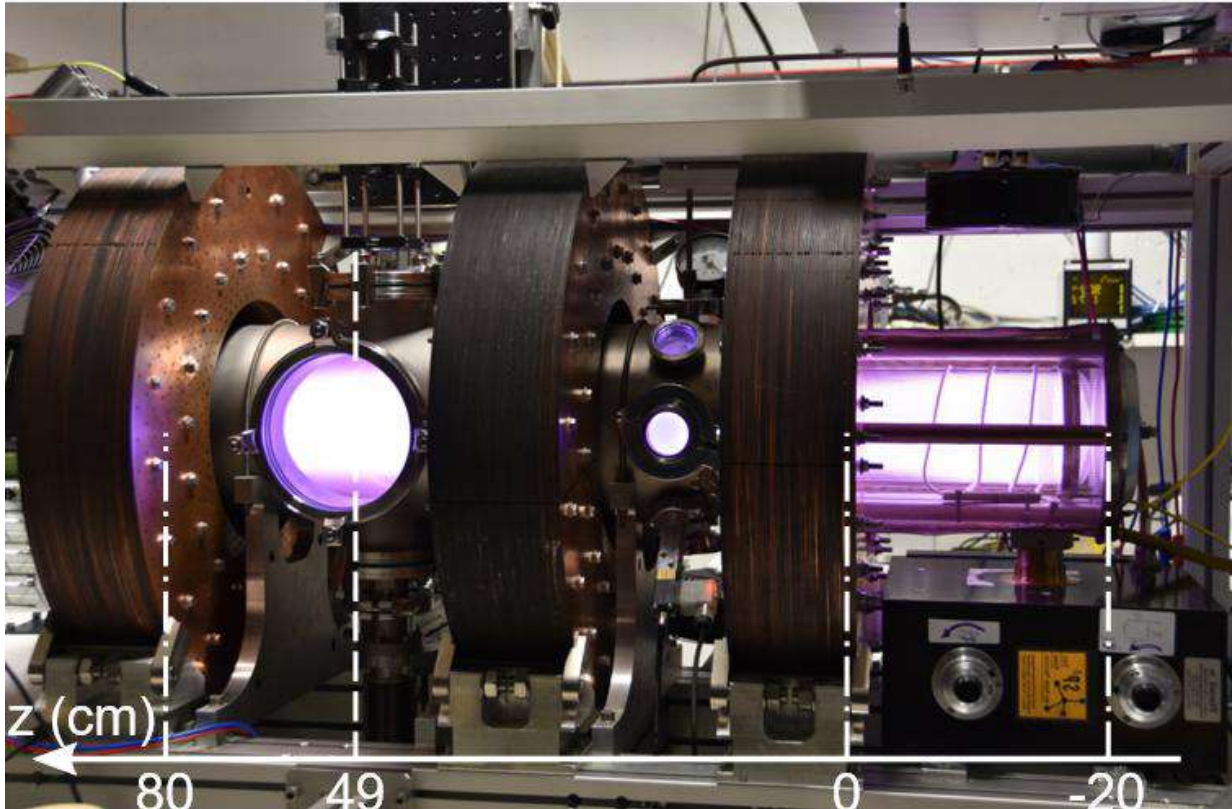


Gaz : argon

3 paramètres :

- $p_0 = 0.1 - 0.5 \text{ Pa}$
- $P_w = 0.5 - 2 \text{ kW}$
- $B = 100 - 1000 \text{ G}$

Le dispositif Von-Kármán Plasma (VKP)



Gaz : argon

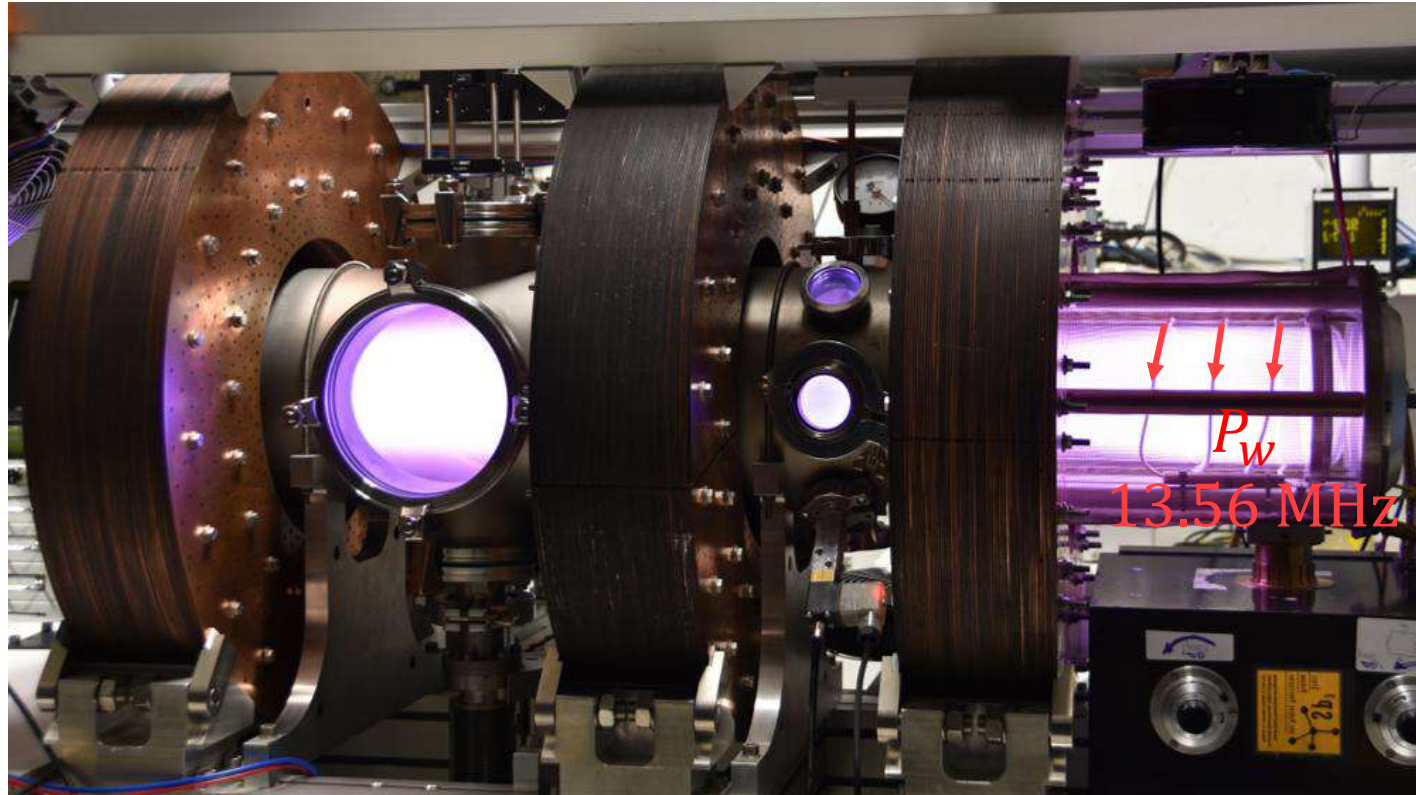
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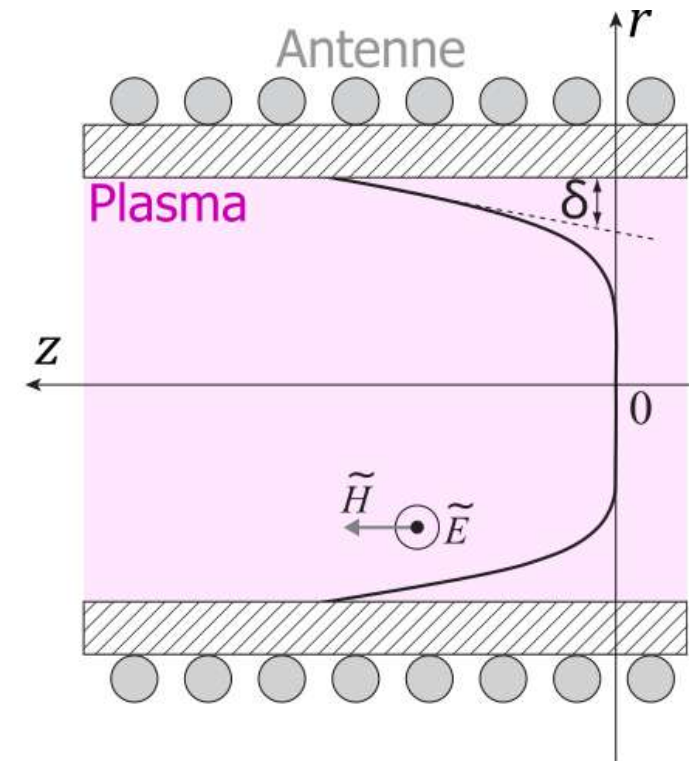
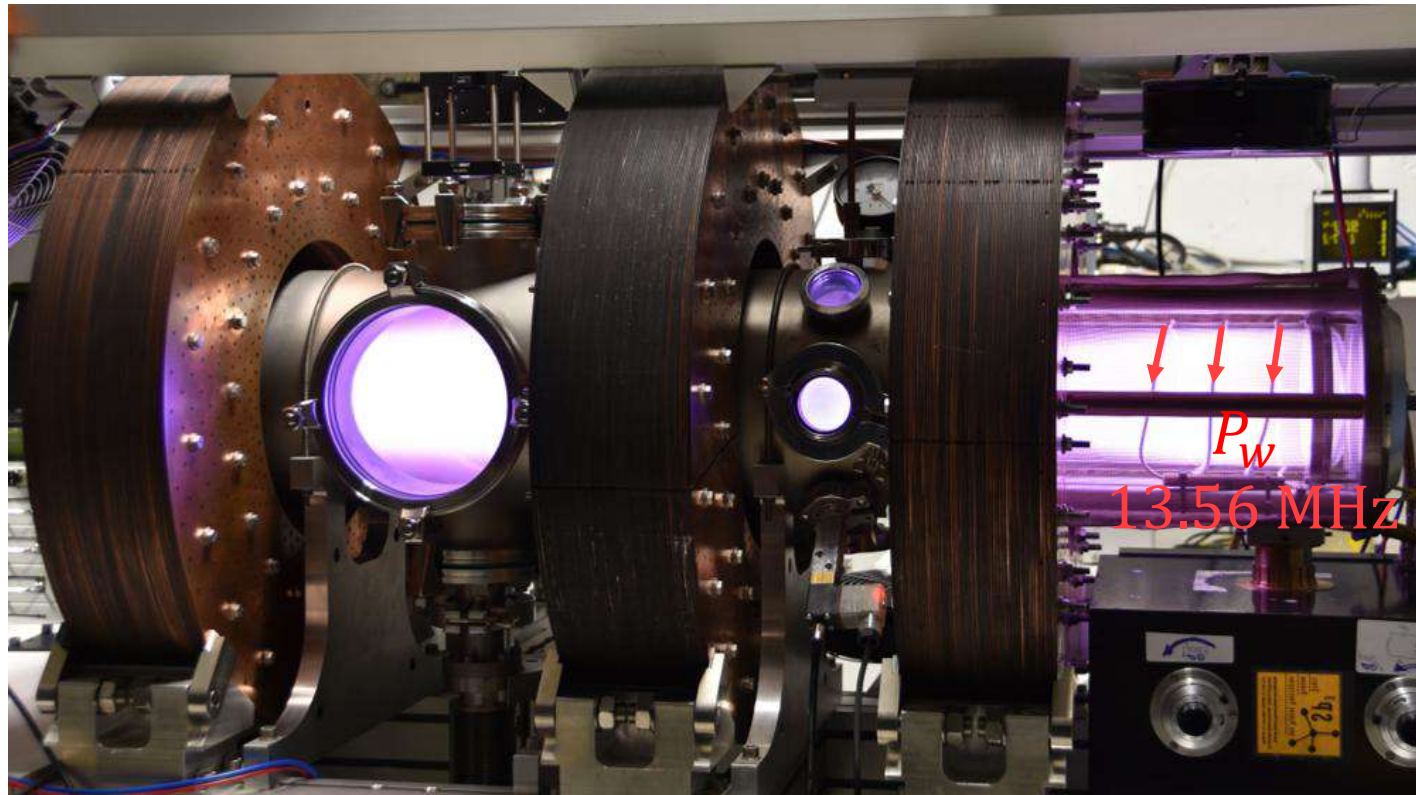
$B \text{ (G = } 10^{-4} \text{ T)}$	$n \text{ (} 10^{18} \text{ m}^{-3}\text{)}$	$T_e \text{ (eV)}$	$T_i \text{ (eV)}$	$\gamma \text{ (\%)}$
170	~ 1	~ 4	~ 0.2	~ 3

Note : 1 eV = 11 600 K

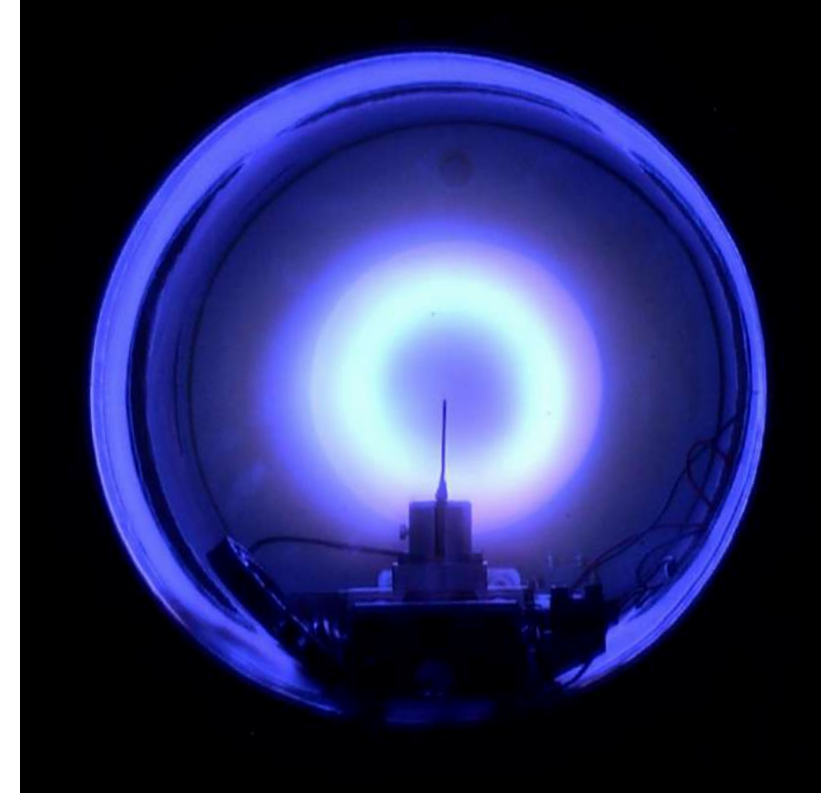
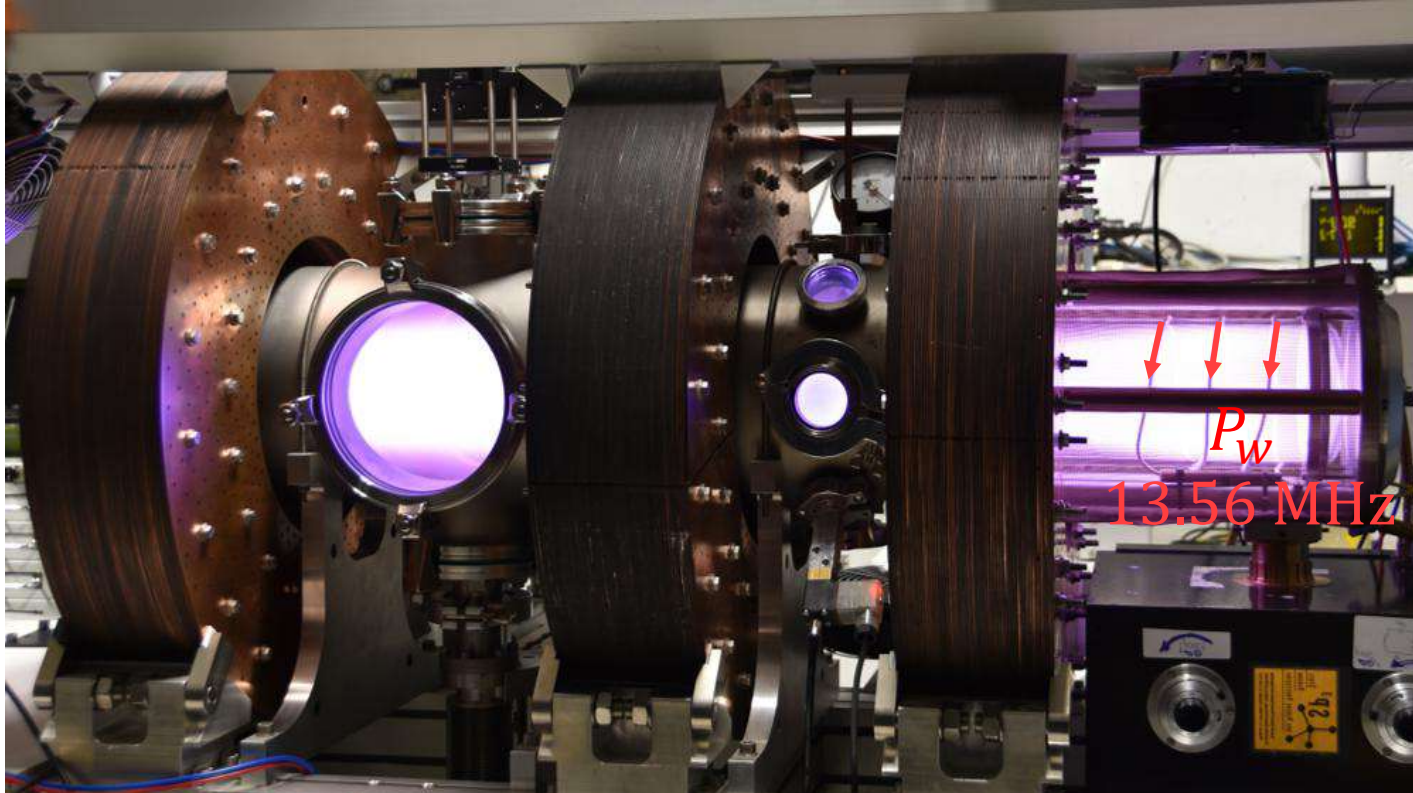
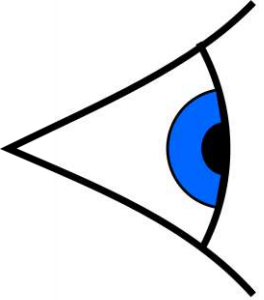
Génération du plasma



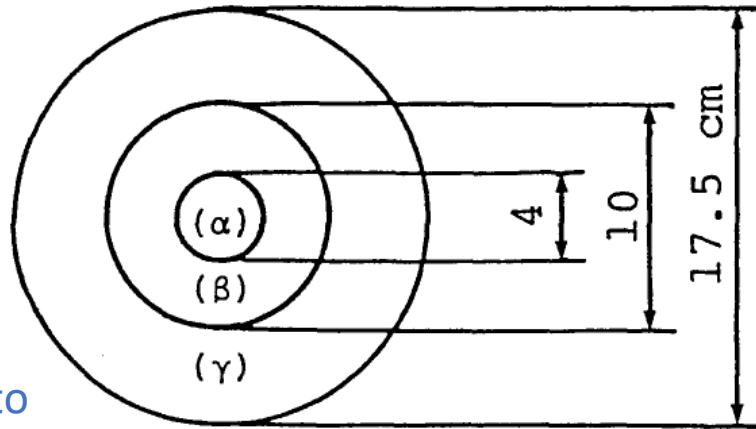
Génération du plasma



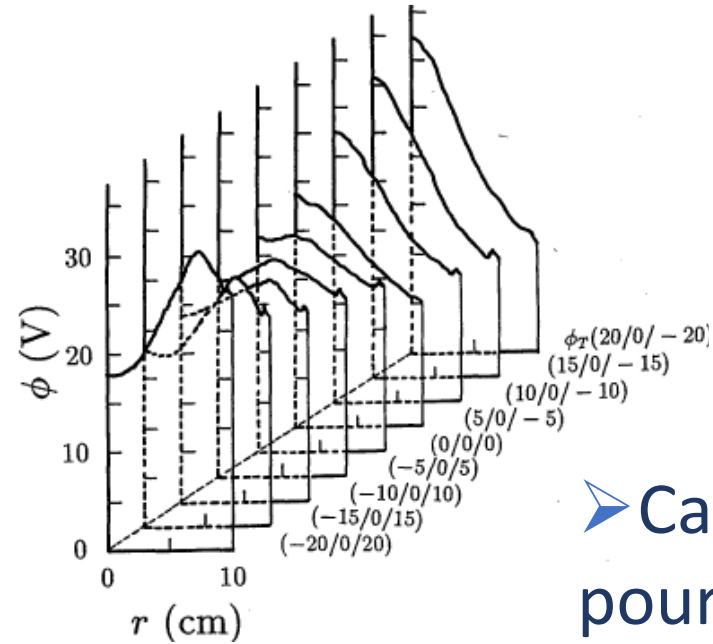
Génération du plasma



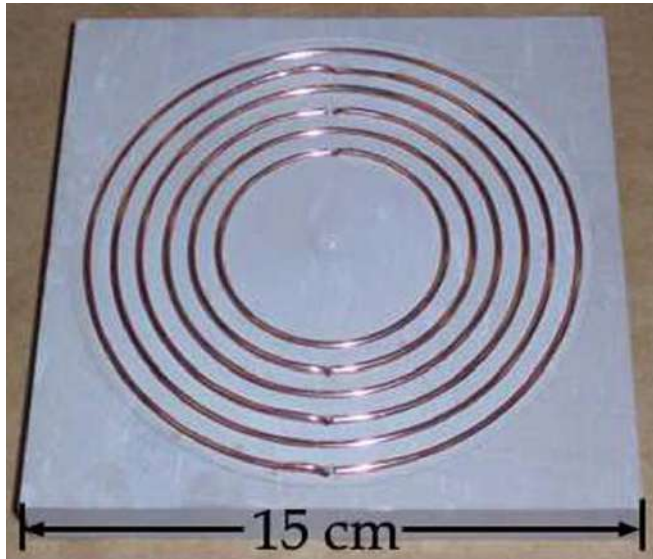
Cathode froide comme paramètre de contrôle



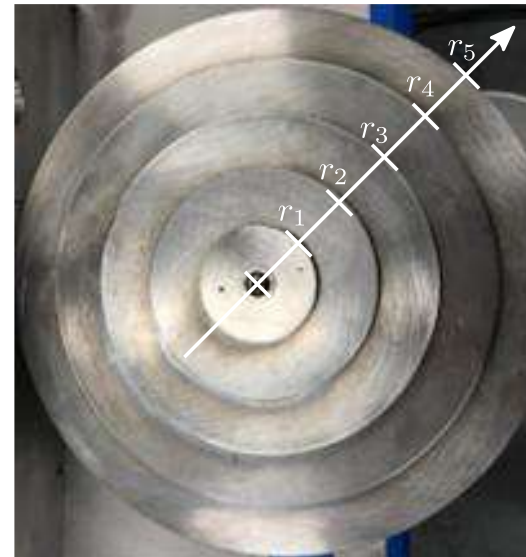
Tsushima, Sato
J. Phys. Soc. Japan (1991)



➤ Cathodes larges
pour un contrôle fin de ϕ_p

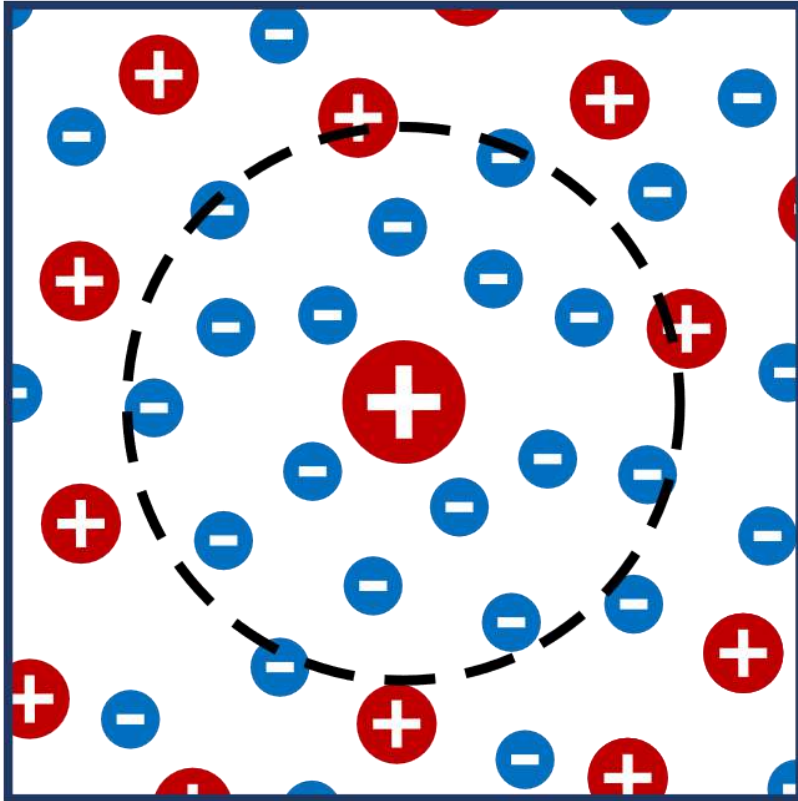


Desjardins, Gilmore
Phys. Plasmas (2018)



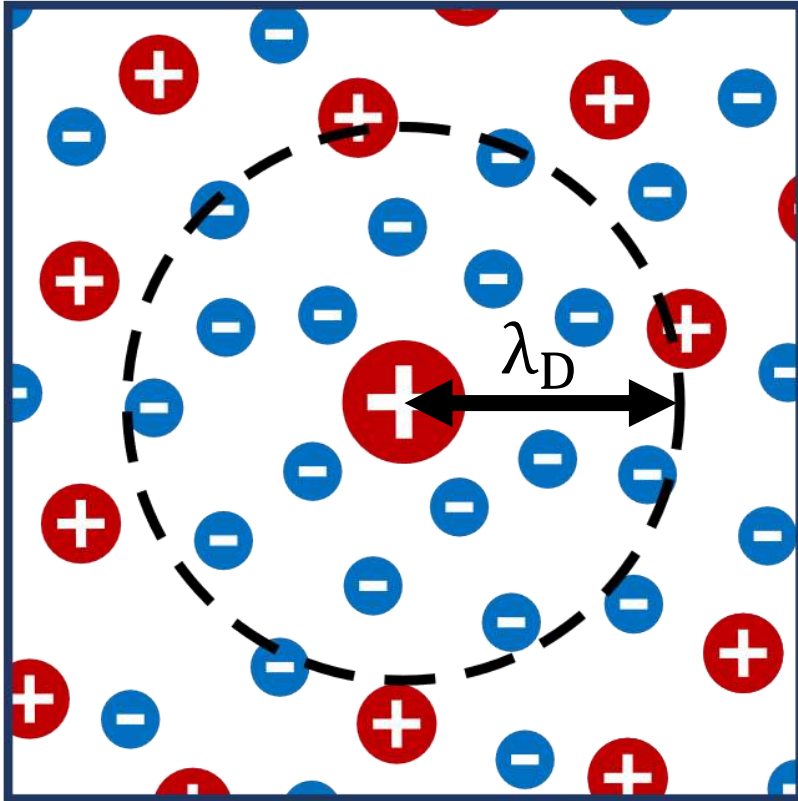
Gueroult *et al.*
ArXiv (2024)

Écrantage électrostatique



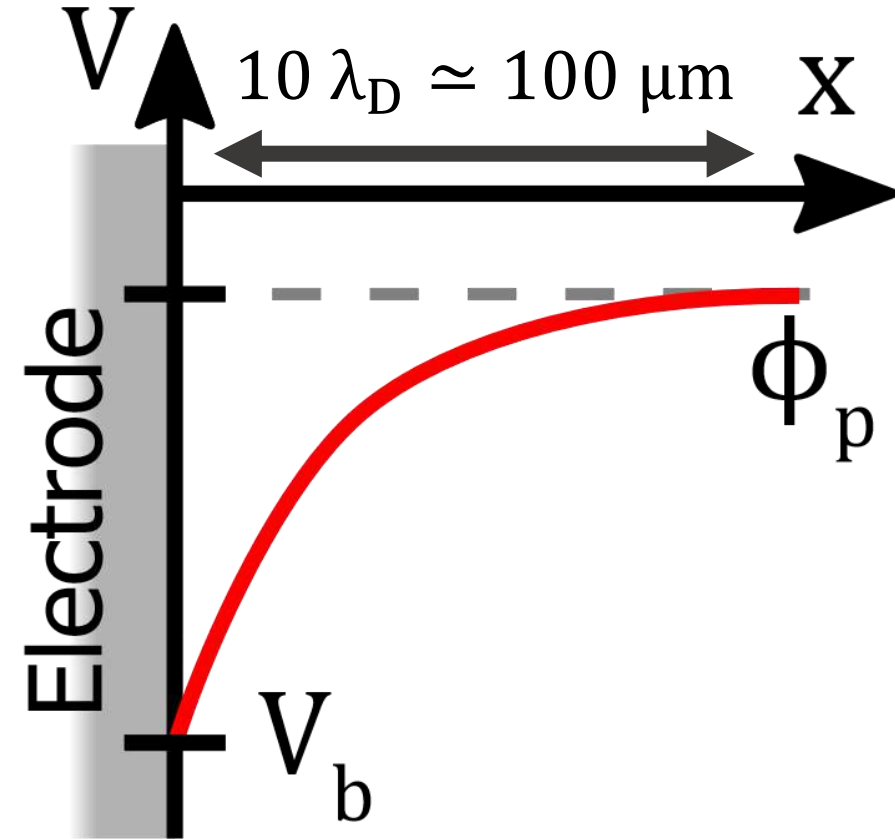
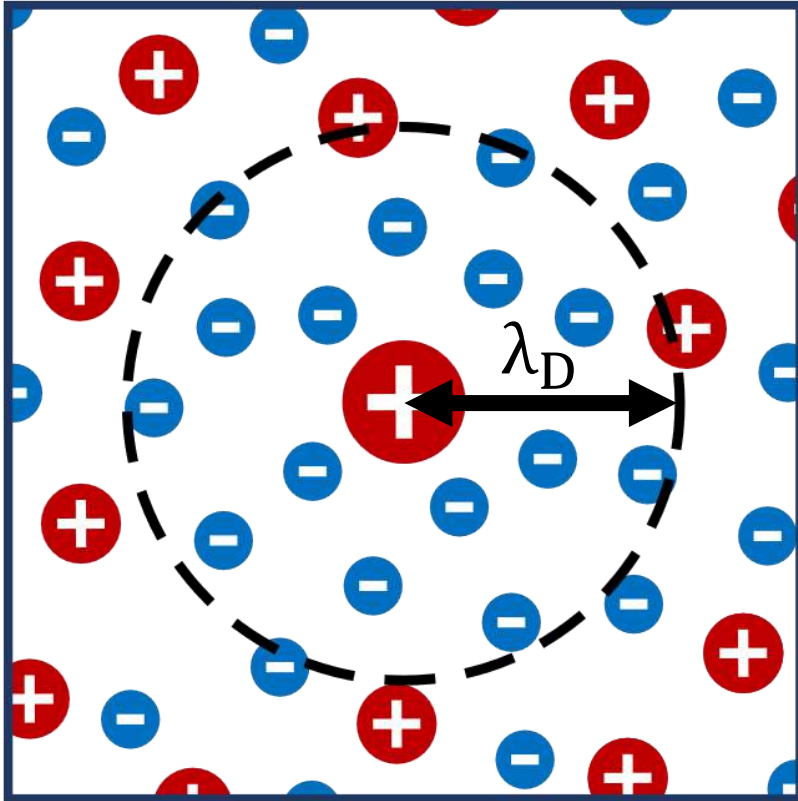
Écrantage électrostatique

$$\lambda_D = \sqrt{\frac{\epsilon T_e}{ne}}$$



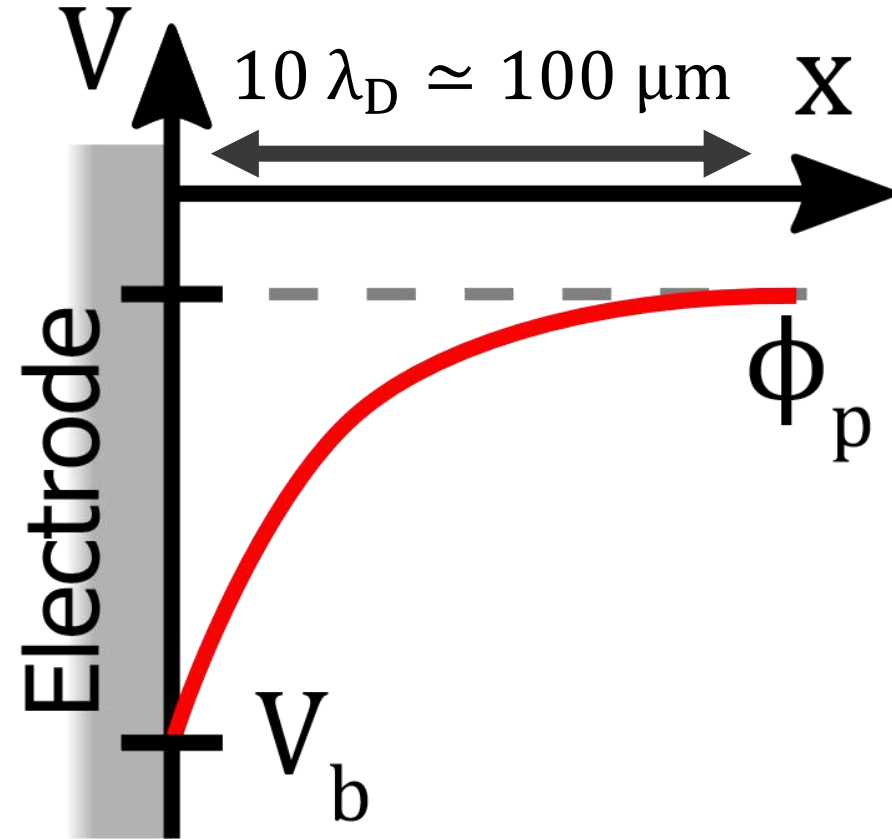
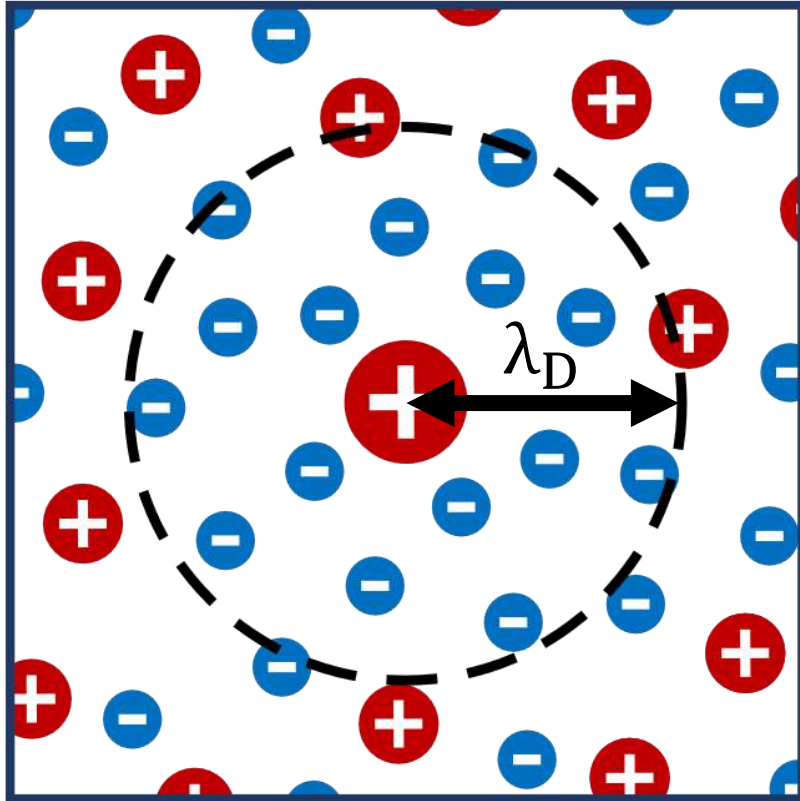
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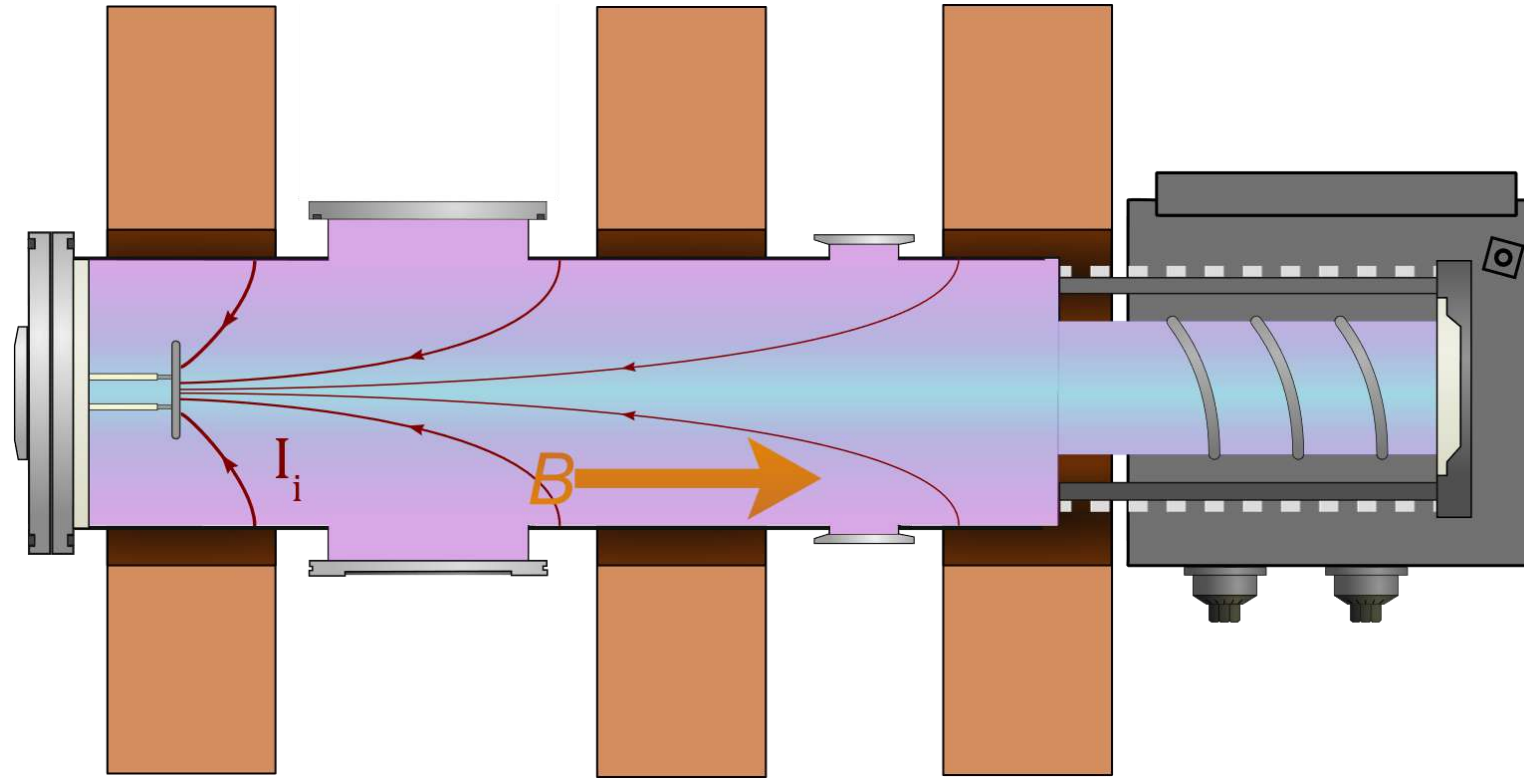
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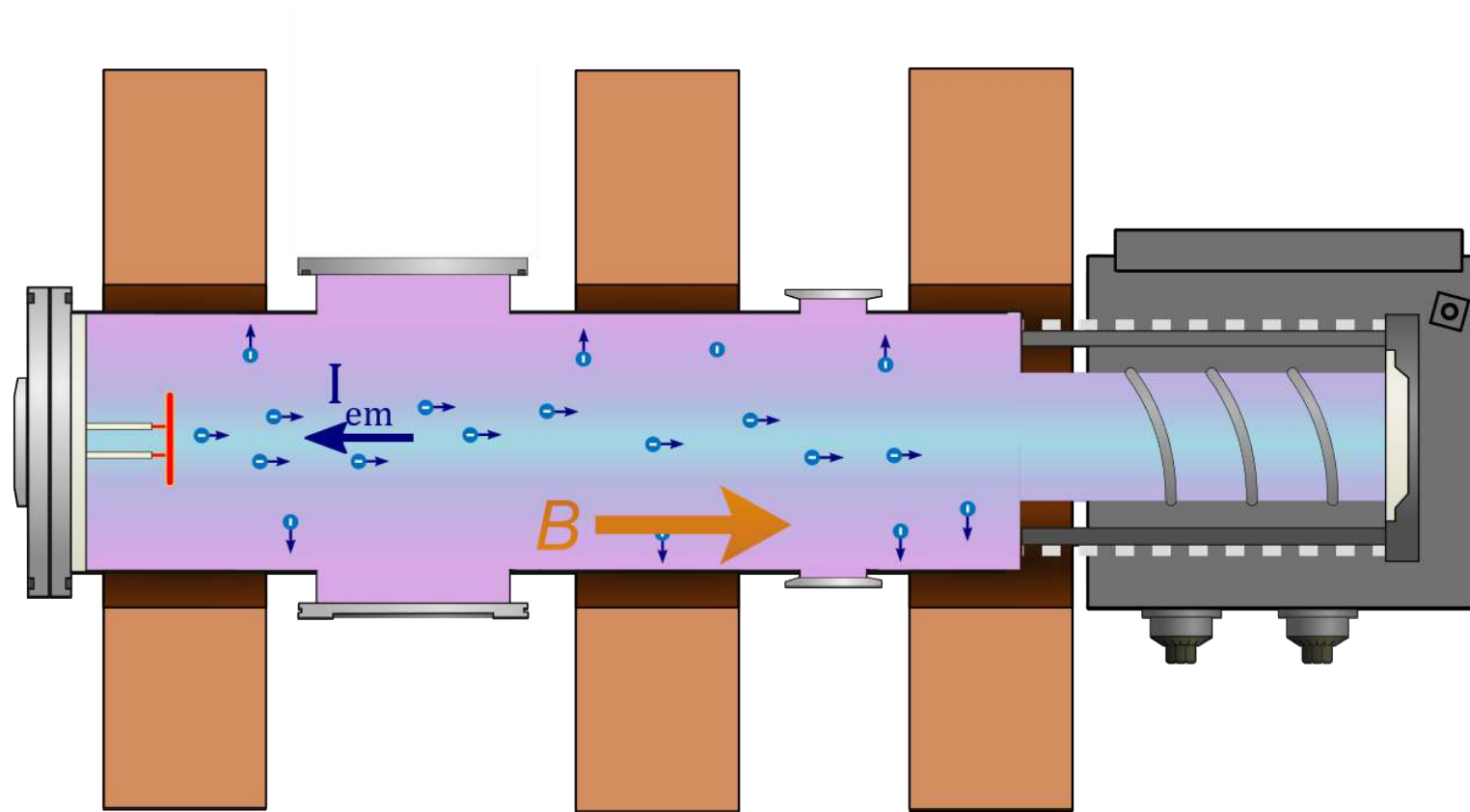


- Écrantage par les charges du plasma
- Impossibilité de modifier le potentiel électrostatique à longue portée

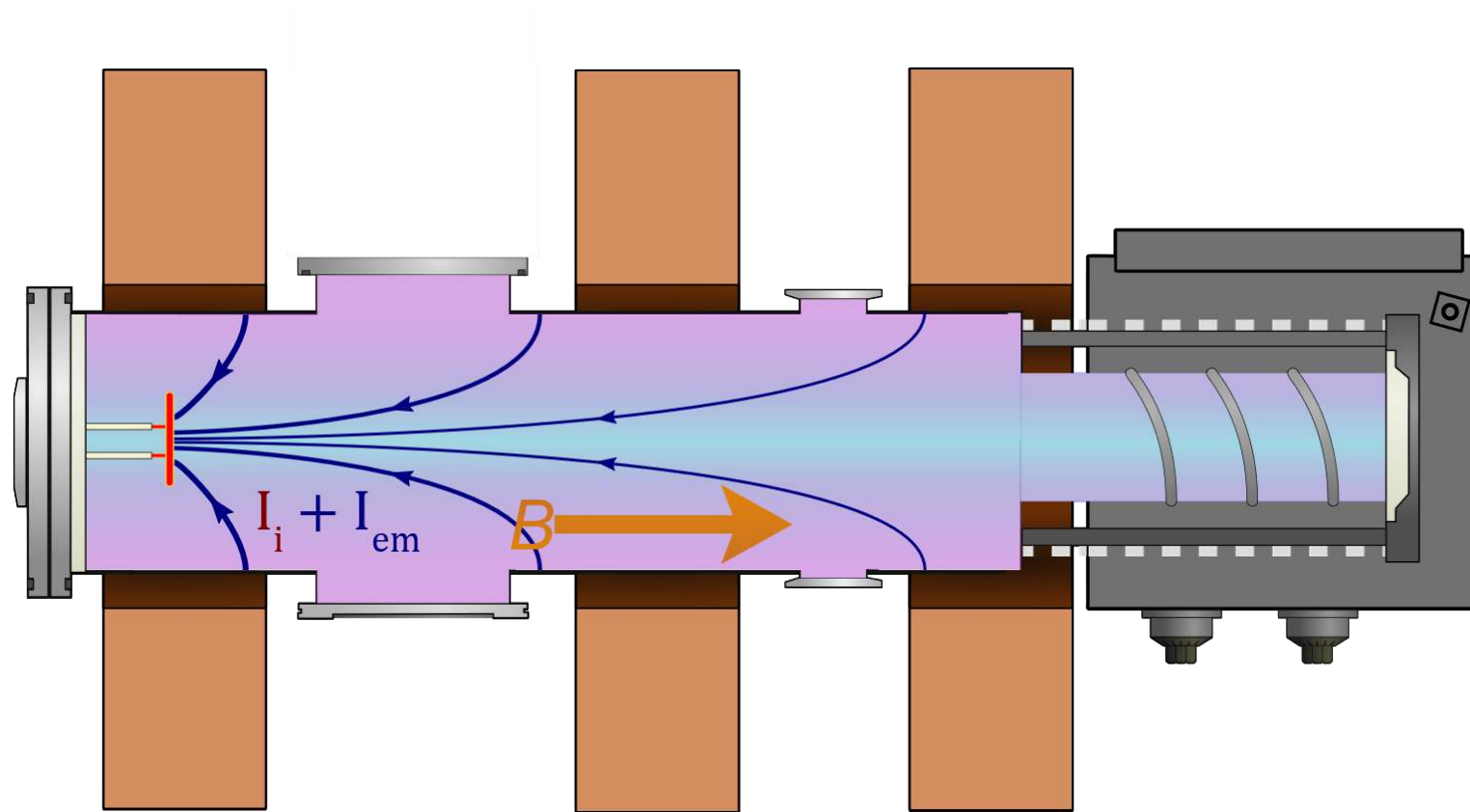
Effet de la cathode émissive



Effet de la cathode émissive



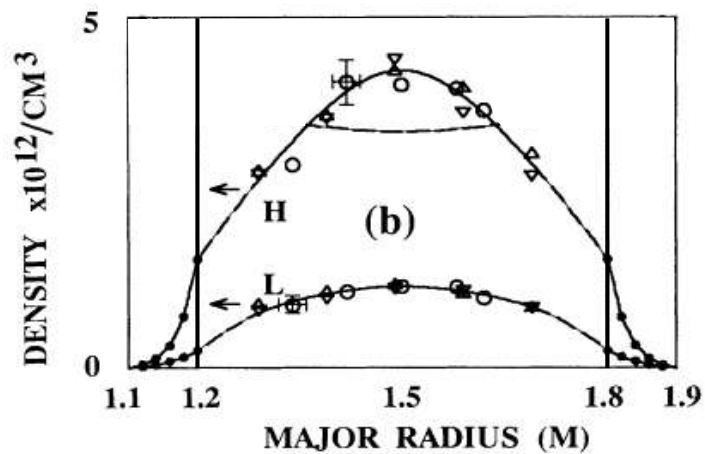
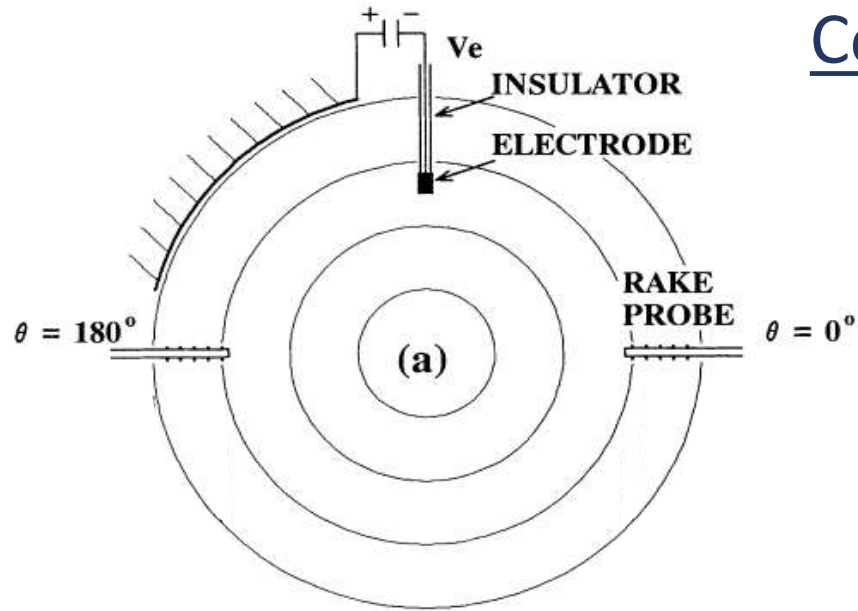
Effet de la cathode émissive



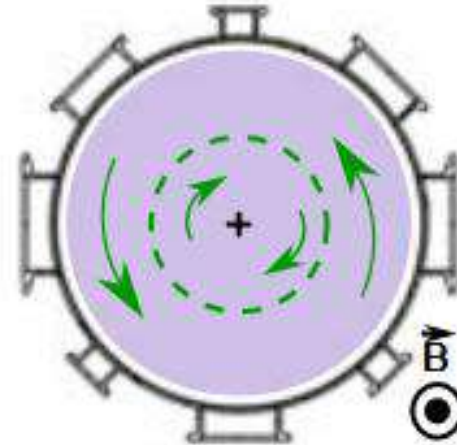
- Modification des flux de courants au sein du plasma
- Paramètre de contrôle supplémentaire : $\Xi = I_{em}/I_i \simeq 10$

Contrôle empirique via cathode émissive

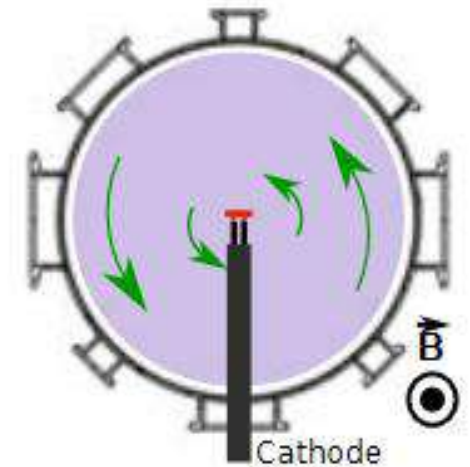
Contrôle de l'écoulement



Pas de cathode

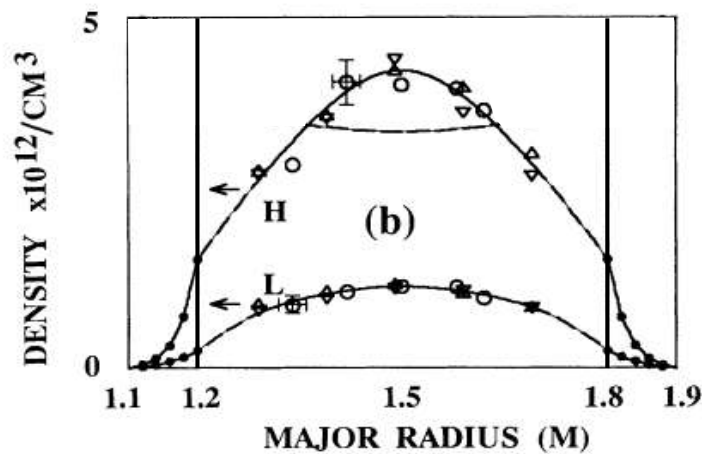
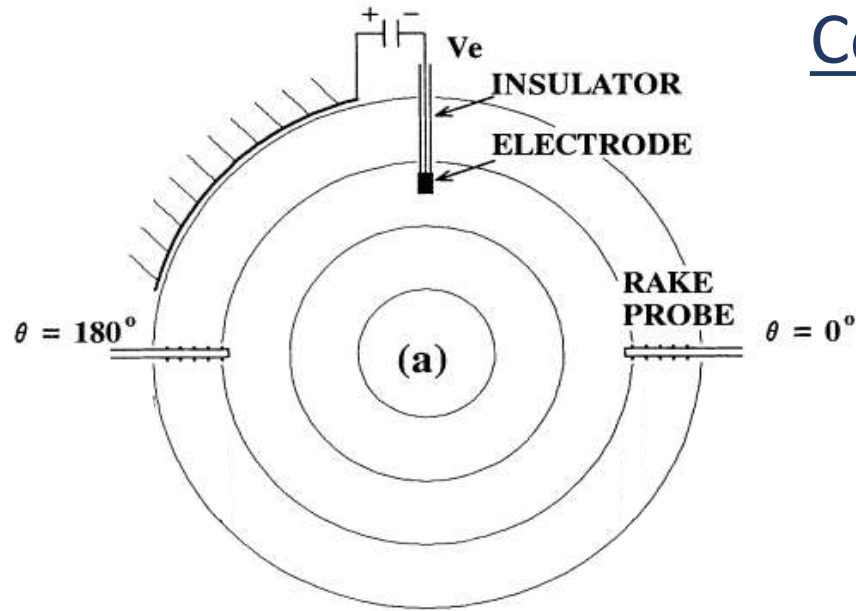


Cathode au centre

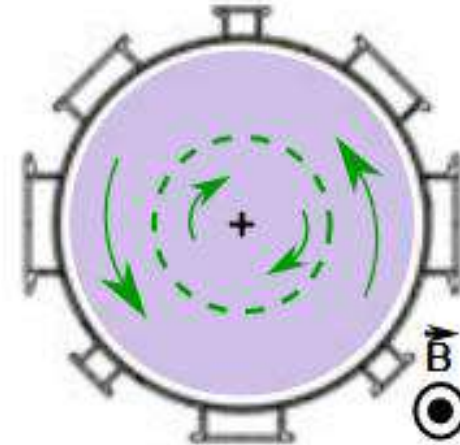


Contrôle empirique via cathode émissive

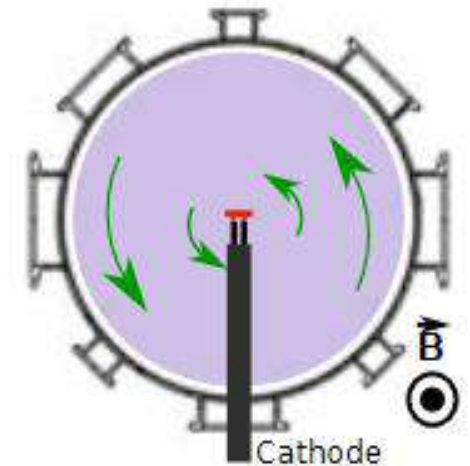
Contrôle de l'écoulement



Pas de cathode



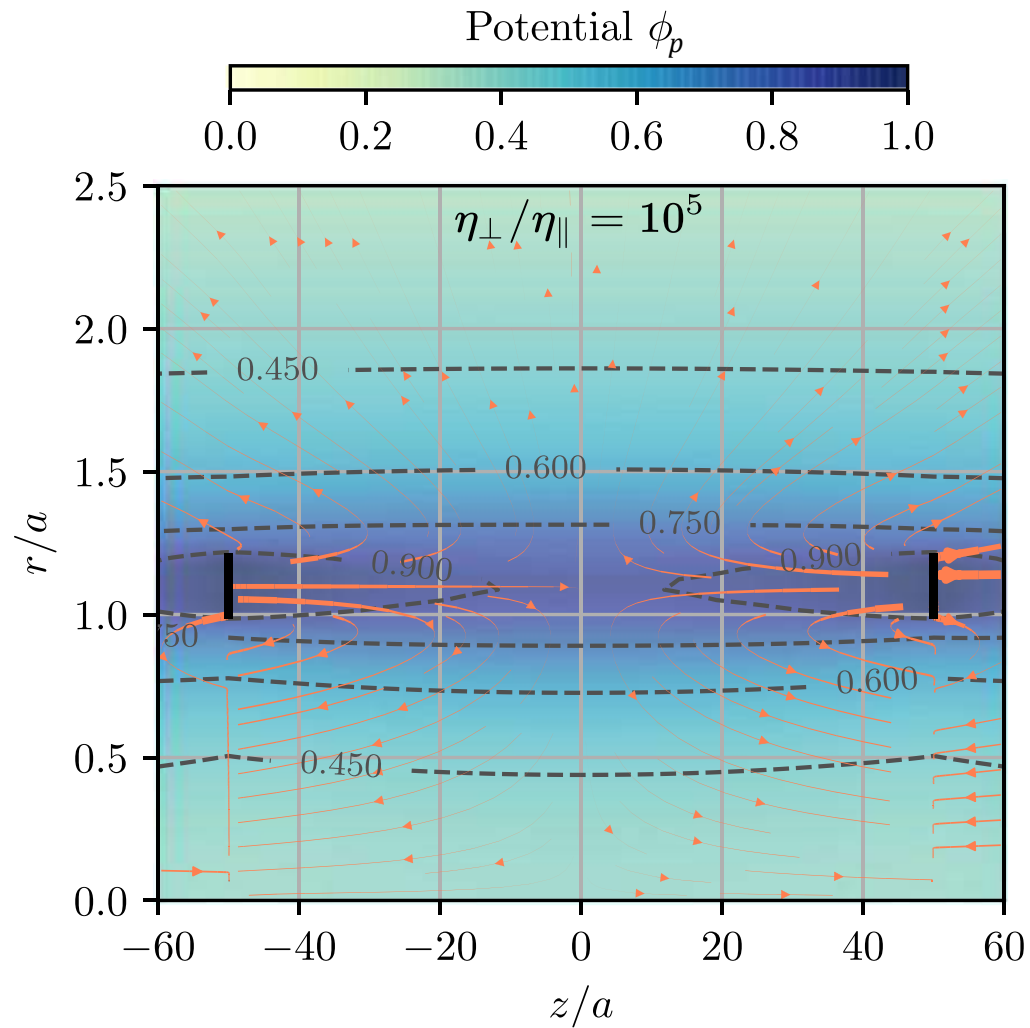
Cathode au centre



Motivation :

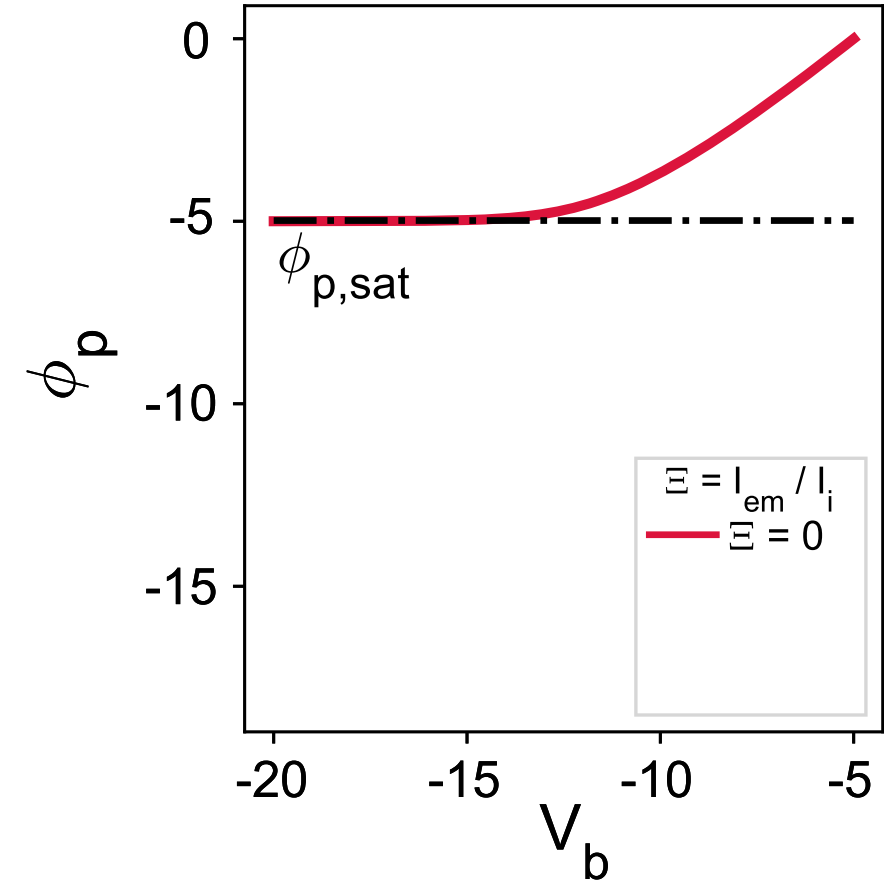
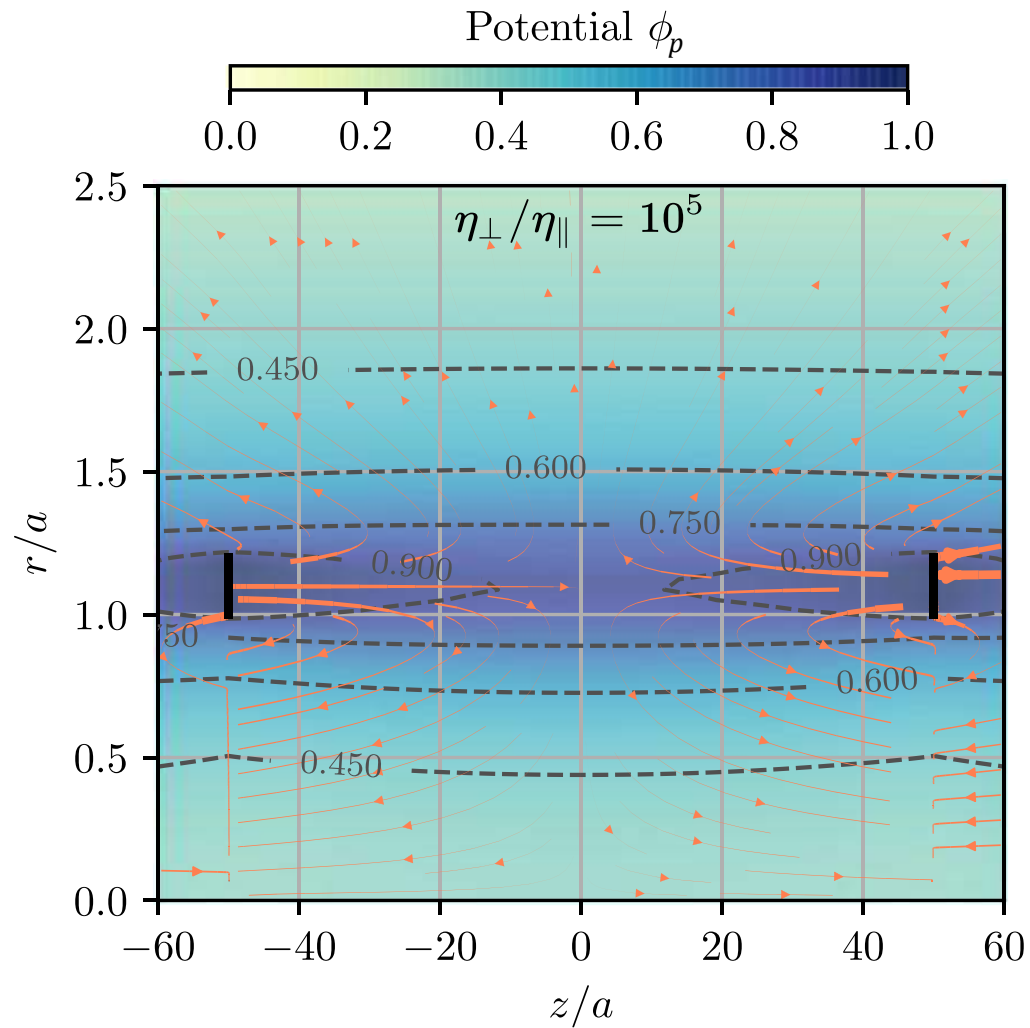
Comprendre le contrôle du potentiel plasma

Modèle du contrôle du potentiel plasma



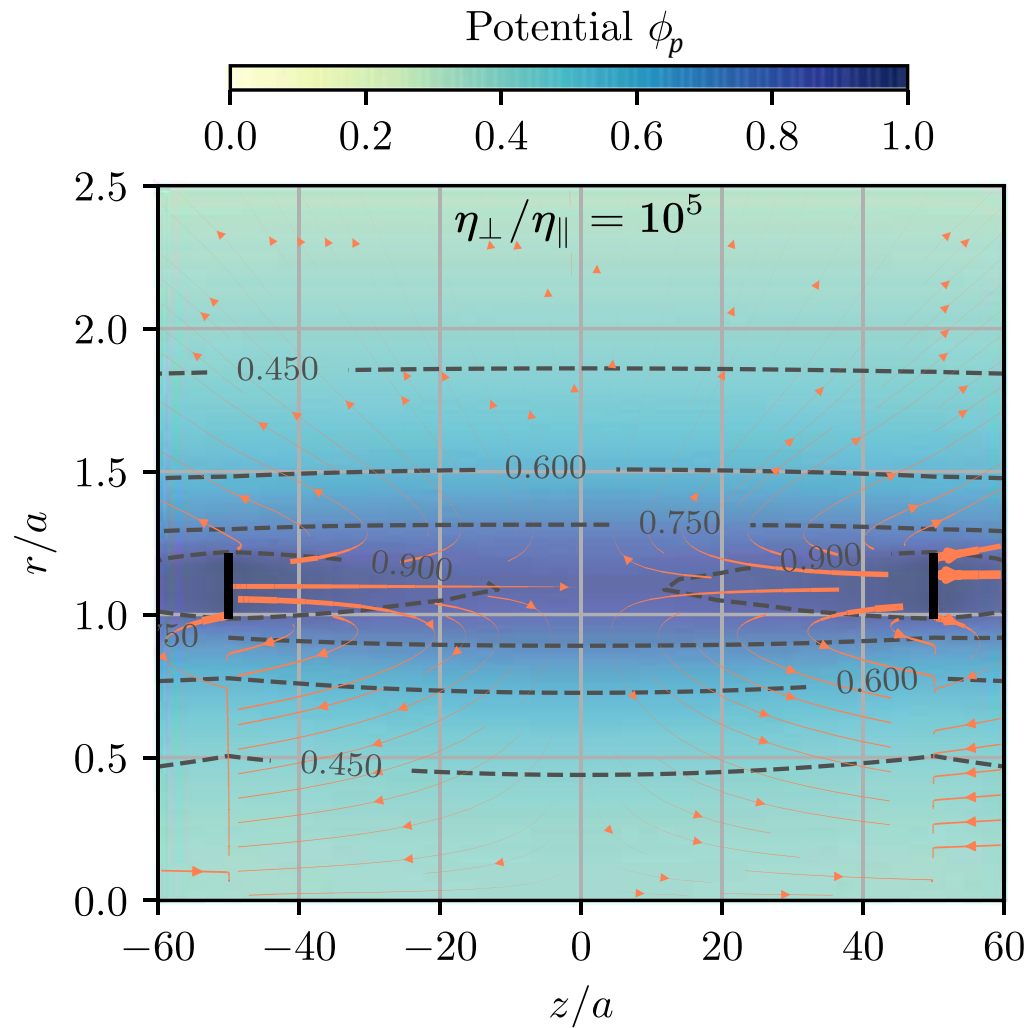
➤ Chute locale de potentiel par la loi d'Ohm

Modèle du contrôle du potentiel plasma

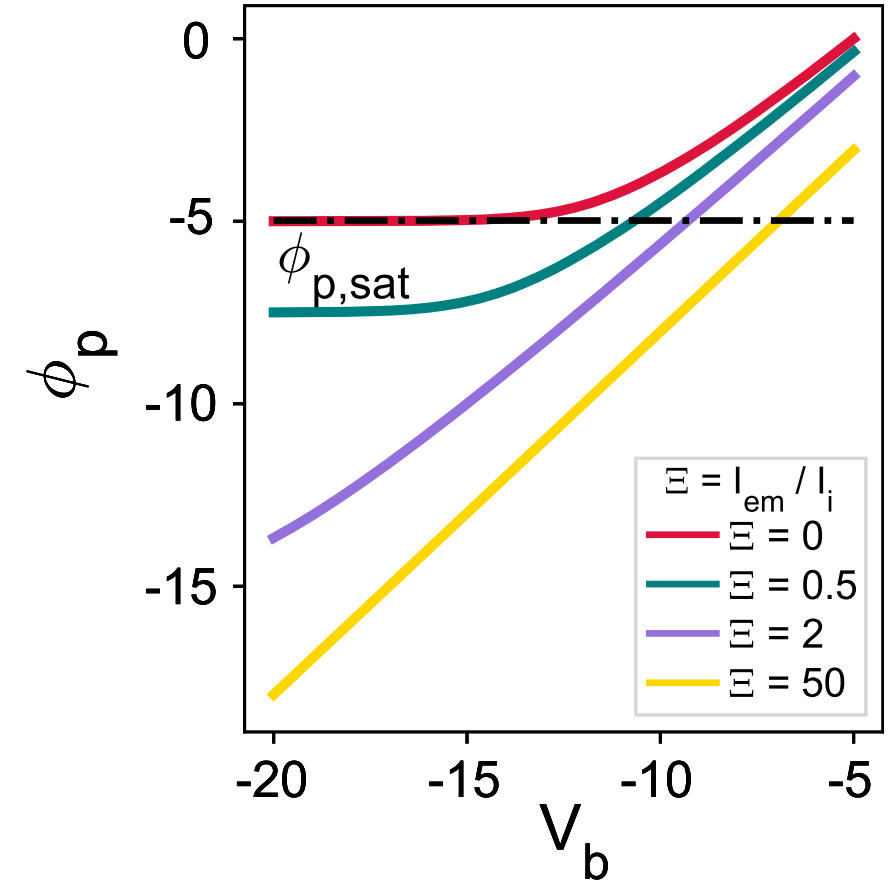


➤ Chute locale de potentiel par la loi d'Ohm

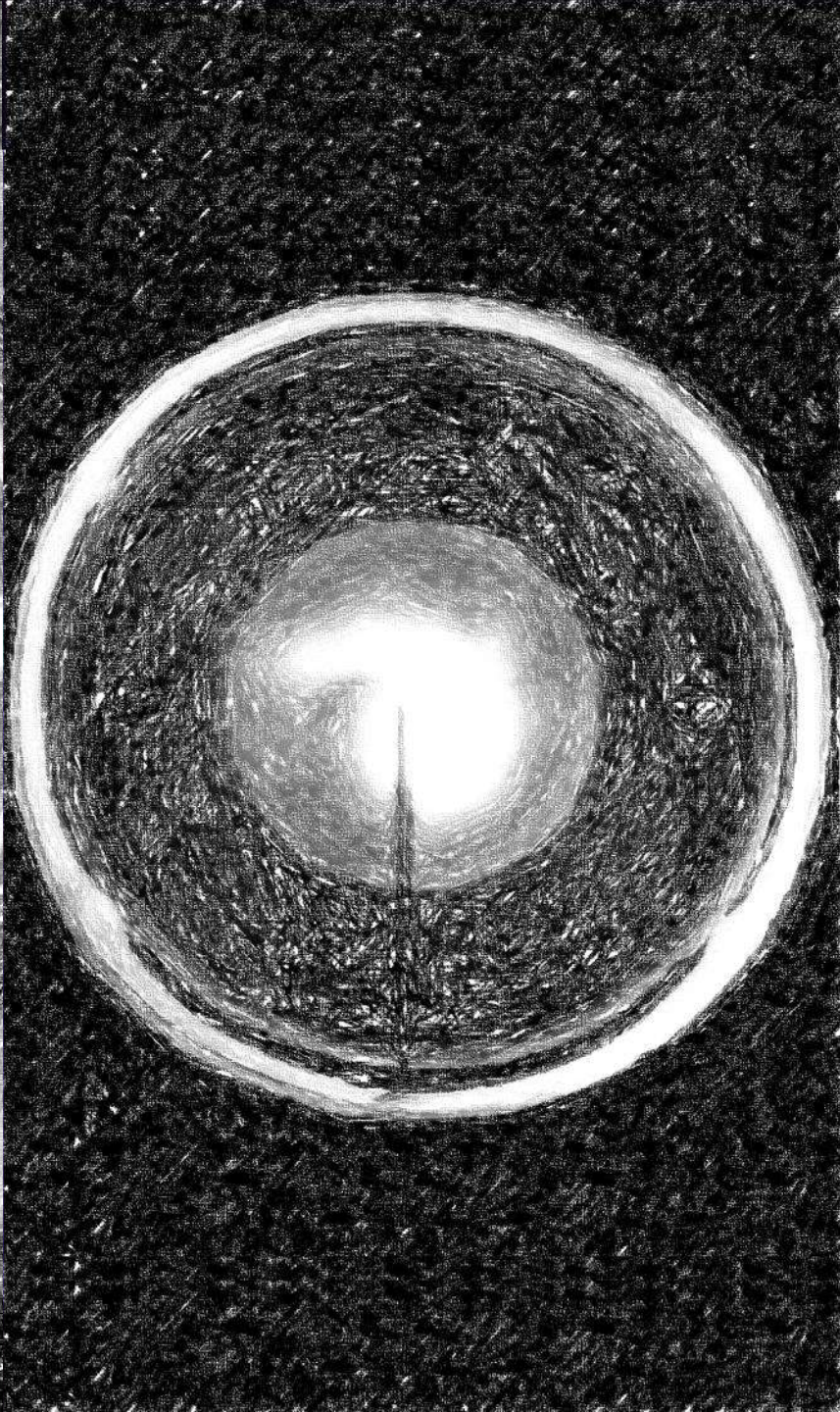
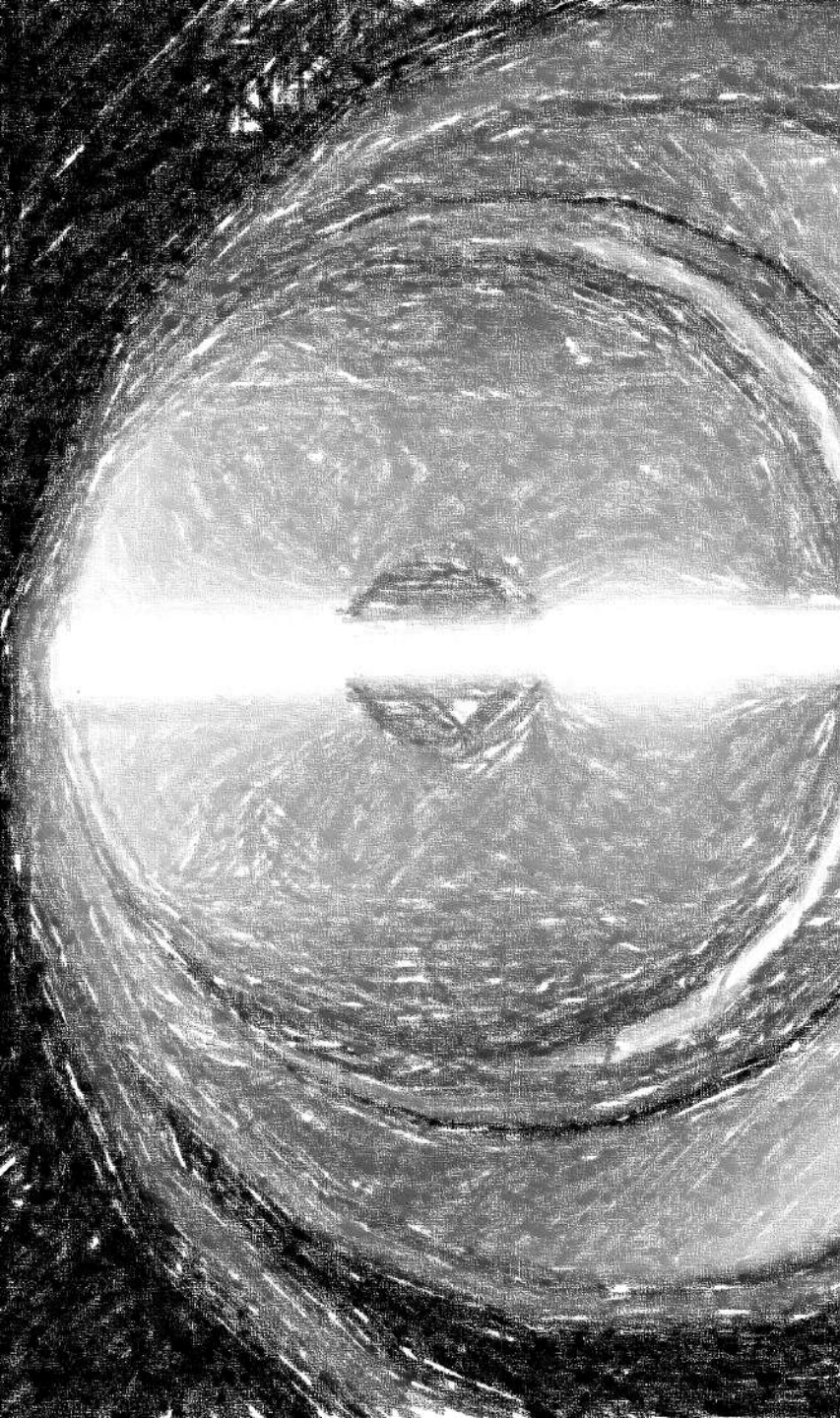
Modèle du contrôle du potentiel plasma



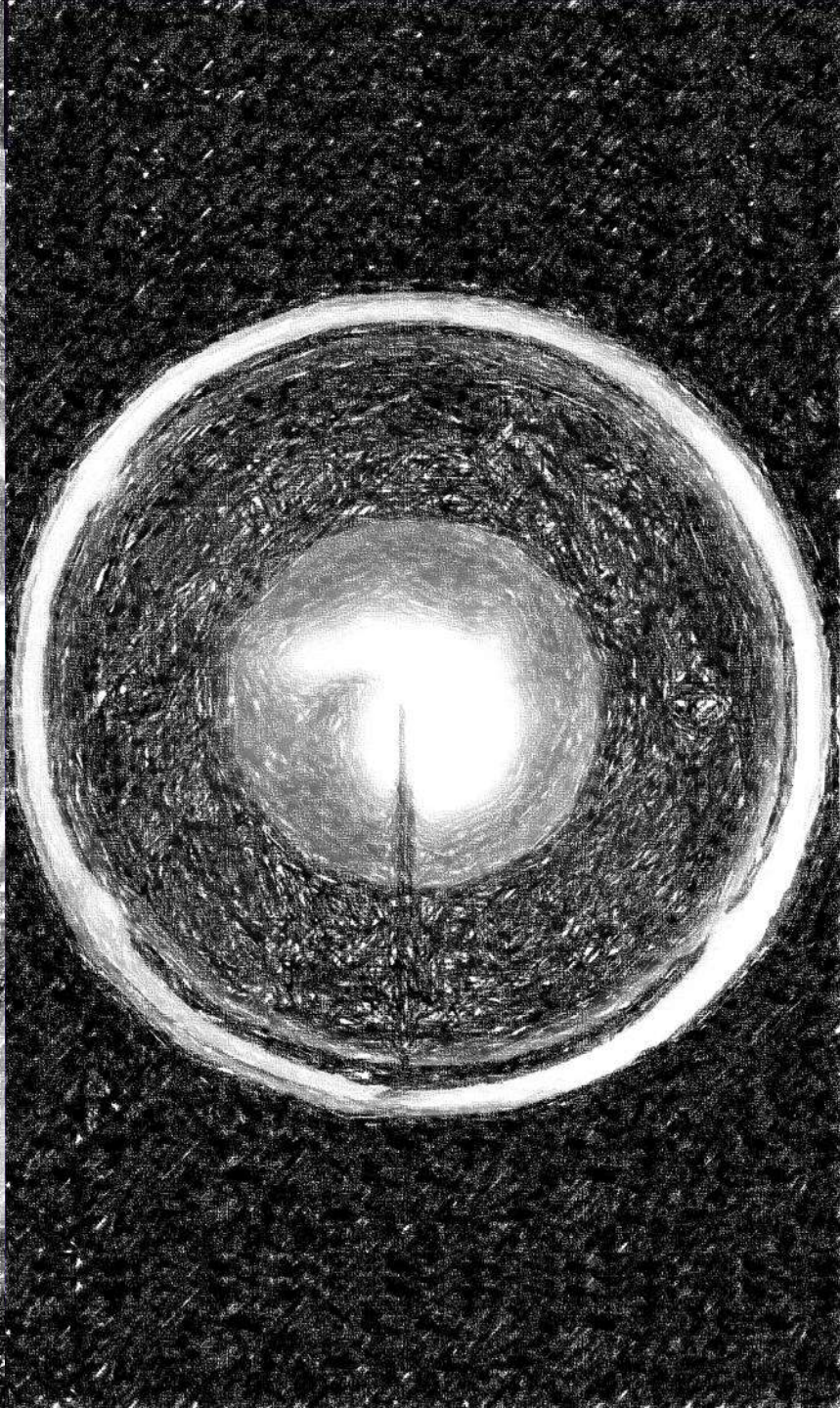
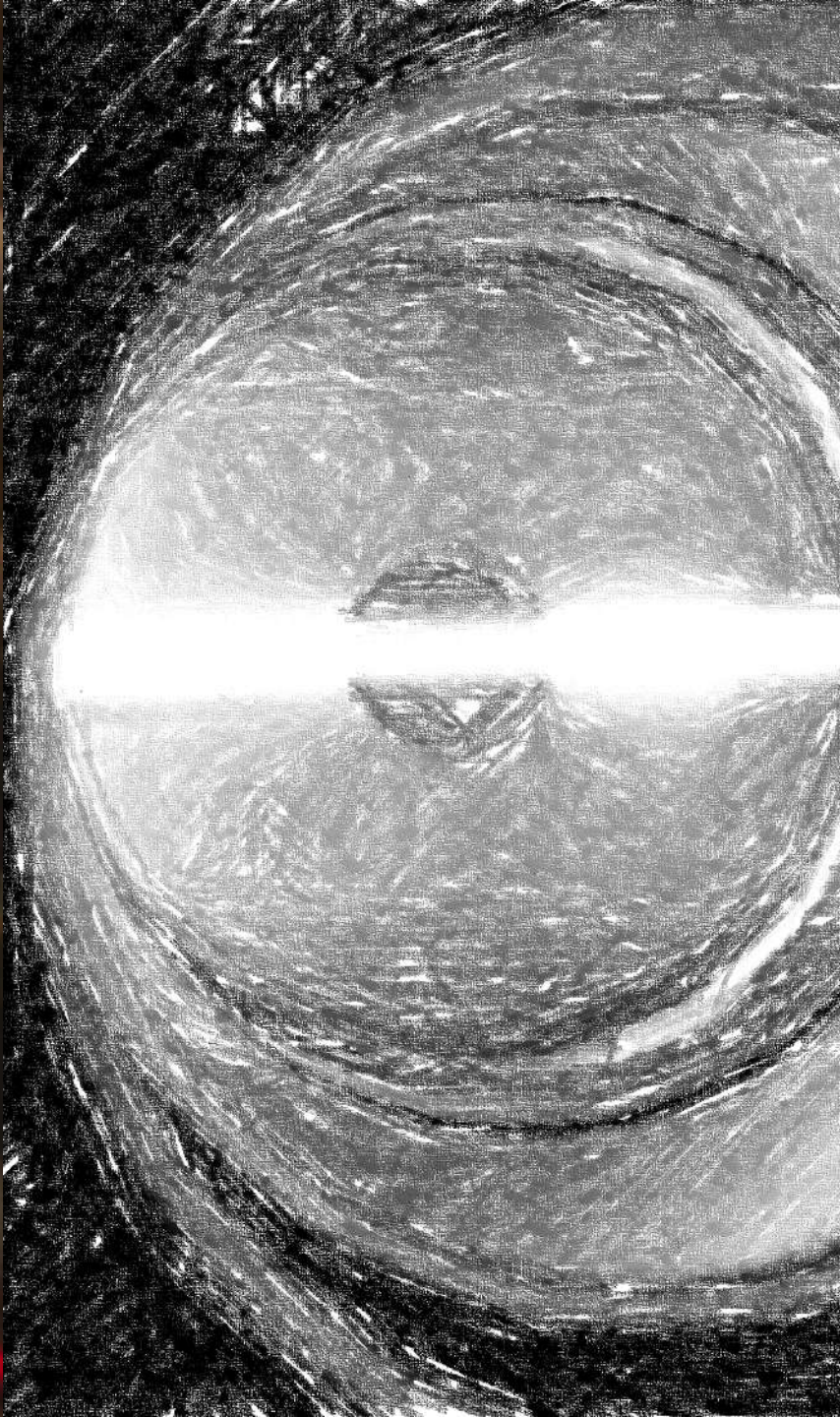
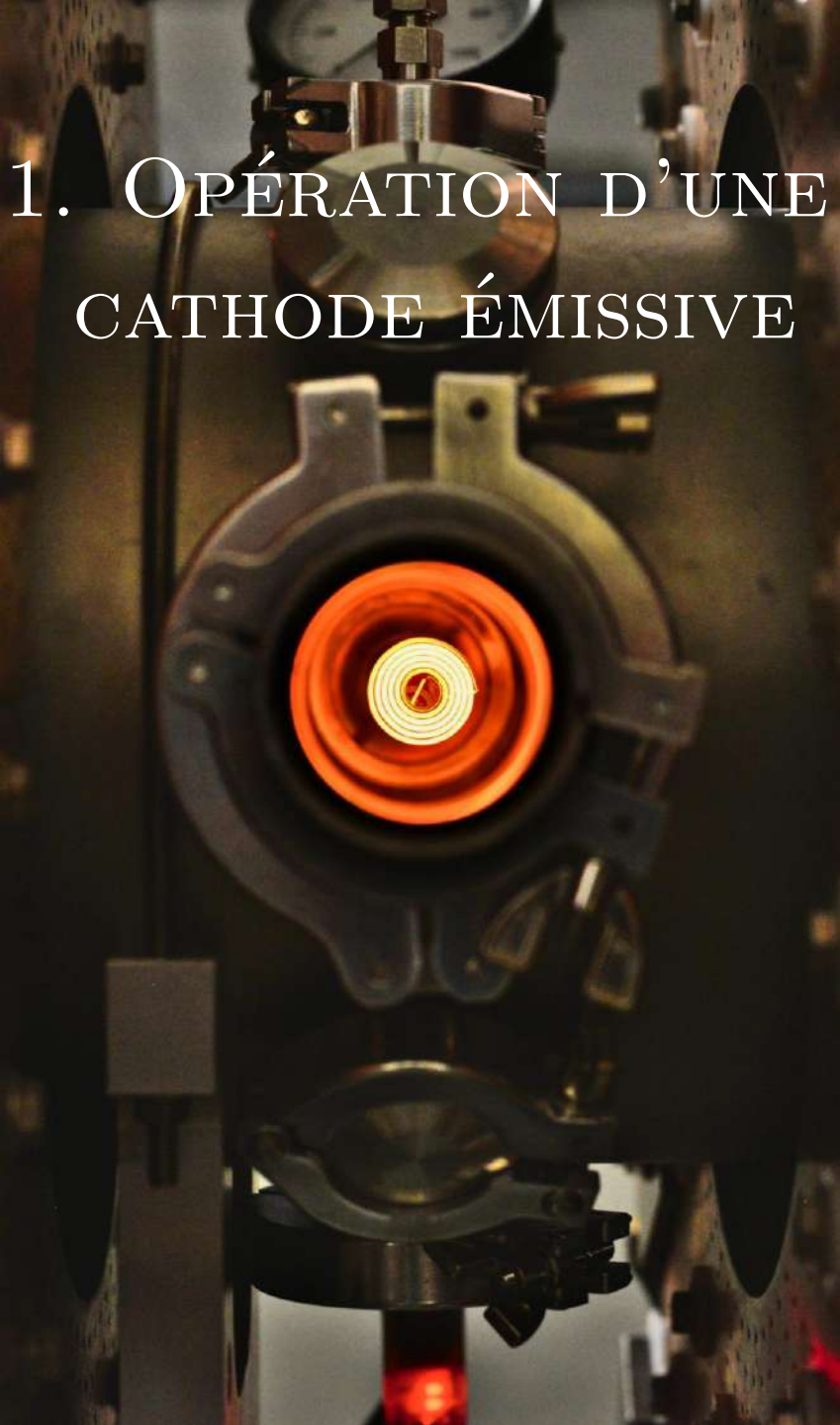
➤ Chute locale de potentiel par la loi d'Ohm



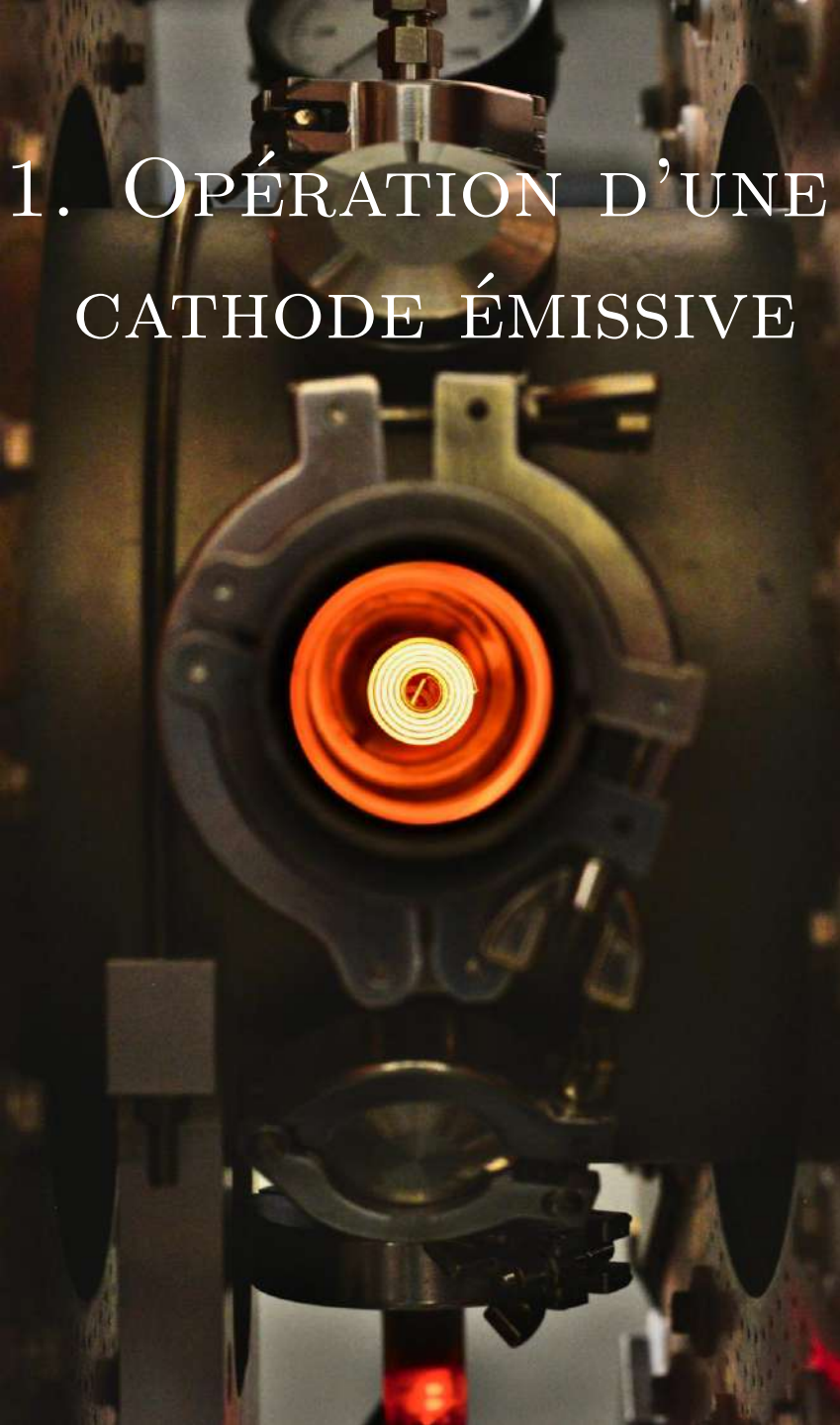
➤ Modèle de contrôle par cathode émissive



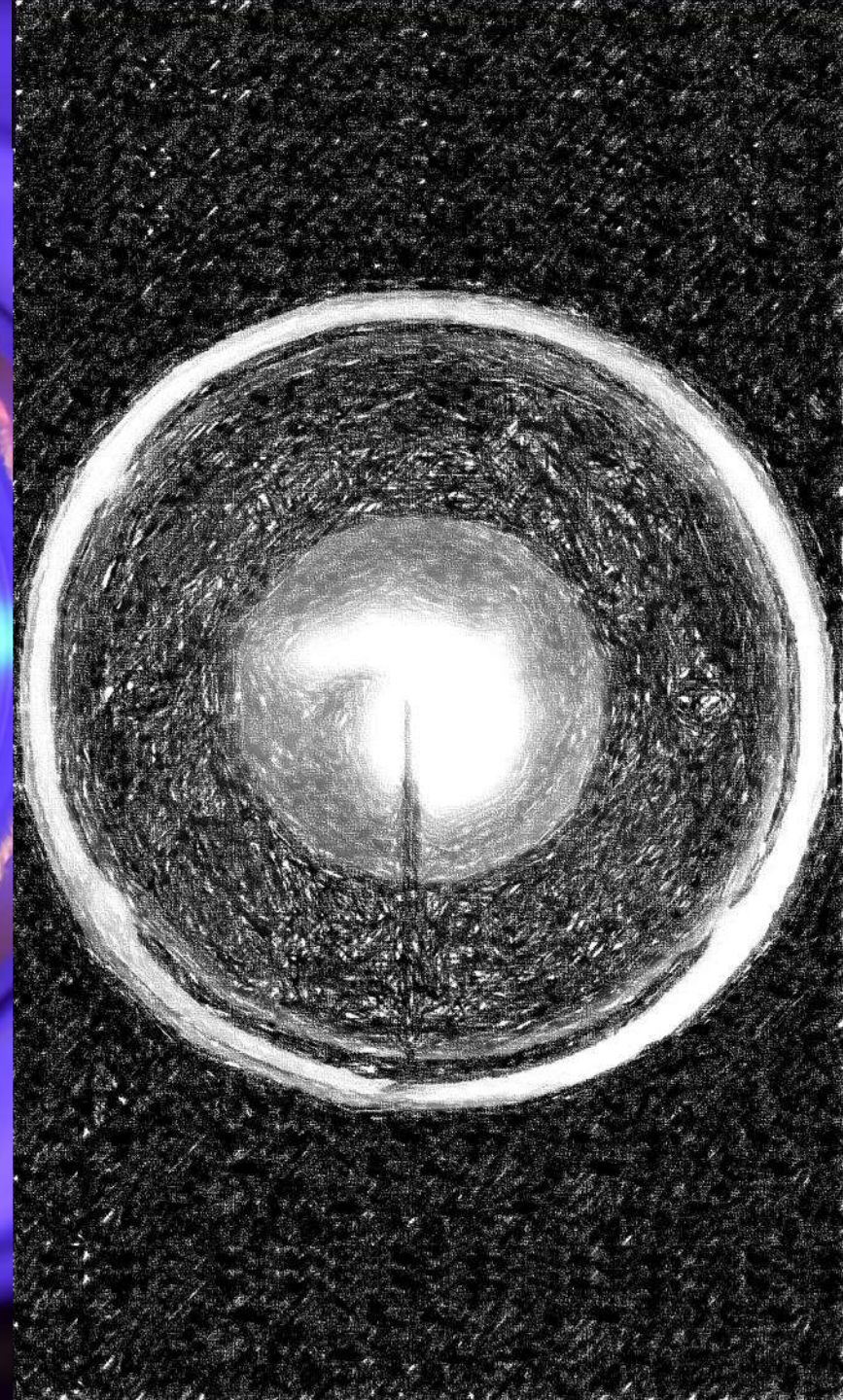
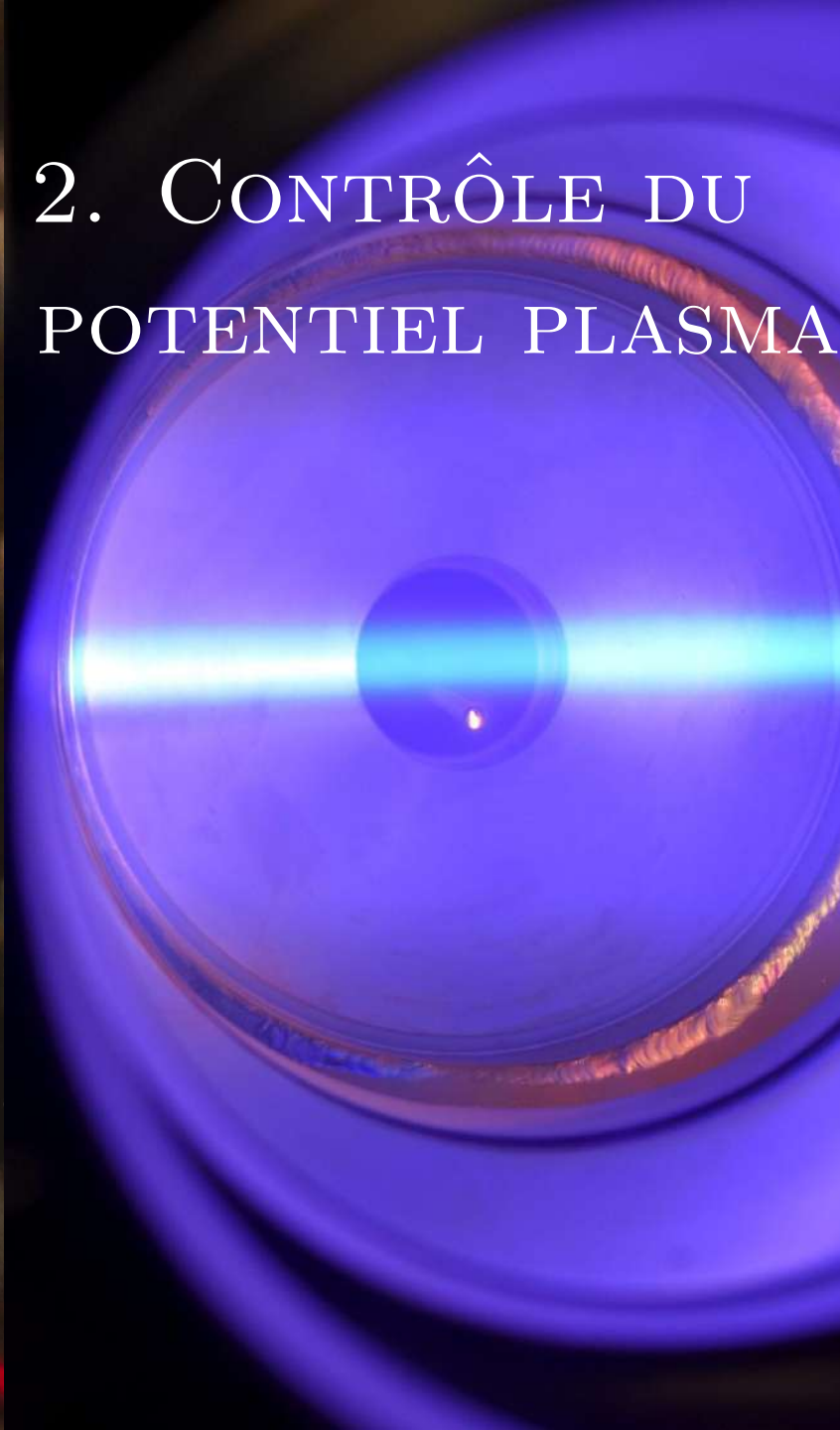
1. OPÉRATION D'UNE CATHODE ÉMISSIVE



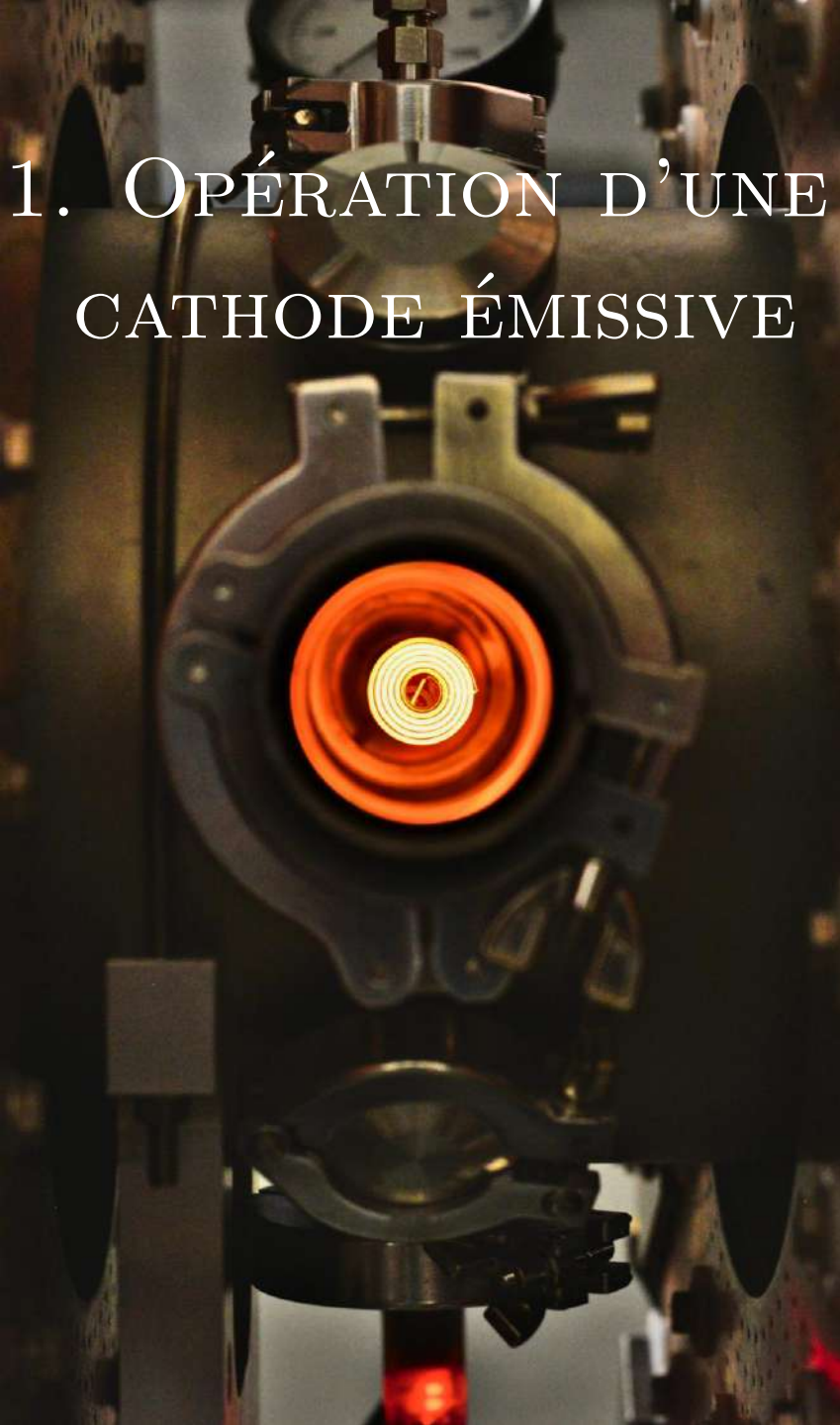
1. OPÉRATION D'UNE
CATHODE ÉMISSIVE



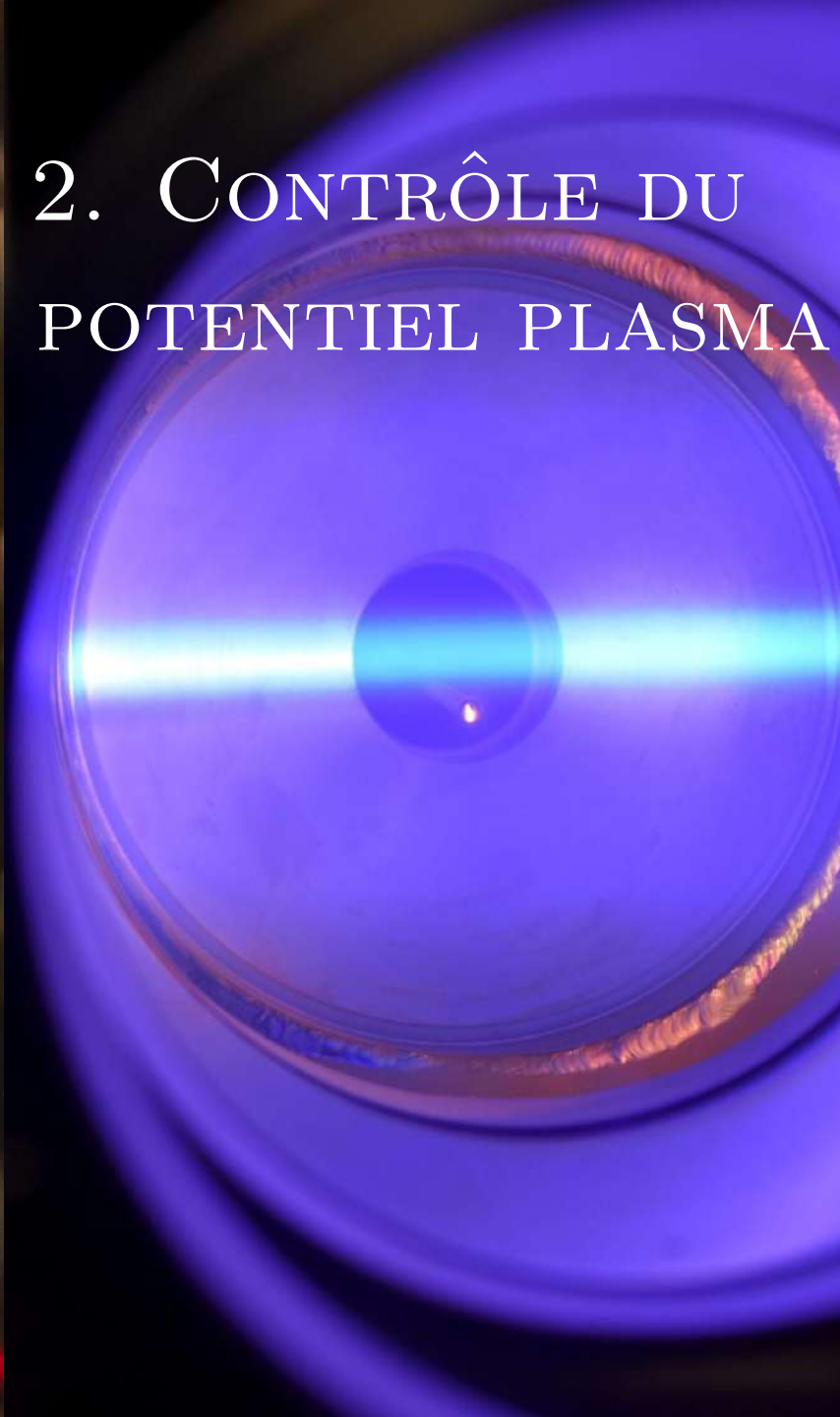
2. CONTRÔLE DU
POTENTIEL PLASMA



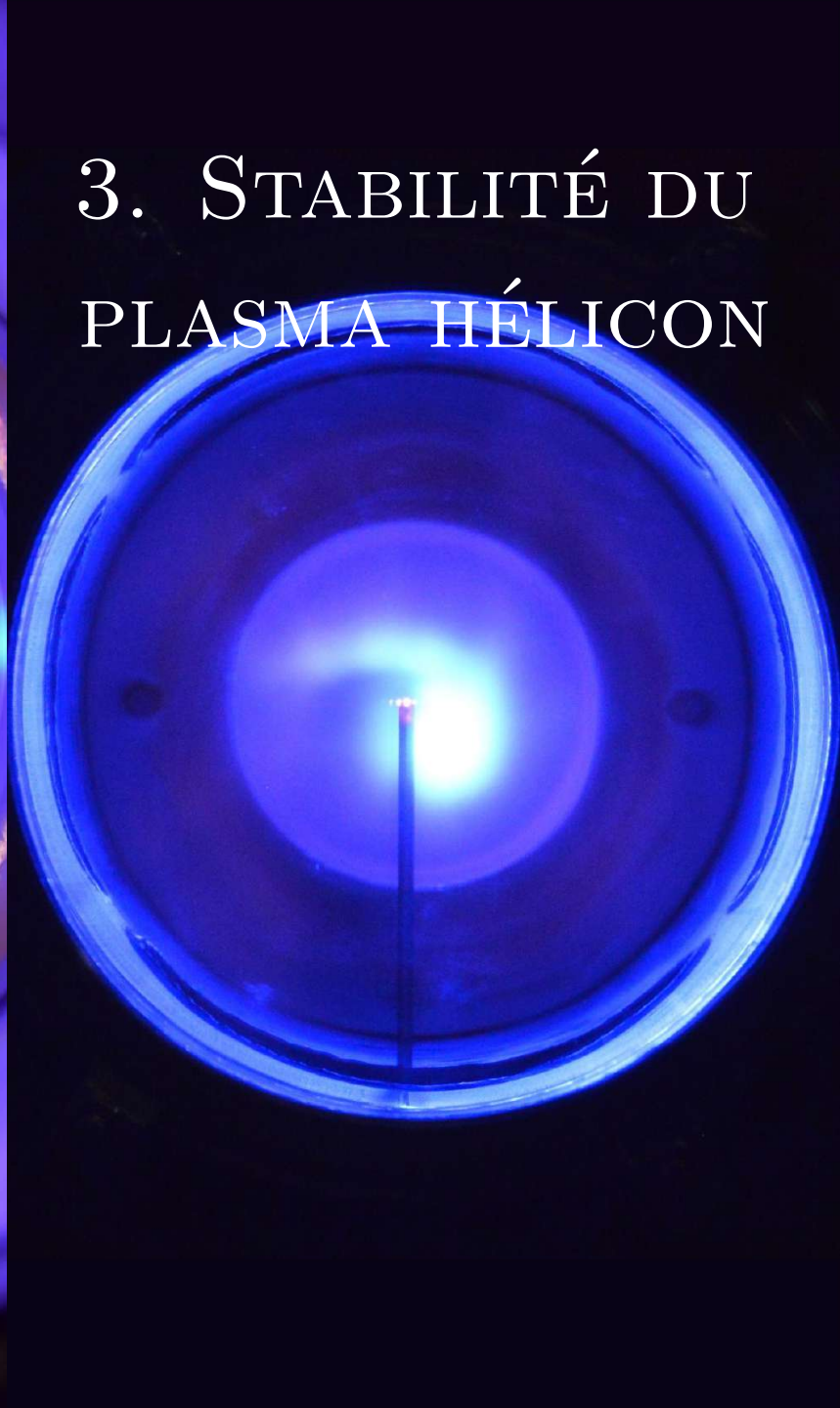
1. OPÉRATION D'UNE
CATHODE ÉMISSIVE



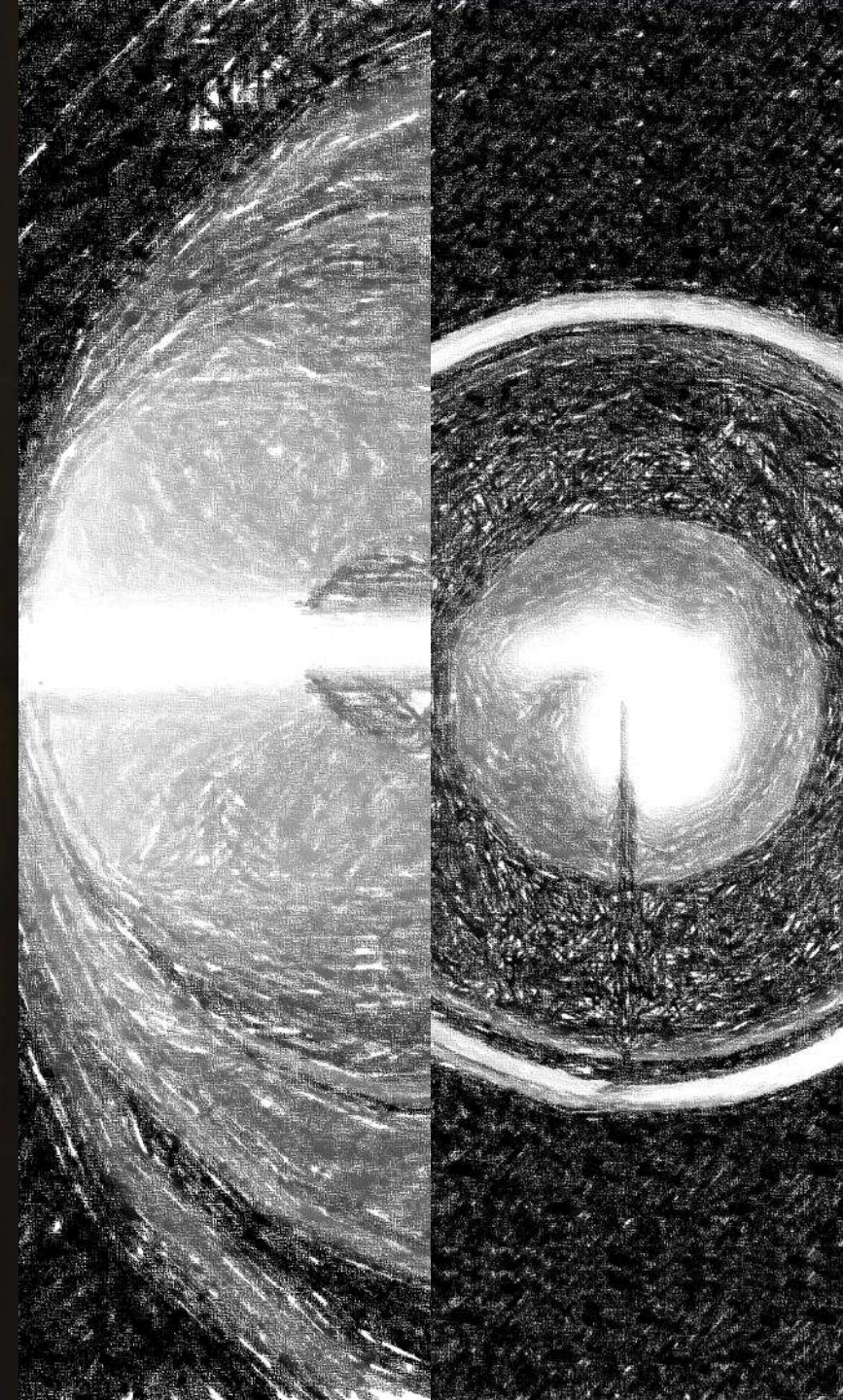
2. CONTRÔLE DU
POTENTIEL PLASMA



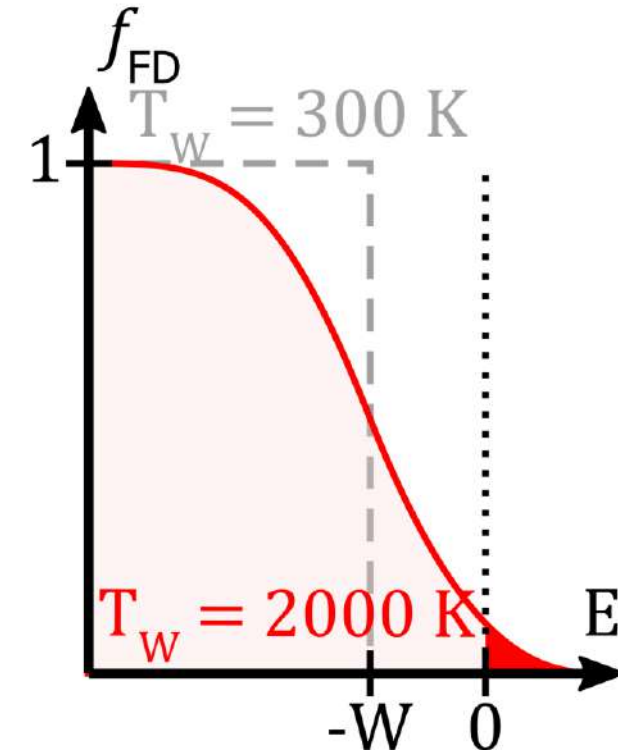
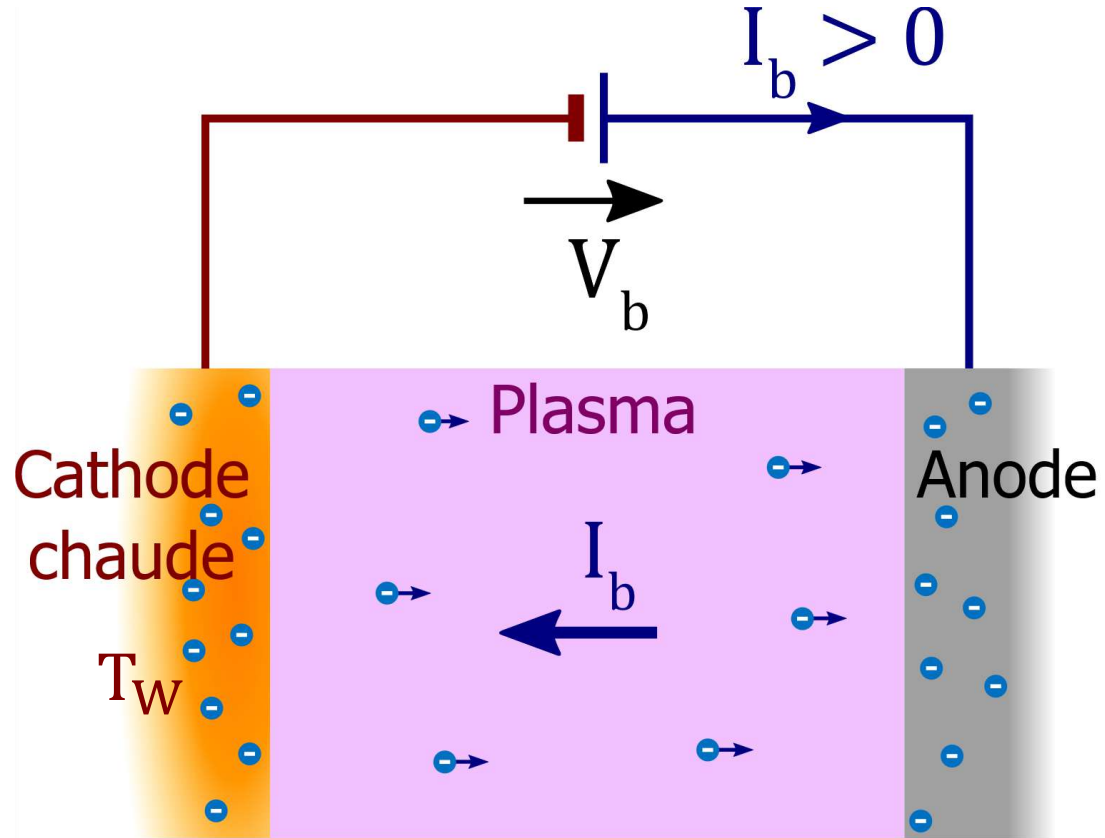
3. STABILITÉ DU
PLASMA HÉLICON



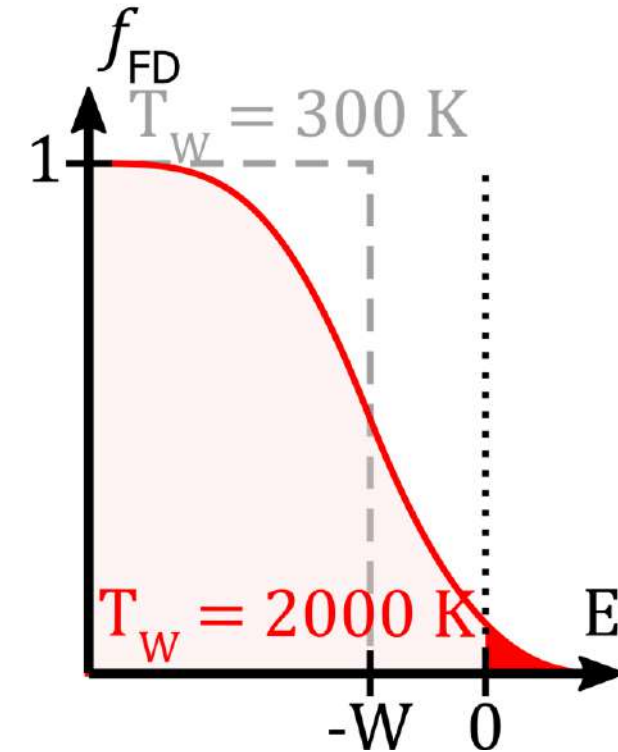
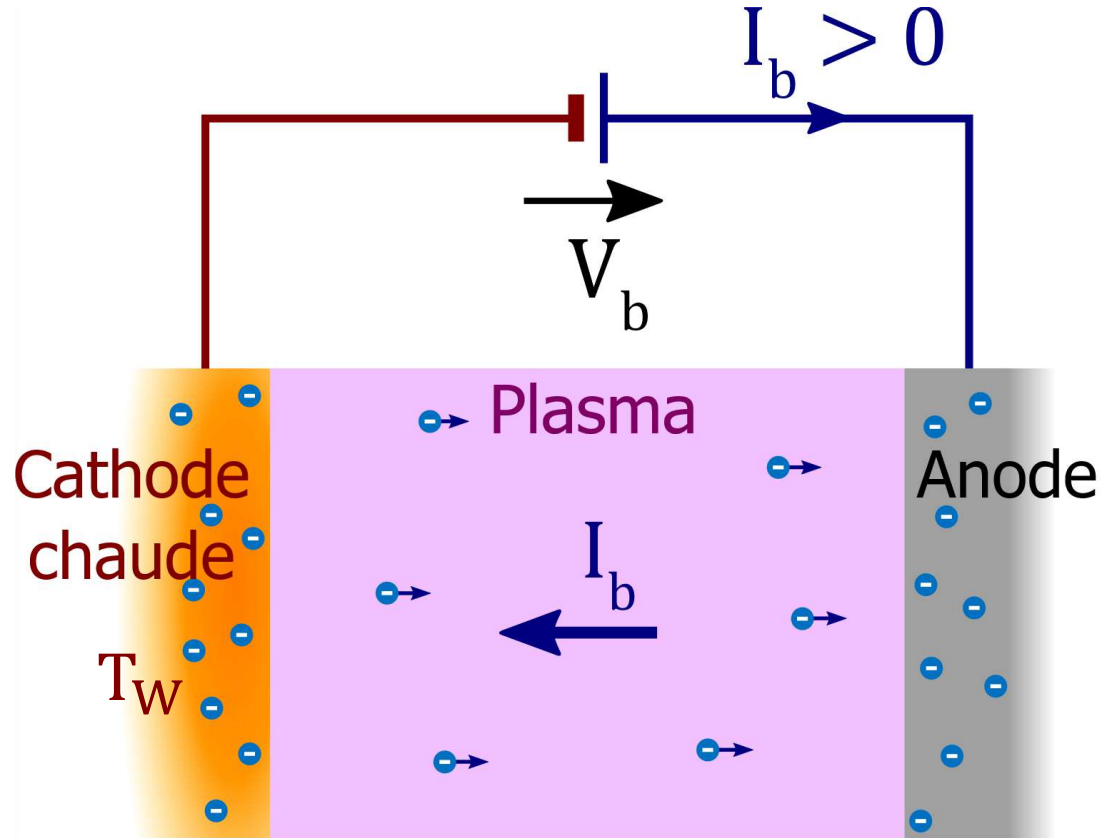
1. OPÉRATION D'UNE CATHODE ÉMISSIVE



Principe d'une cathode émissive



Principe d'une cathode émissive



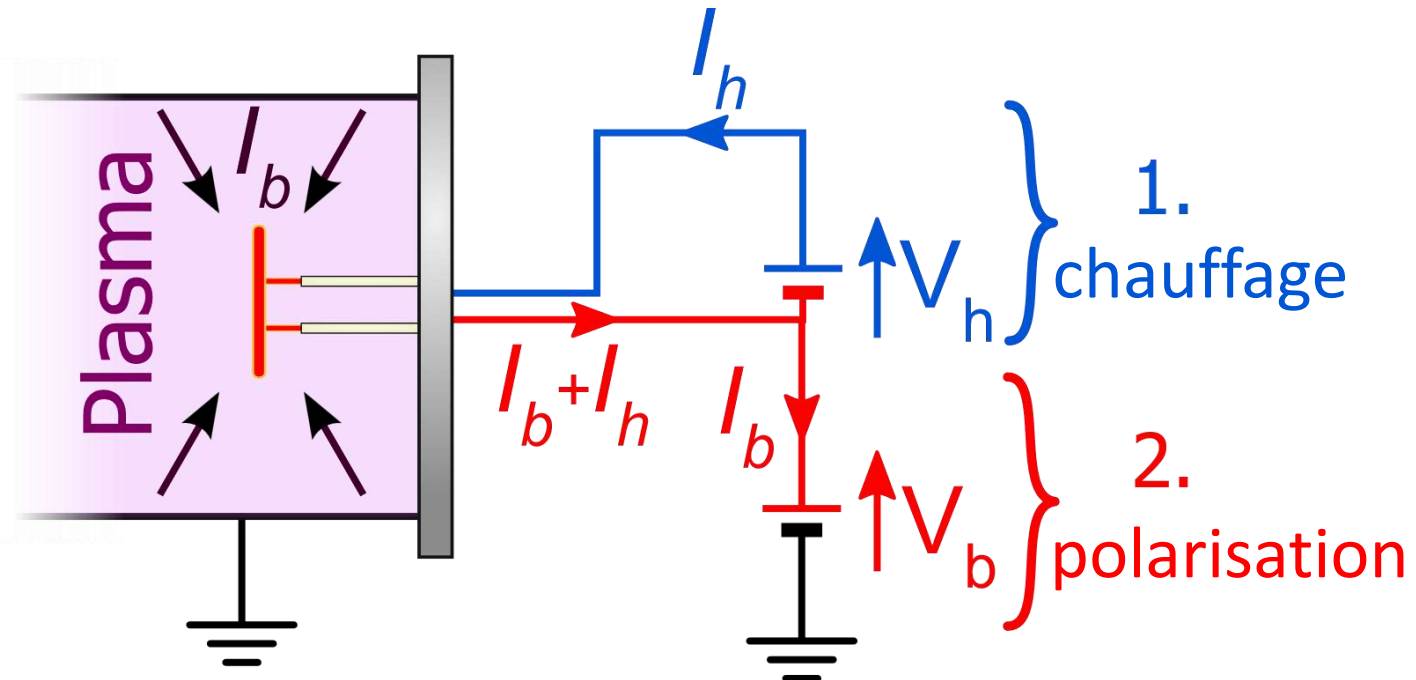
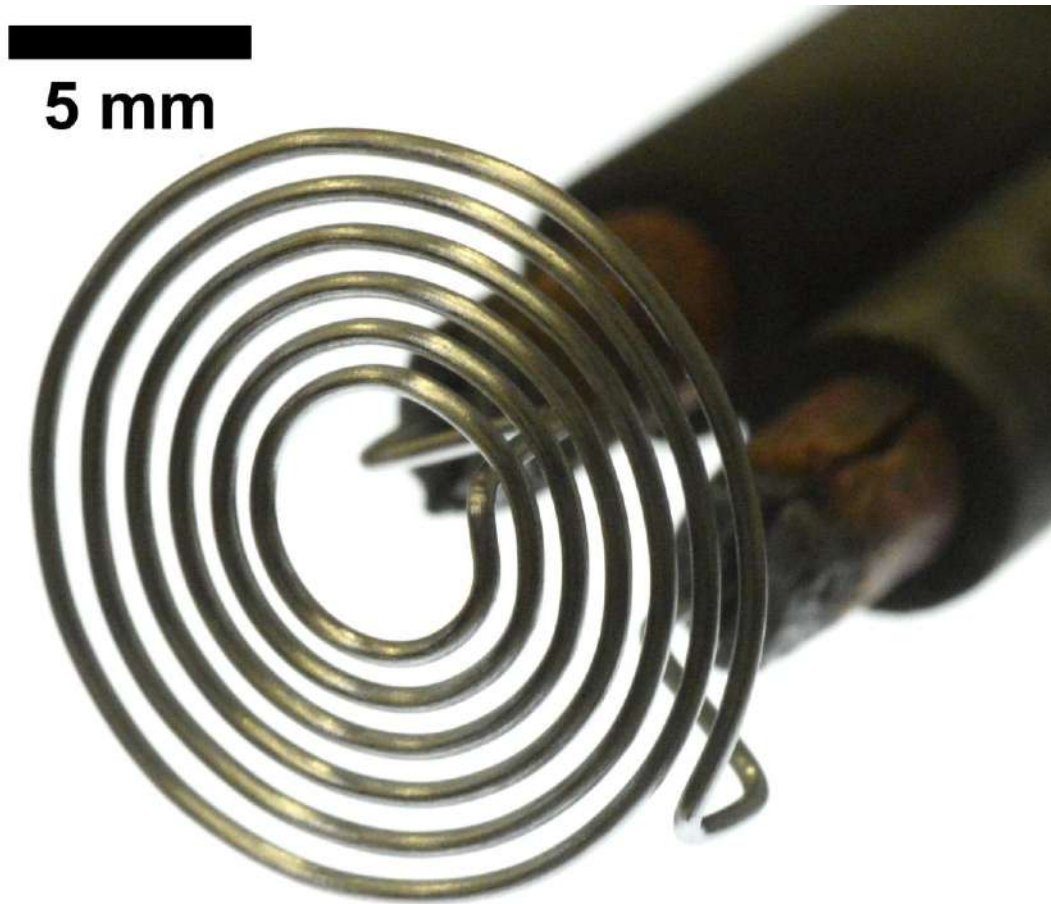
Courant de Richardson

$$I_{em} = A_G S T_W^2 \exp \left(-\frac{eW}{k_B T_W} \right)$$

Herring, Nichols, Rev. Modern Phys. (1949)

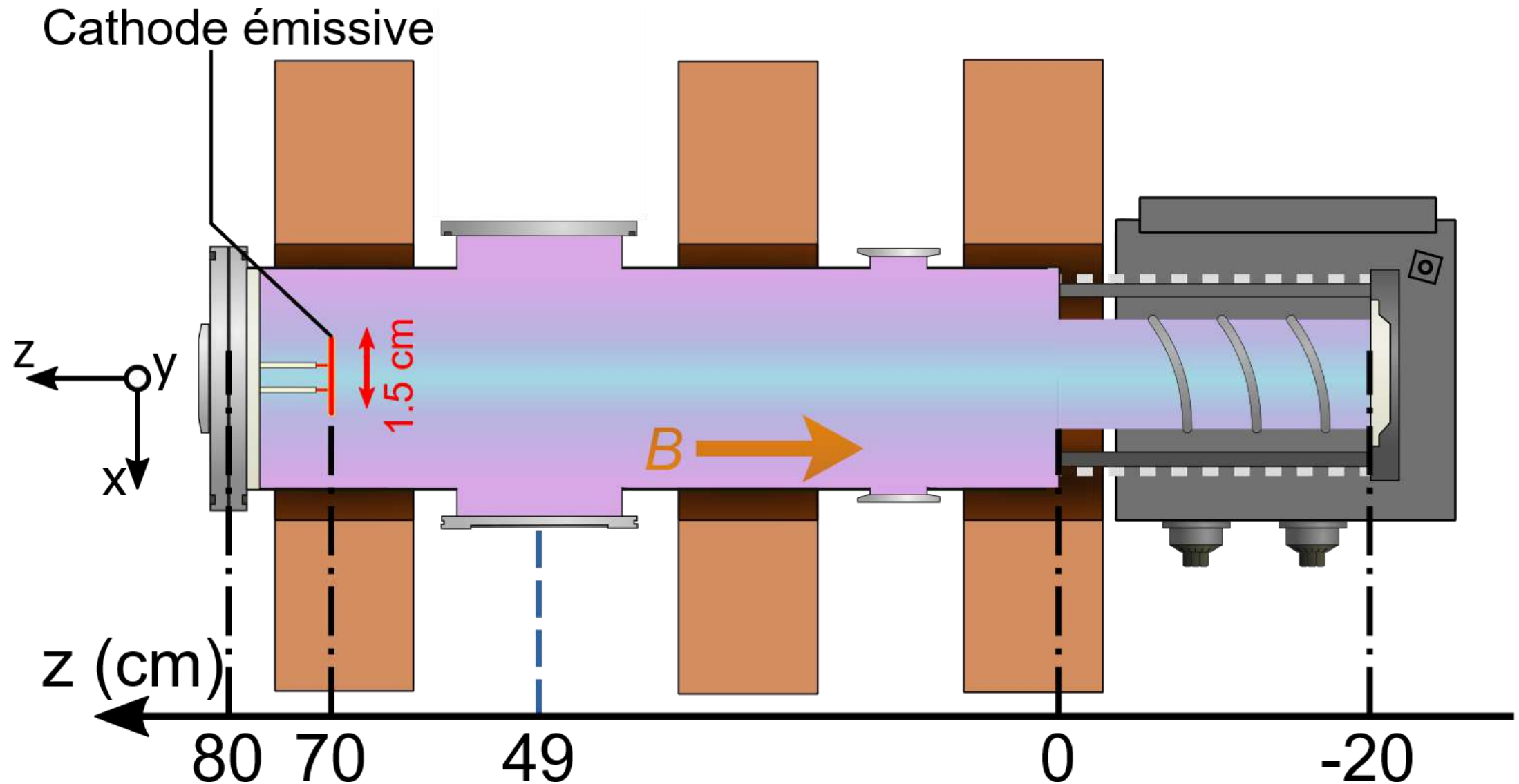
- Cathode chaude + plasma = injection d'électrons

Cathode émissive en tungstène



- Cathode à température T_W
- Courant significatif à $T_W > 2500$ K

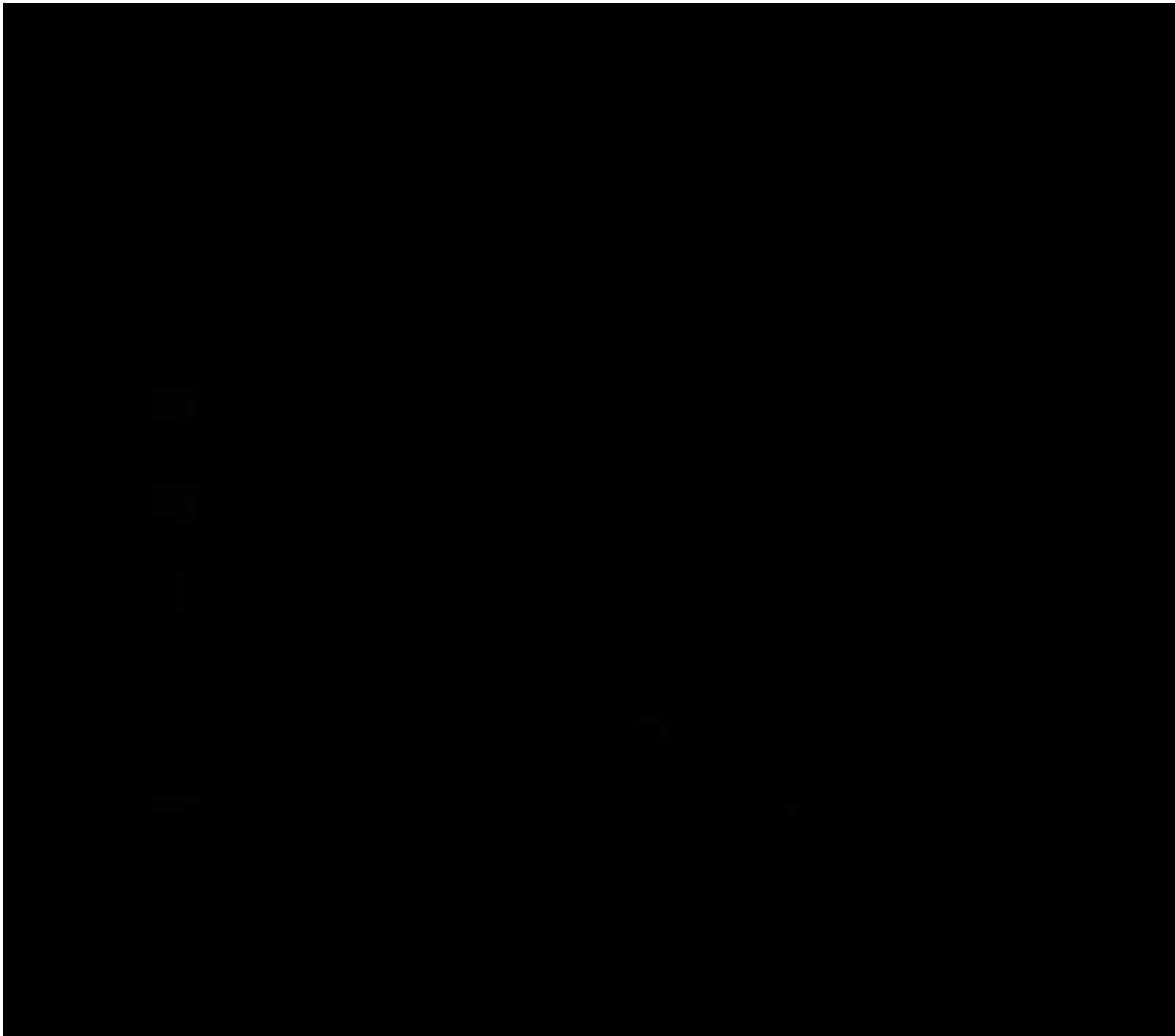
Configuration expérimentale



Protocole opératoire

- Opération pulsée
 - Haute reproductibilité
 - Automatisation via Labview

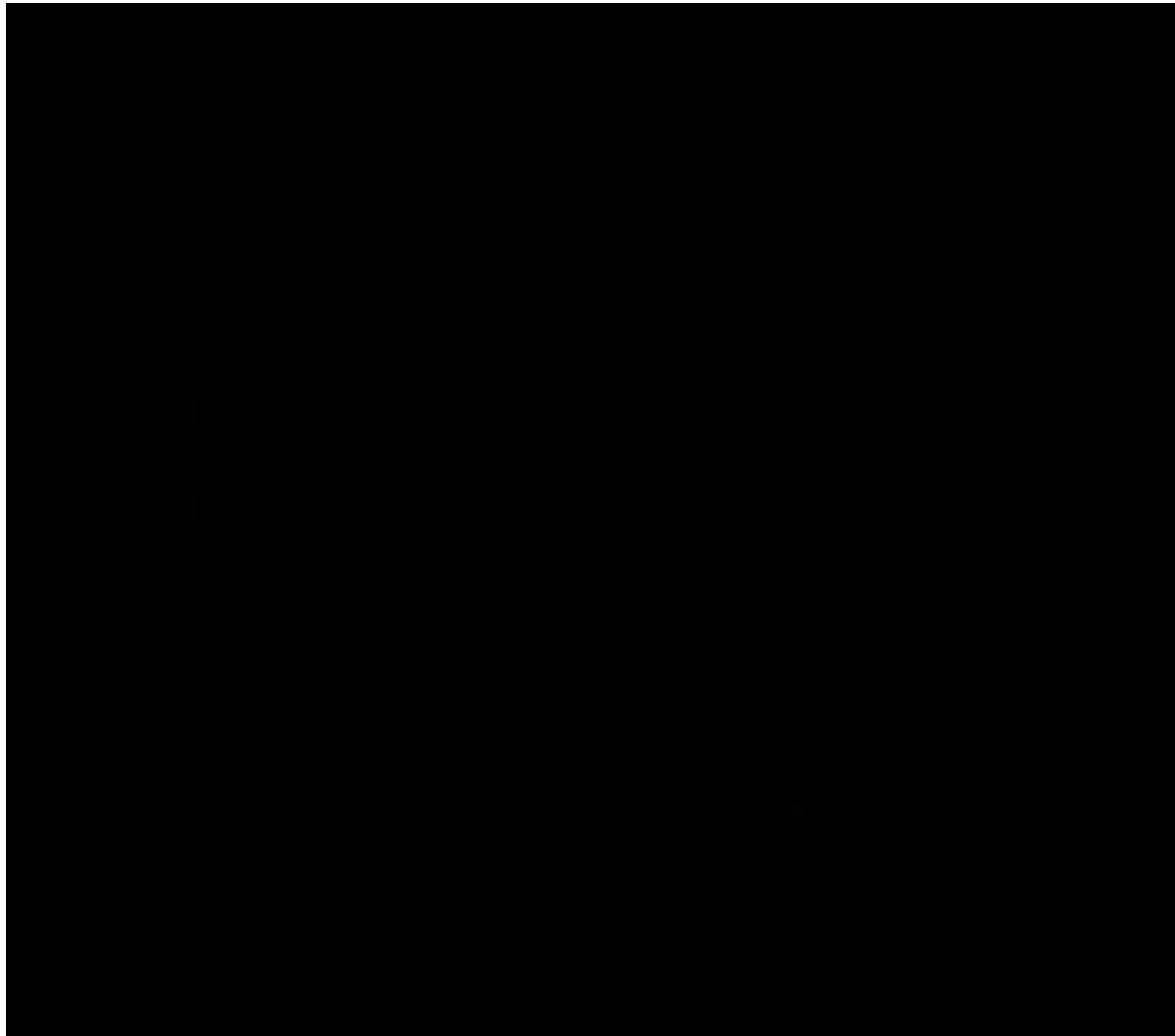
Protocole opératoire



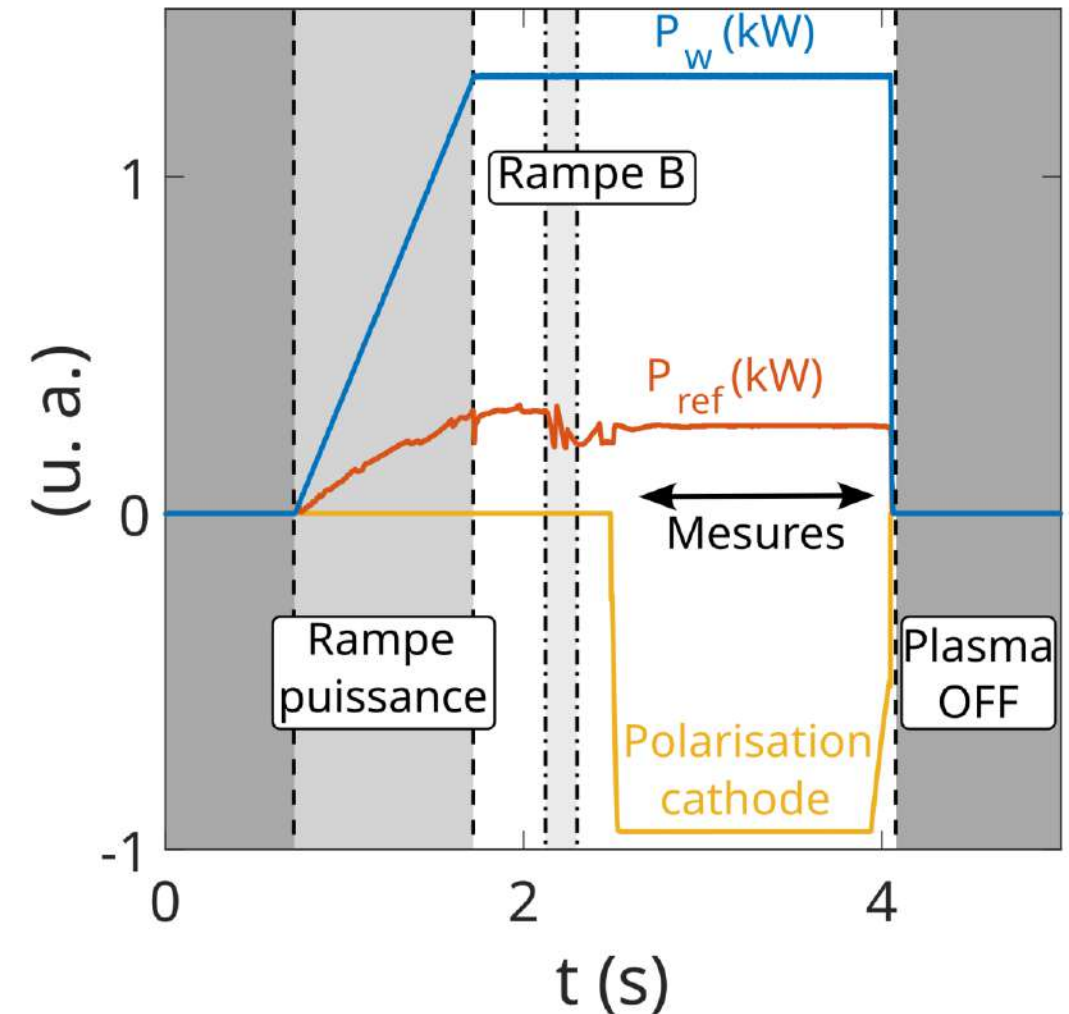
- Opération pulsée

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Protocole opératoire



- Opération pulsée

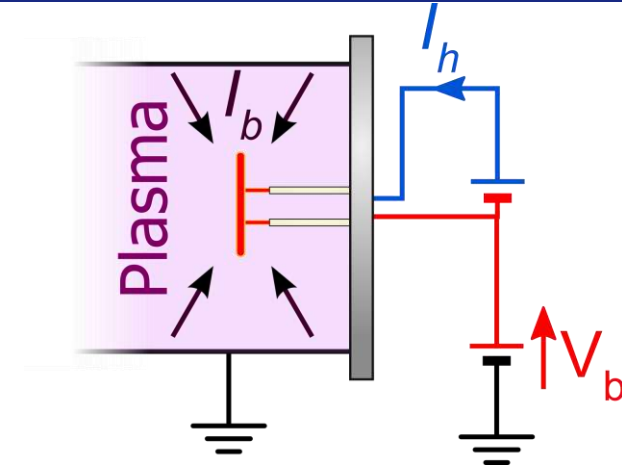
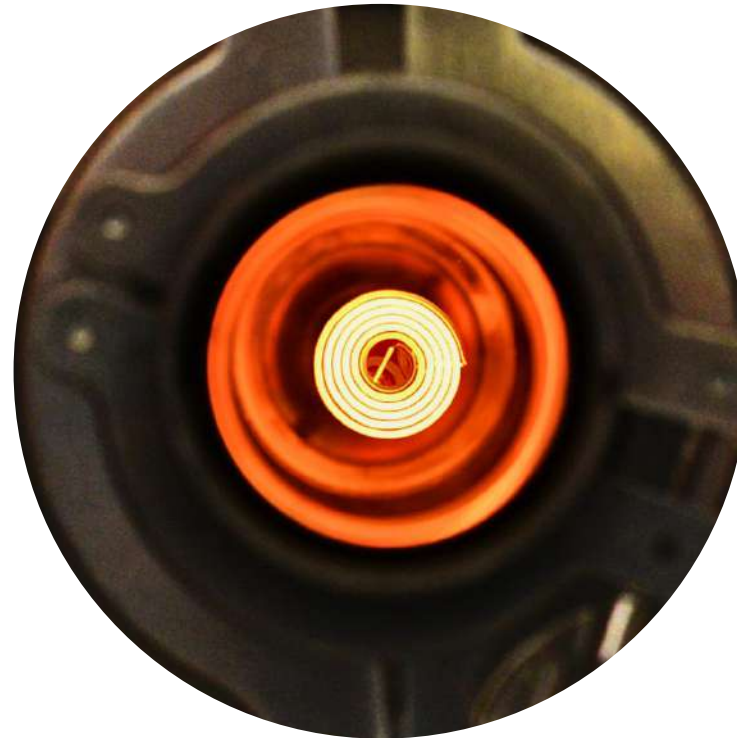


- Haute reproductibilité
- Automatisation via Labview

Courant émis : analyse préliminaire

$$I_b = I_{em} + \cancel{I_i} - \cancel{I_e}$$

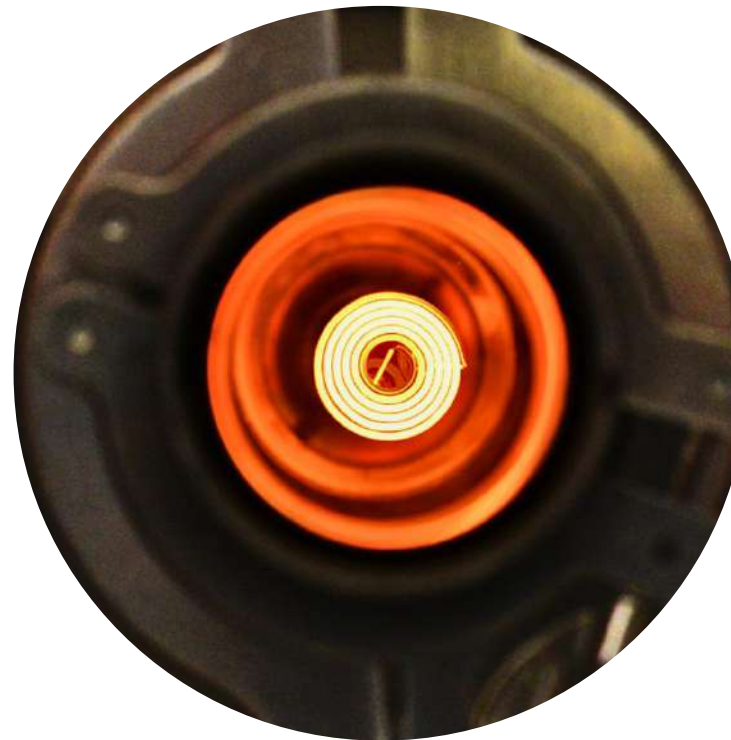
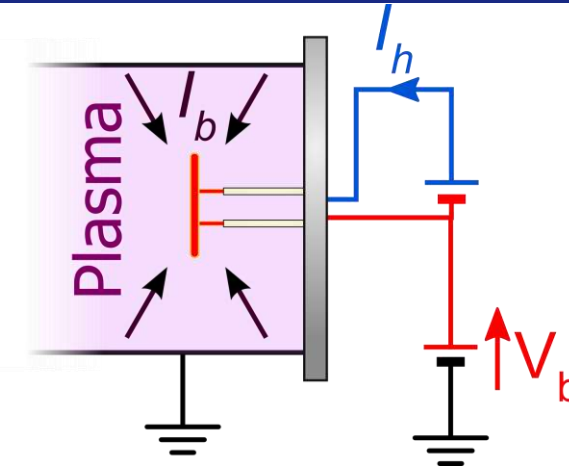
$\Xi = I_{em}/I_i \gg 1$



Courant émis : analyse préliminaire

$$I_b = I_{em} + \cancel{I_i} - \cancel{I_e} \quad \xrightarrow{\quad} \quad I_{em} = A_G S T_W^2 \exp \left(- \frac{eW}{k_B T_W} \right)$$

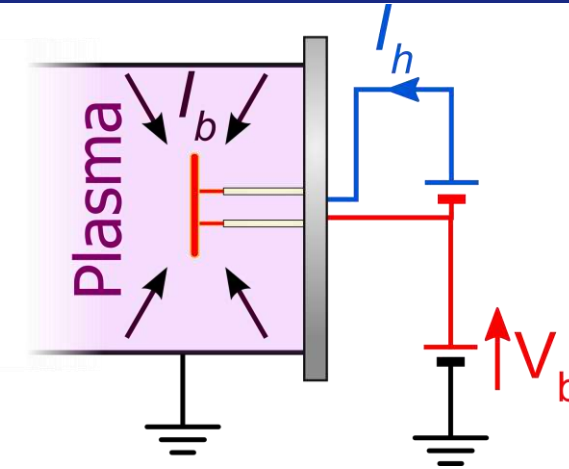
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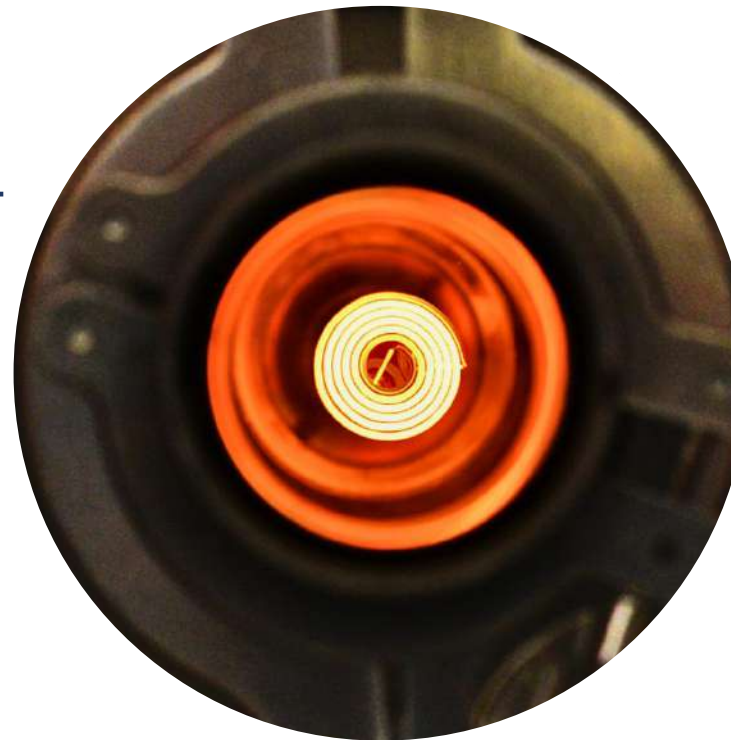


Chauffage par effet Joule :

$$RI_h^2$$

Pertes radiatives:

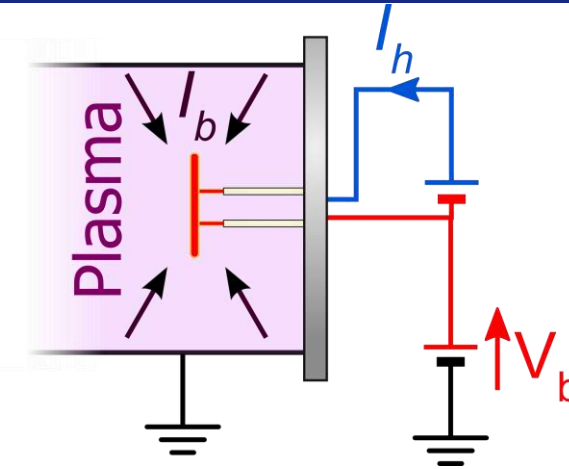
$$\sigma S \epsilon T_W^4$$



Courant émis : analyse préliminaire

$$I_b = I_{em} + \cancel{I_i} - \cancel{I_e} \quad \xrightarrow{\quad} \quad I_{em} = A_G S T_W^2 \exp \left(-\frac{eW}{k_B T_W} \right)$$

$\Xi = I_{em}/I_i \gg 1$

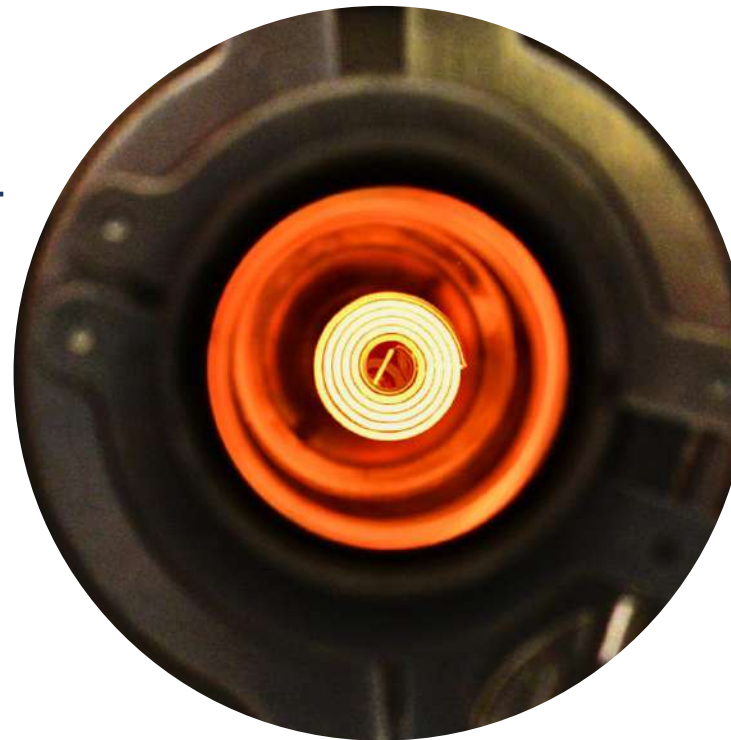


Chauffage par effet Joule :

$$RI_h^2$$

Pertes radiatives:

$$\sigma S \epsilon T_W^4$$



$$T_W \propto \sqrt{I_h}$$

$$I_b \propto I_h \exp \left(-1/\sqrt{I_h} \right)$$

Courant émis : analyse préliminaire

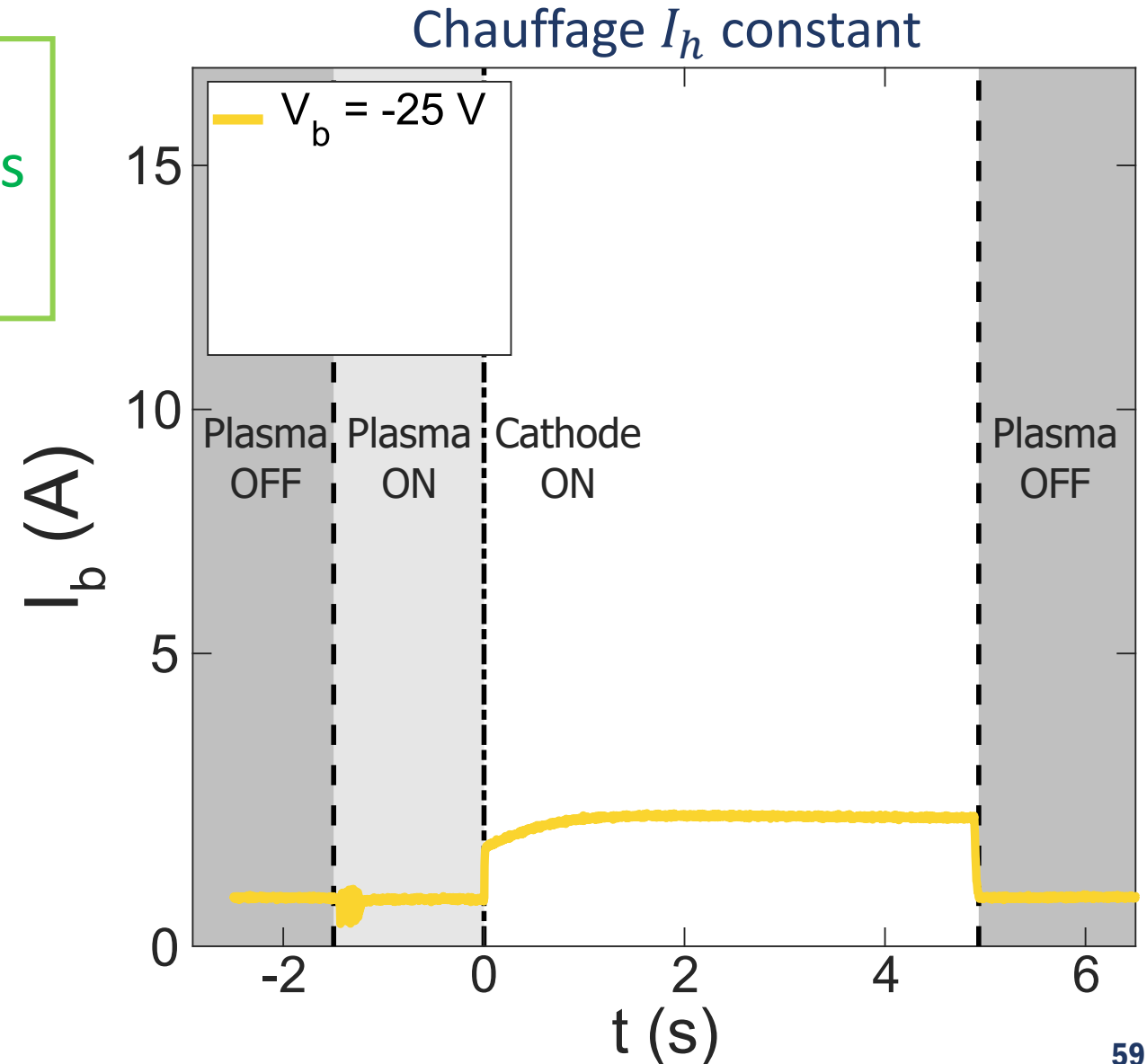
Interprétation naïve

- Le chauffage contrôle le courant émis
- Pas d'impact de la polarisation

Courant émis : analyse préliminaire

Interprétation naïve

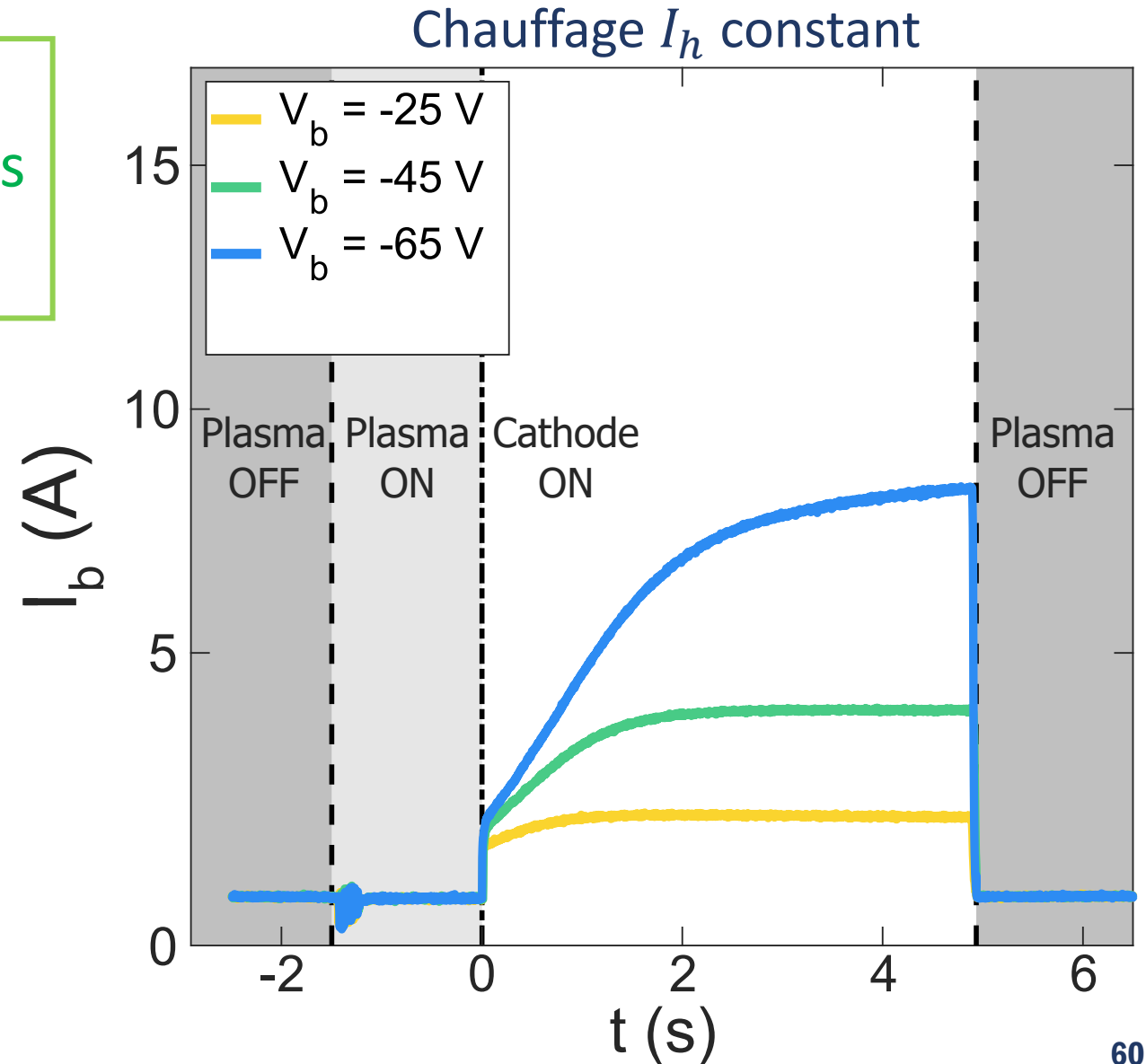
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Courant émis : analyse préliminaire

Interprétation naïve

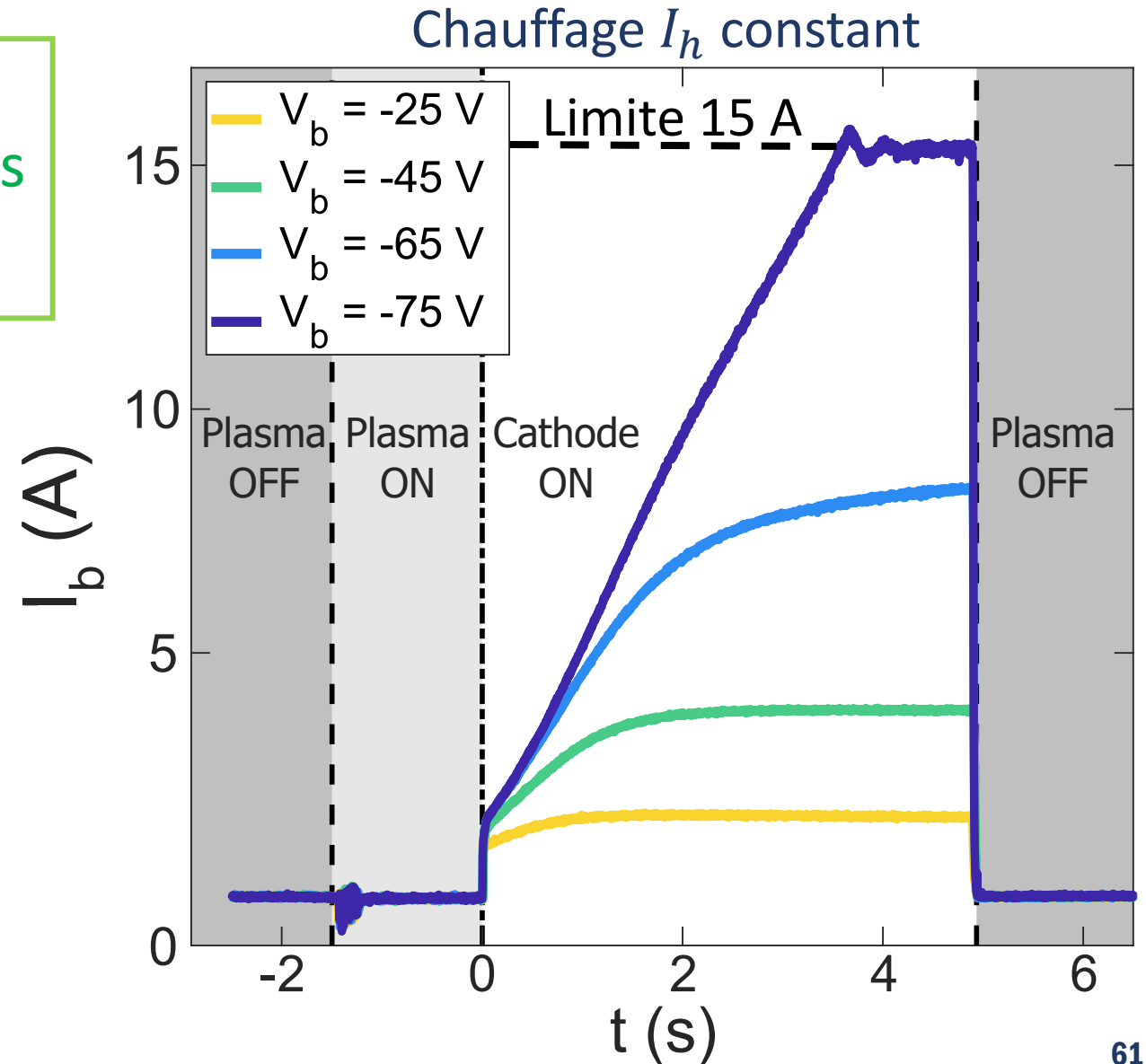
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Courant émis : analyse préliminaire

Interprétation naïve

- Le chauffage contrôle le courant émis
- Pas d'impact de la polarisation



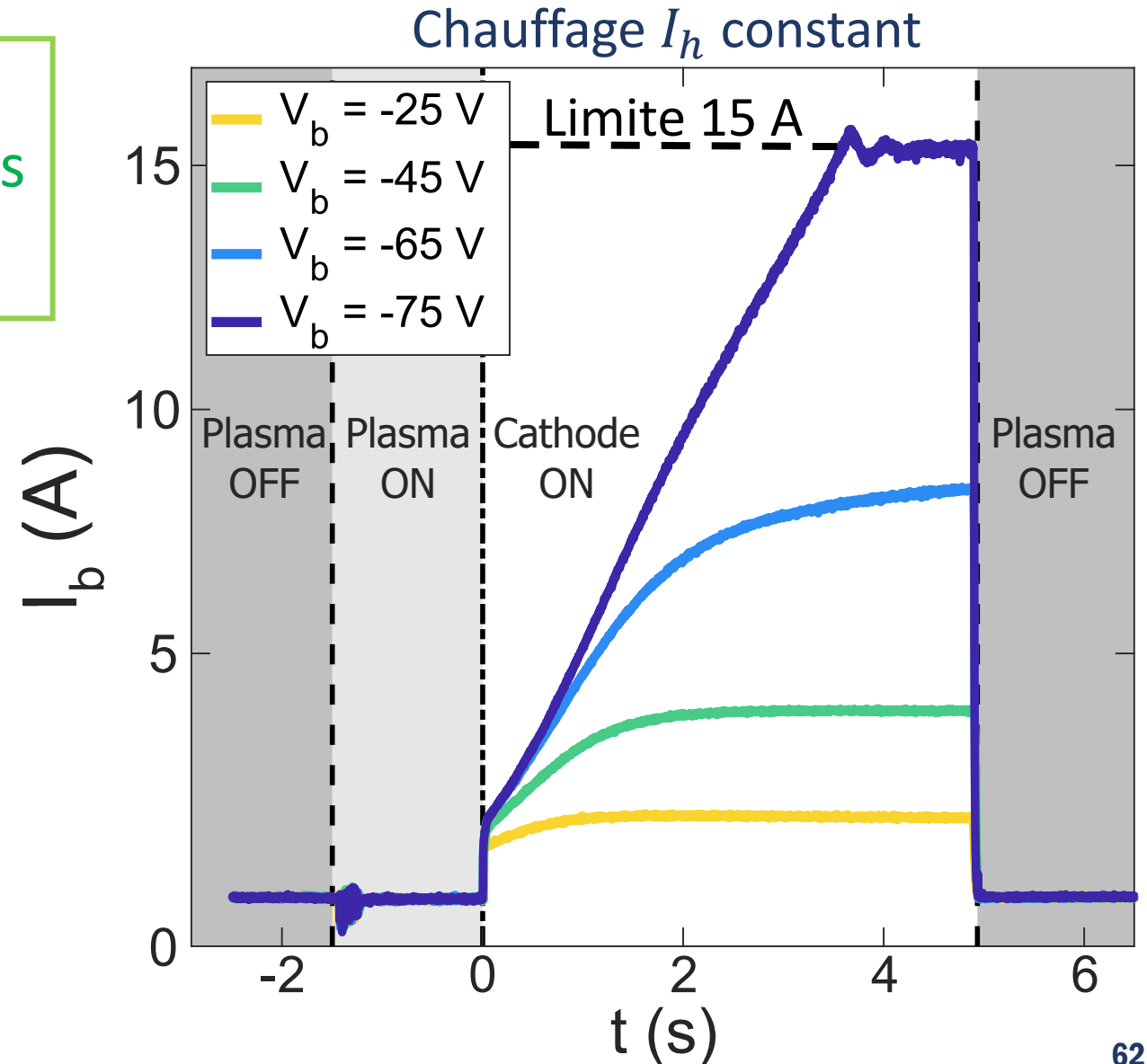
Courant émis : analyse préliminaire

Interprétation naïve

- Le chauffage contrôle le courant émis
- Pas d'impact de la polarisation

Observation expérimentale :

- Fort impact de la polarisation
- Divergence à $V_b \ll \phi_p$



Courant émis : analyse préliminaire

Interprétation naïve

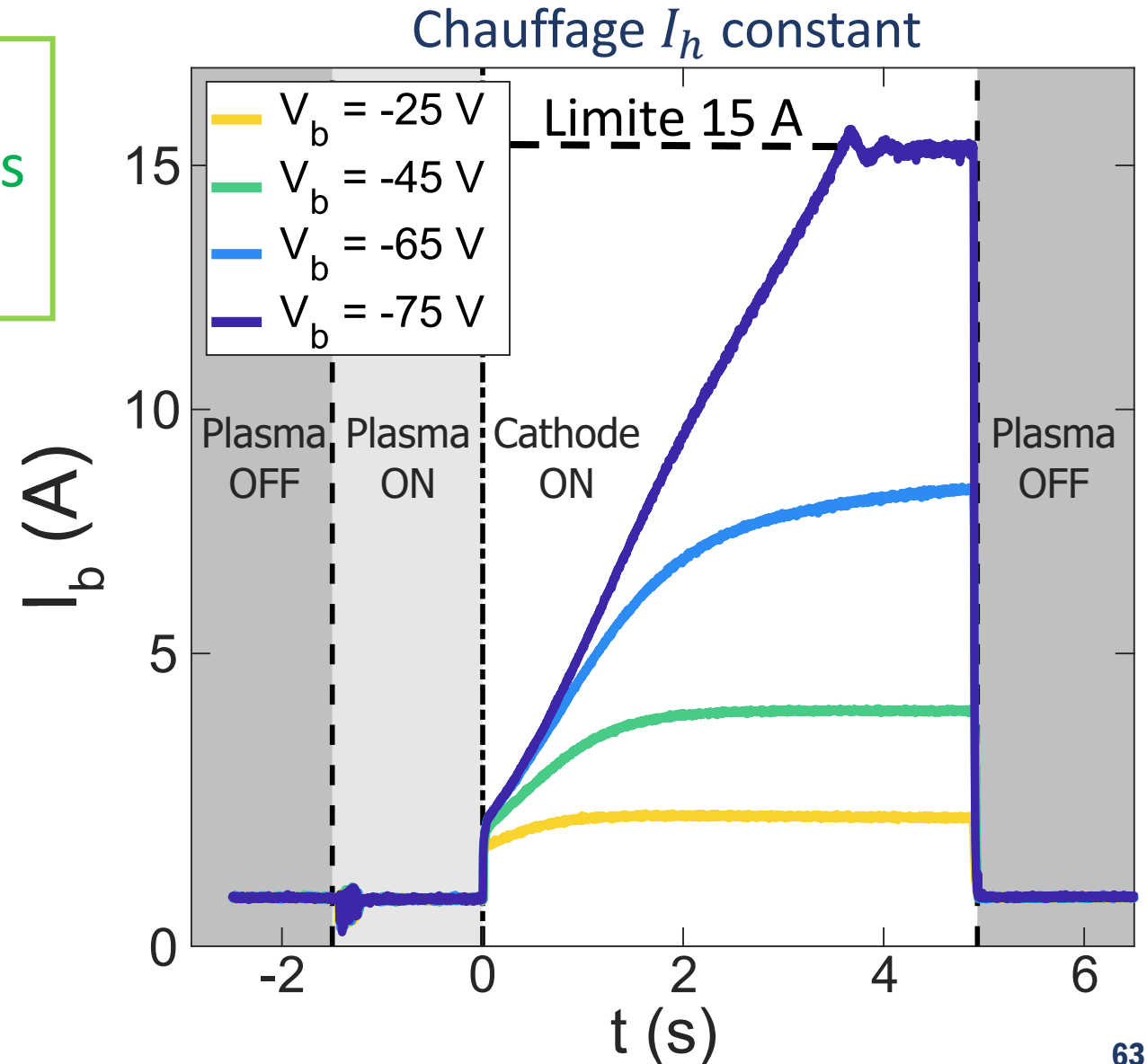
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Observation expérimentale :

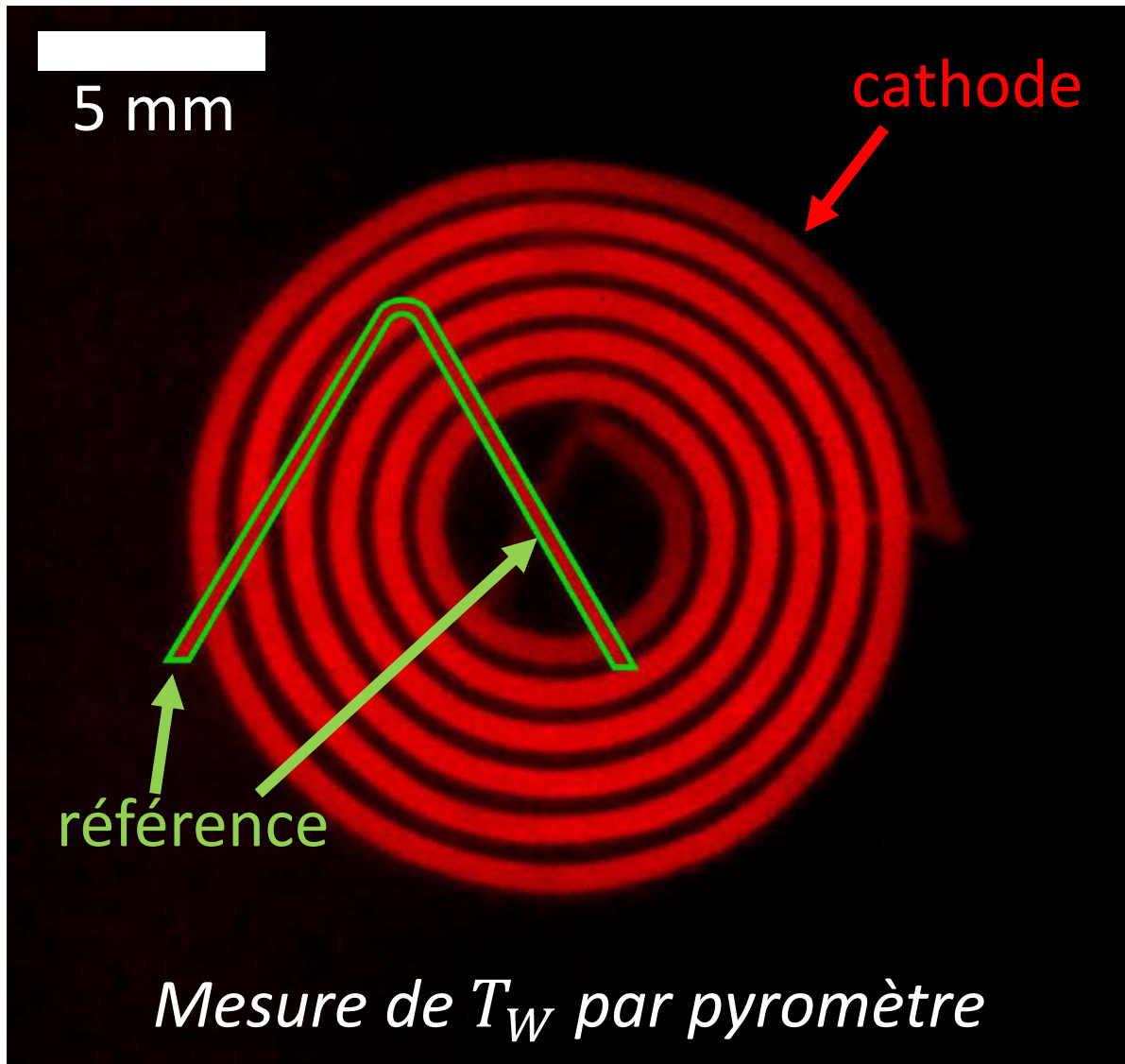
- Fort impact de la polarisation
- Divergence à $V_b \ll \phi_p$

Evolution de $I_{em}(I_h, V_b, \dots)$?

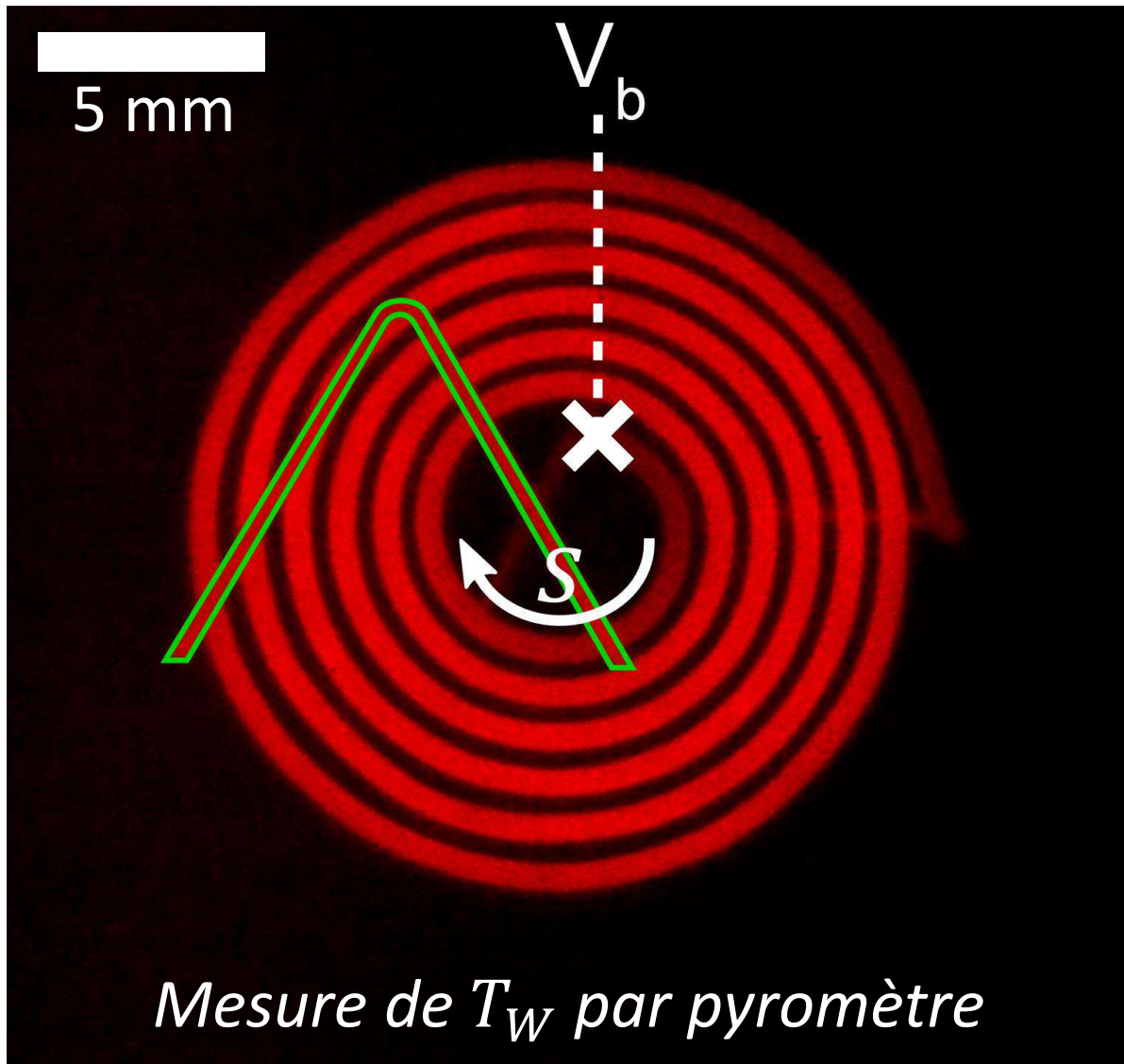
Quel ratio $\Xi = I_{em}/I_i$?



Mesures de température



Mesures de température



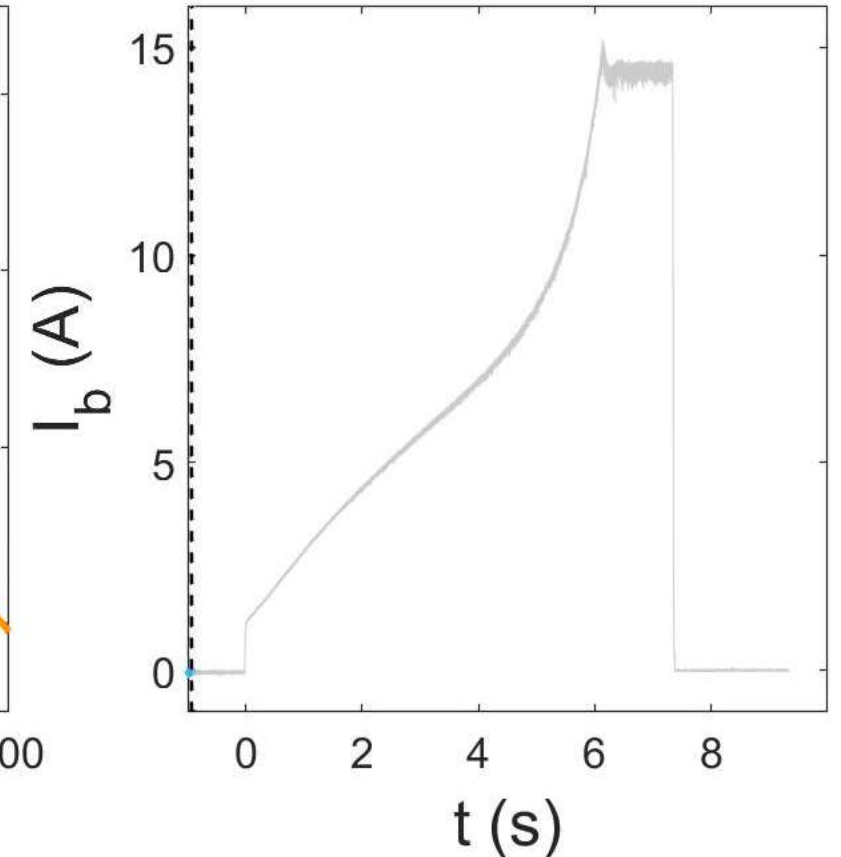
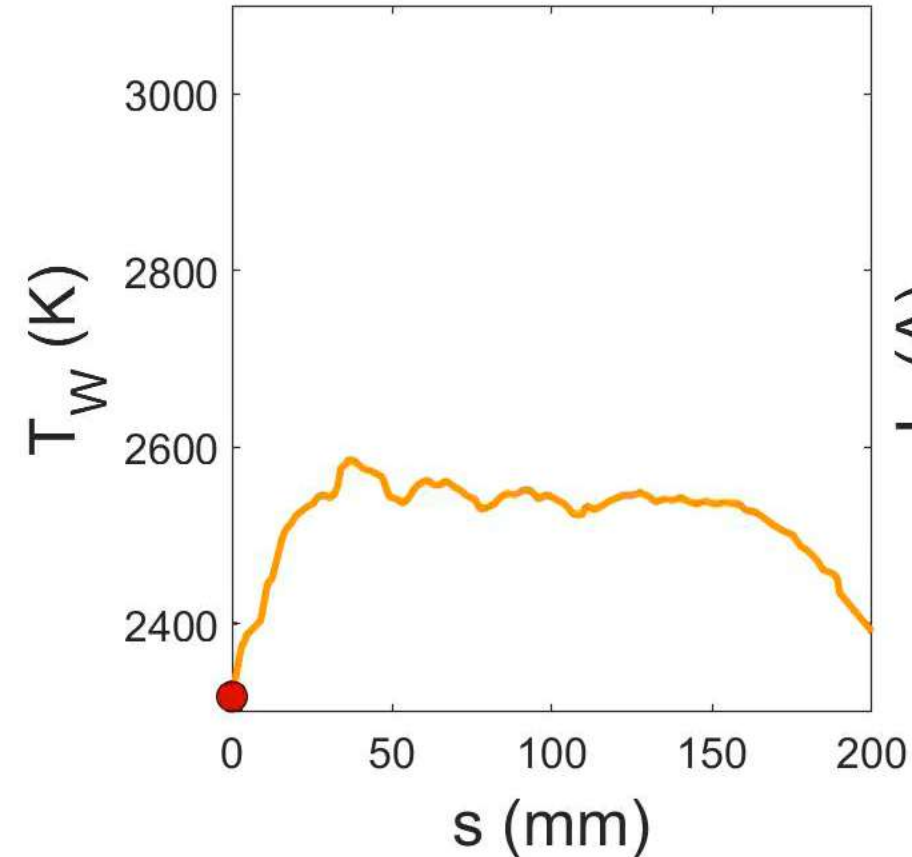
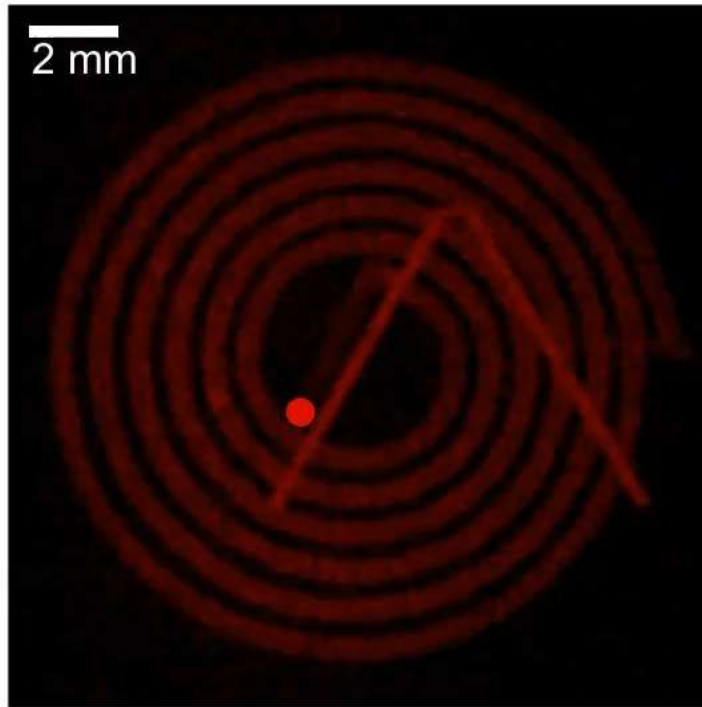
- Comparaison de l'intensité lumineuse à 650 nm.
- **Mesure spatialement résolue de T_W .**
- Abscisse curviligne : s

Profil hétérogène de température

- Chauffage hétérogène de la cathode lors de l'émission d'électrons
 - Divergence de I_b = cathode à 3000 K

Profil hétérogène de température

$t = -1.0$ s



- Chauffage hétérogène de la cathode lors de l'émission d'électrons
 - Divergence de I_b = cathode à 3000 K

Calcul du courant à la cathode

$$I_b = I_{em} + \cancel{I_i} - \cancel{I_e}$$

$\Xi = I_{em}/I_i \gg 1$

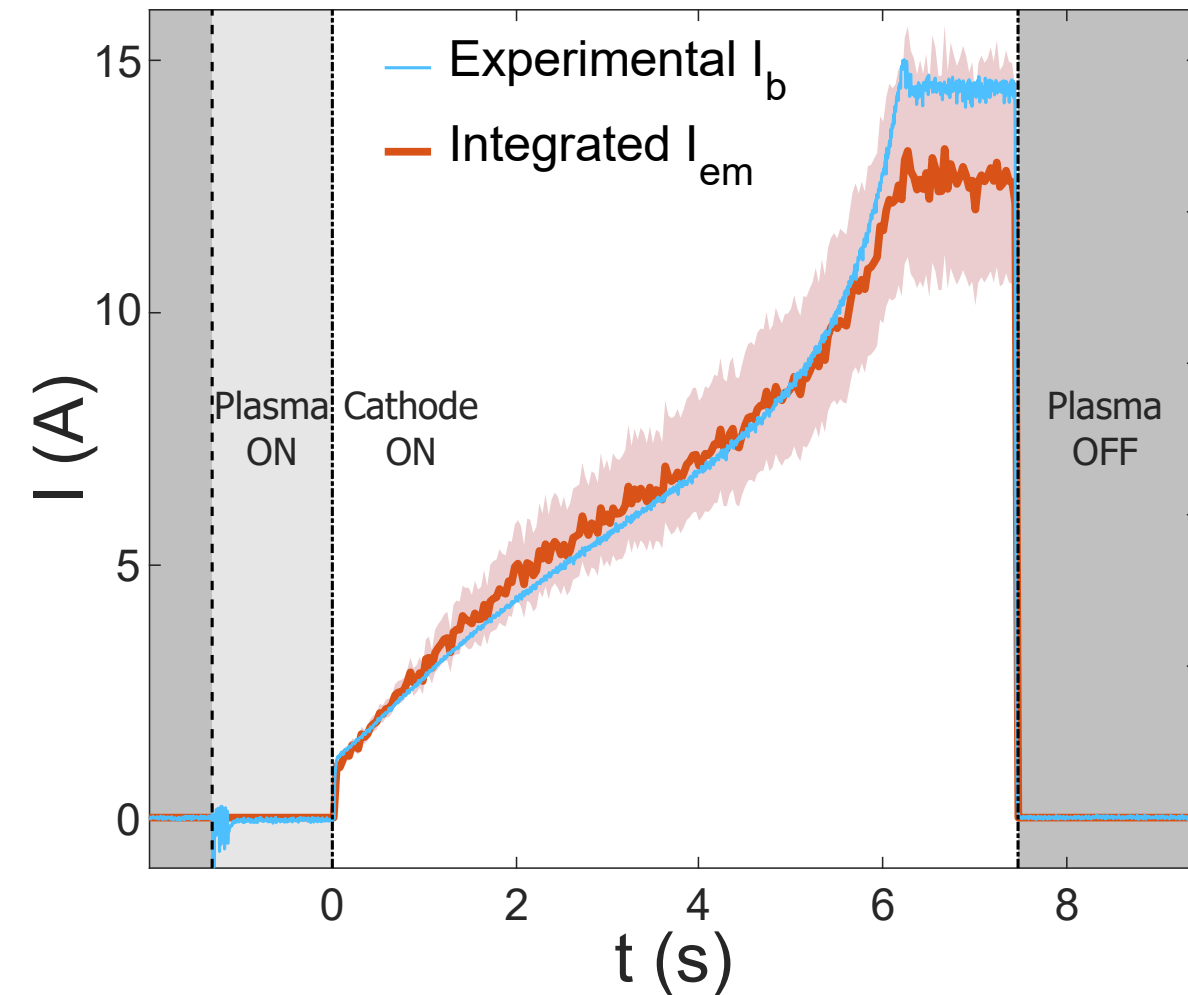
$$I_{em} = \int_0^s A_G T_W^2 \exp \left(- \frac{eW}{k_B T_W} \right) 2\pi r \, dx$$

Calcul du courant à la cathode

$$I_b = I_{em} + \cancel{I_i} - \cancel{I_e}$$

$E = I_{em}/I_i \gg 1$

$$I_{em} = \int_0^s A_G T_W^2 \exp\left(-\frac{eW}{k_B T_W}\right) 2\pi r dx$$

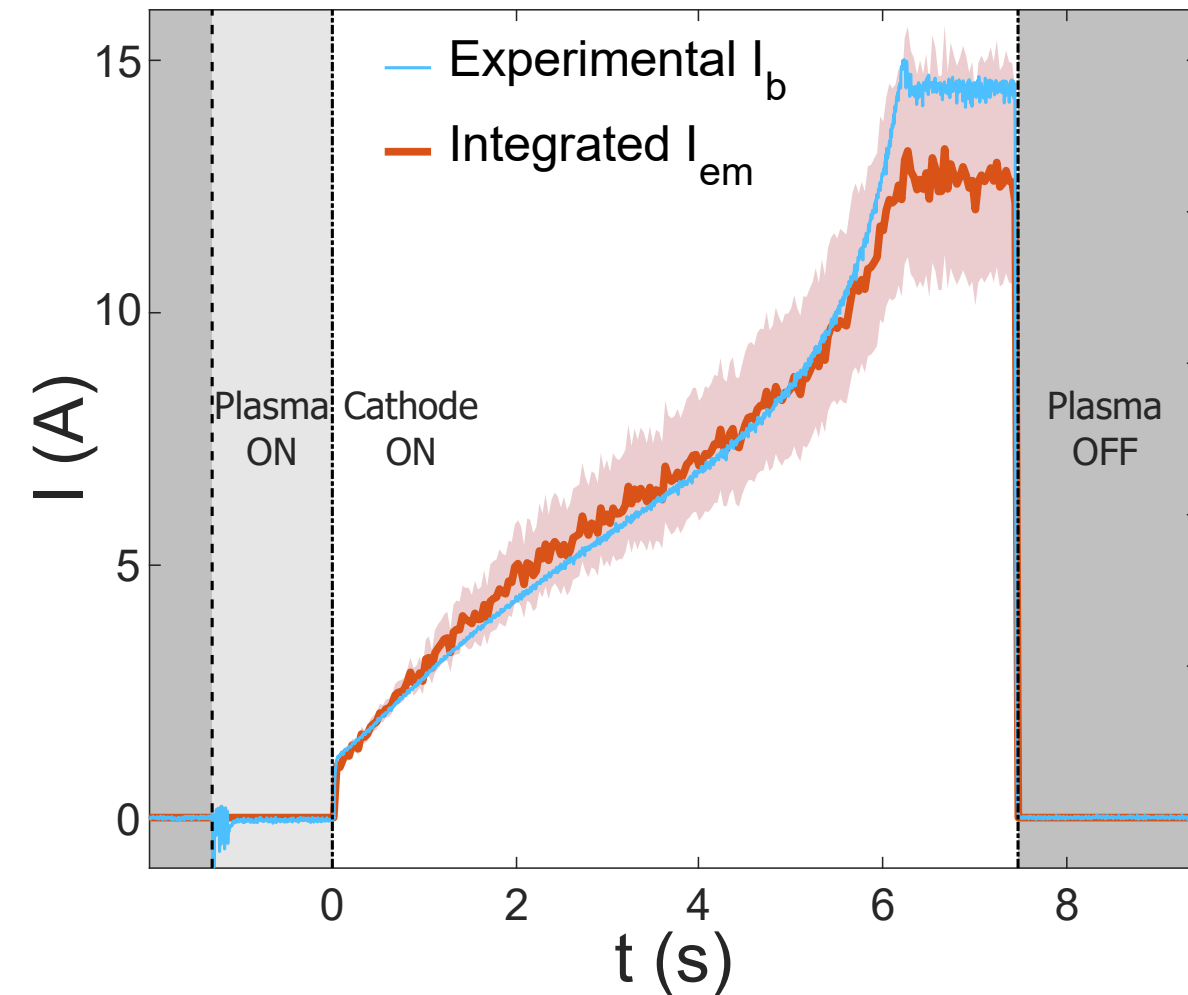


Calcul du courant à la cathode

$$I_b = I_{em} + \cancel{I_i} - \cancel{I_e} \quad \text{with } \cancel{I_i} = I_i \text{ and } \cancel{I_e} = I_e$$

$\Xi = I_{em}/I_i \gg 1$

$$I_{em} = \int_0^s A_G T_W^2 \exp\left(-\frac{eW}{k_B T_W}\right) 2\pi r dx$$



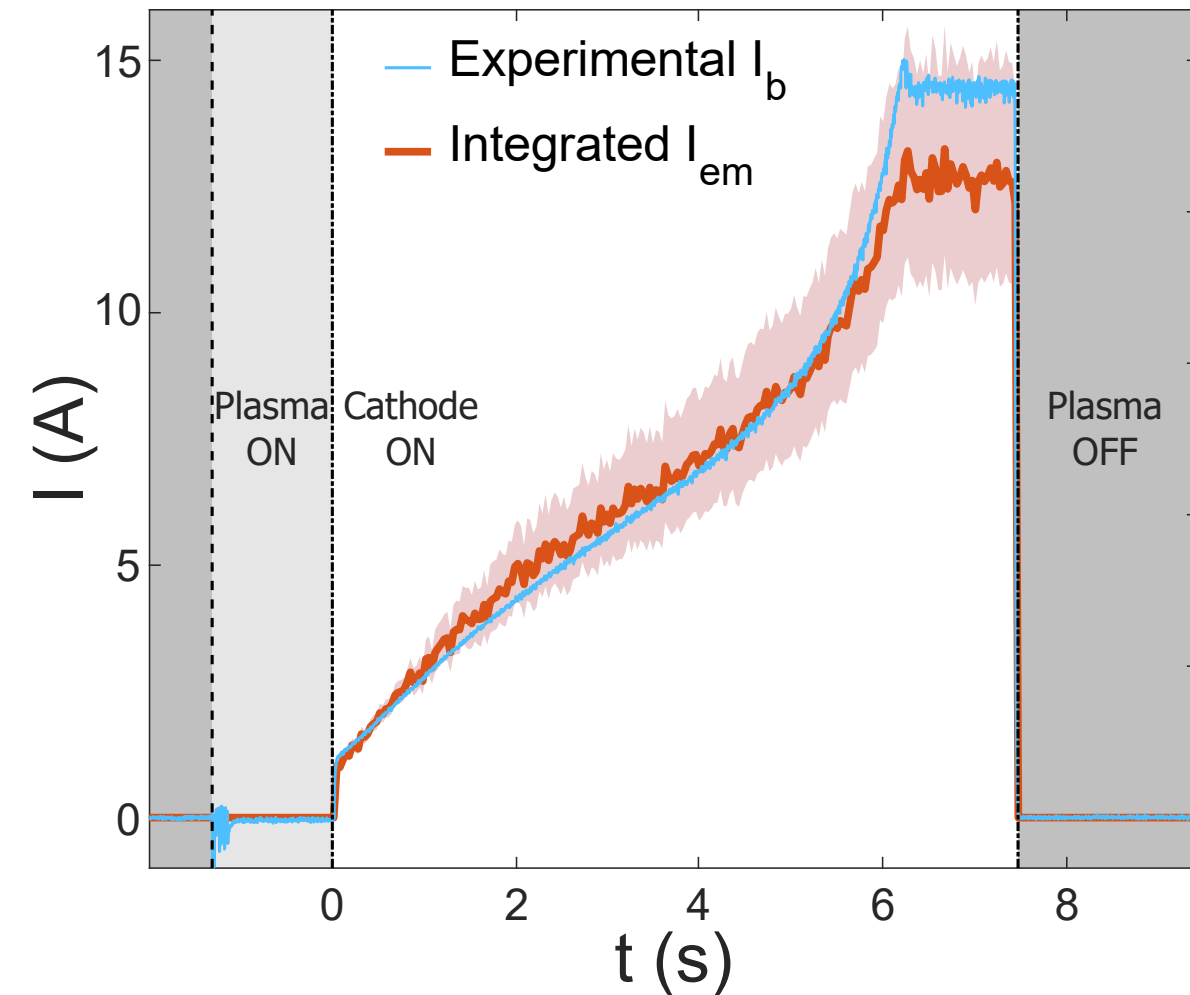
➤ La mesure de T_W permet de connaître I_b

Quel est le profil de température ?

Calcul du courant à la cathode

$$I_b = I_{em} + \cancel{I_i} - \cancel{I_e} \quad \text{with } \cancel{E = I_{em}/I_i \gg 1}$$

$$I_{em} = \int_0^s A_G T_W^2 \exp\left(-\frac{eW}{k_B T_W}\right) 2\pi r dx$$



➤ La mesure de T_W permet de connaître I_b

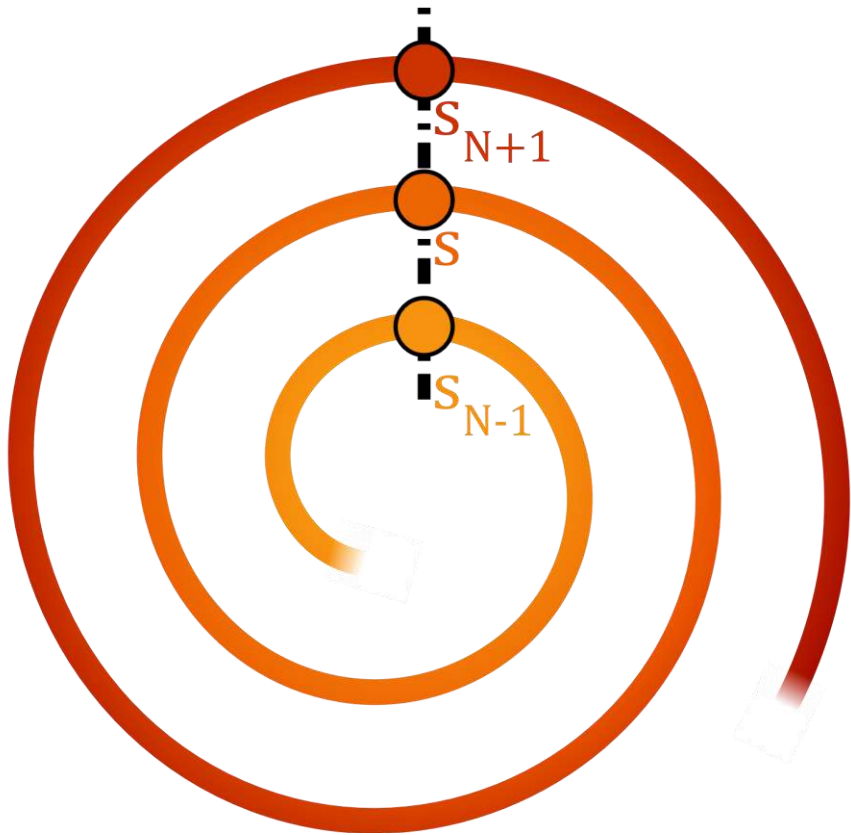
➔ *Quel est le profil de température ?*

Modèle thermique de la cathode

En l'absence de plasma

$\dot{H} =$

Inertie thermique : $\dot{H} = C_p \rho V \frac{\partial T_w}{\partial t}$



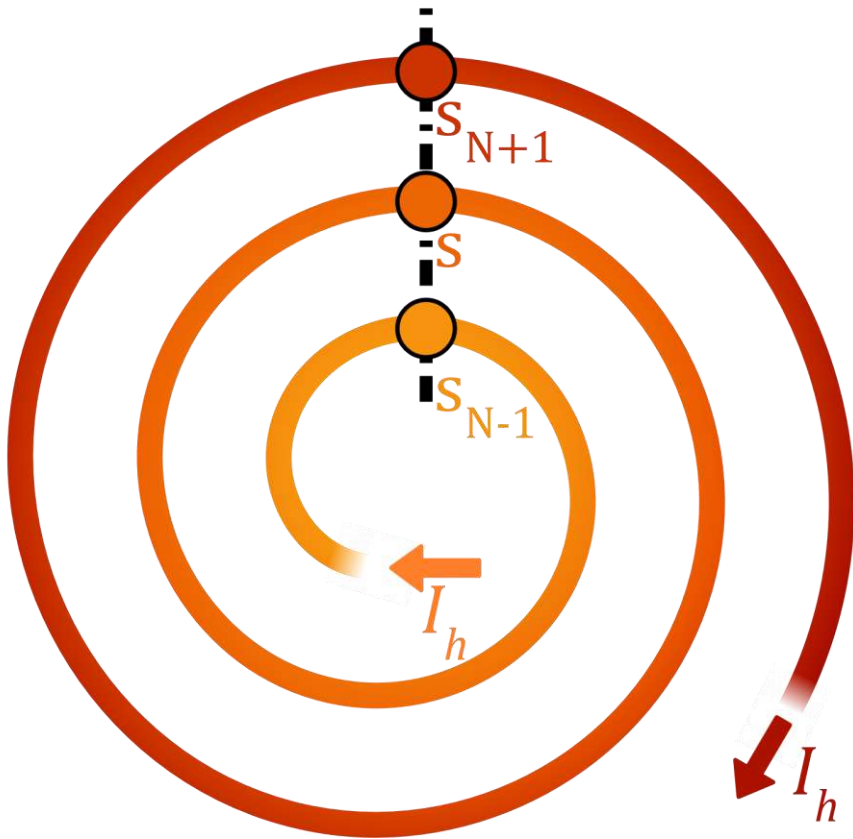
Modèle thermique de la cathode

En l'absence de plasma

$$\dot{H} = \dot{Q}_\Omega$$

Inertie thermique : $\dot{H} = C_p \rho V \frac{\partial T_w}{\partial t}$

Effet Joule : $\dot{Q}_\Omega = R(I_h + I_b)^2$



Modèle thermique de la cathode

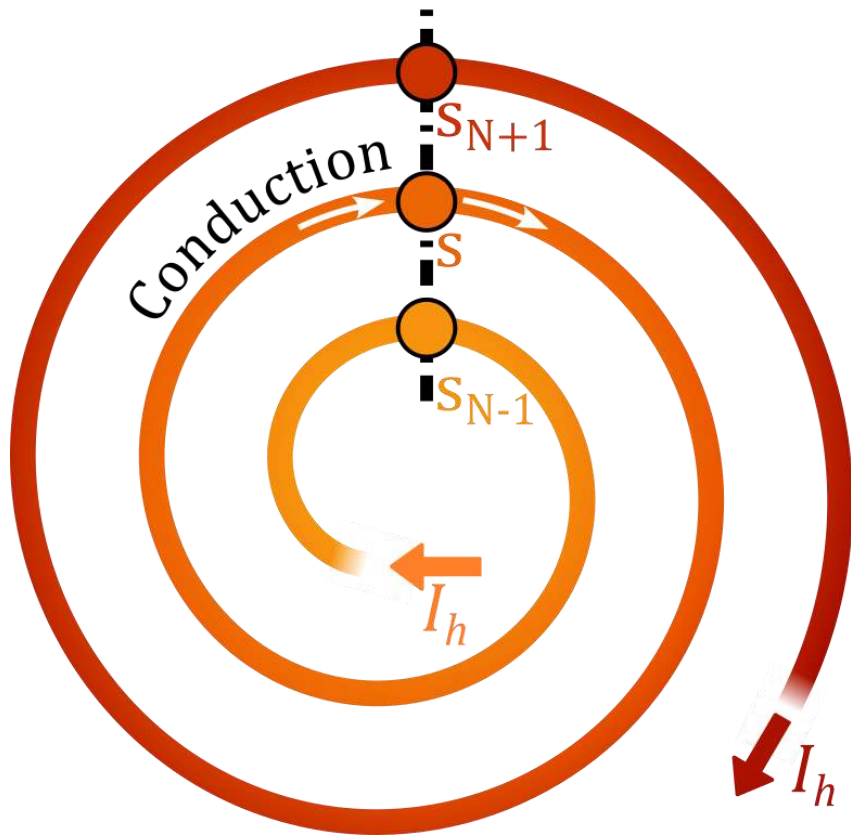
En l'absence de plasma

$$\dot{H} = \dot{Q}_\Omega + \dot{Q}_c$$

Inertie thermique : $\dot{H} = C_p \rho V \frac{\partial T_w}{\partial t}$

Effet Joule : $\dot{Q}_\Omega = R(I_h + I_b)^2$

Conduction thermique : $\dot{Q}_c = \lambda \pi r^2 \frac{\partial^2 T_w}{\partial s^2}$



Modèle thermique de la cathode

En l'absence de plasma

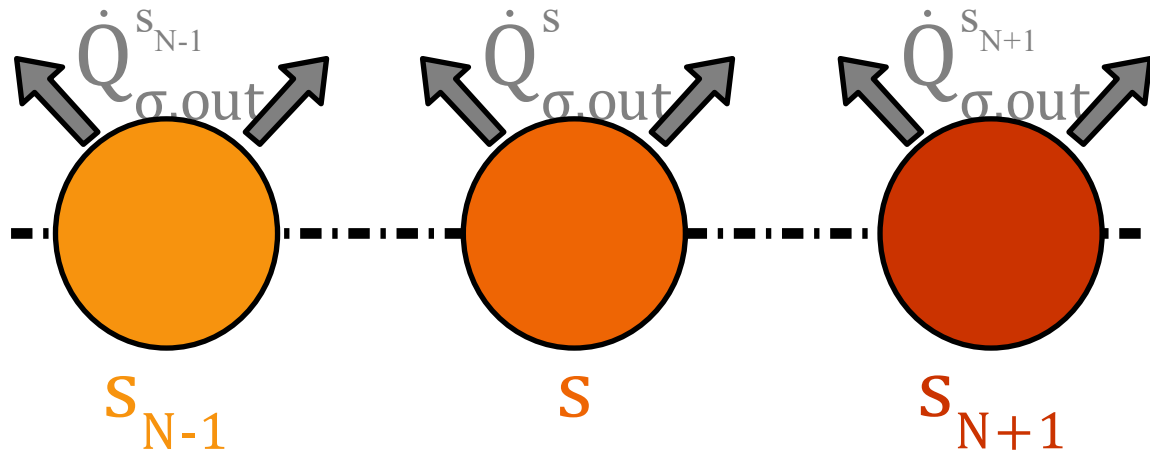
$$\dot{H} = \dot{Q}_\Omega + \dot{Q}_c - \dot{Q}_{\sigma, out}$$

Inertie thermique : $\dot{H} = C_p \rho V \frac{\partial T_W}{\partial t}$

Effet Joule : $\dot{Q}_\Omega = R(I_h + I_b)^2$

Conduction thermique : $\dot{Q}_c = \lambda \pi r^2 \frac{\partial^2 T_W}{\partial s^2}$

Dissipation radiative : $\dot{Q}_{\sigma, out} = \sigma S \epsilon T_W^4$



Modèle thermique de la cathode

En l'absence de plasma

$$\dot{H} = \dot{Q}_\Omega + \dot{Q}_c - \dot{Q}_{\sigma,out} + \dot{Q}_{\sigma,in}$$

Inertie thermique : $\dot{H} = C_p \rho V \frac{\partial T_W}{\partial t}$

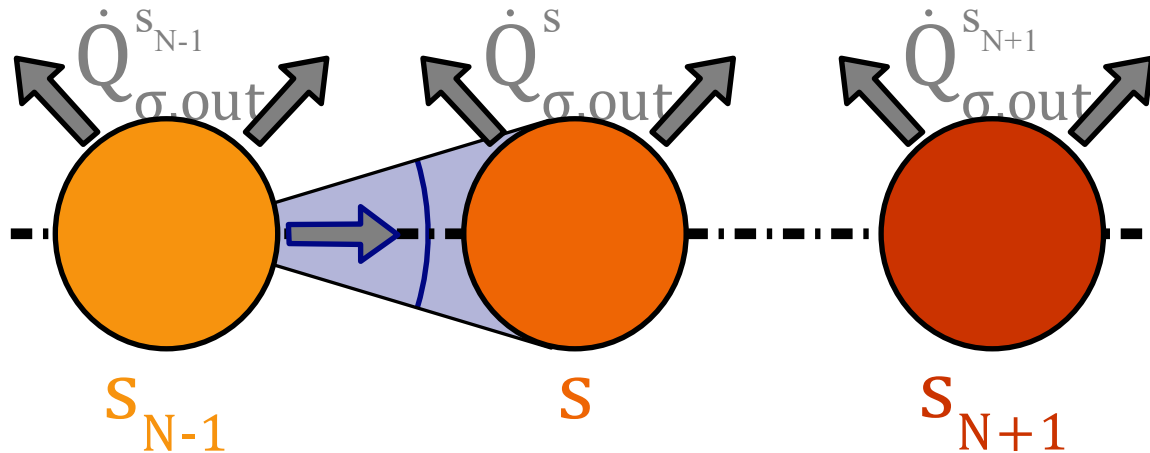
Effet Joule : $\dot{Q}_\Omega = R(I_h + I_b)^2$

Conduction thermique : $\dot{Q}_c = \lambda \pi r^2 \frac{\partial^2 T_W}{\partial s^2}$

Dissipation radiative : $\dot{Q}_{\sigma,out} = \sigma S \epsilon T_W^4$

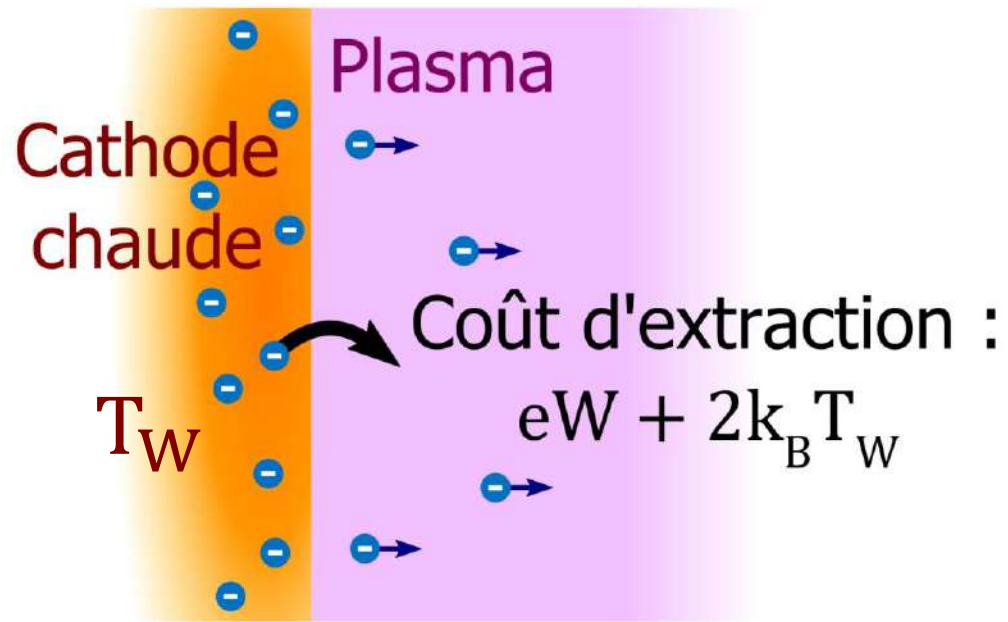
Géométrie spiralaire : $\dot{Q}_{\sigma,in} = C_1 (Q_{\sigma,out}^{N-1} + Q_{\sigma,out}^{N+1})$

Facteur géométrique : $C_1 = 0.15$



Interactions plasma-cathode

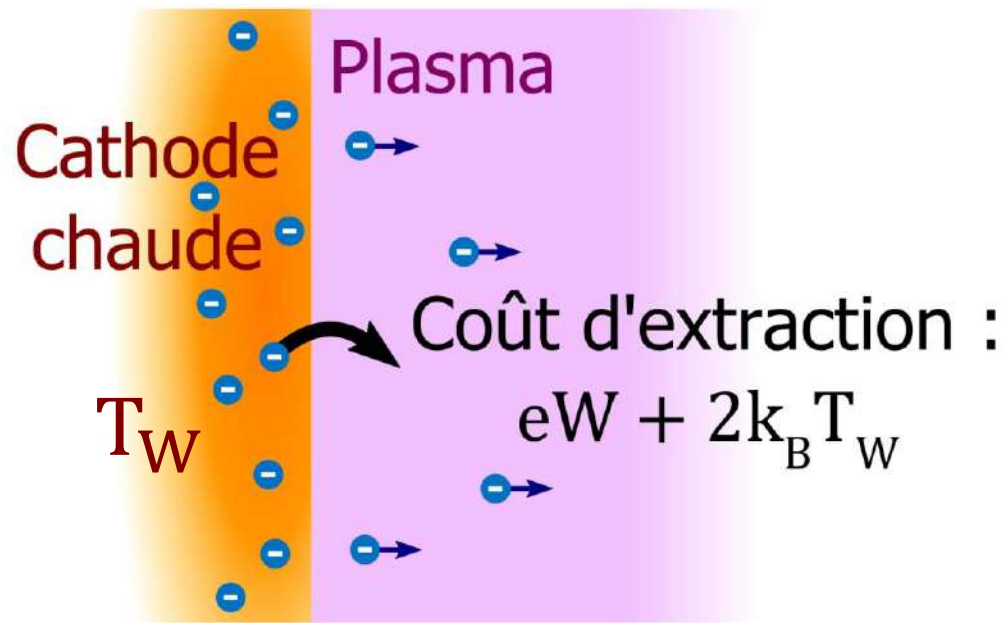
Refroidissement par
émission d'électrons



$$\dot{Q}_e = \frac{I_{em}}{e} (eW + 2k_B T_W)$$

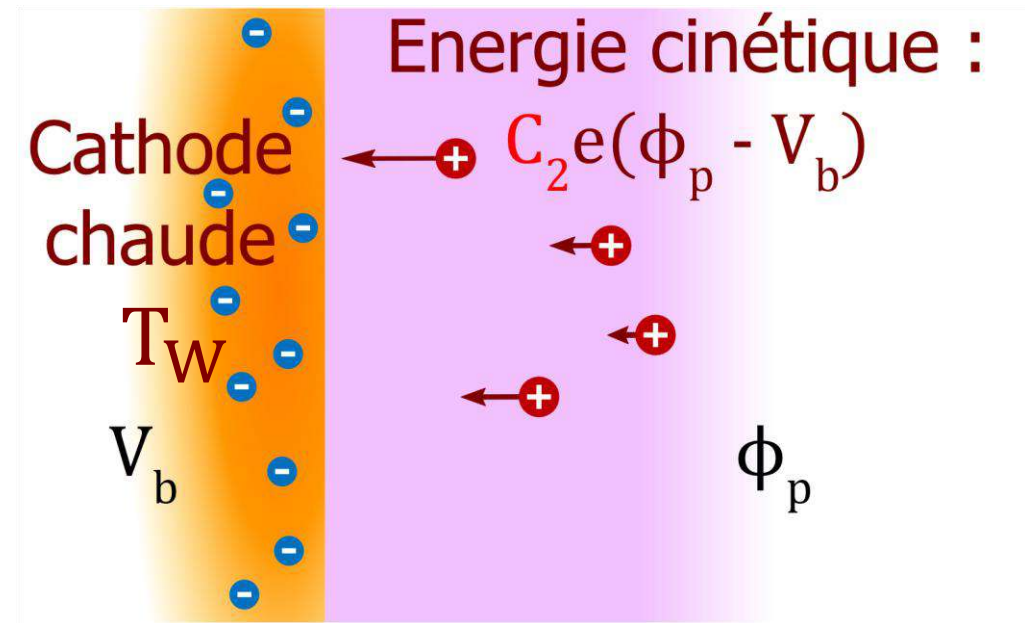
Interactions plasma-cathode

Refroidissement par émission d'électrons



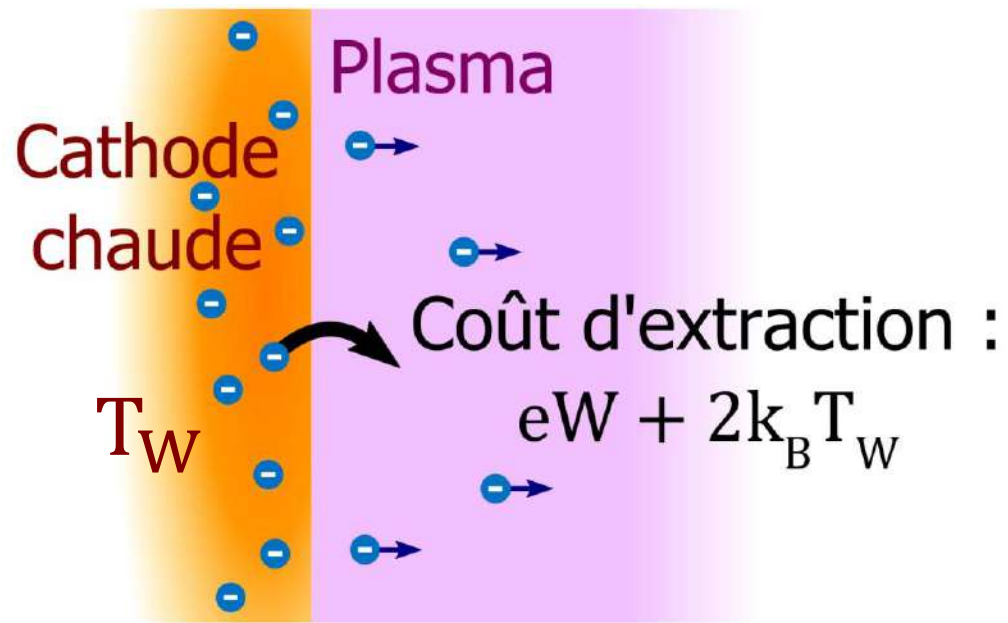
$$\dot{Q}_e = \frac{I_{em}}{e} (eW + 2k_B T_W)$$

Chauffage par bombardement ionique



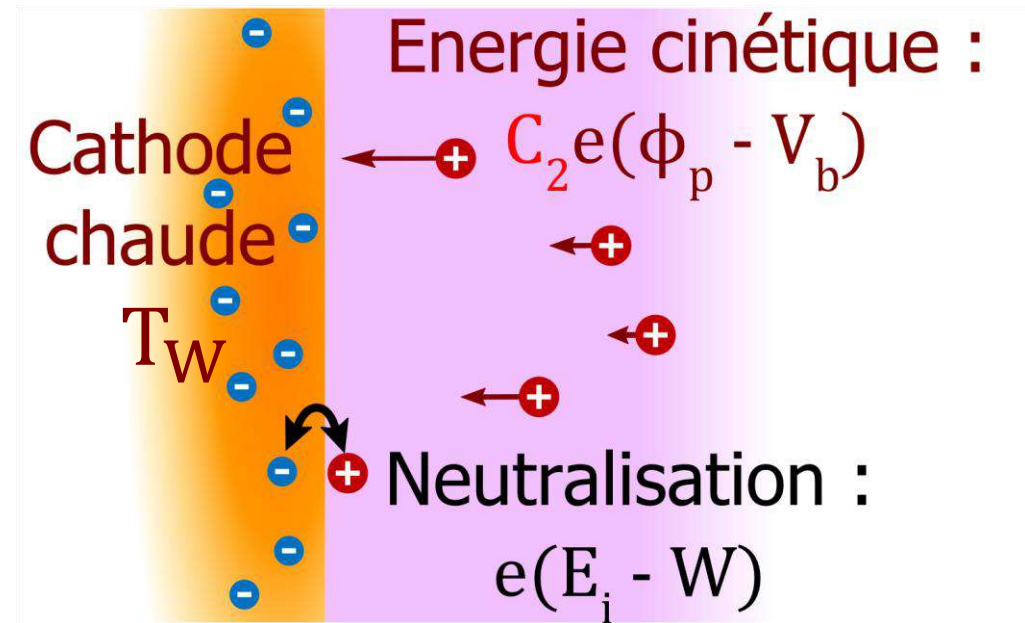
Interactions plasma-cathode

Refroidissement par émission d'électrons



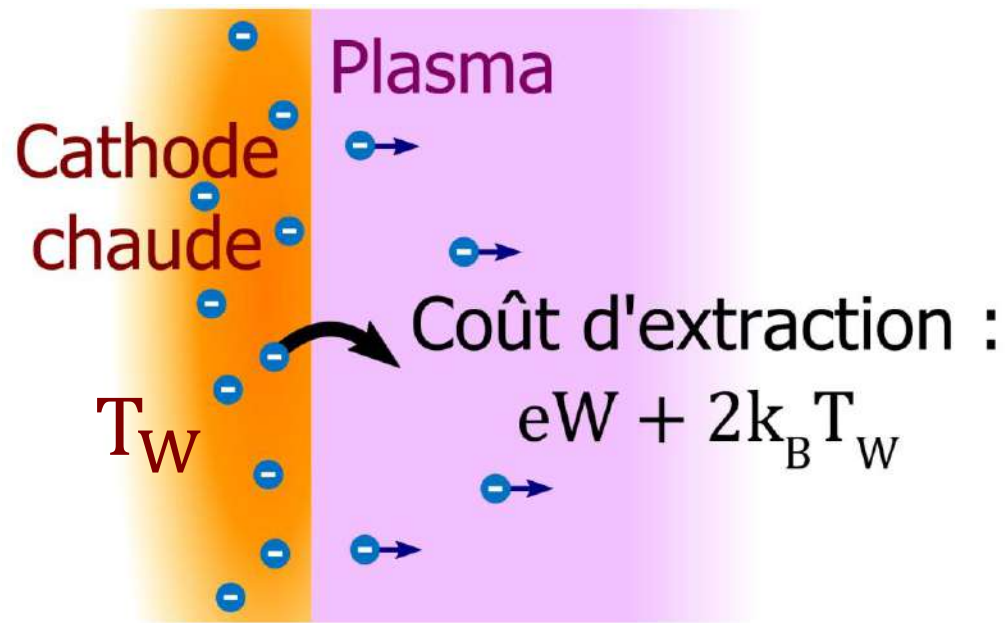
$$\dot{Q}_e = \frac{I_{em}}{e} (eW + 2k_B T_W)$$

Chauffage par bombardement ionique



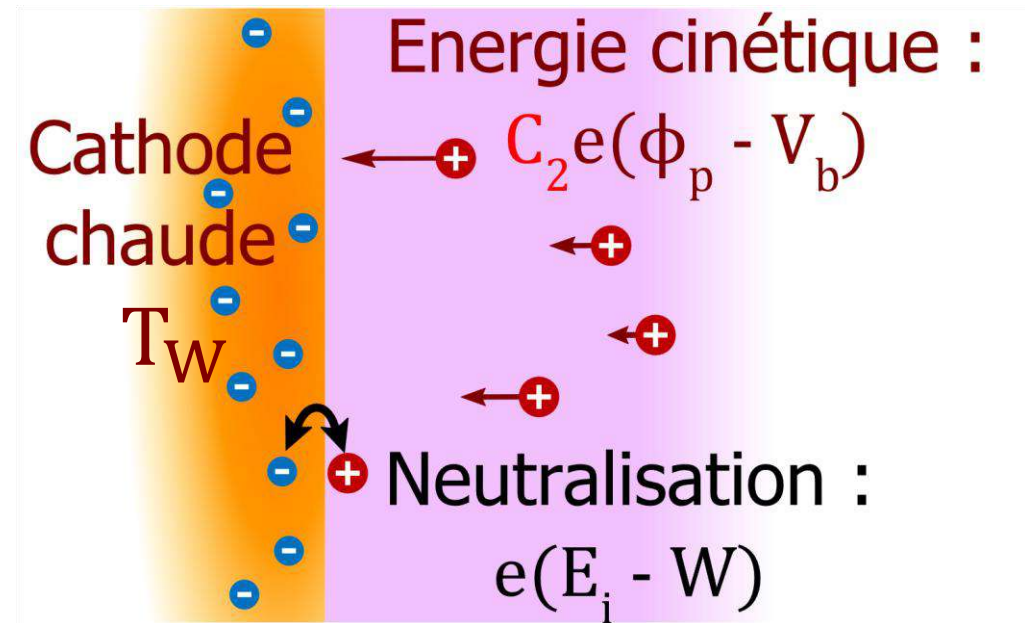
Interactions plasma-cathode

Refroidissement par émission d'électrons



$$\dot{Q}_e = \frac{I_{em}}{e} (eW + 2k_B T_W)$$

Chauffage par bombardement ionique

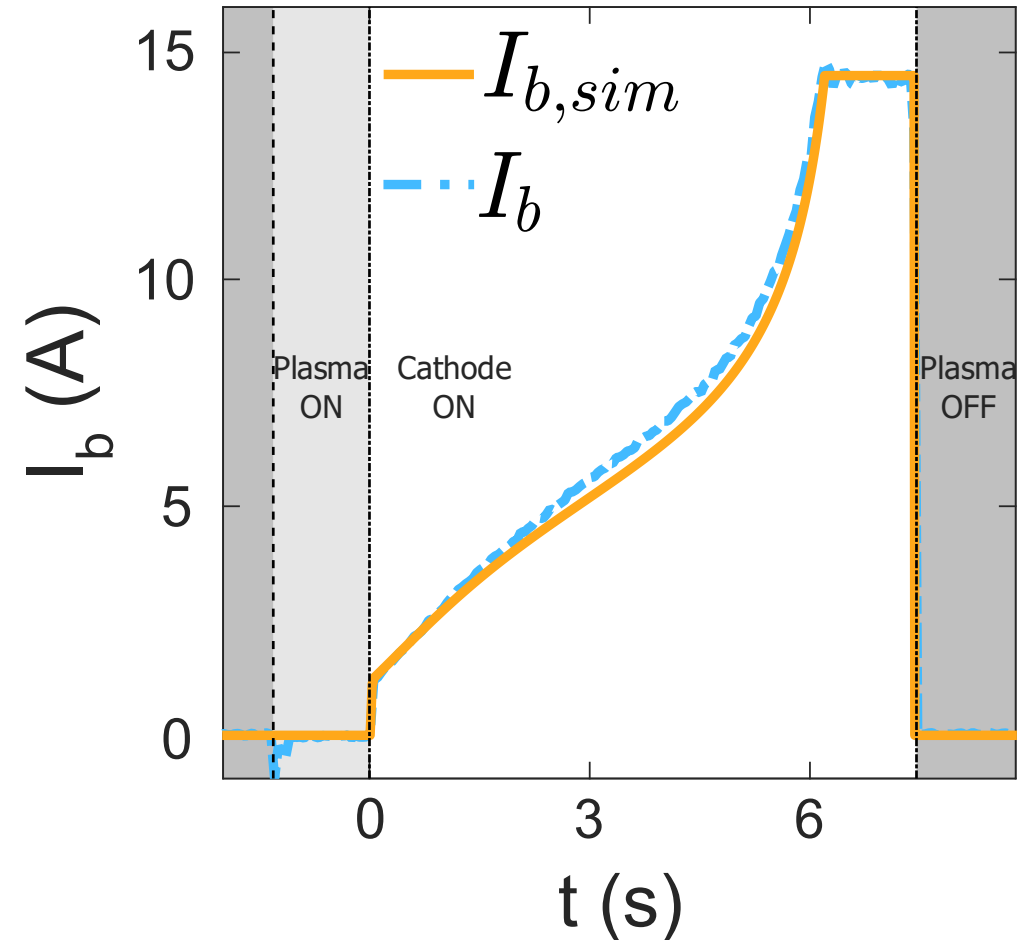


$$\dot{Q}_i = \frac{I_i}{e} e [E_i - W + C_2 (\phi_p - V_b)]$$

Résolution du modèle thermique

Résolution numérique : $\dot{H} = \dot{Q}_\Omega + \dot{Q}_c - \dot{Q}_{\sigma,out} + \dot{Q}_{\sigma,in} + \dot{Q}_i - \dot{Q}_e$

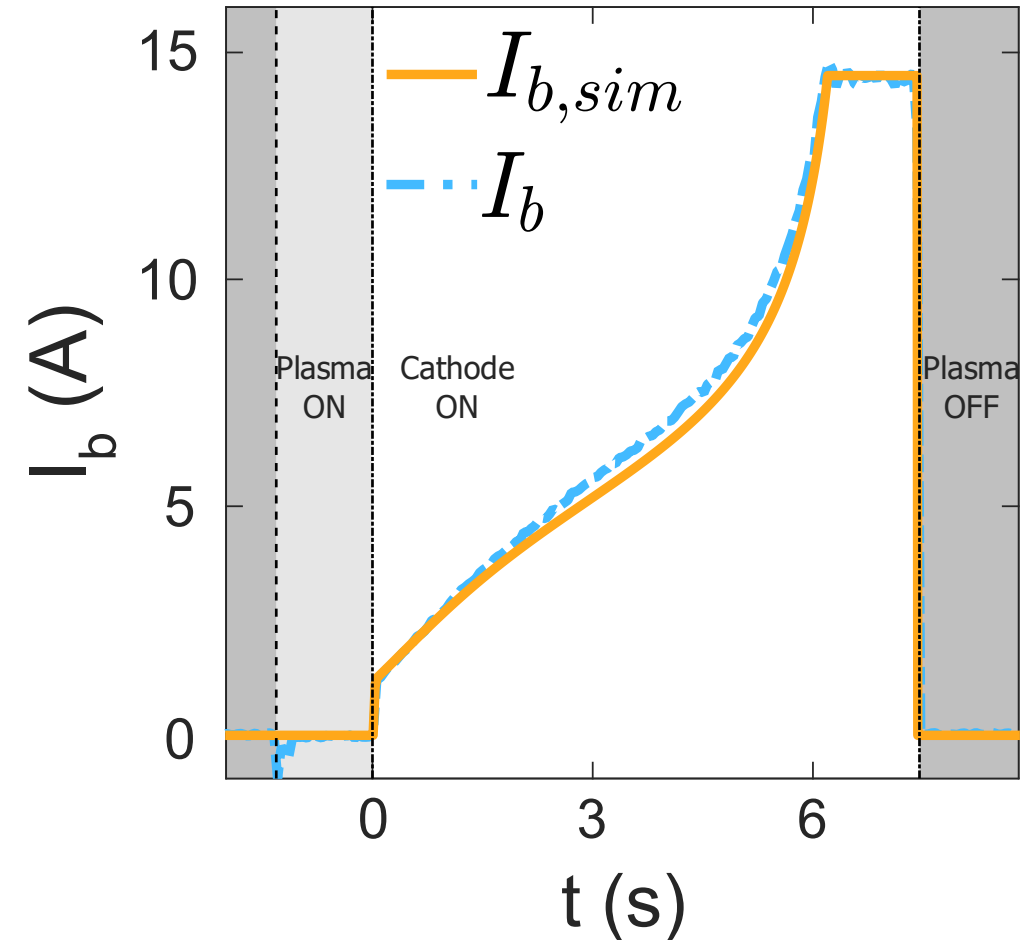
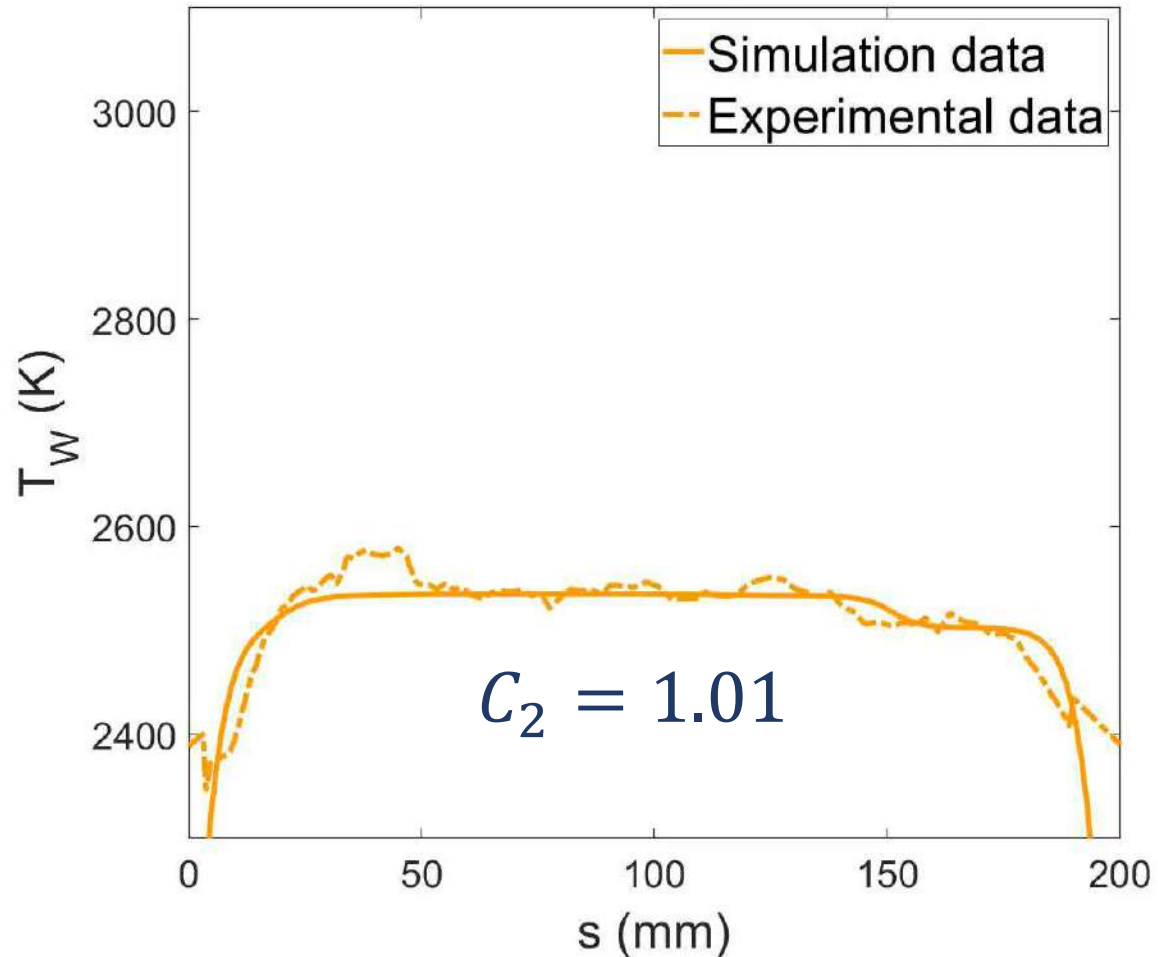
$$C_2 = 1.01$$



Résolution du modèle thermique

Résolution numérique : $\dot{H} = \dot{Q}_\Omega + \dot{Q}_c - \dot{Q}_{\sigma,out} + \dot{Q}_{\sigma,in} + \dot{Q}_i - \dot{Q}_e$

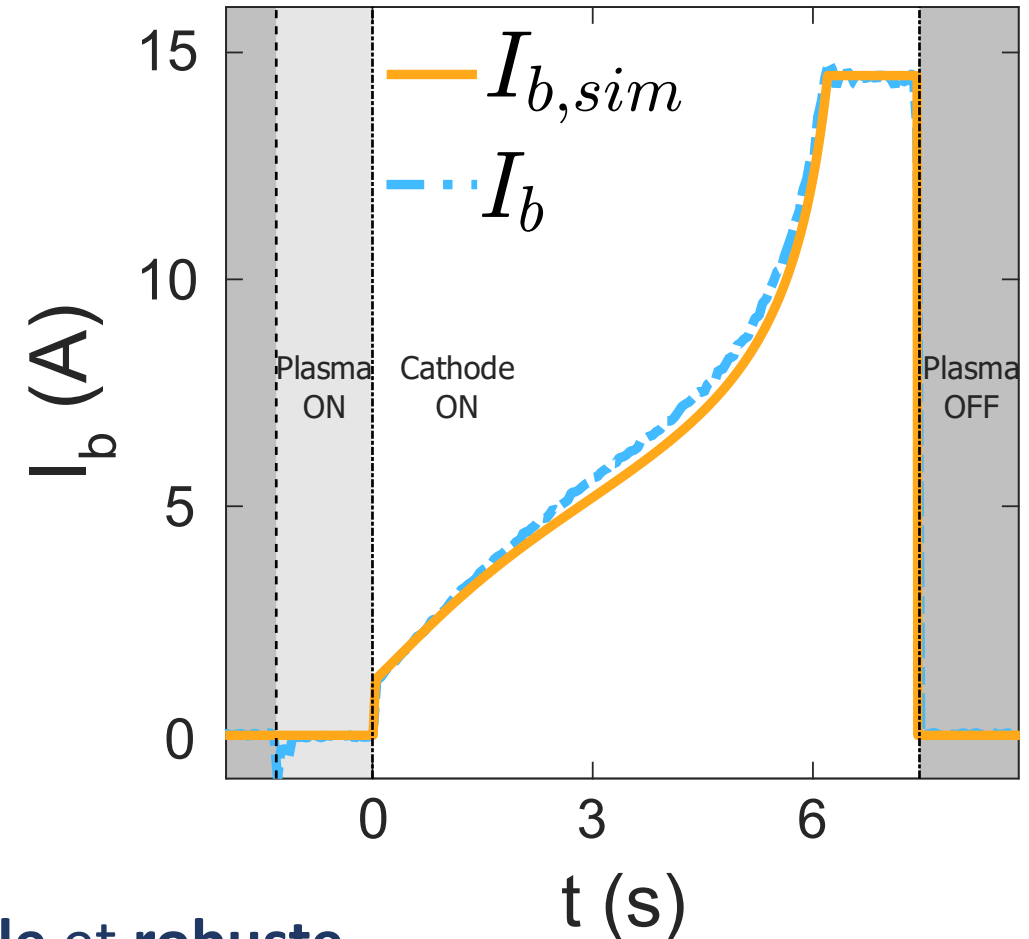
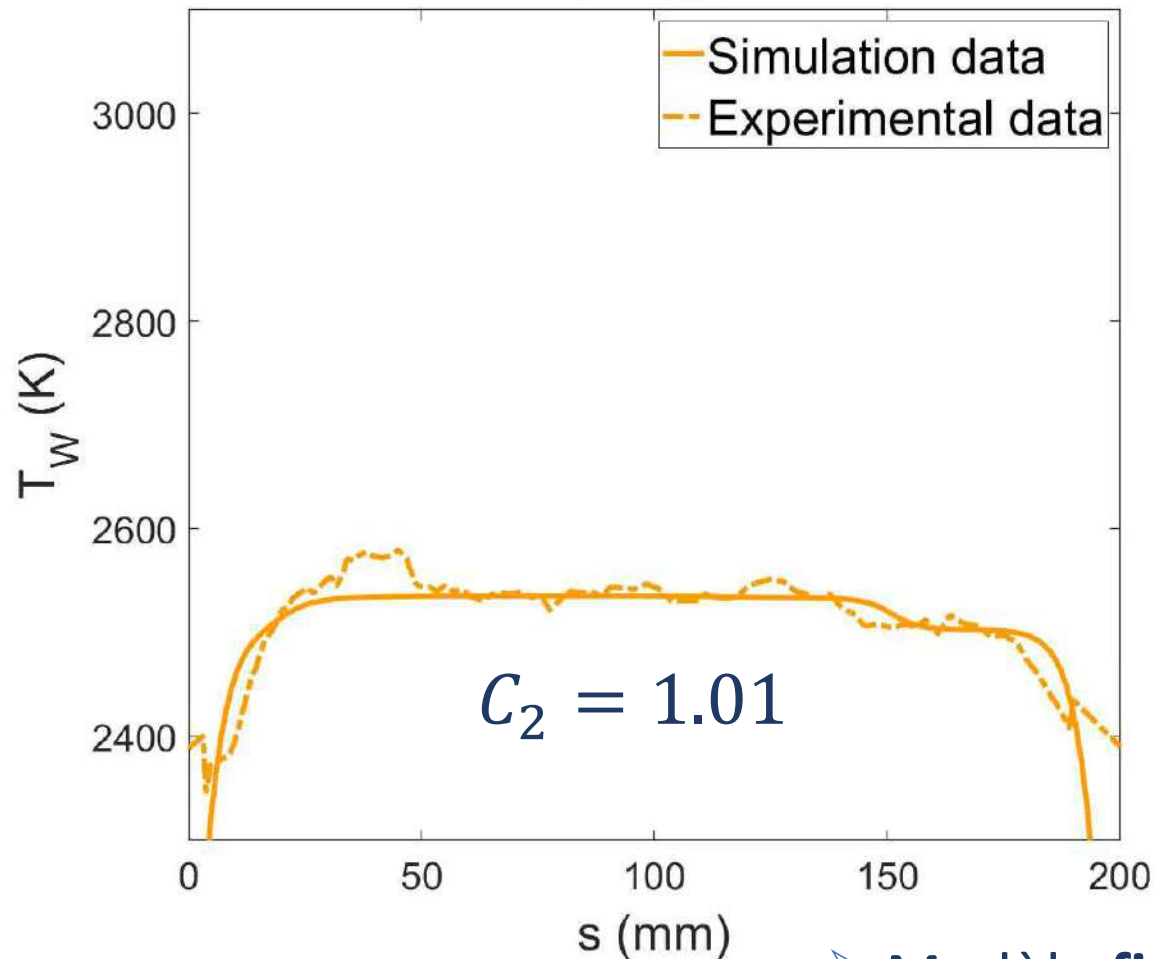
$t = -1.0 \text{ s}$



Résolution du modèle thermique

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$t = -1.0 \text{ s}$



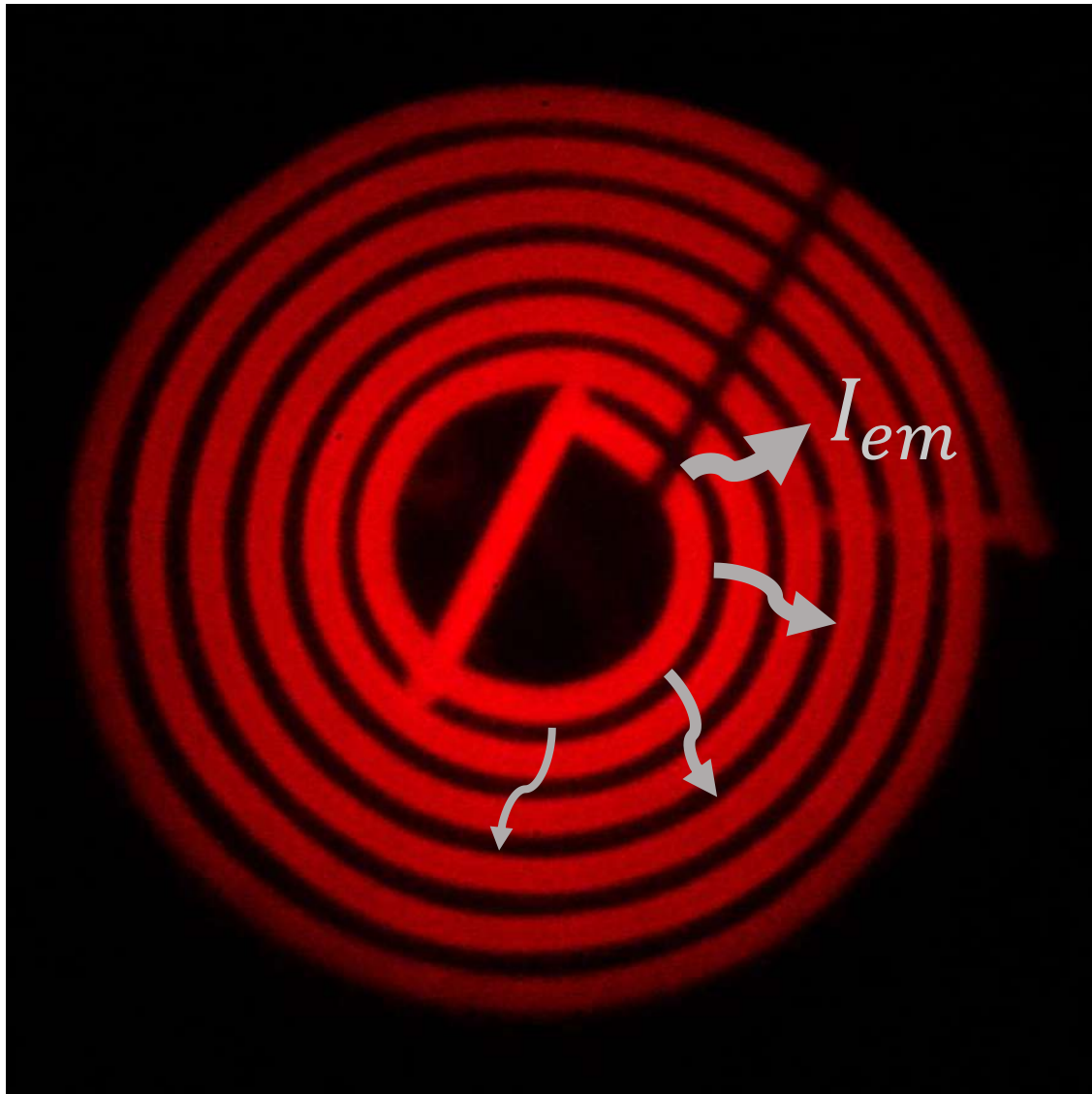
➤ Modèle **fidèle** et **robuste**

➤ Raffinements : modèle amélioré, expériences plus précises

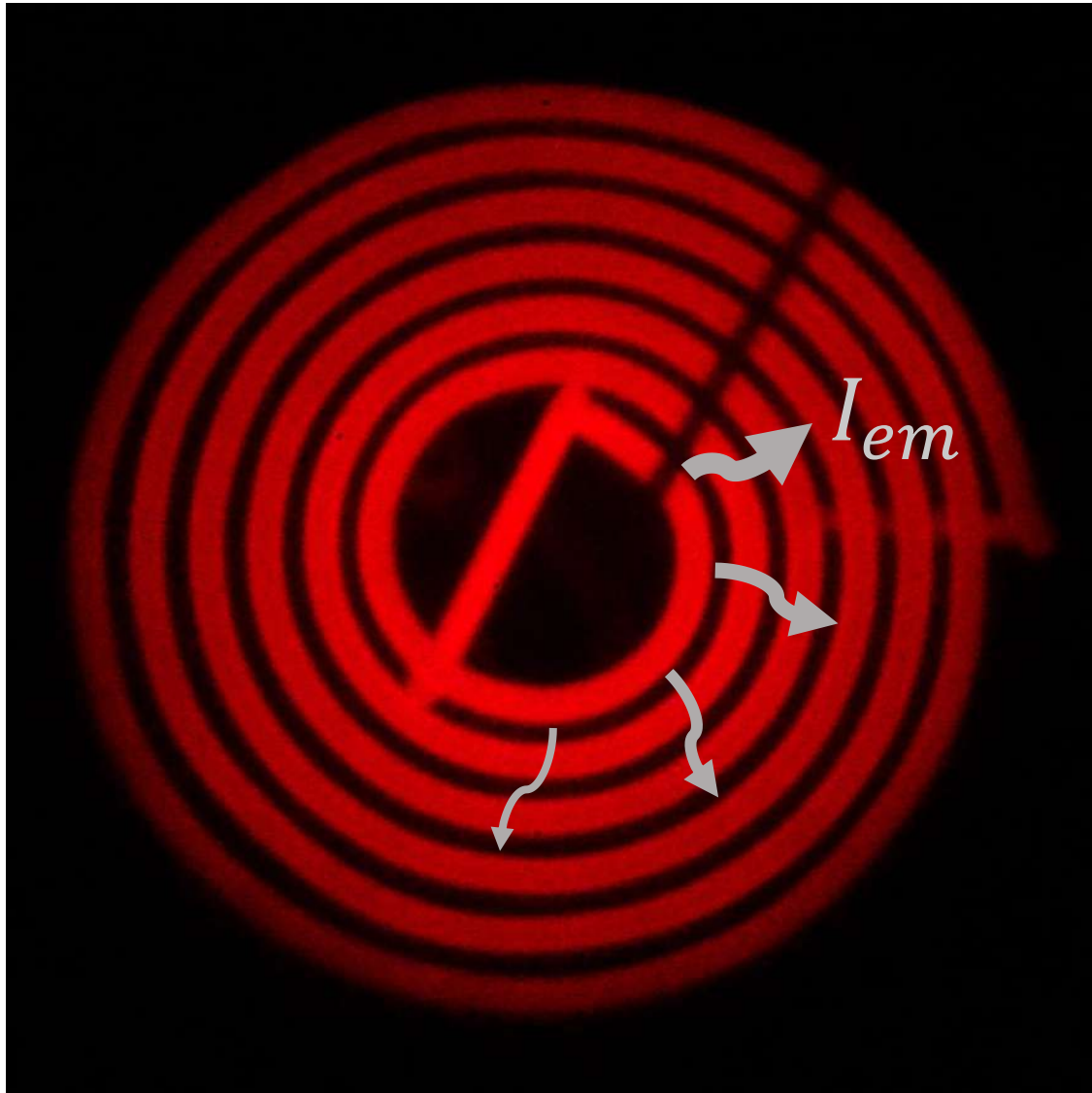
Rôles des interactions plasma-cathode



Rôles des interactions plasma-cathode



Rôles des interactions plasma-cathode

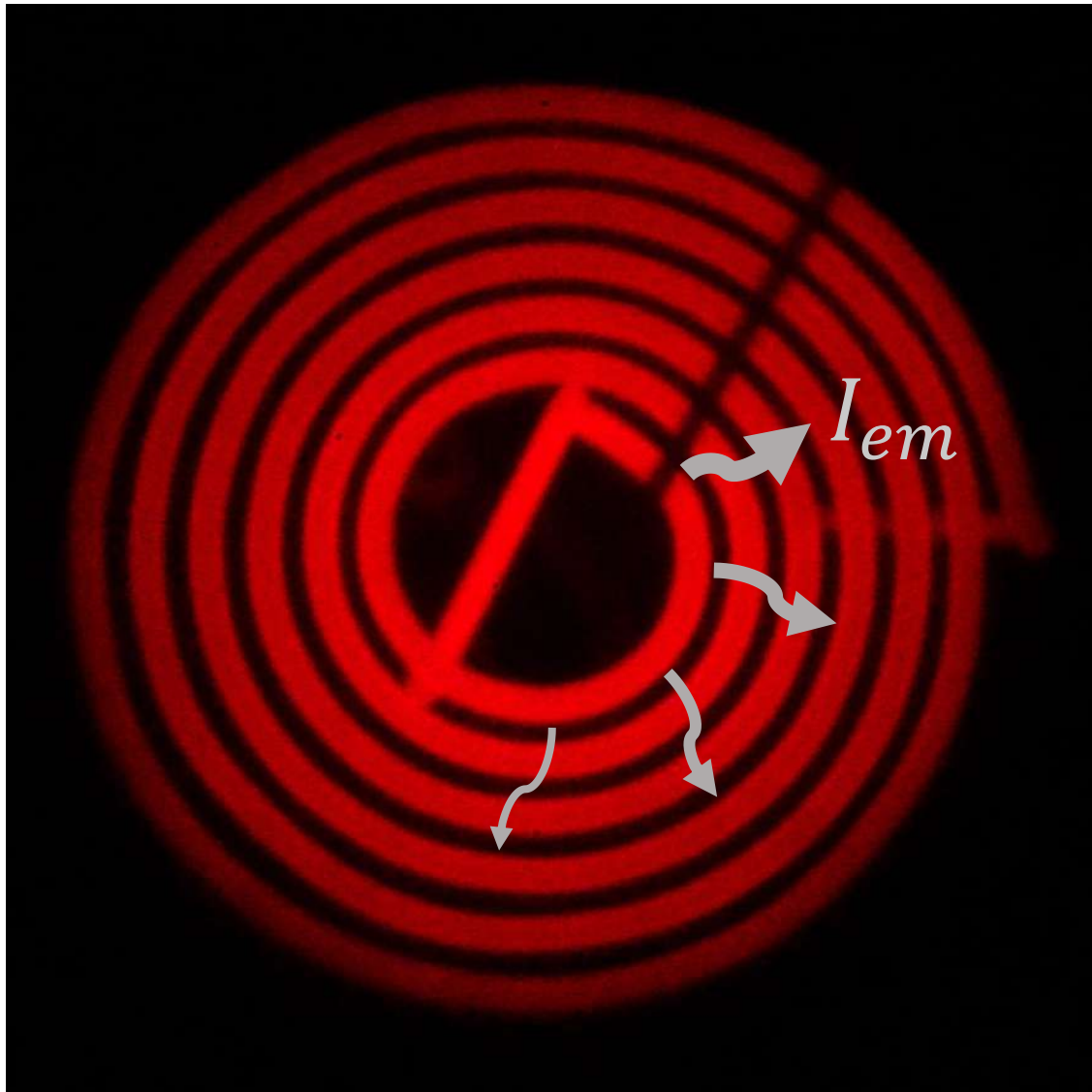


Chauffage par effet Joule :

$$\dot{Q}_{\Omega} = R \left(I_h + \int_s^{L_W} i_{em}(u) du \right)^2$$

➤ Chauffage hétérogène, terme non-linéaire

Rôles des interactions plasma-cathode



Chauffage par effet Joule :

$$\dot{Q}_{\Omega} = R \left(I_h + \int_s^{L_W} i_{em}(u) du \right)^2$$

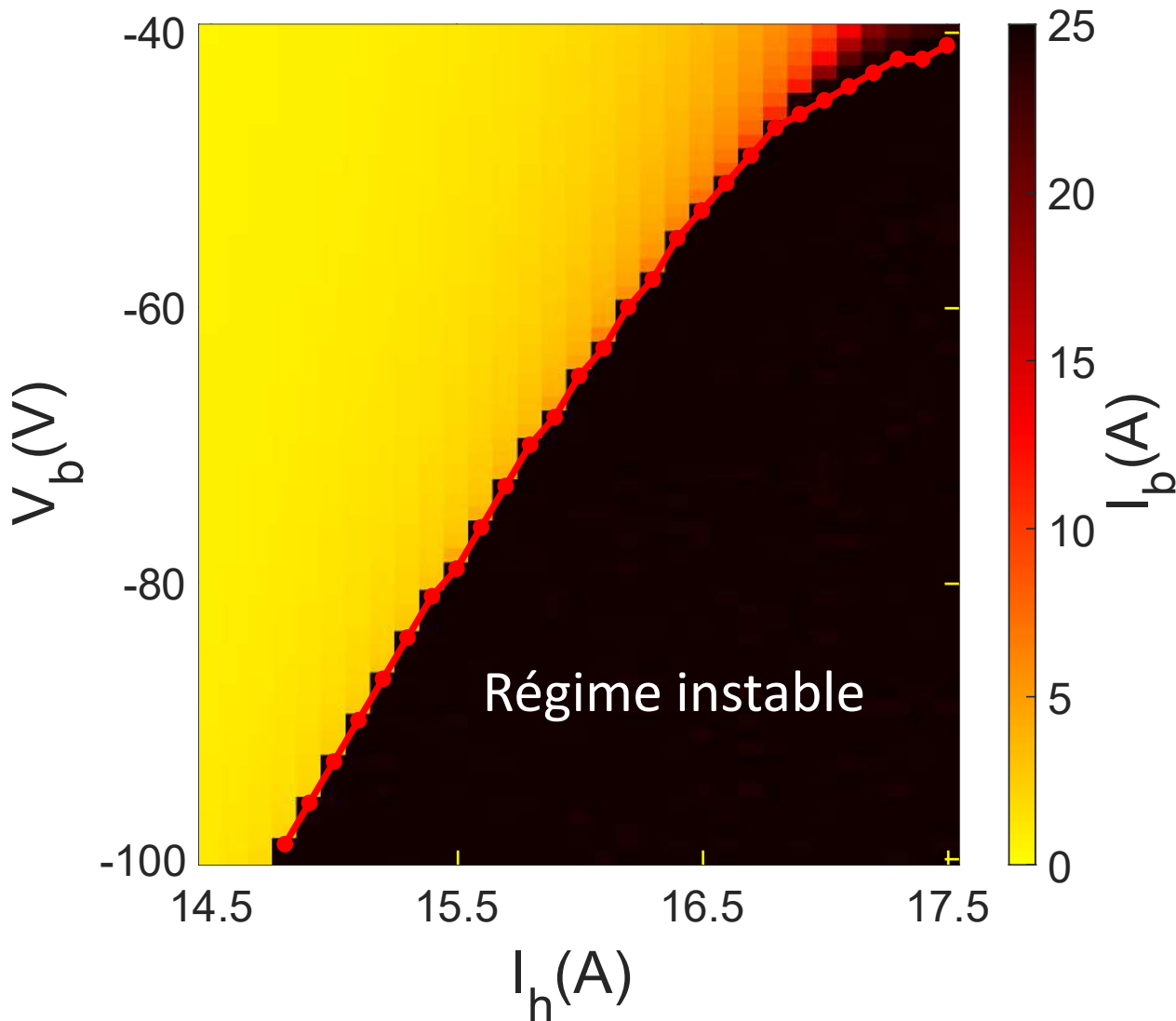
➤ Chauffage hétérogène, terme non-linéaire

Bombardement ionique :

$$\dot{Q}_i = I_i [E_i - W + C_2(\phi_p - V_b)]$$

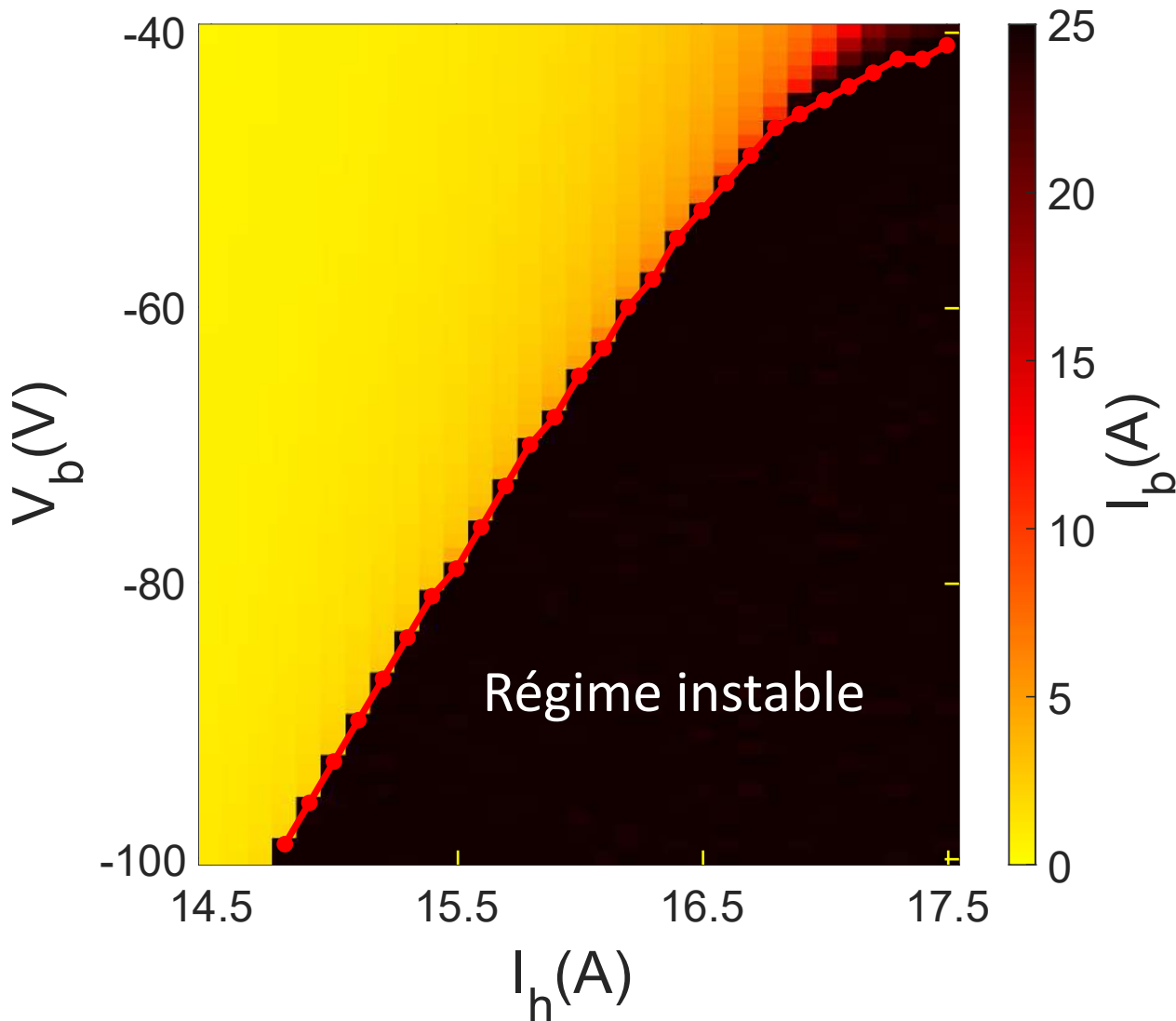
➤ Influence de V_b + couplage avec le plasma
 ➤ 5% du chauffage

Prédiction de I_b et domaine de stabilité



- Simulations dans le plan $(I_h; V_b)$
- Prédiction de I_b fiable
- Instabilité pouvant endommager la cathode

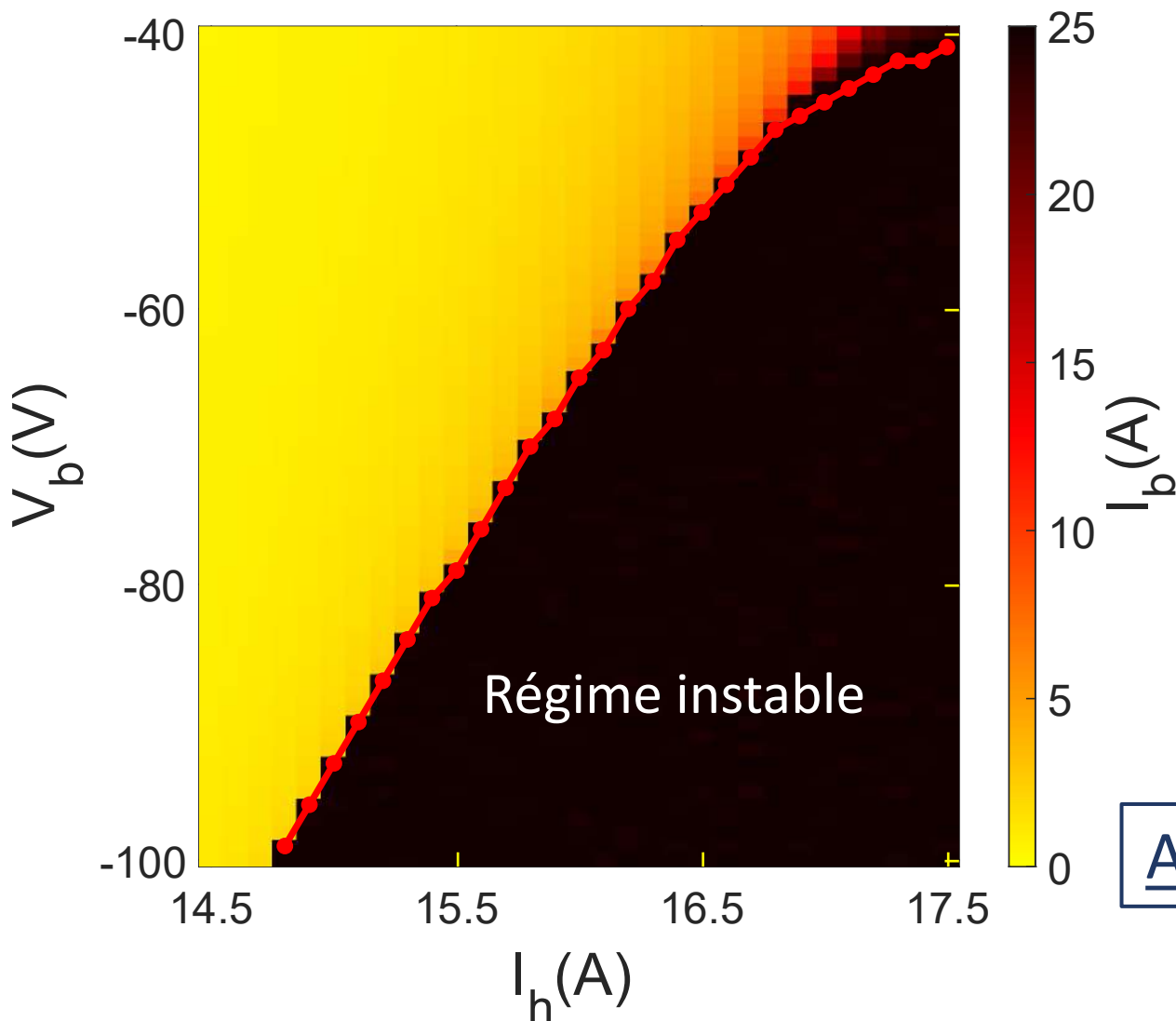
Prédiction de I_b et domaine de stabilité



- Simulations dans le plan ($I_h; V_b$)
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*Article: Pagaud et al.,
Plasma Sources Sci. Technol. (2023)*

Prédiction de I_b et domaine de stabilité

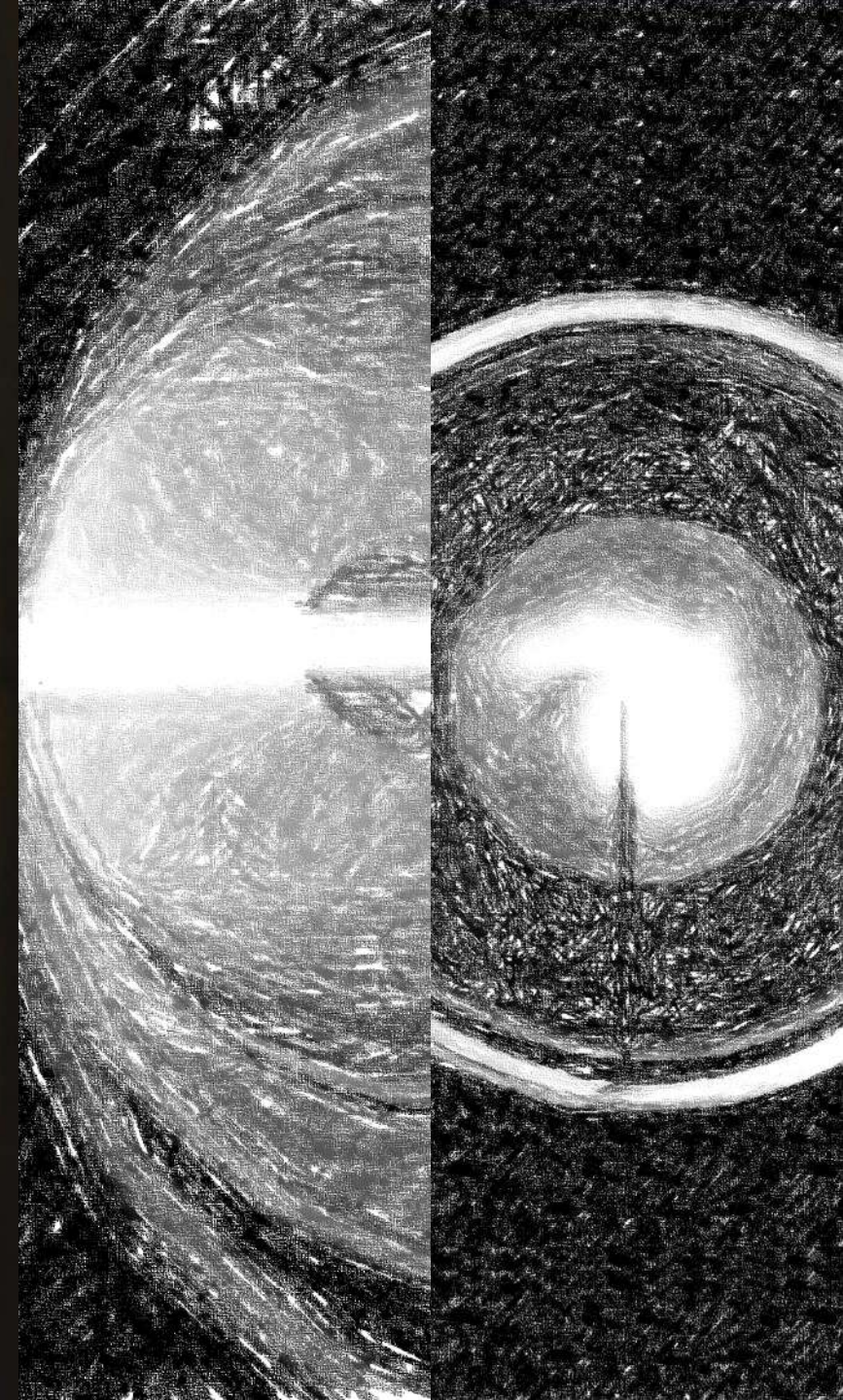


- Simulations dans le plan $(I_h; V_b)$
- Prédiction de I_b fiable
- Instabilité pouvant endommager la cathode

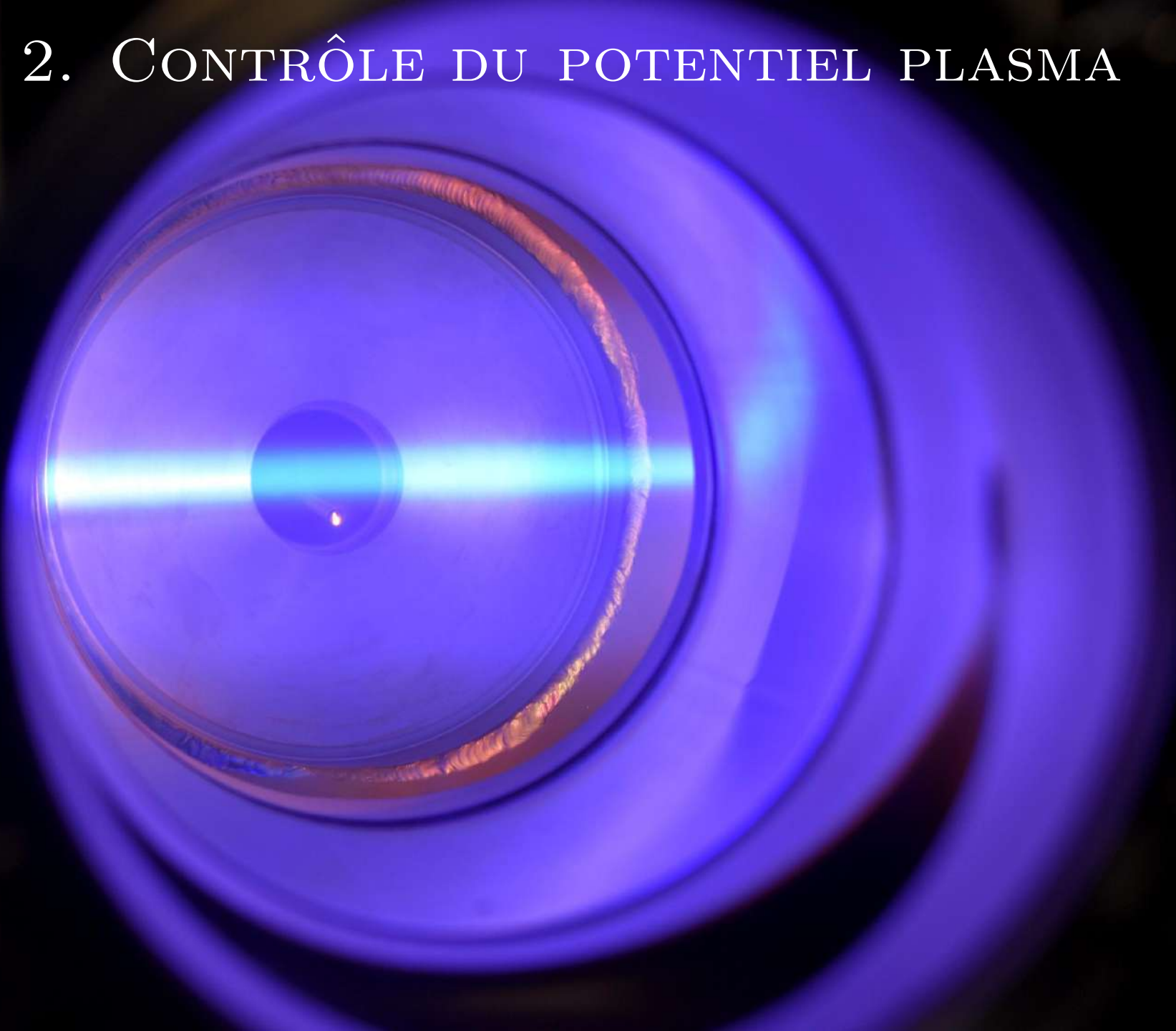
*Article: Pagaud et al.,
Plasma Sources Sci. Technol. (2023)*

Application : Contrôle du potentiel plasma

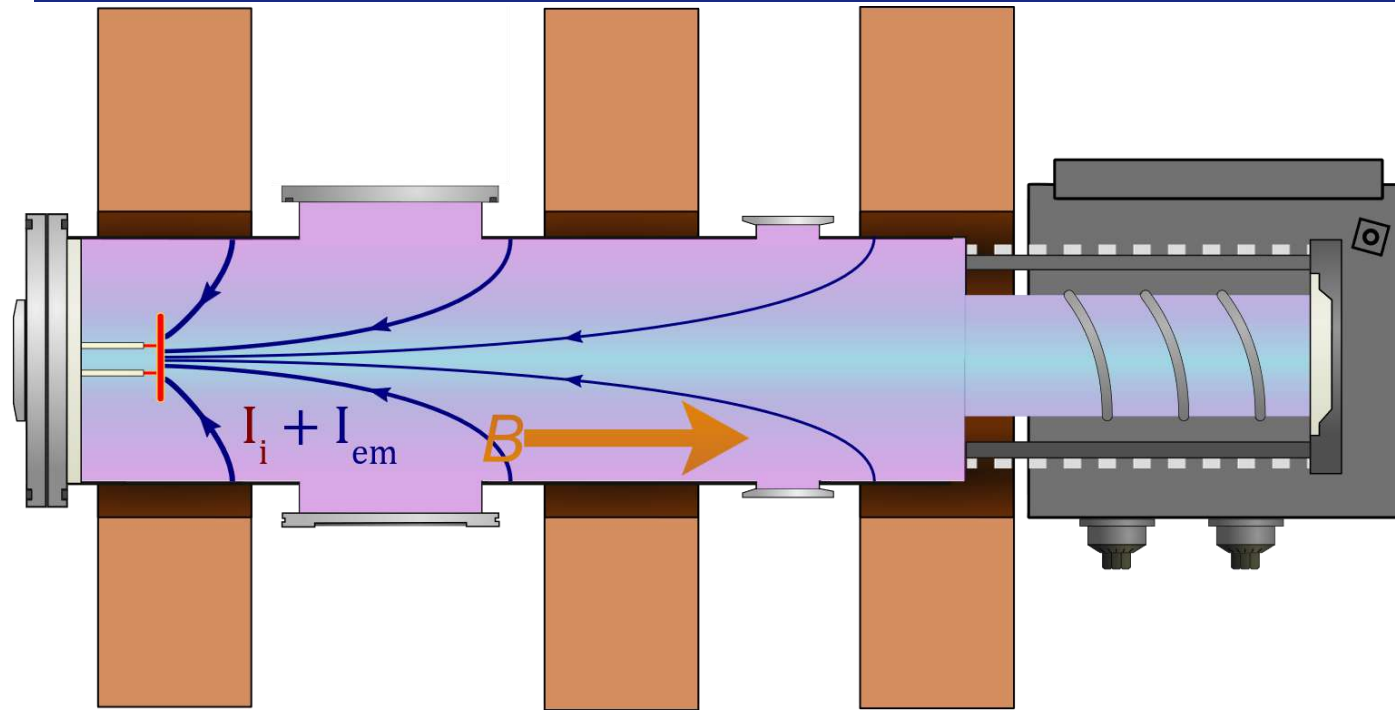
1. OPÉRATION D'UNE CATHODE ÉMISSIVE



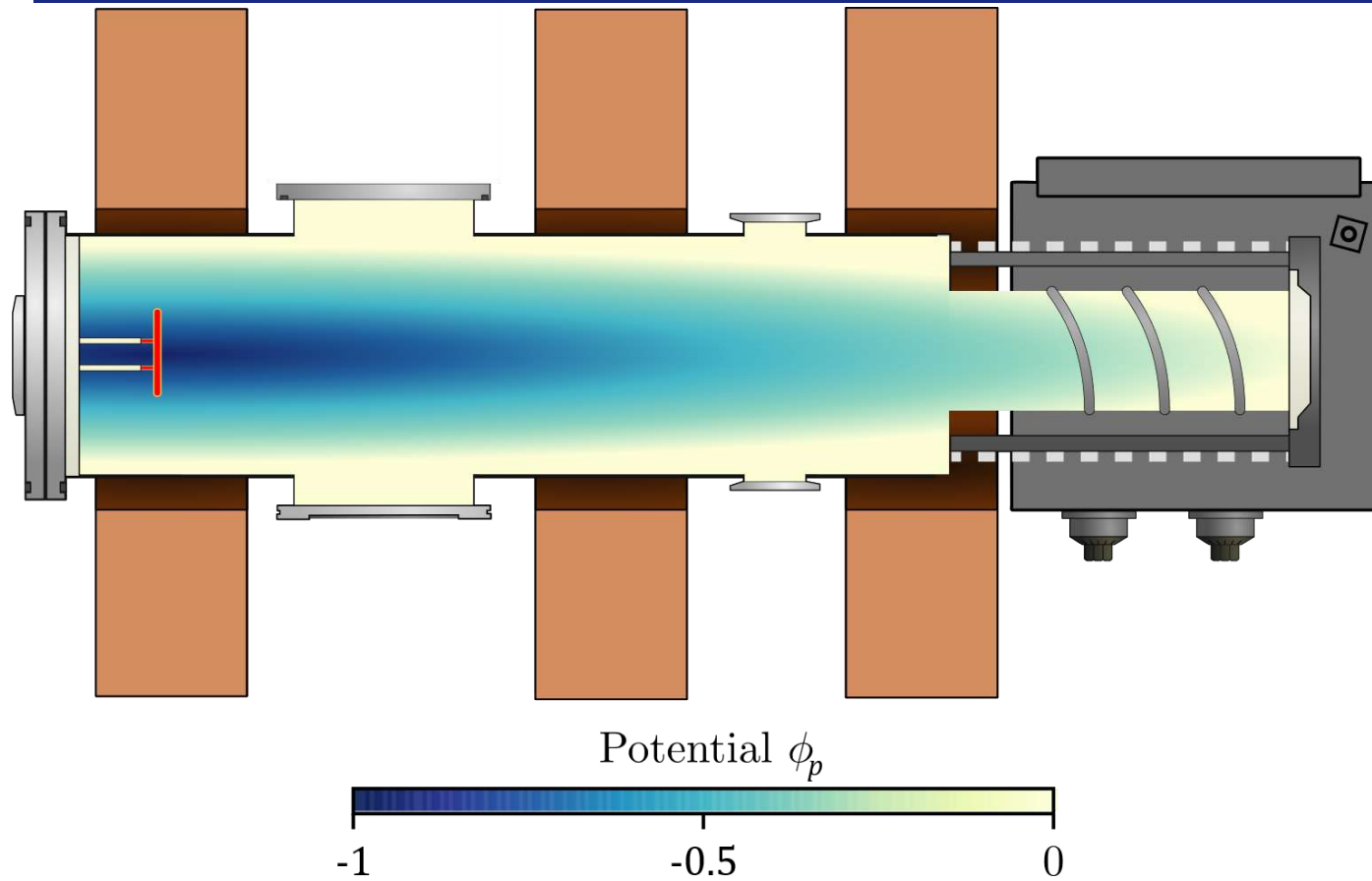
2. CONTRÔLE DU POTENTIEL PLASMA



Abaissement du potentiel plasma

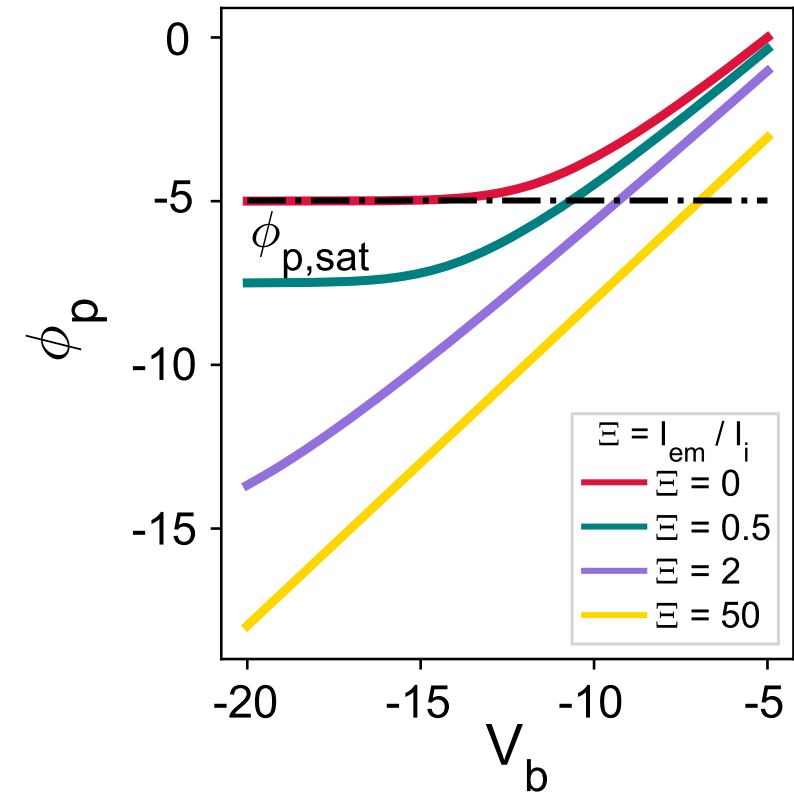


Abaissement du potentiel plasma

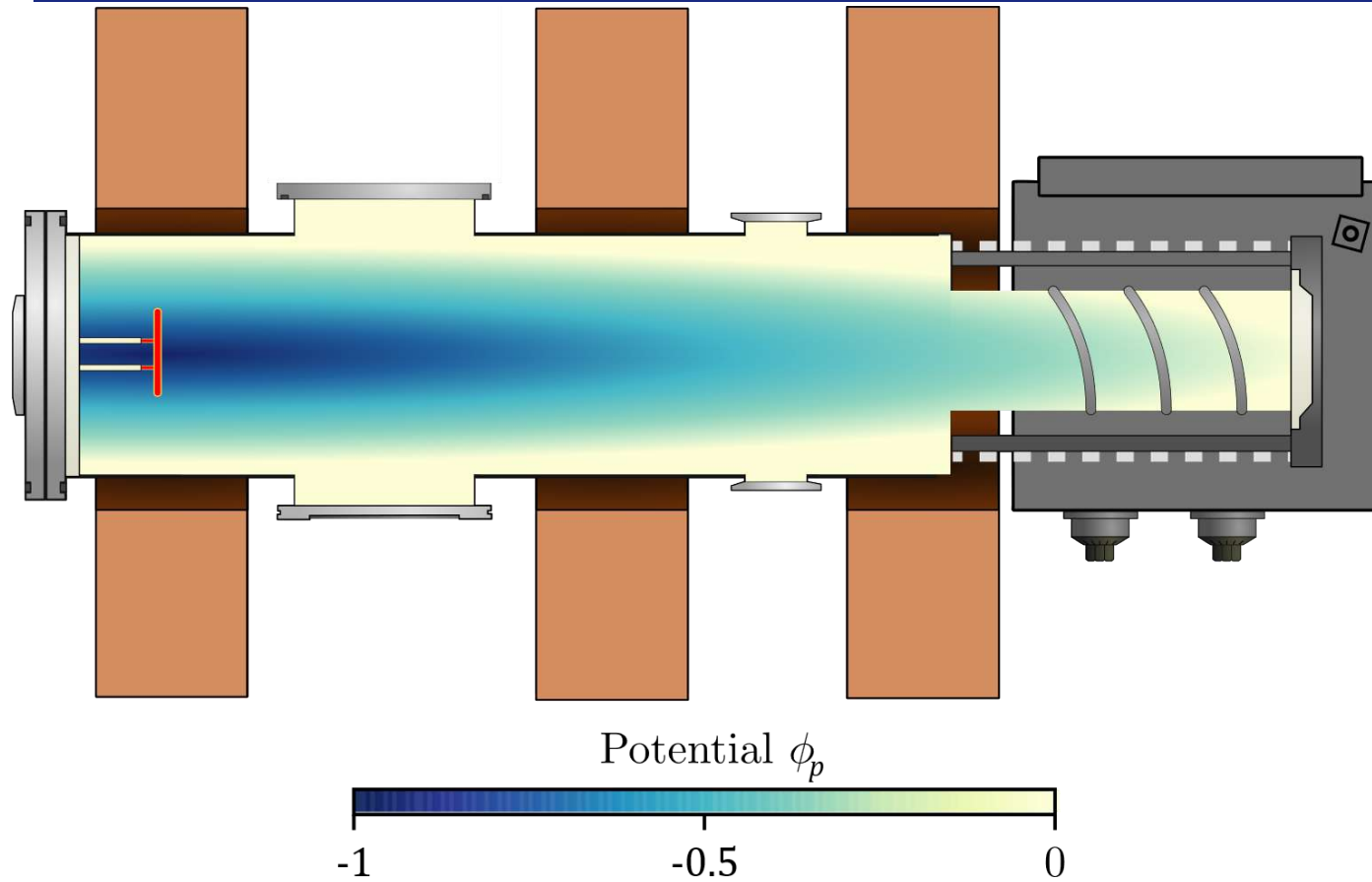


Conductivité anisotrope :

$$\Delta_{\perp} \phi_p \gg \Delta_{\parallel} \phi_p$$

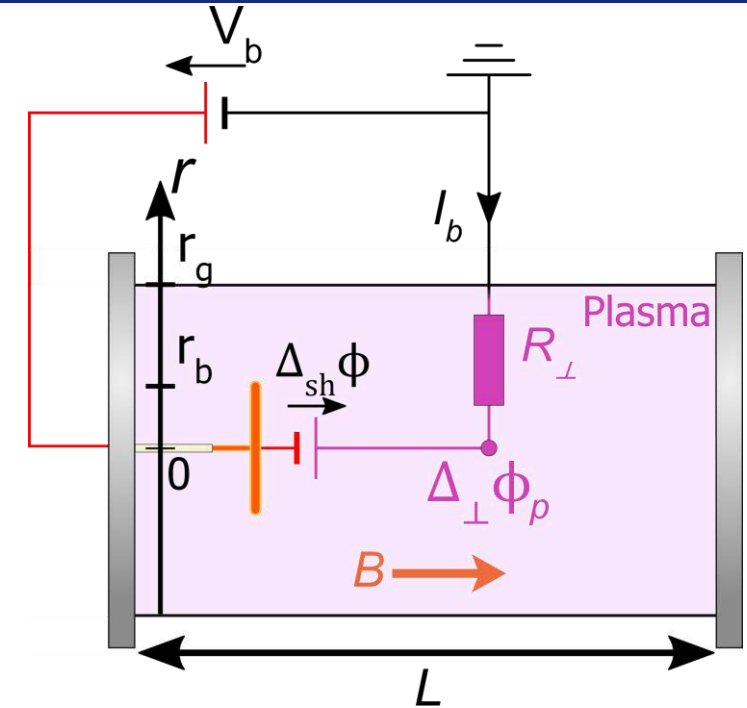


Abaissement du potentiel plasma



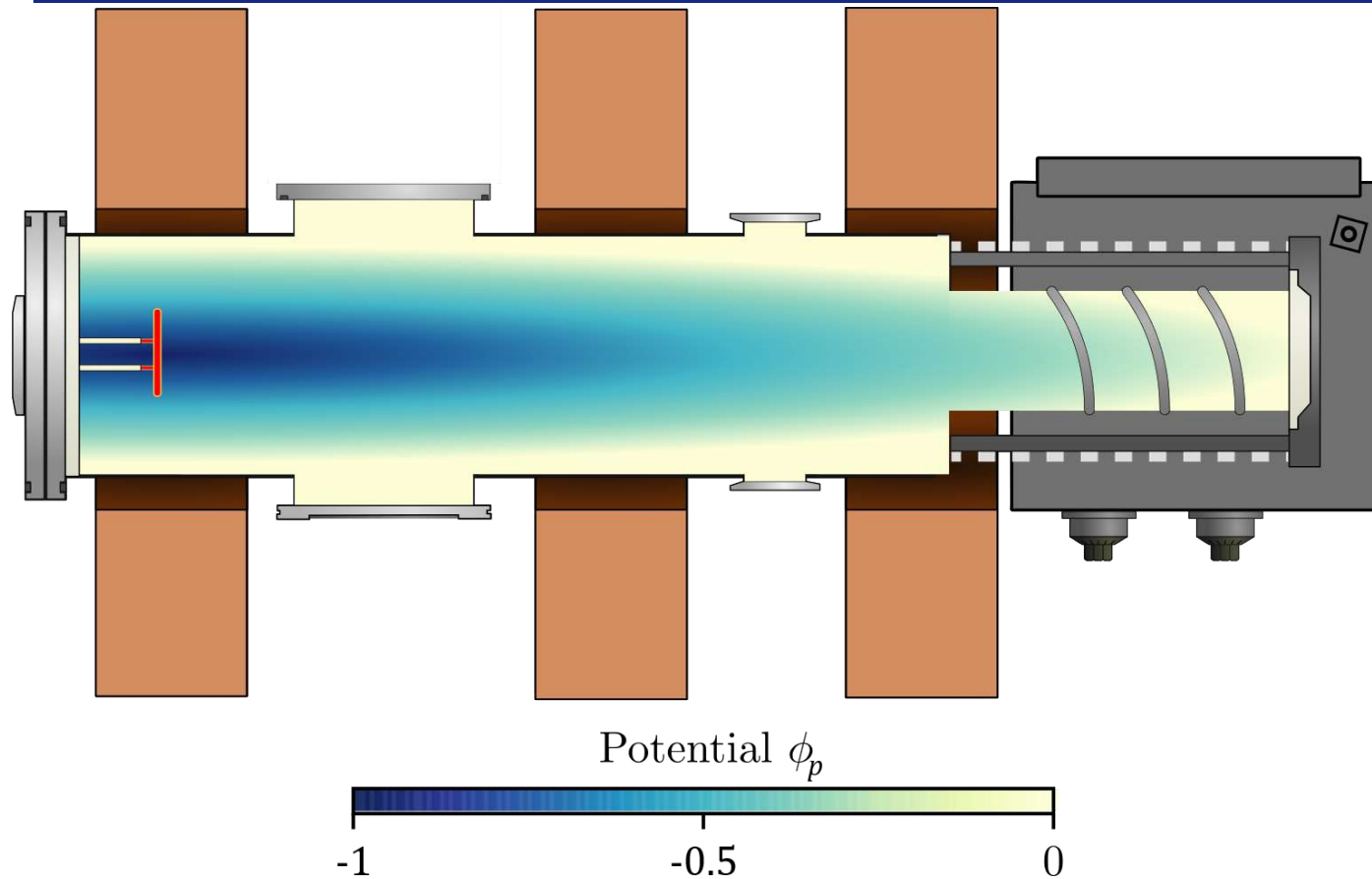
Conductivité anisotrope :

$$\Delta_{\perp} \phi_p \gg \Delta_{\parallel} \phi_p$$



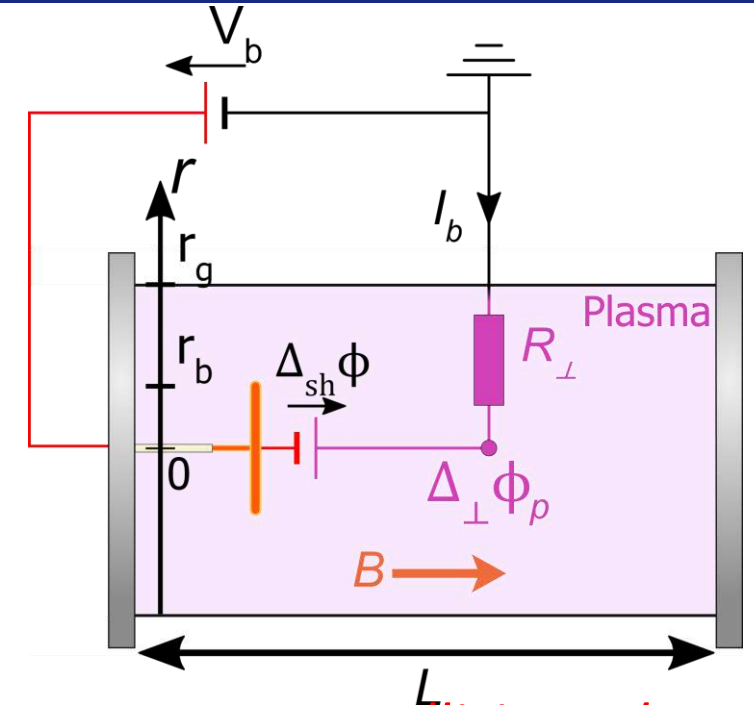
$$R_{\perp} = \int_{r_b}^{r_g} \frac{\eta_{\perp}(r)}{2\pi r L} dr$$

Abaissement du potentiel plasma



Conductivité anisotrope :

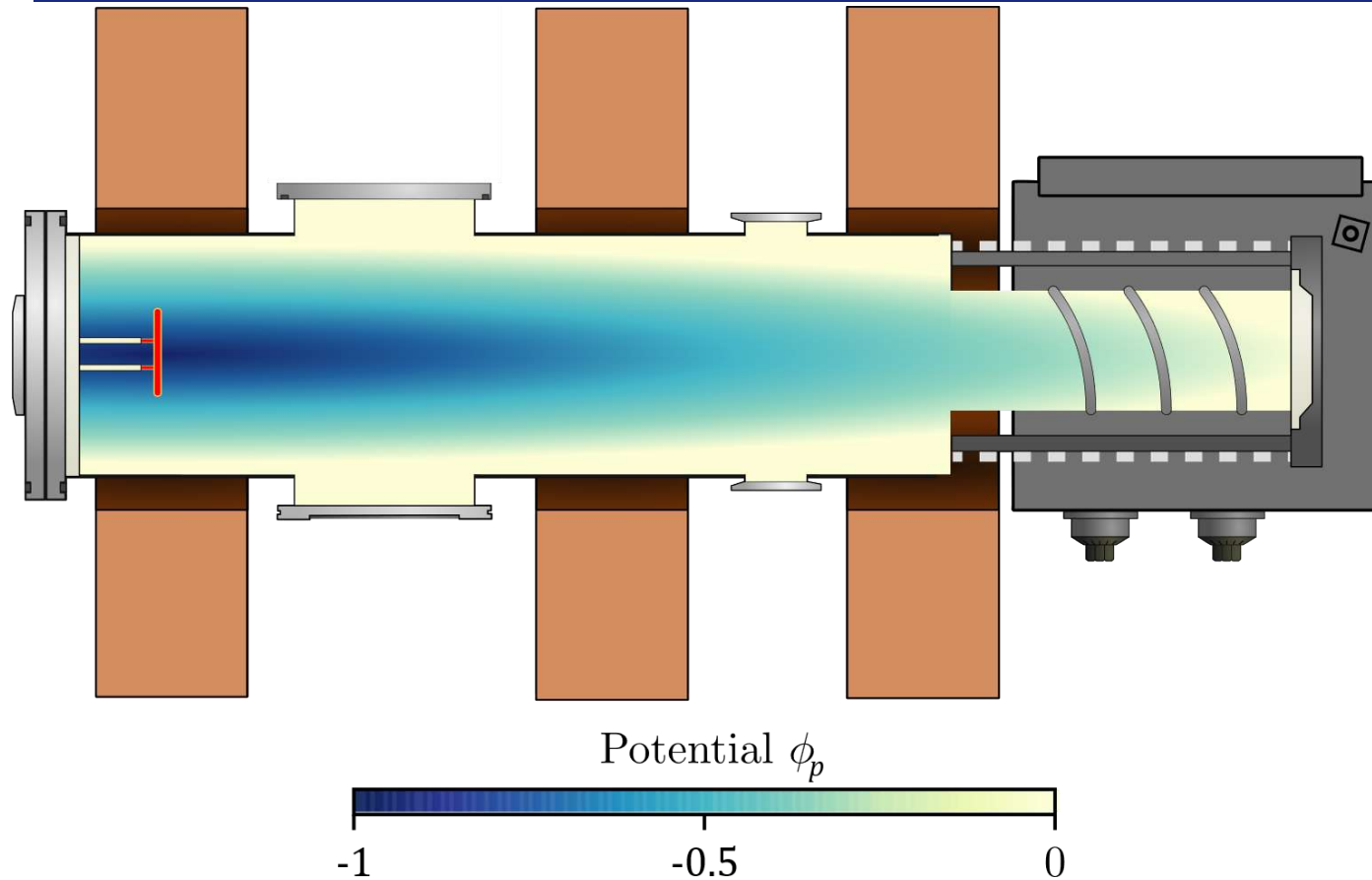
$$\Delta_{\perp} \phi_p \gg \Delta_{\parallel} \phi_p$$



Processus collisionnels : $\omega_{c,i}/\nu_{in}$

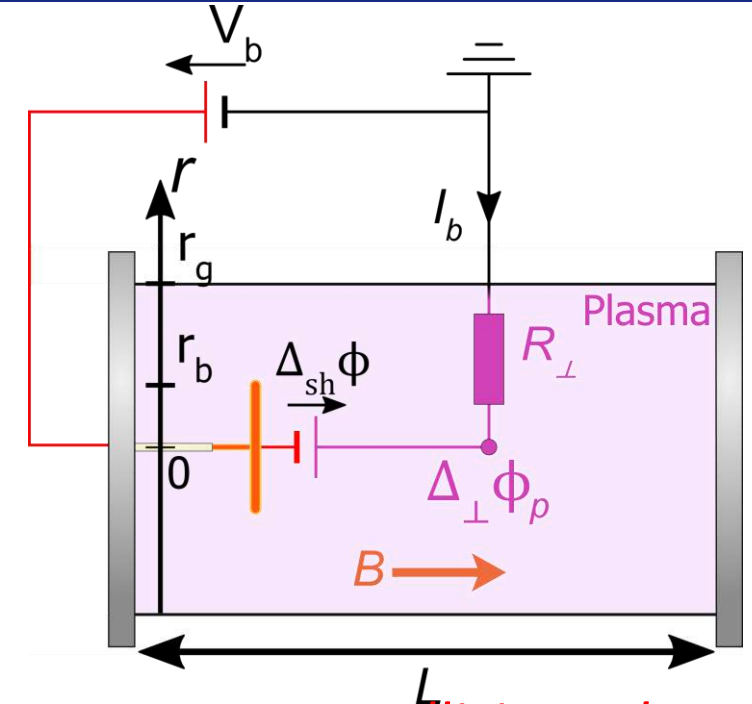
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Abaissement du potentiel plasma



Conductivité anisotrope :

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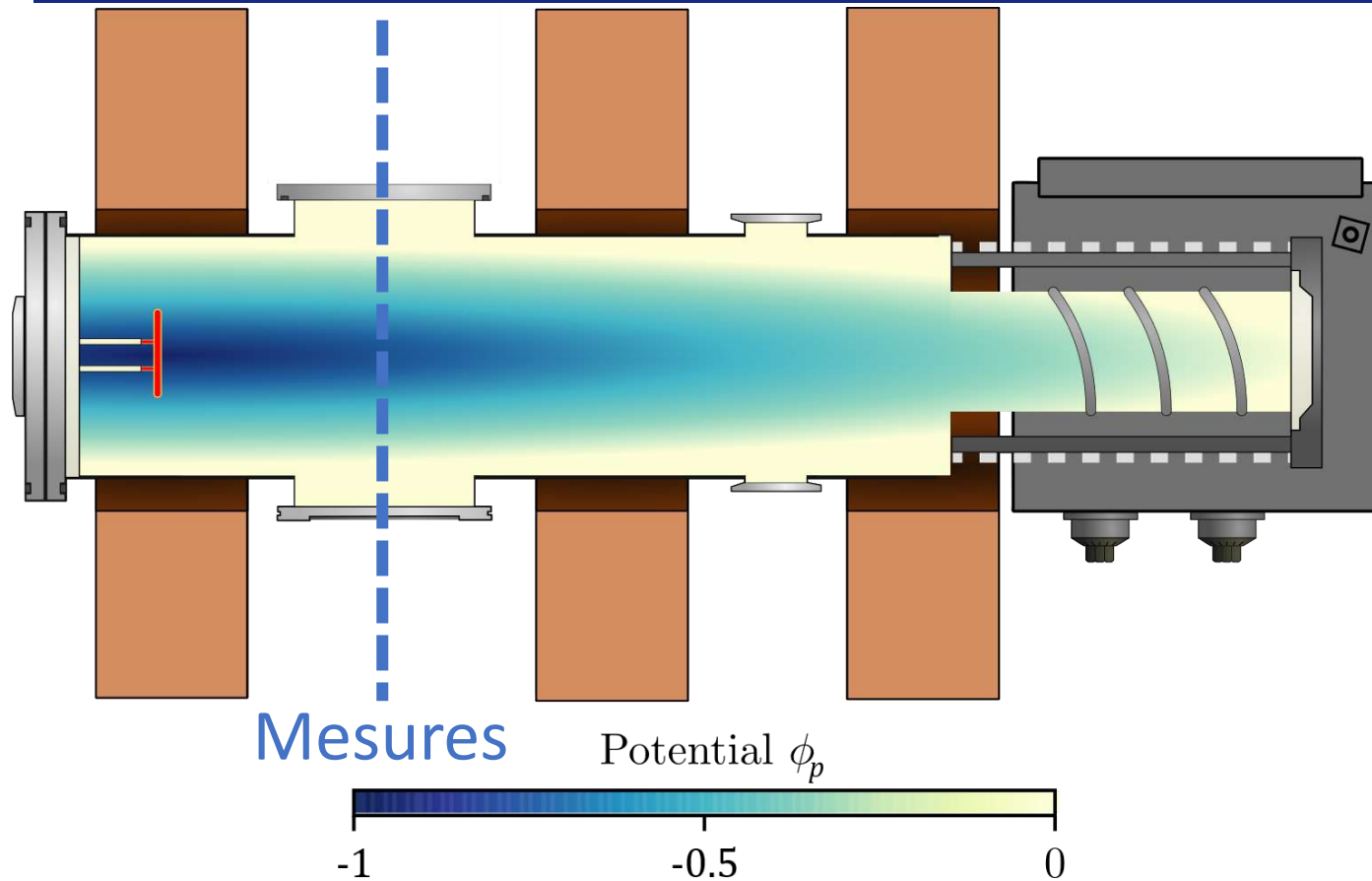
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$$R_{\perp} = \int_{r_b}^{r_g} \frac{\eta_{\perp}(r)}{2\pi r L} dr$$

$$\Delta_{\perp} \phi_p = R_{\perp} I_b$$

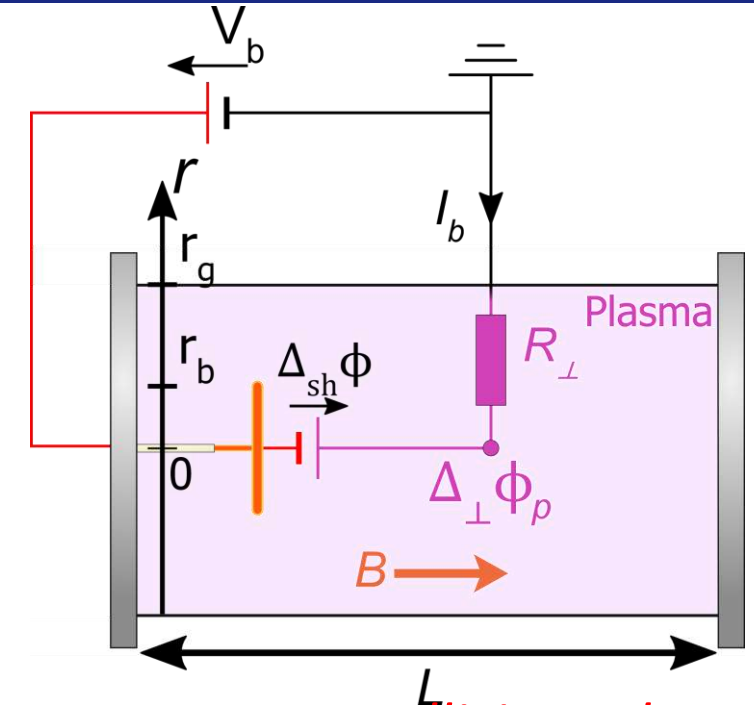
Collaboration : B. Trotabas, R. Gueroult,
LAPLACE (Toulouse) 97

Abaissement du potentiel plasma



Conductivité anisotrope :

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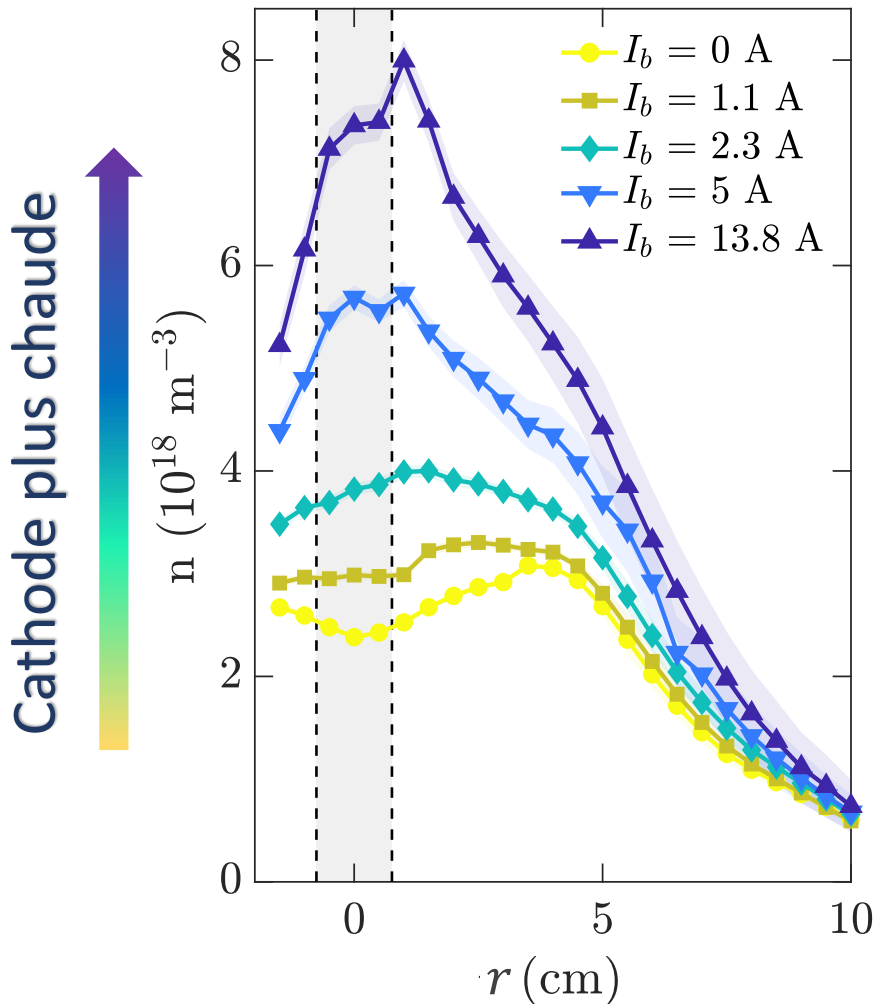
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Collaboration : B. Trotabas, R. Guerout,
LAPLACE (Toulouse) 98

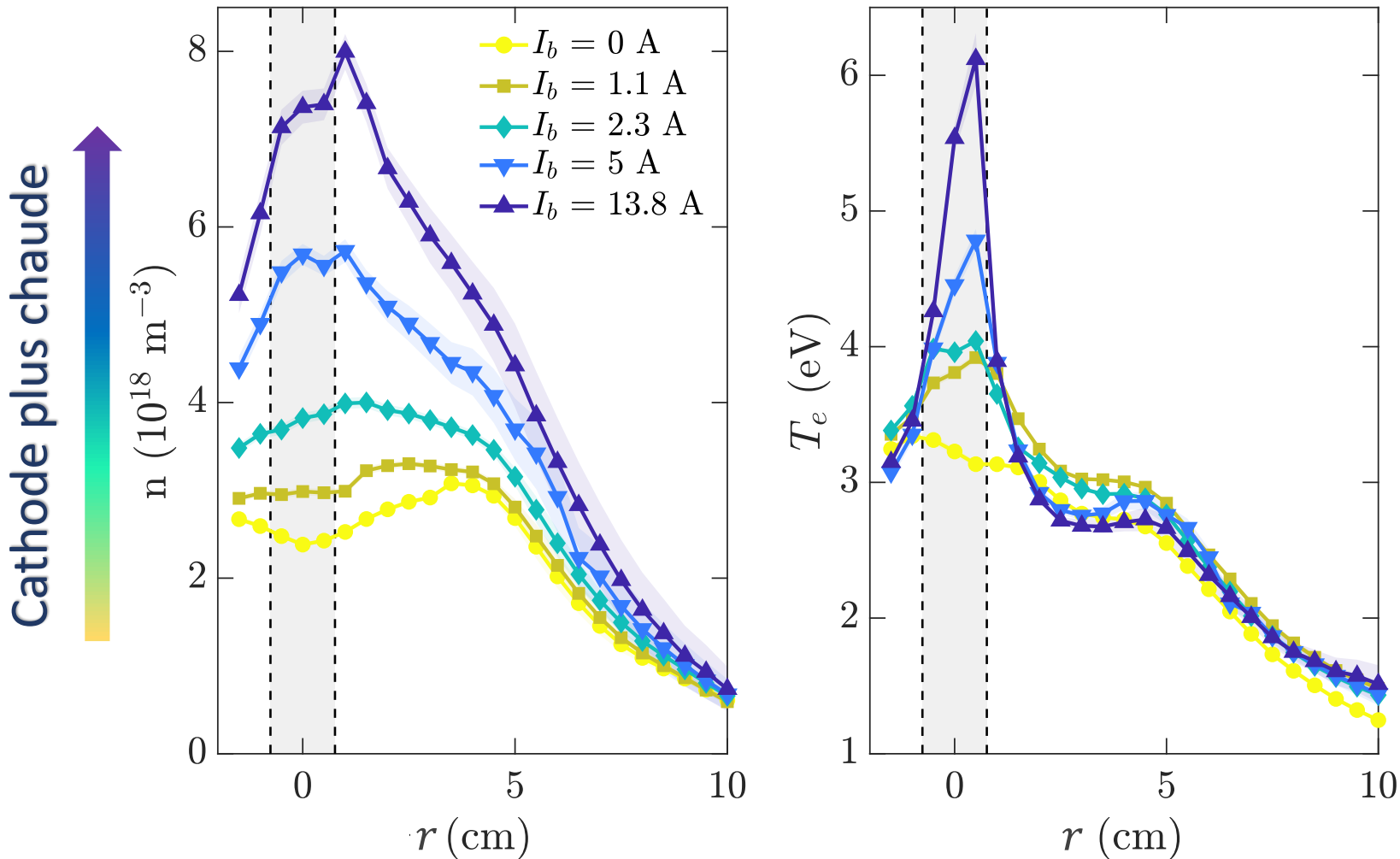
Modification des propriétés plasma

Paramètres expérimentaux : 240 G, 0.27 Pa, 1 kW, $V_b = -60$ V



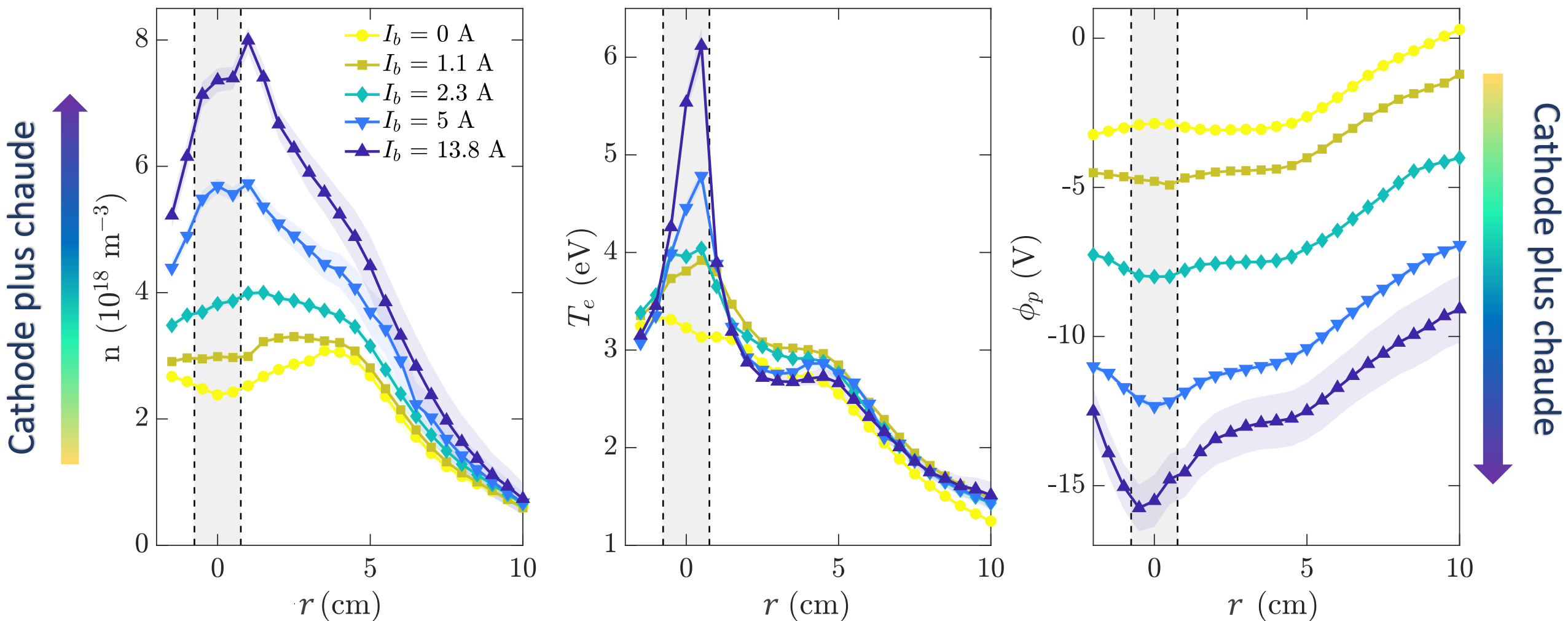
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Modification des propriétés plasma

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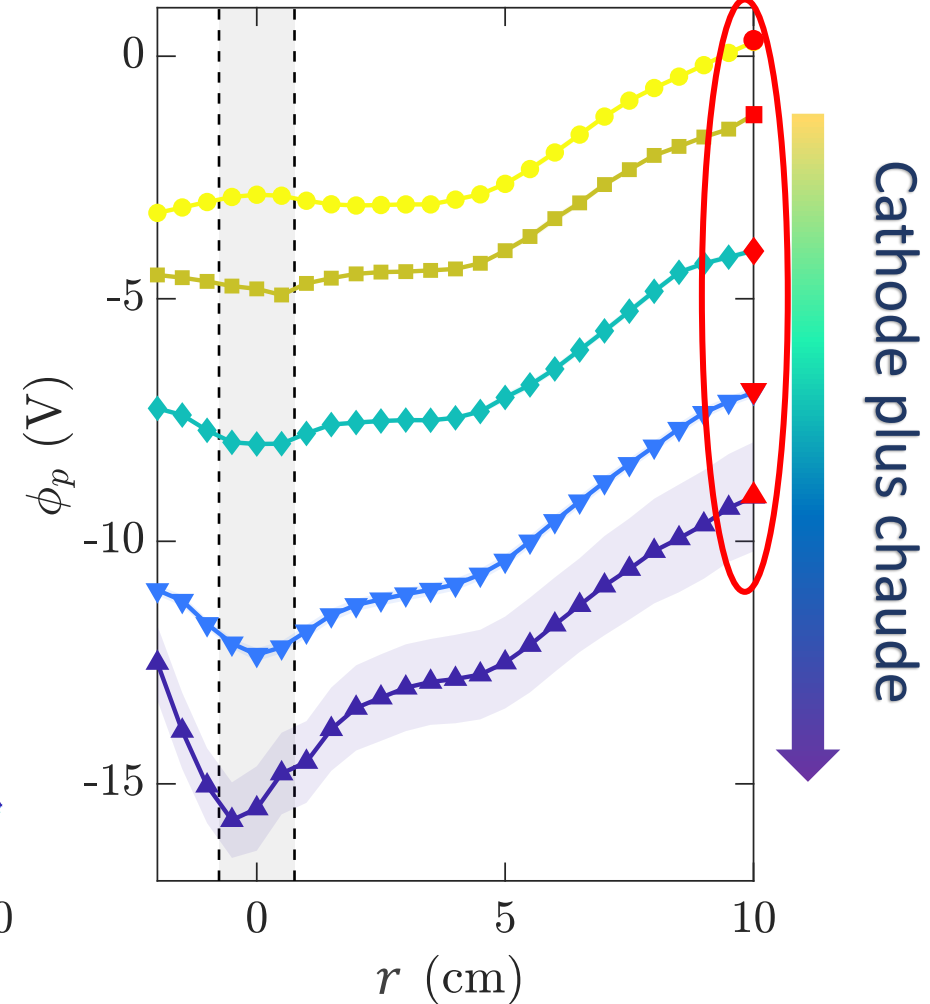
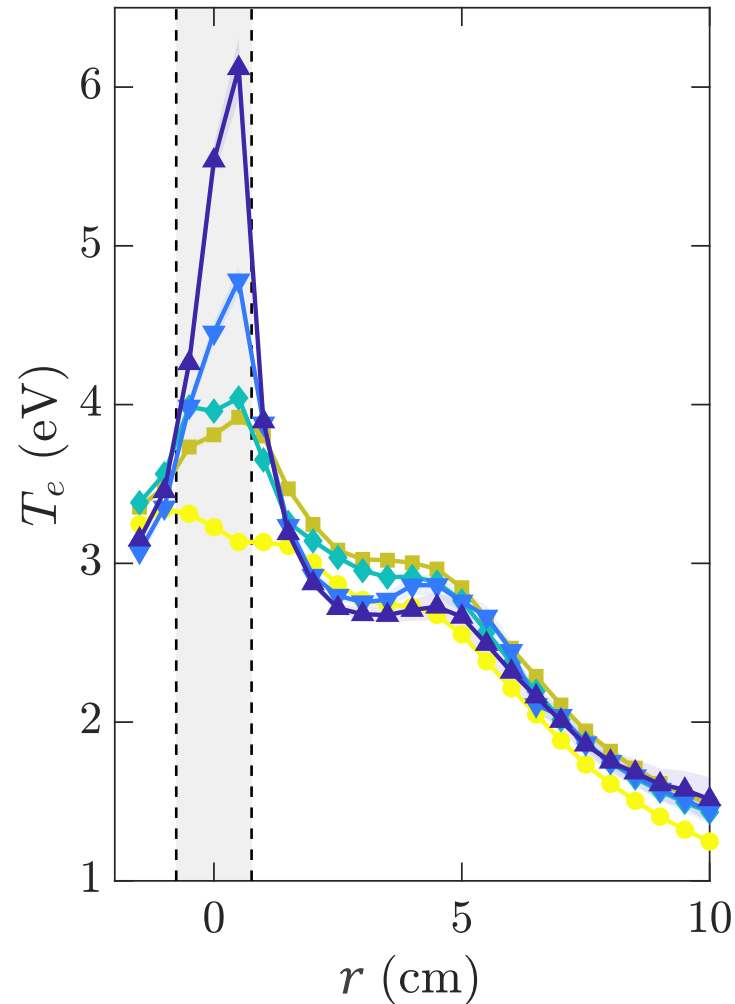
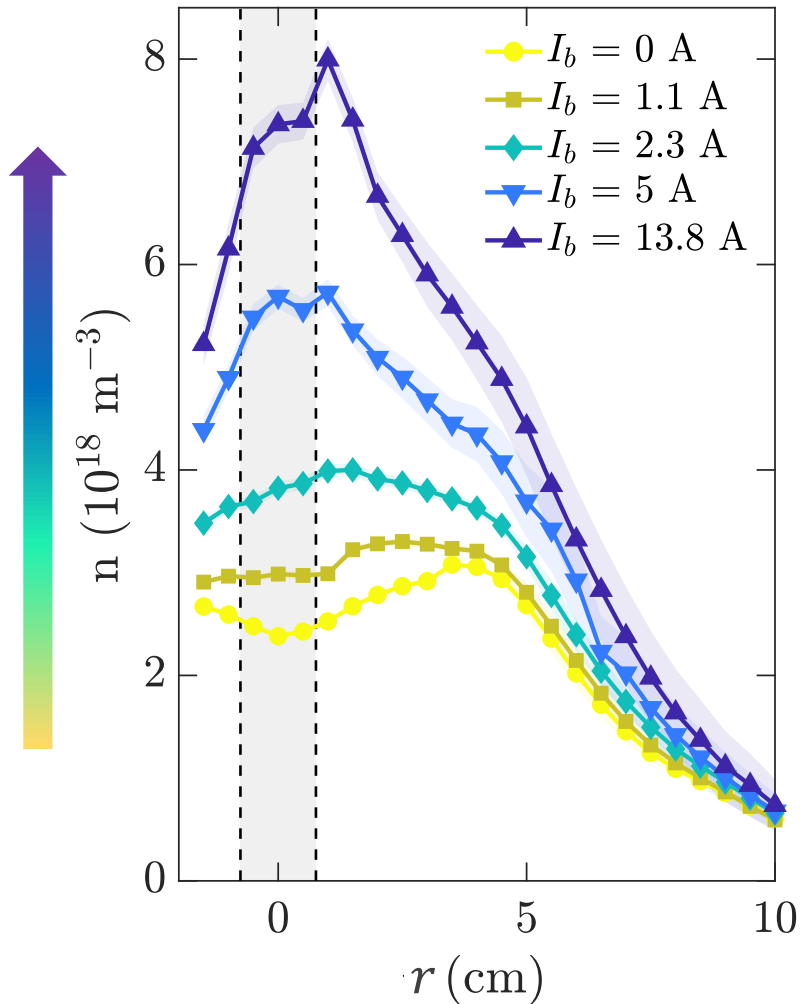


Modification des propriétés plasma

Paramètres expérimentaux : 240 G, 0.27 Pa, 1 kW, $V_b = -60$ V

$$\phi_p(x = r_g) \neq 0$$

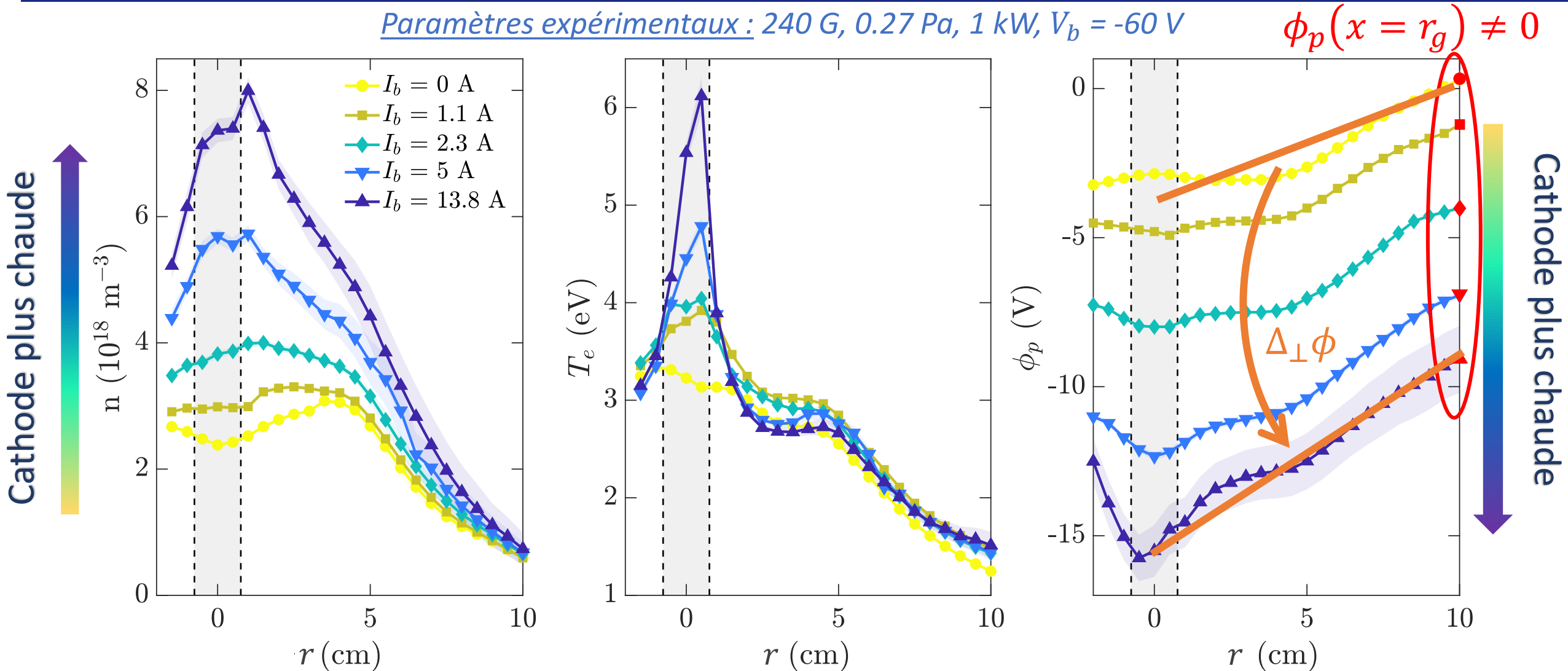
Cathode plus chaude



Cathode plus chaude

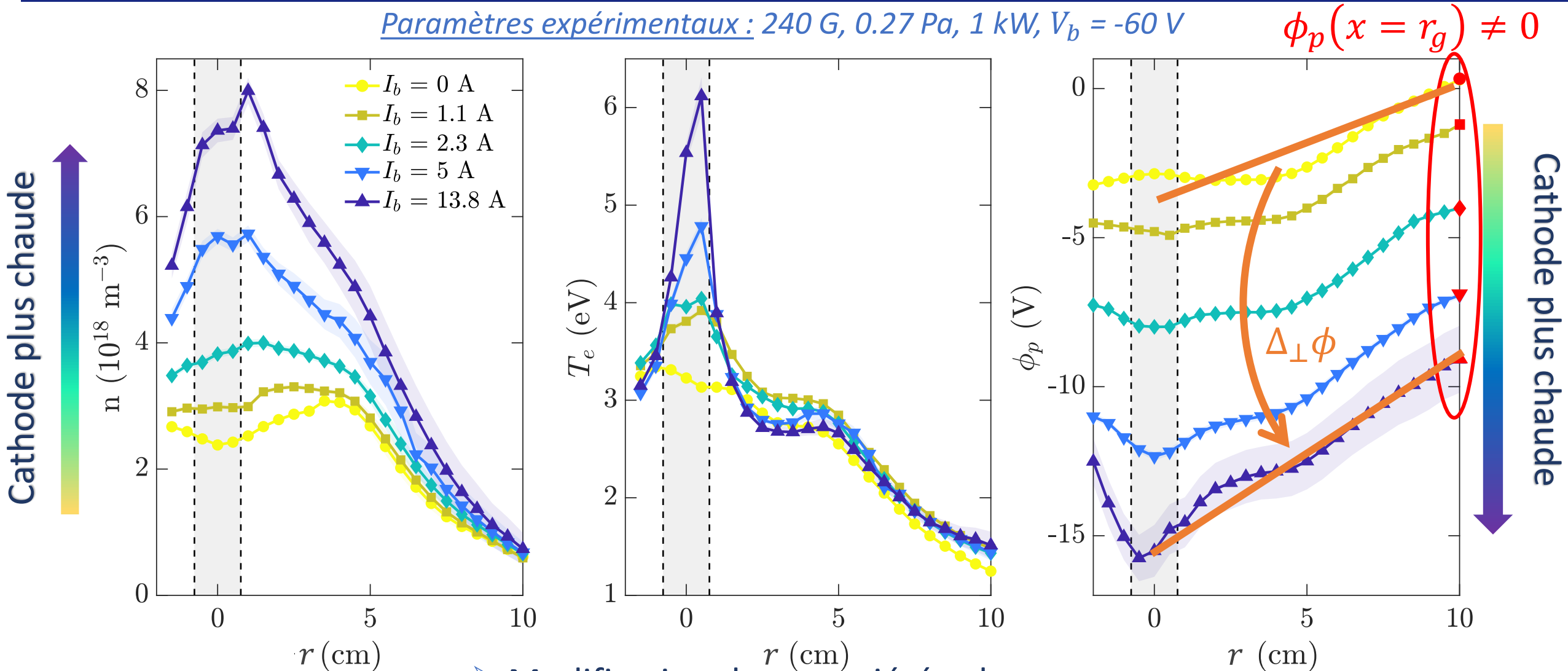
Modification des propriétés plasma

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Modification des propriétés plasma

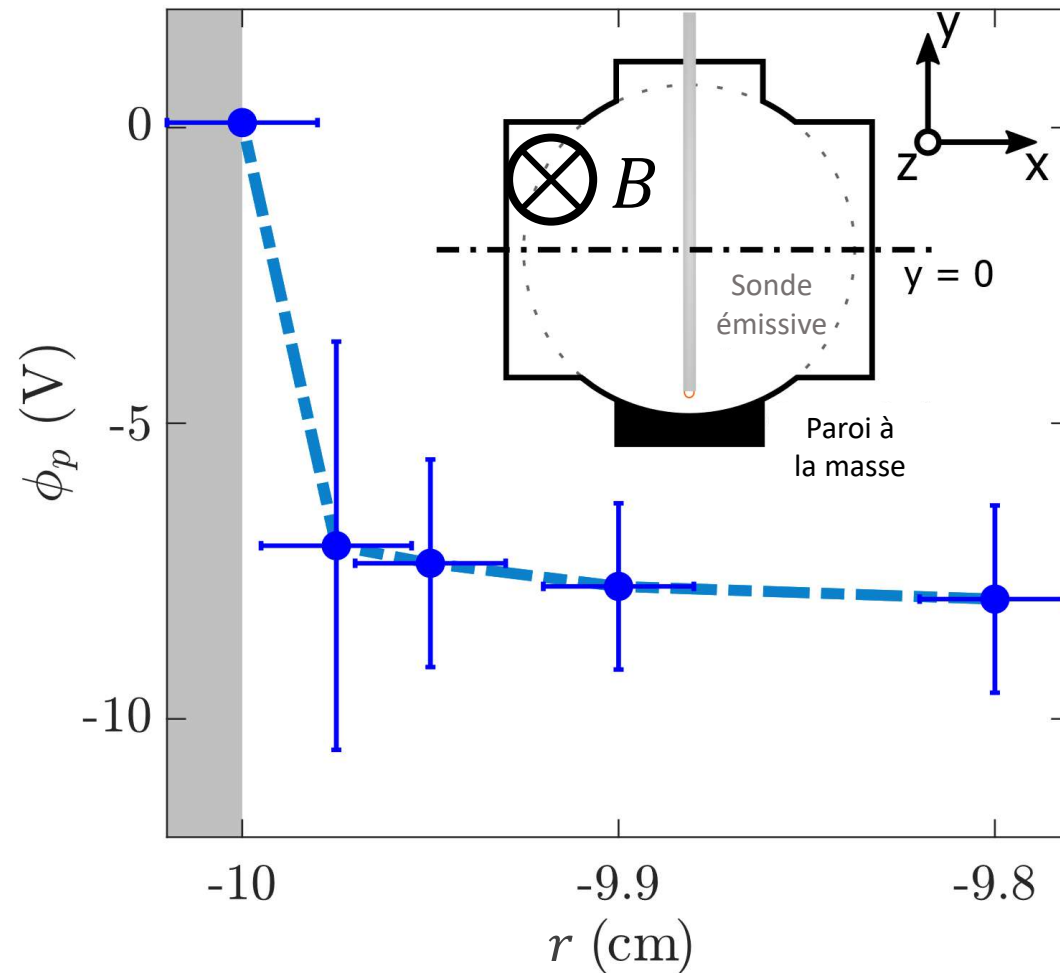
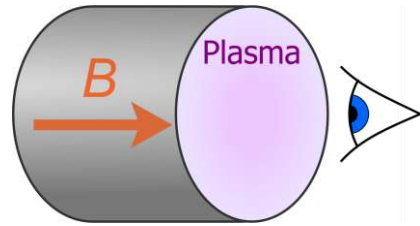
Paramètres expérimentaux : 240 G, 0.27 Pa, 1 kW, $V_b = -60$ V



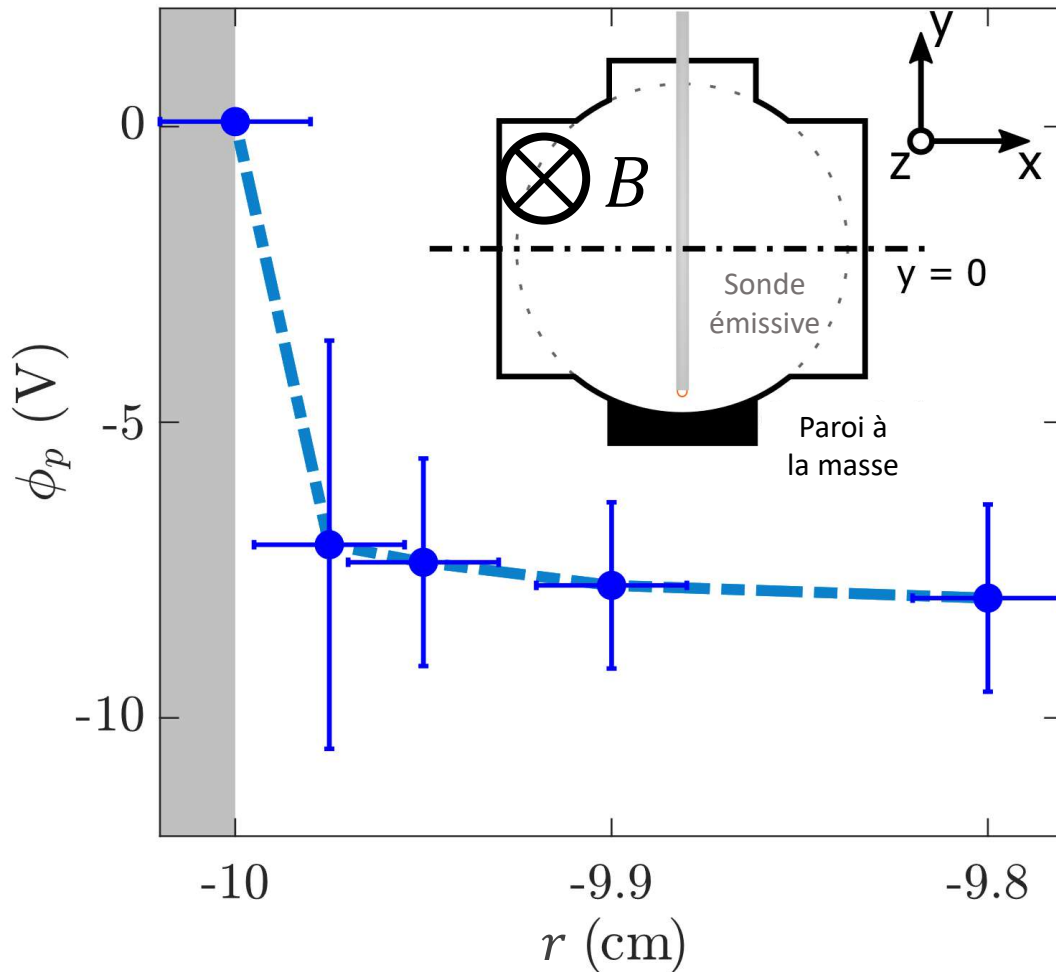
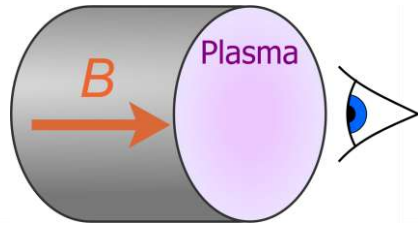
➤ Modification des propriétés plasma

➤ Gaine anodique $\phi_p(x = r_g)$, chute radiale de potentiel $\Delta_{\perp} \phi_p$

Existence d'une gaine anodique

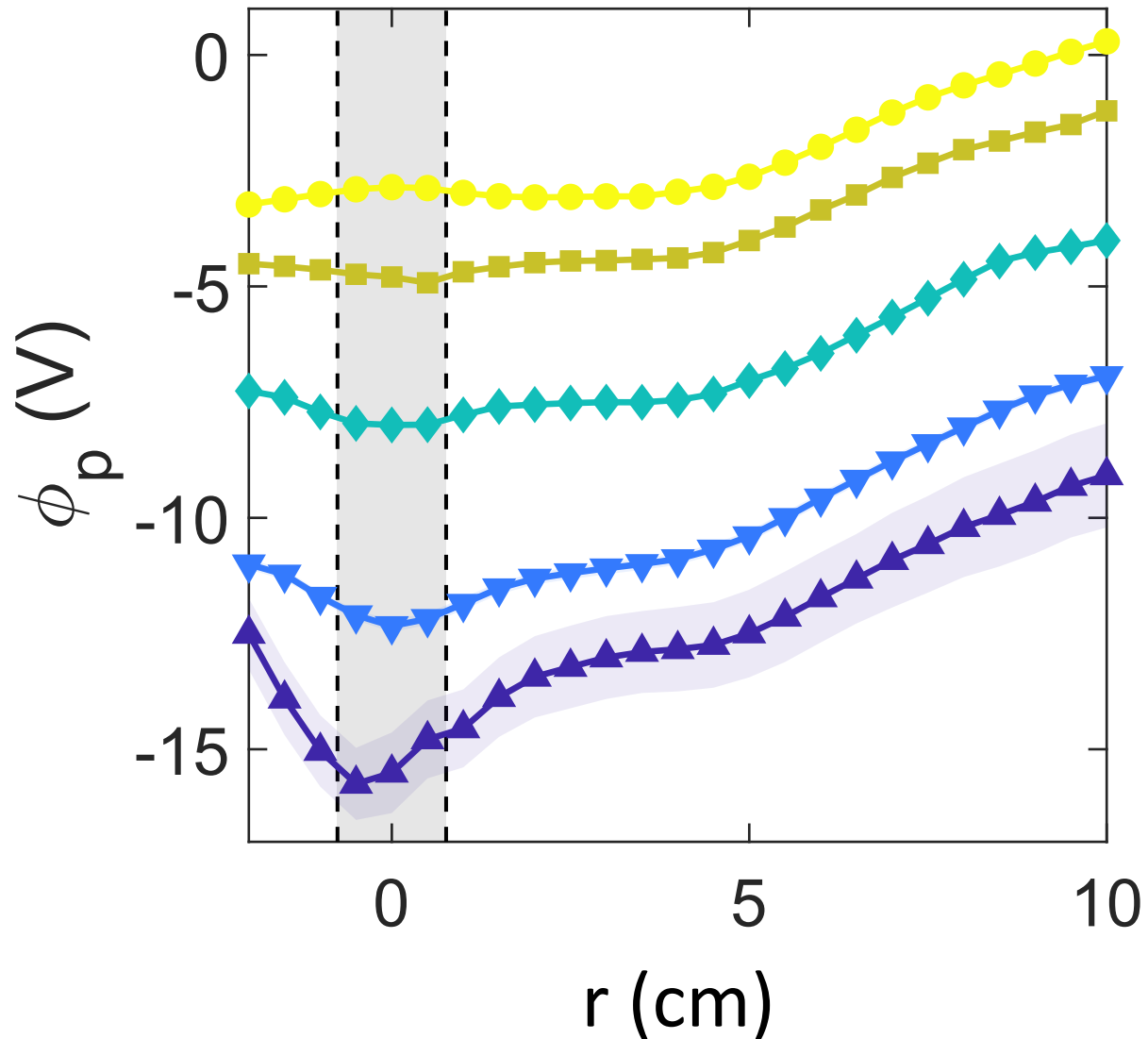


Existence d'une gaine anodique

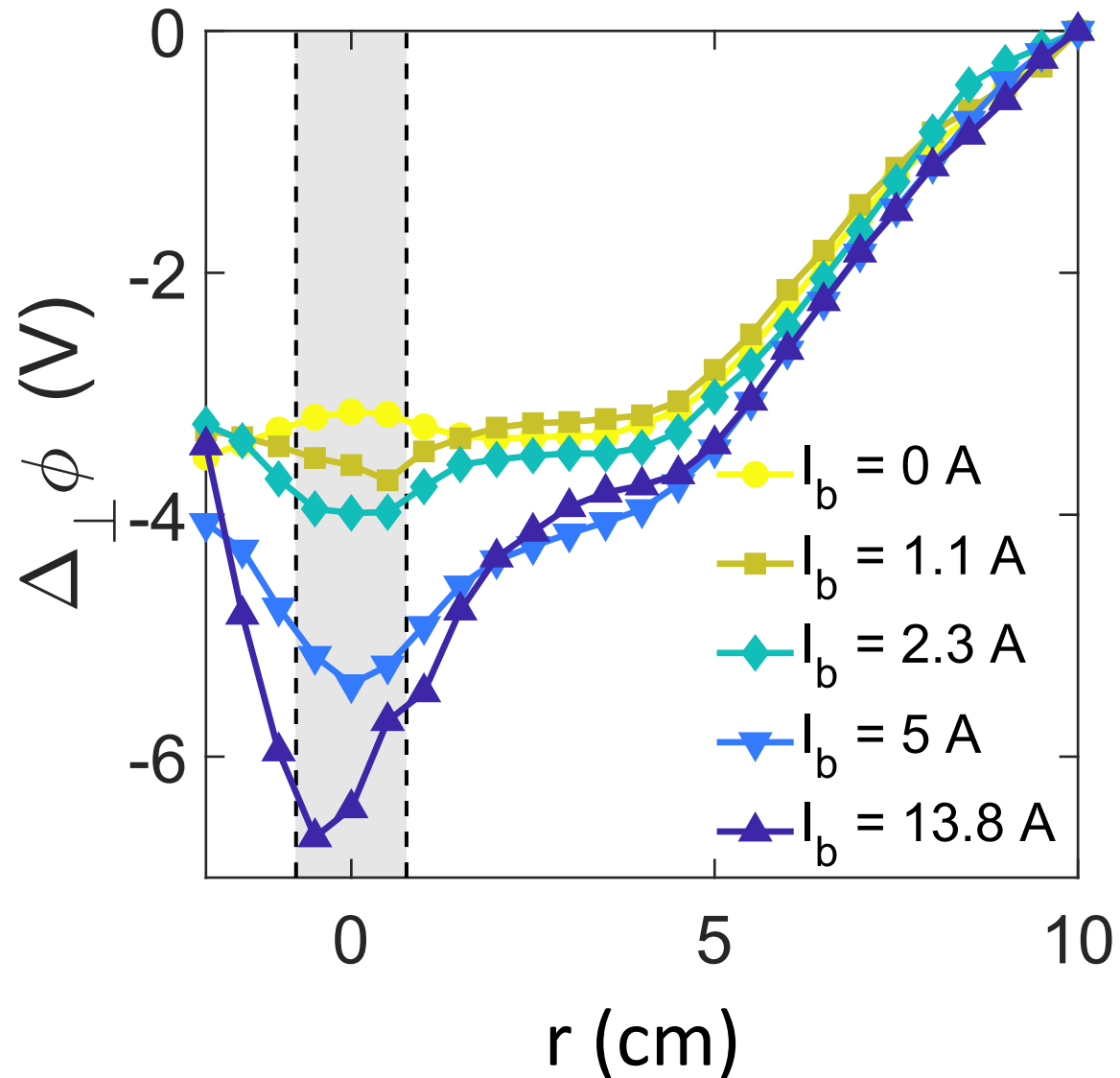


- Gaine anodique lorsque $\Xi > 1$
- Extraction d'électrons à l'anode
- Modification significative de ϕ_p

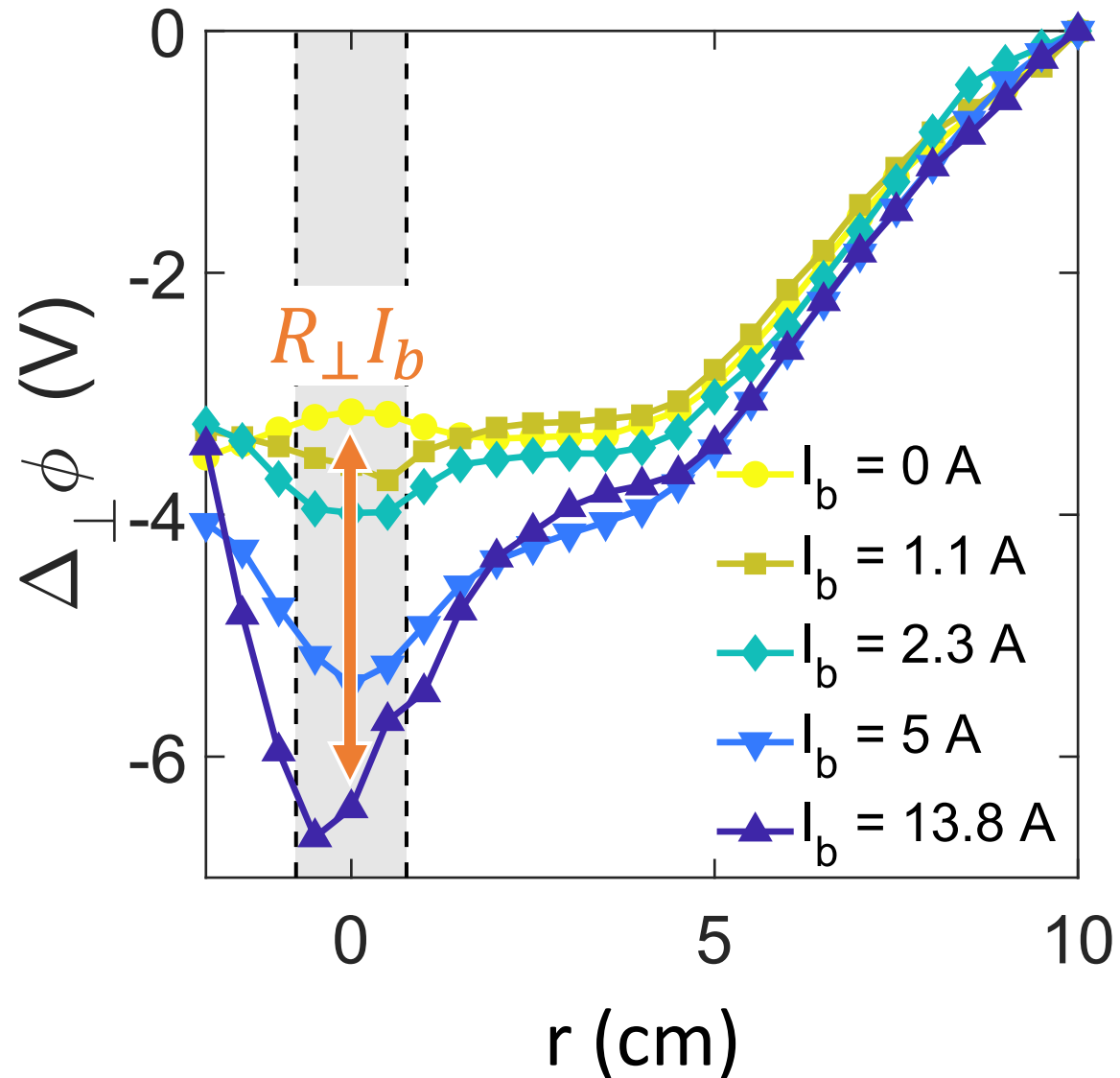
Chute radiale de potentiel



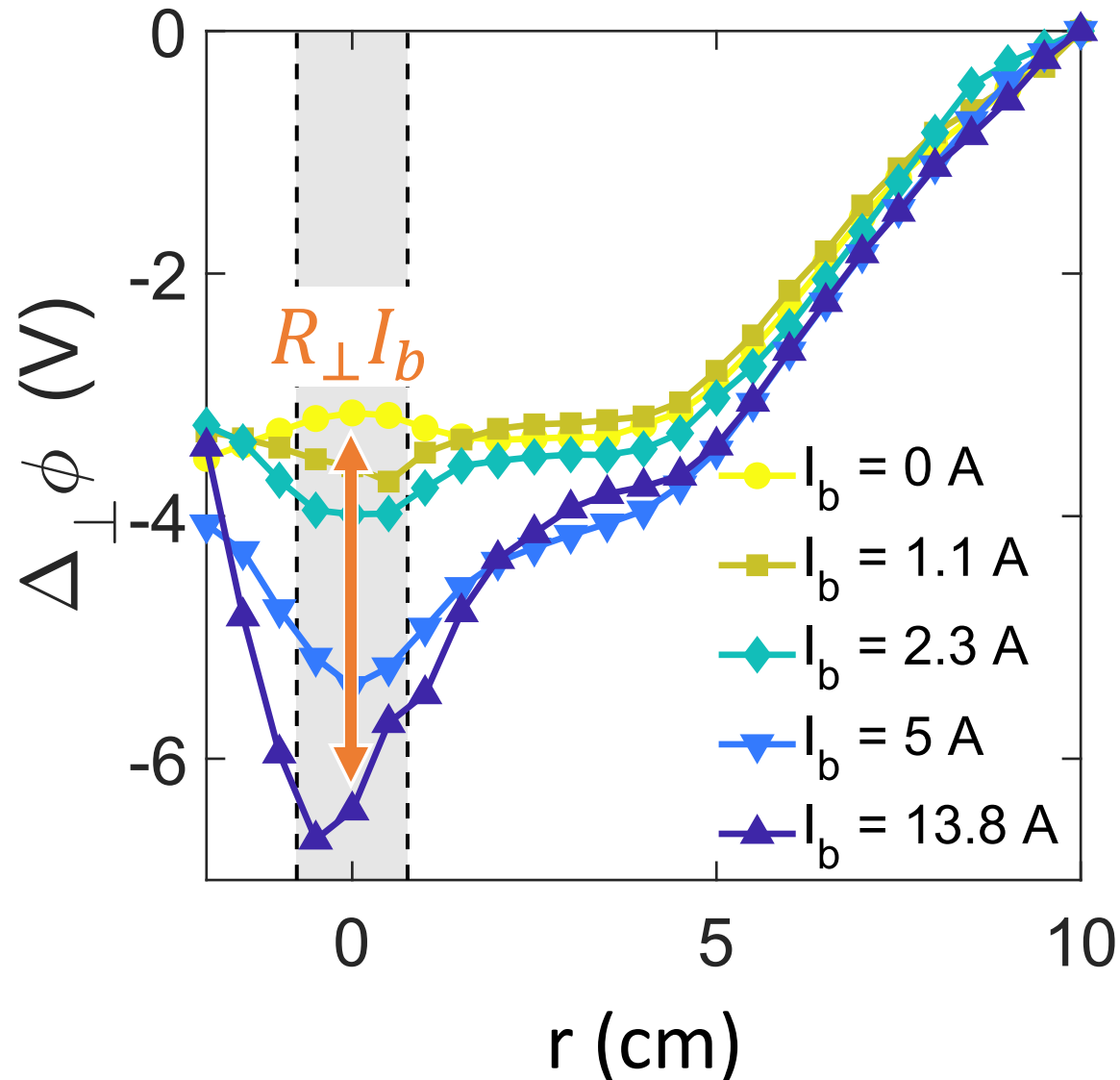
Chute radiale de potentiel



Chute radiale de potentiel



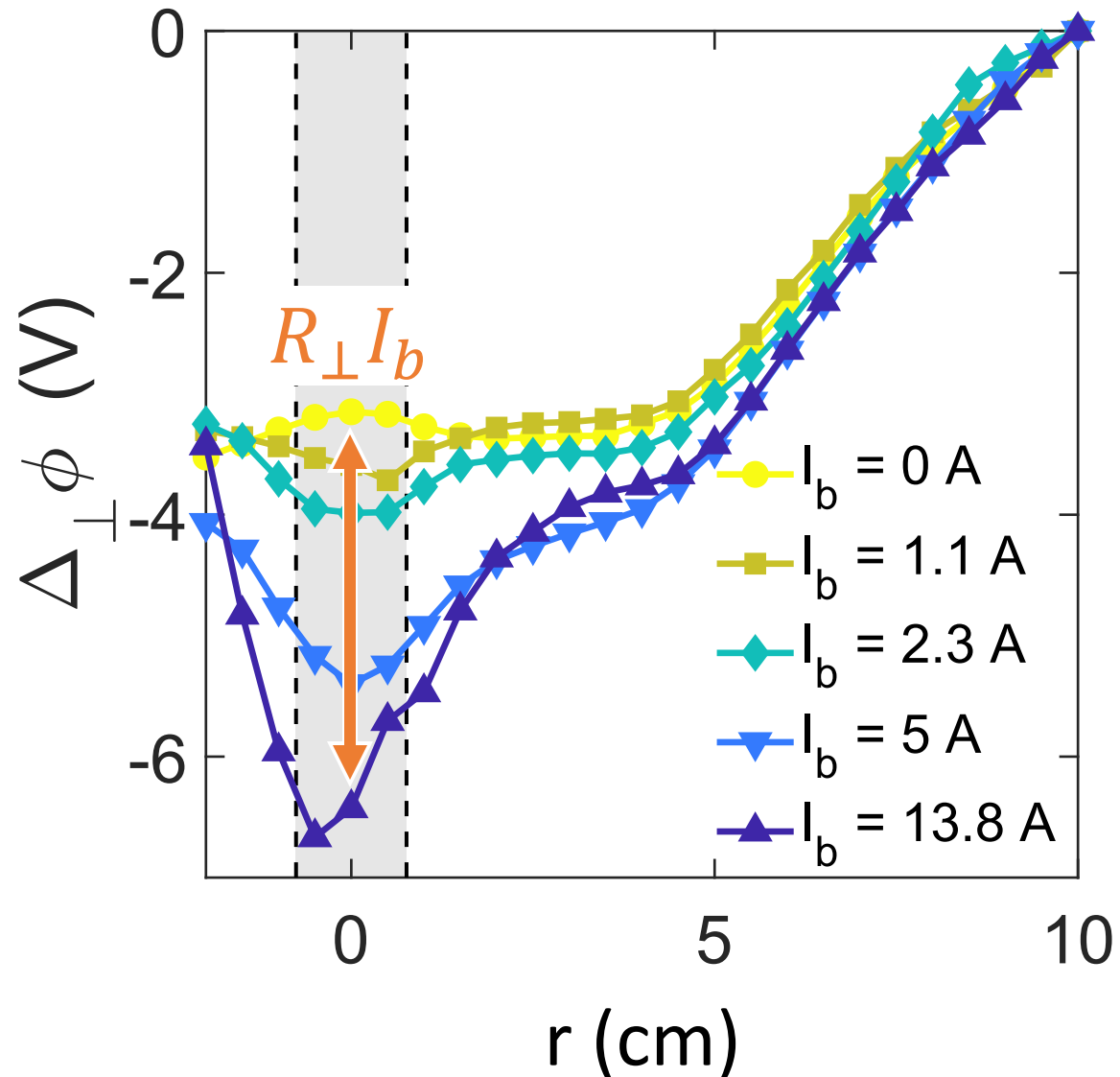
Chute radiale de potentiel



➤ Mesure et calcul de $R_\perp$

$$R_\perp = \int_{r_b}^{r_g} \frac{\eta_\perp(r)}{2\pi r L} dr$$

Chute radiale de potentiel



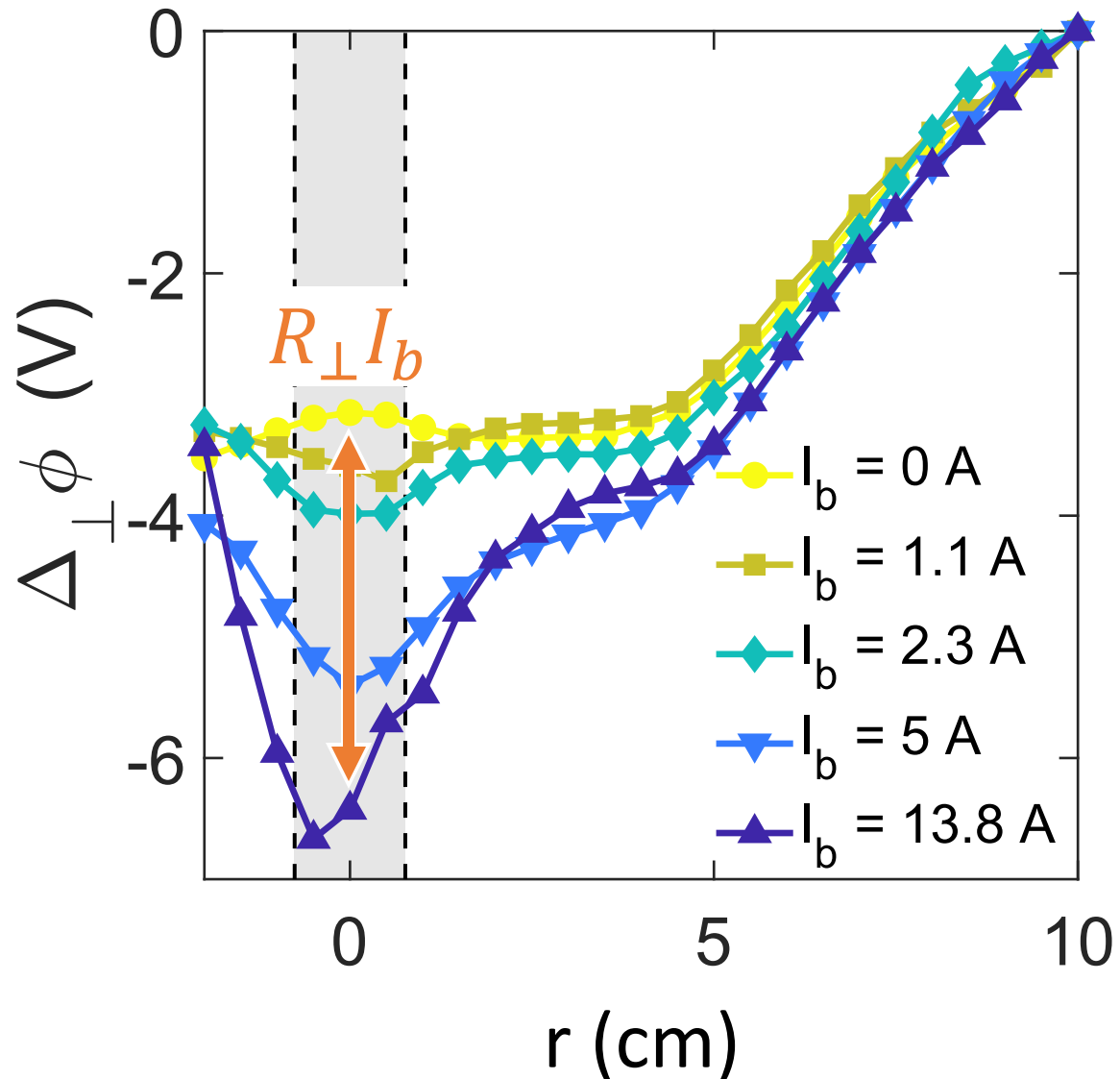
➤ Mesure et calcul de $R_{\perp}$

$$R_{\perp} = \int_{r_b}^{r_g} \frac{\eta_{\perp}(r)}{2\pi r L} dr$$

➤ Dépendance paramétrique :

$$\eta_{\perp} \propto \frac{B^2}{p_0} \frac{1}{n(r)}$$

Chute radiale de potentiel



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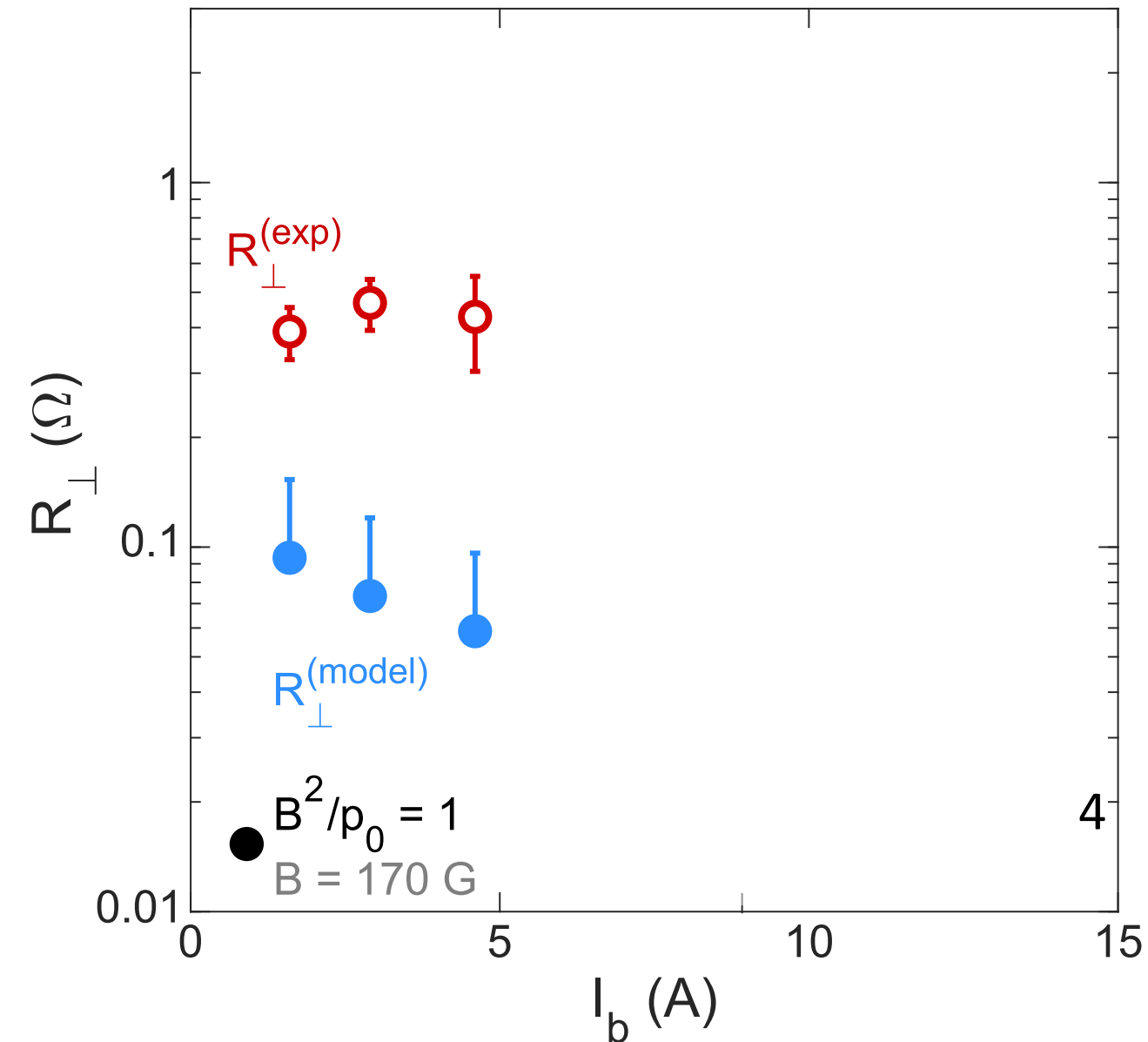
$$\eta_{\perp} \propto \frac{B^2}{p_0} \frac{1}{n(r)}$$

Paramètres

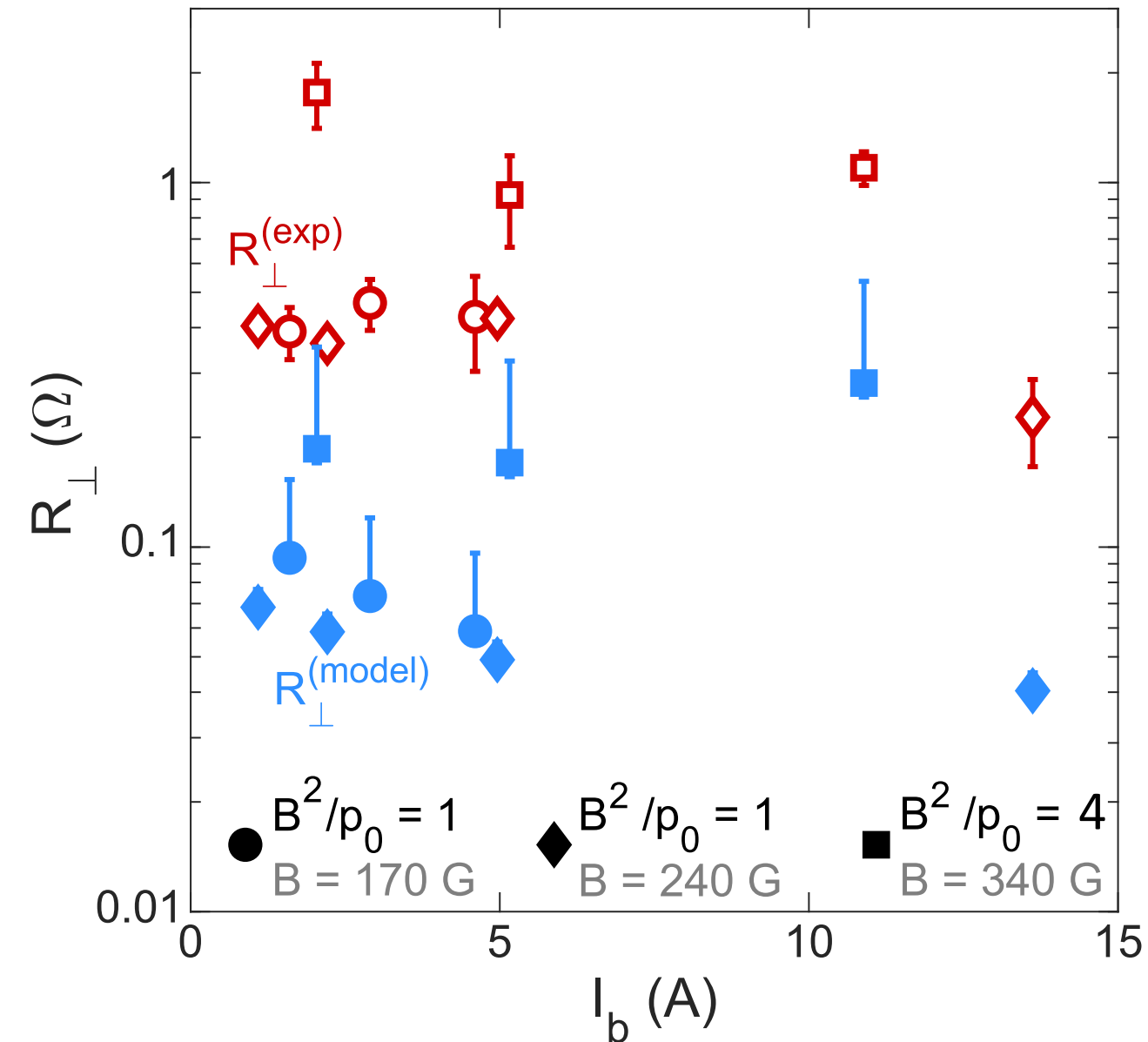
Mesures

exp

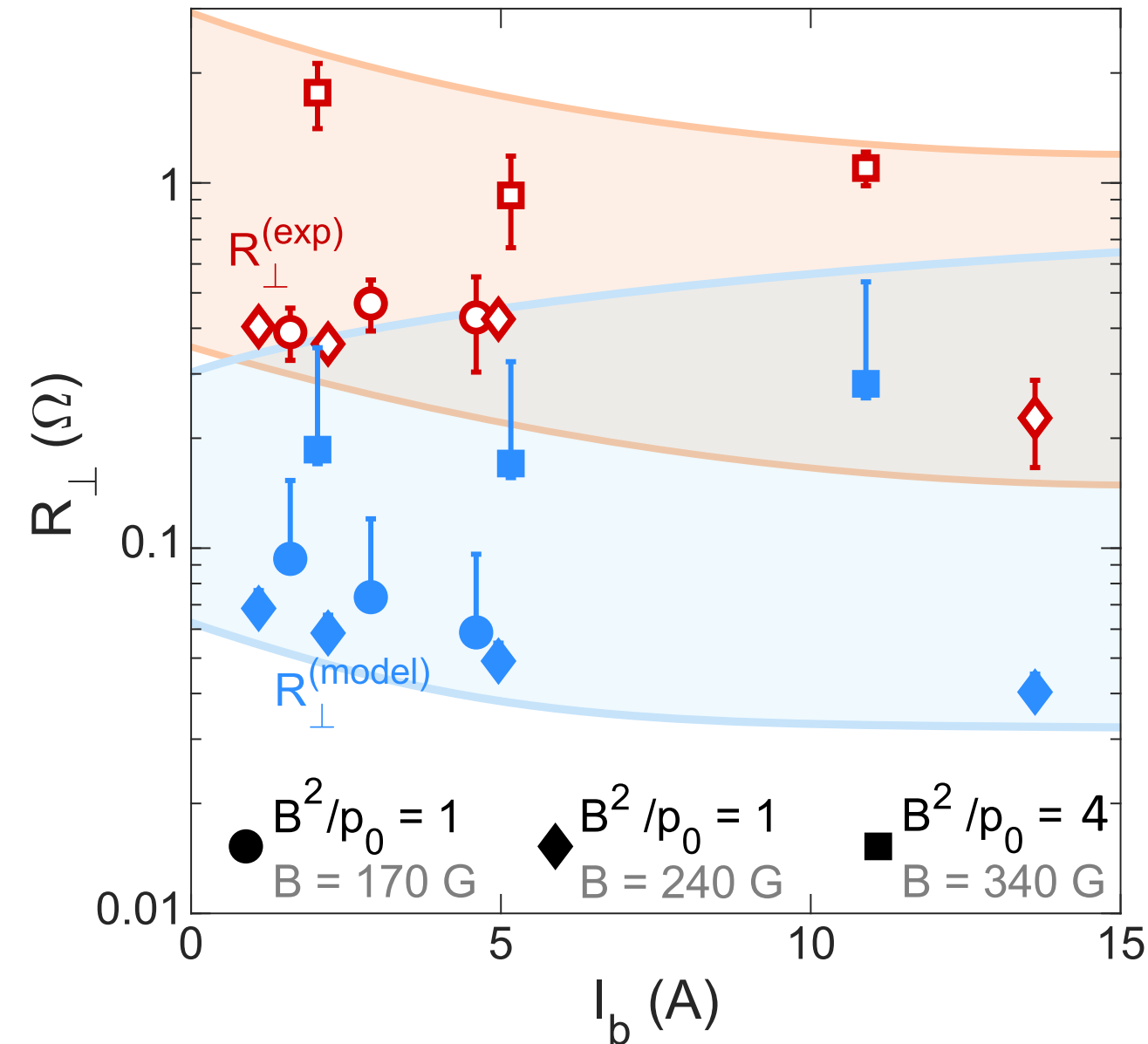
Résistance radiale du plasma



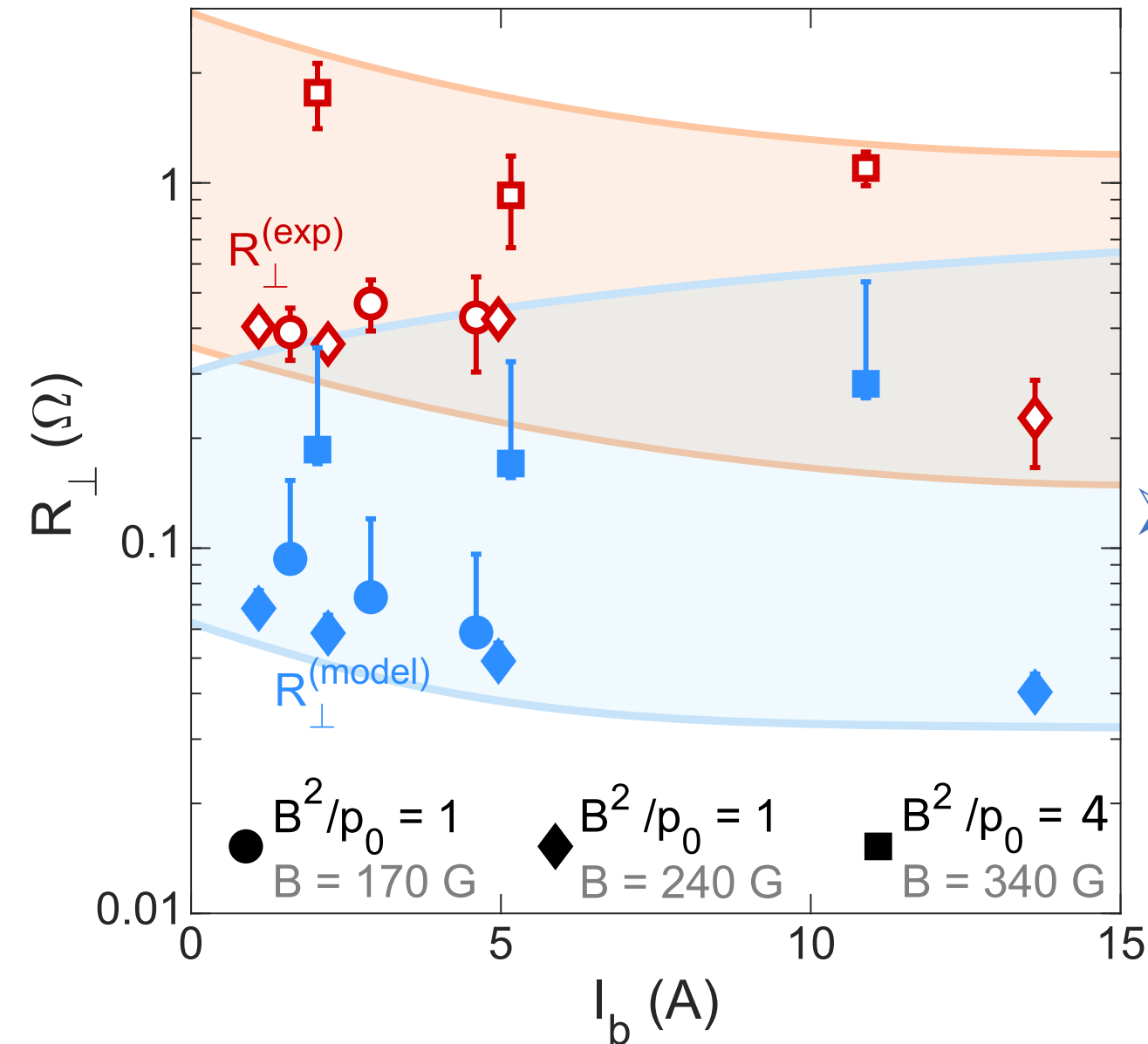
Résistance radiale du plasma



Résistance radiale du plasma



Résistance radiale du plasma

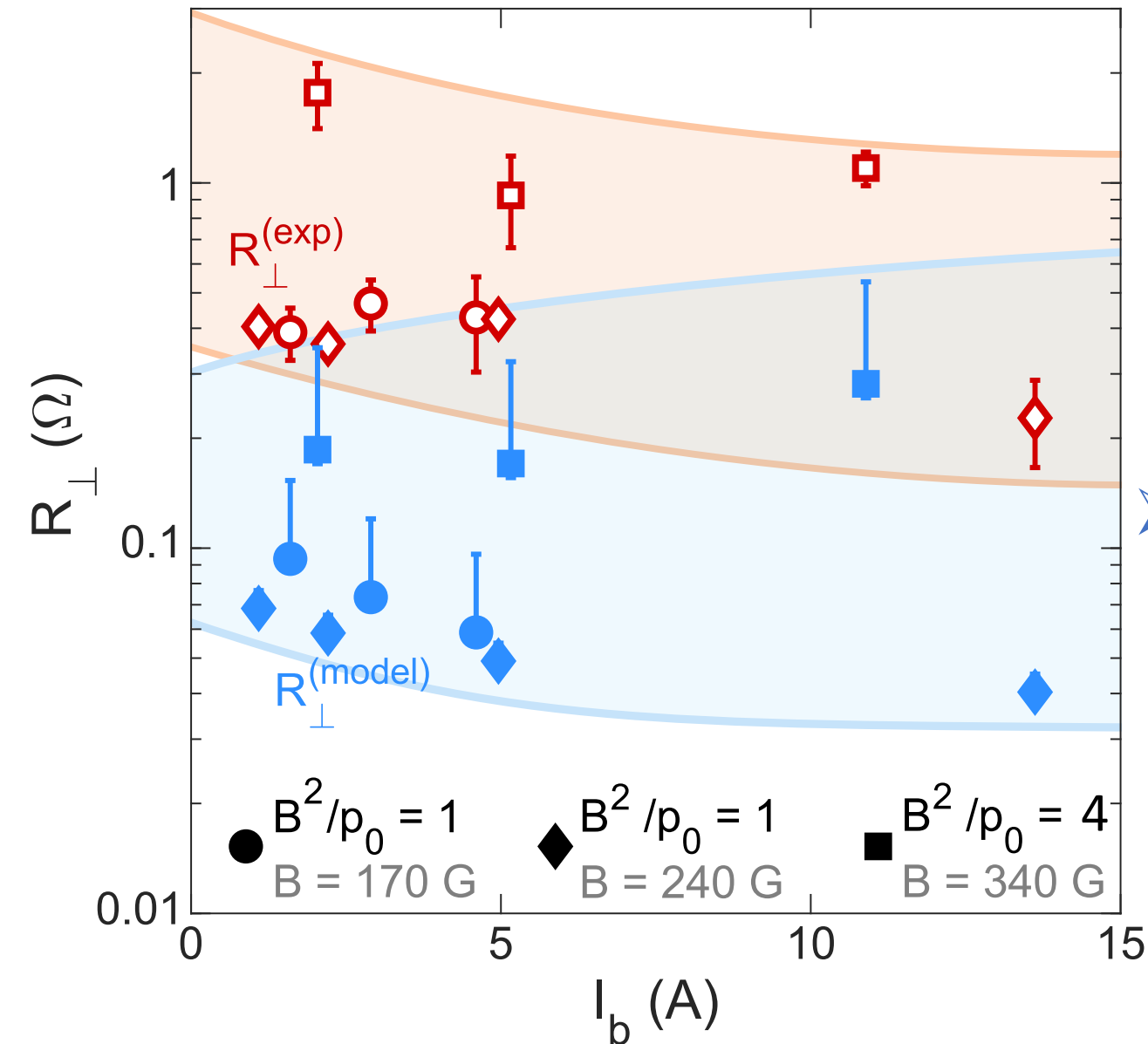


➤ Accord qualitatif : $R_{\perp} \propto B^2/p_0$

➤ Ecart quantitatif : facteur 5-10

➤ Améliorations : processus collisionnels (densité de neutres), modèle raffiné

Résistance radiale du plasma



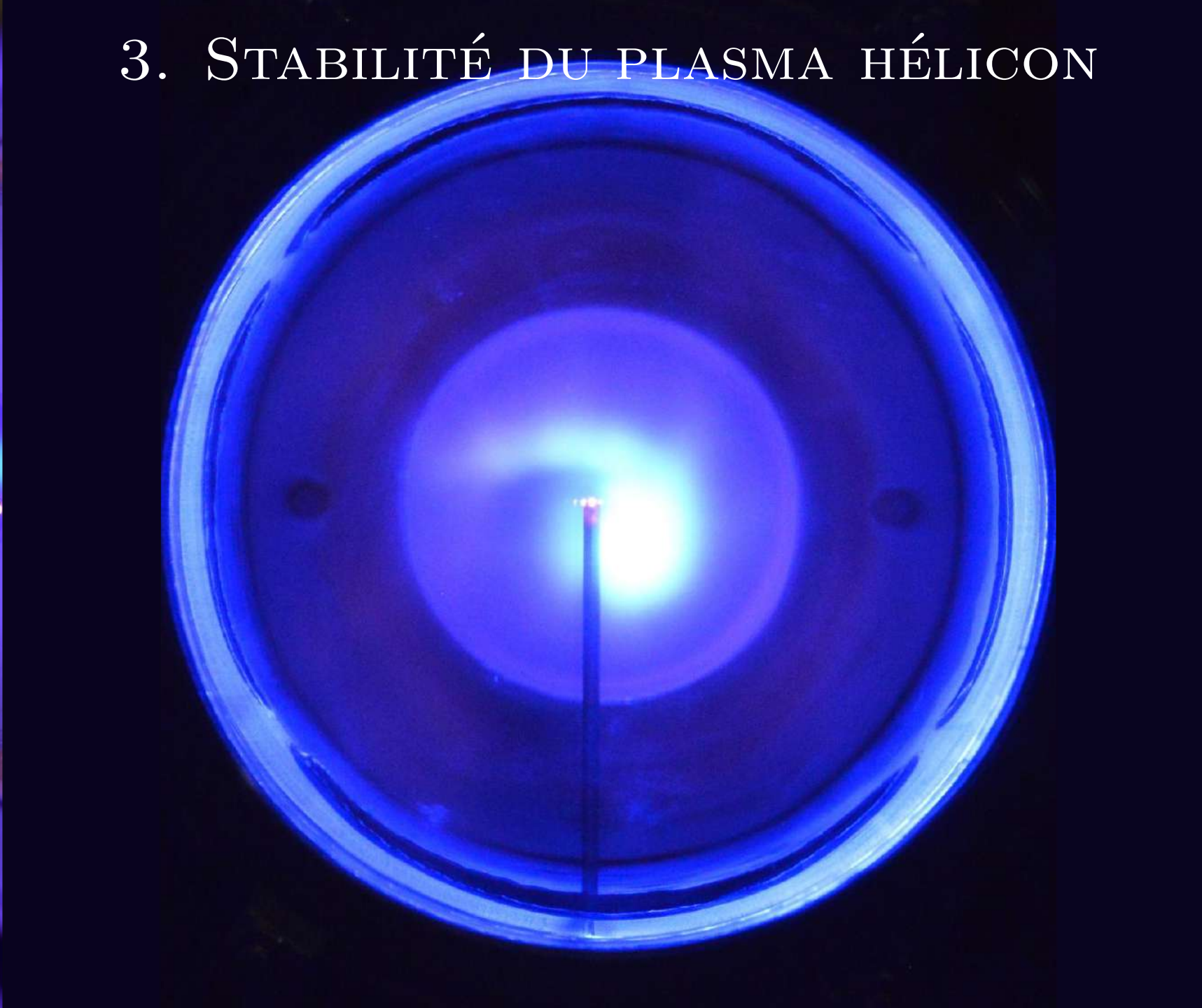
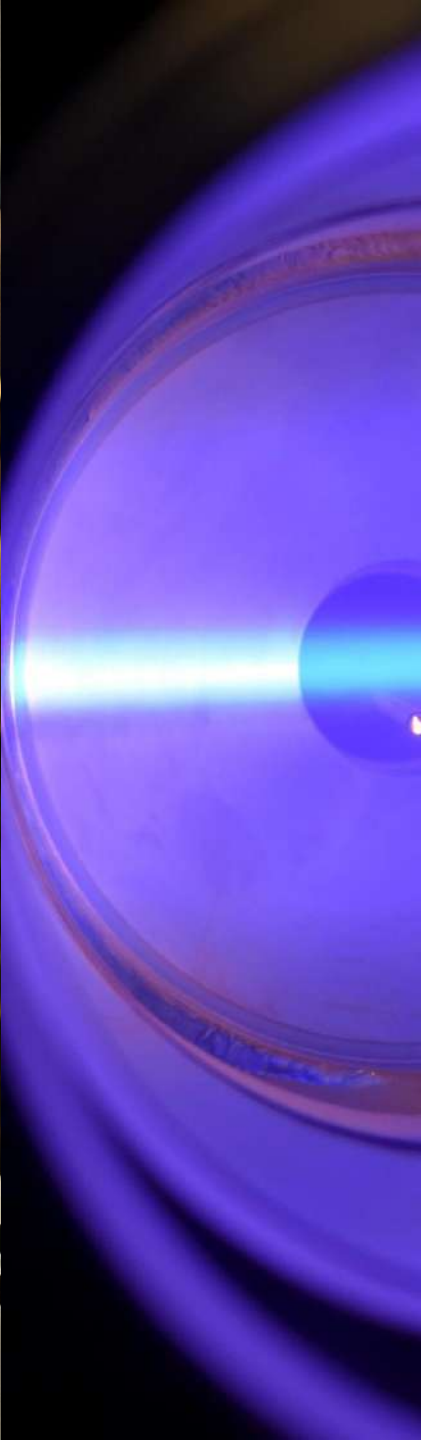
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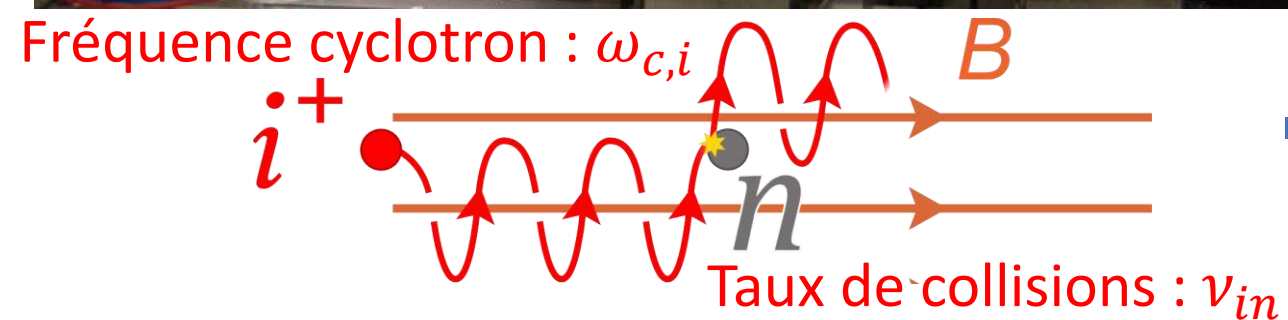
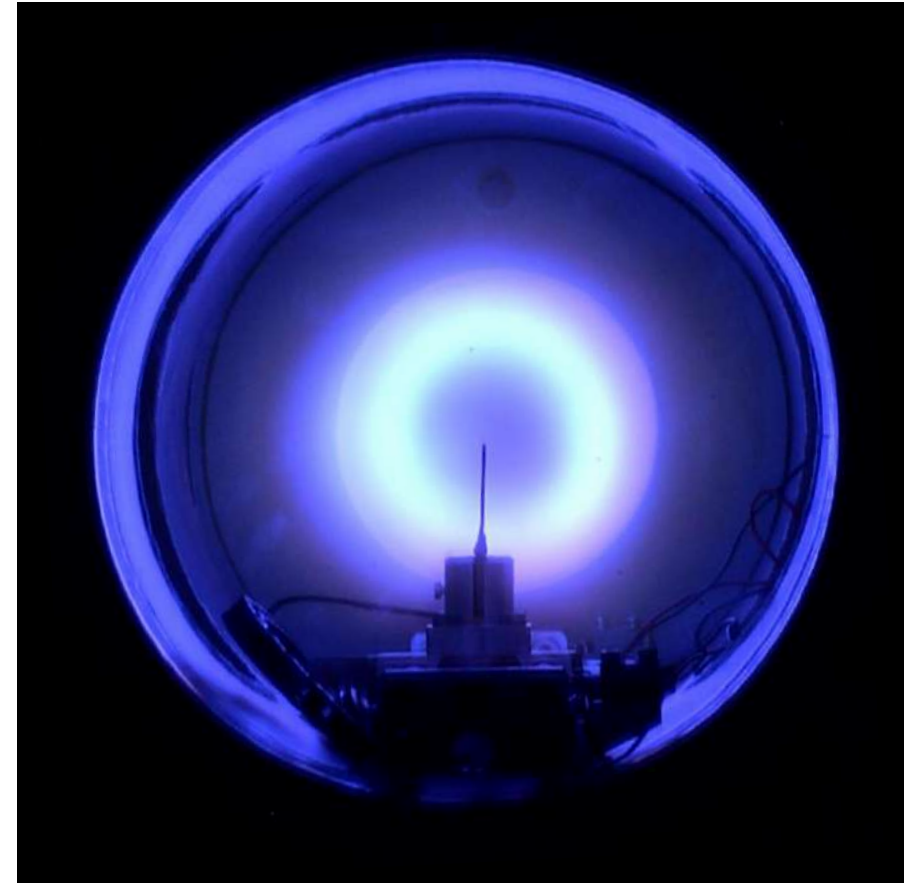
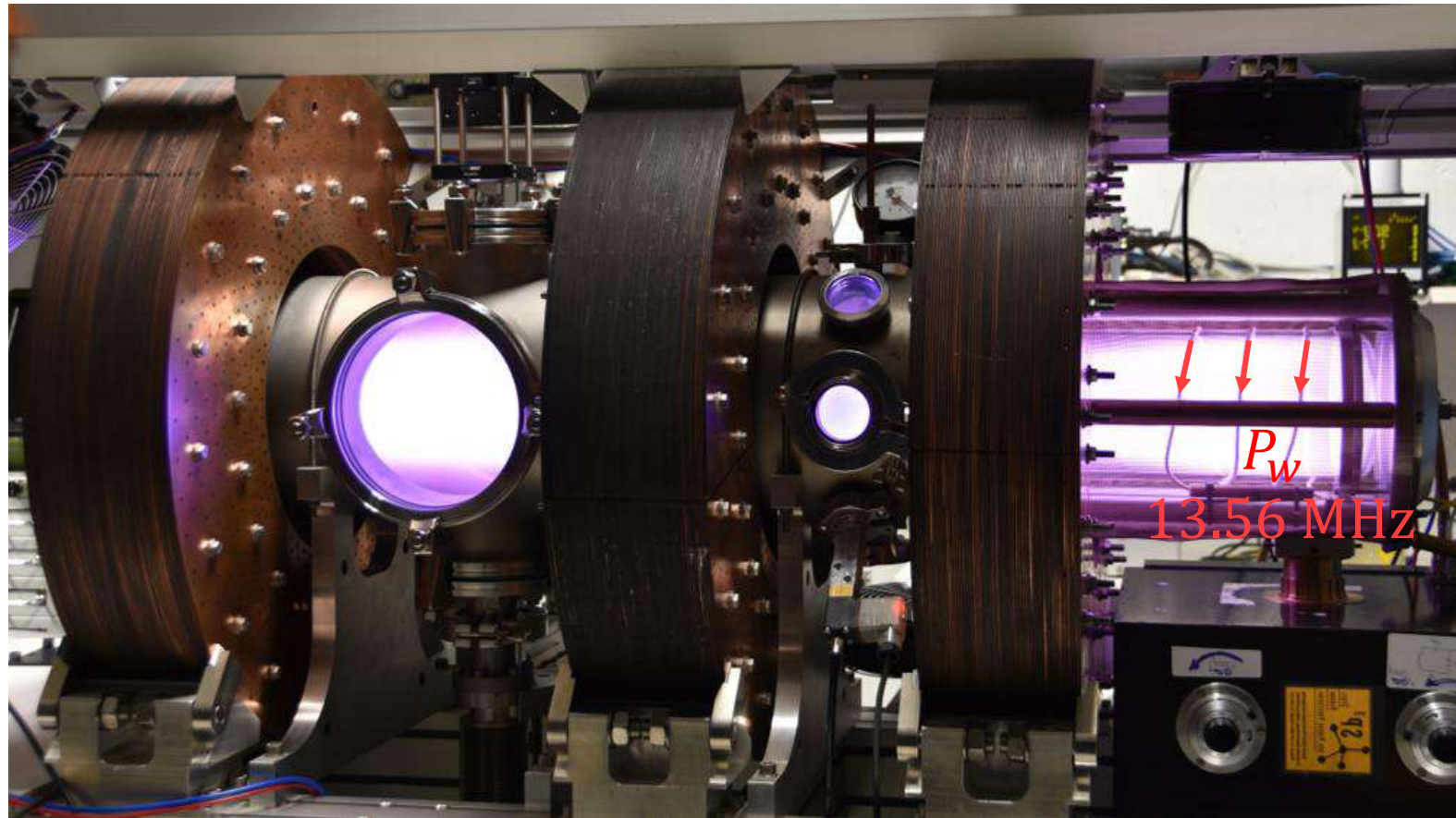
➤ Améliorations : processus collisionnels (densité de neutres), modèle raffiné

Taux de ionisation γ augmenté ?
Hausse de $\omega_{c,i}/\nu_{in}$

3. STABILITÉ DU PLASMA HÉLICON



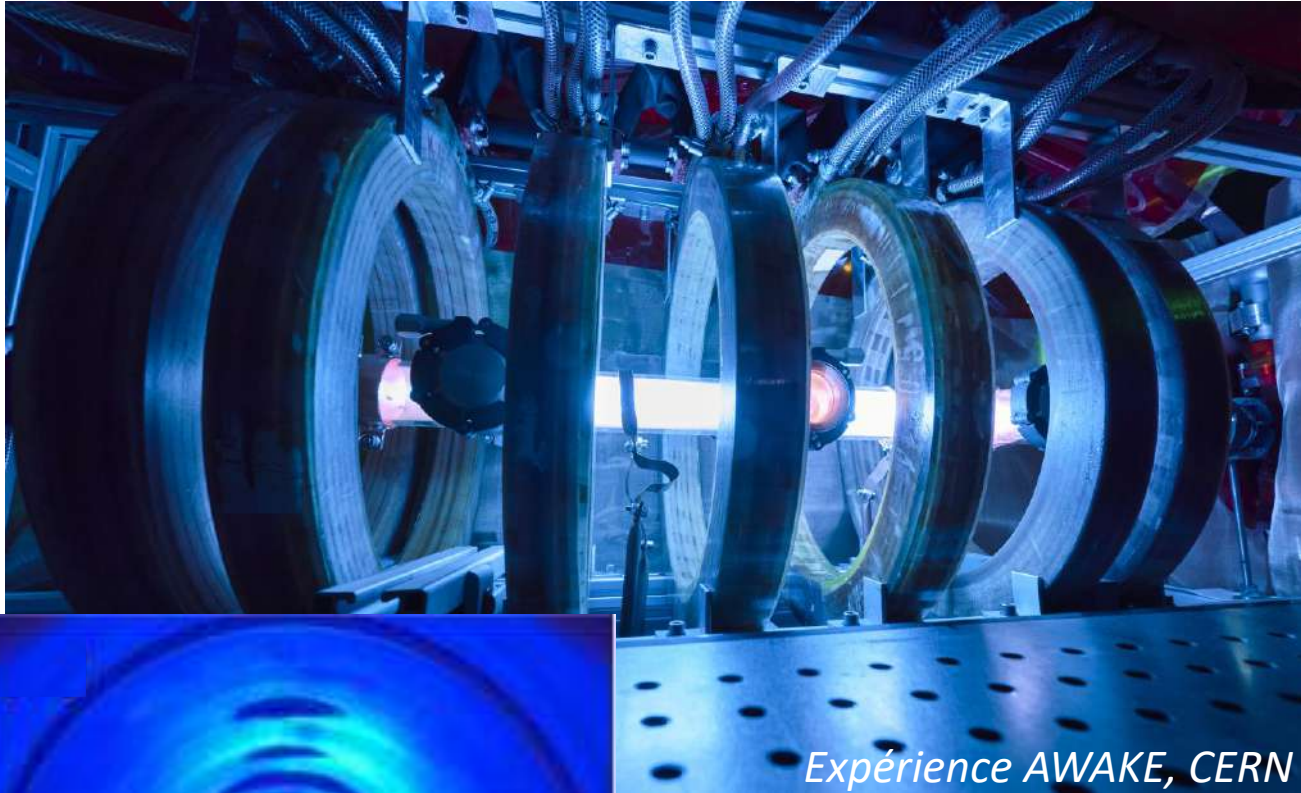
Plasmas inductifs



- Chauffage inductif : limité en densité

$$\omega_{c,i}/\nu_{in} = 0.1 - 2$$

Plasmas hélicons



➤ Plasma magnétisé haute densité

$$\text{➤ } \omega_{c,i}/v_{in} \rightarrow 10$$

Plasma généré par une onde électromagnétique

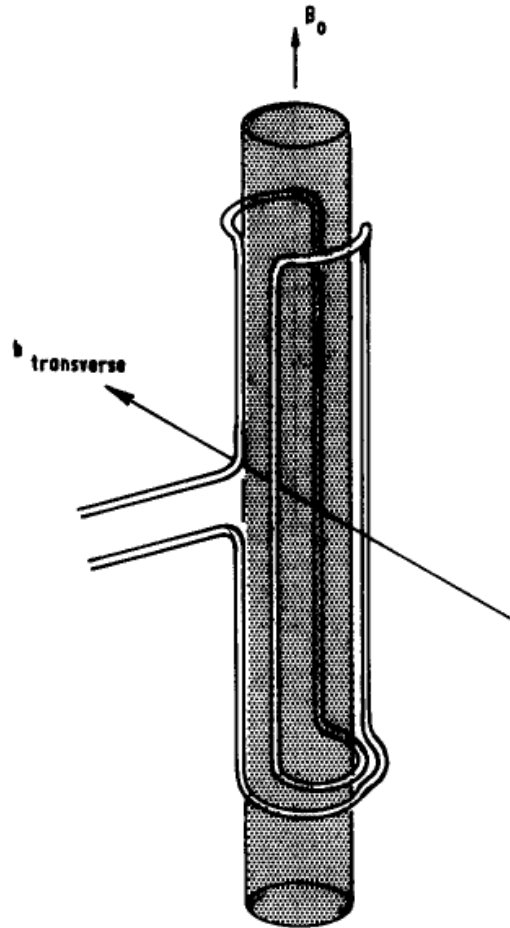
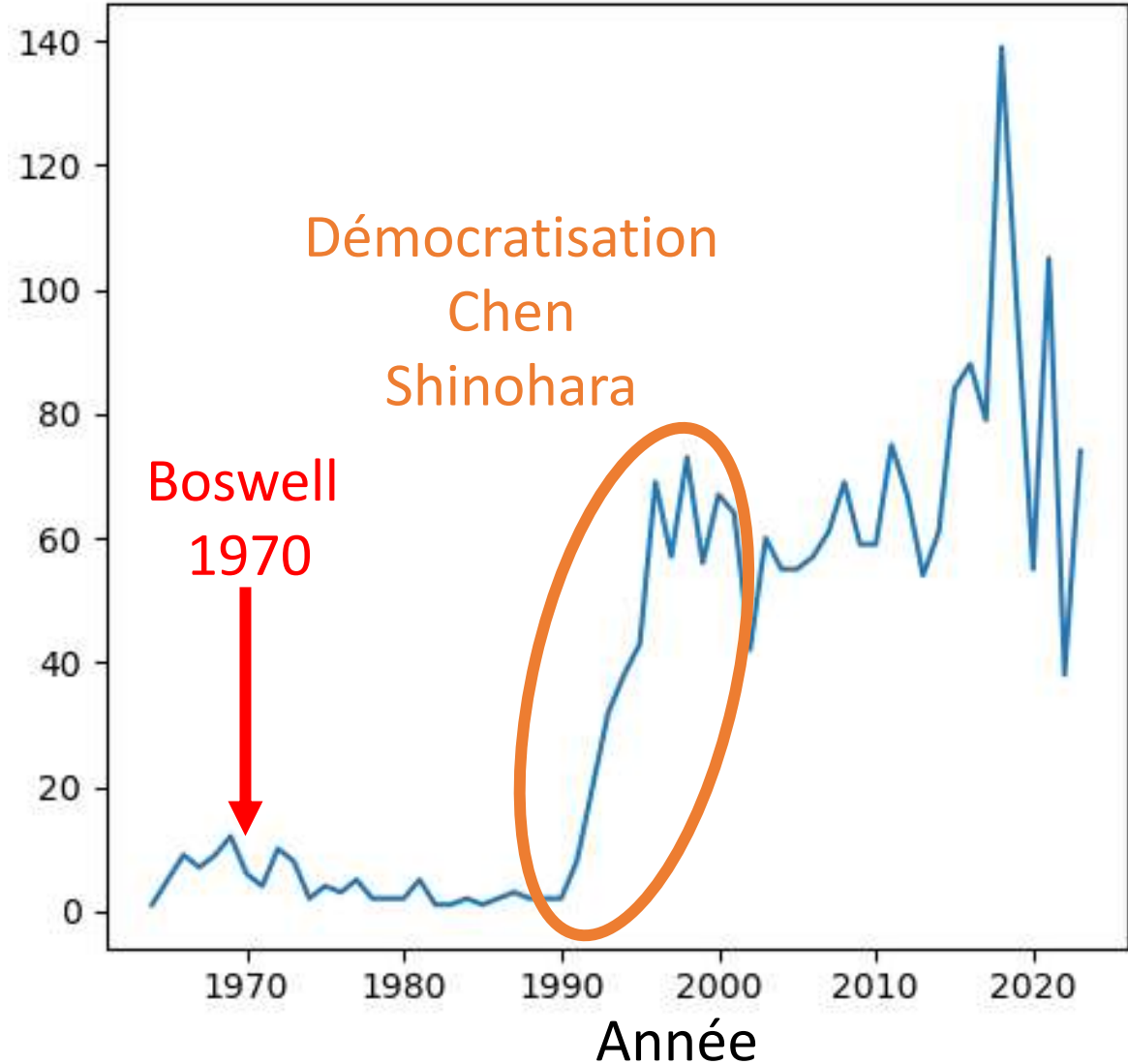


Fig. 1. Coil producing a transverse rf field across the plasma.

Articles « helicon plasma » (aip.org)



Boswell, Phys. Lett. (1970)

Remerciements : Simon Dupouy

Plasma généré par une onde électromagnétique

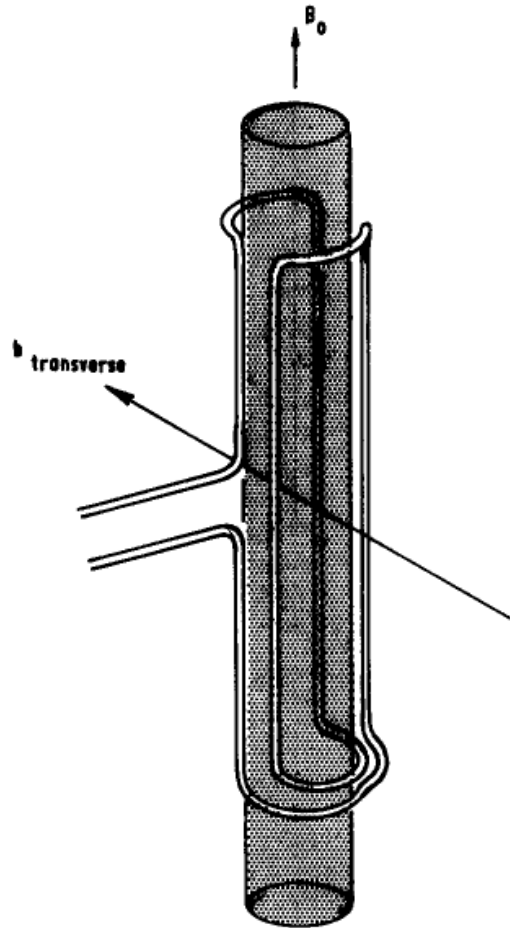
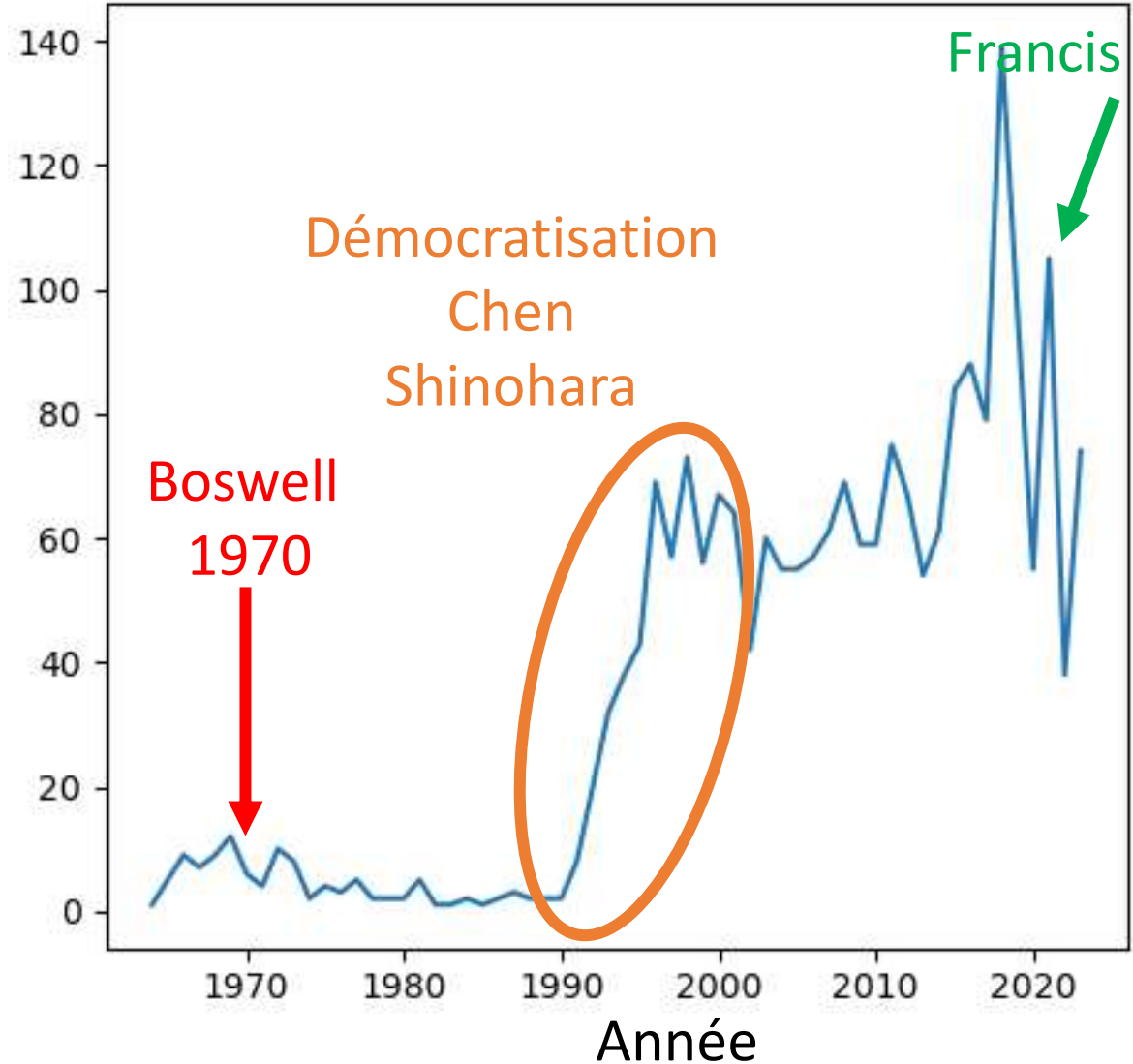


Fig. 1. Coil producing a transverse rf field across the plasma.

Boswell, Phys. Lett. (1970)

Articles « helicon plasma » (aip.org)

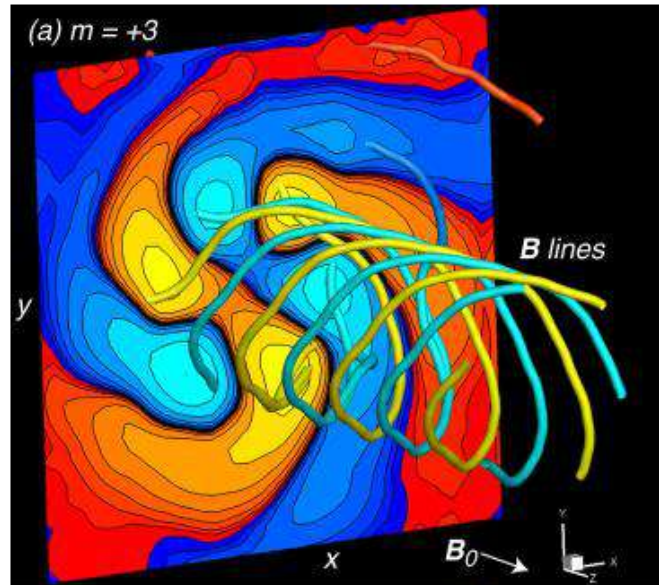
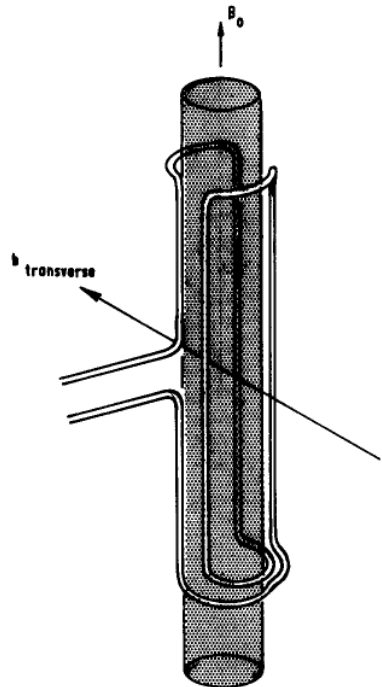


Thèse

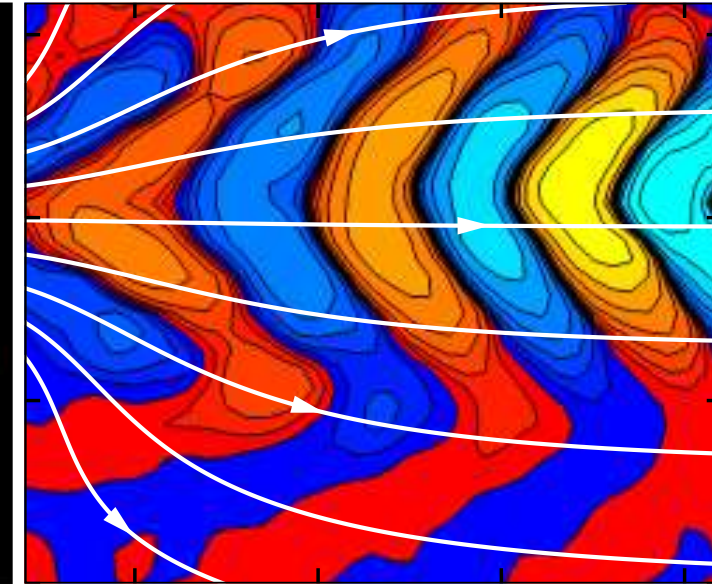
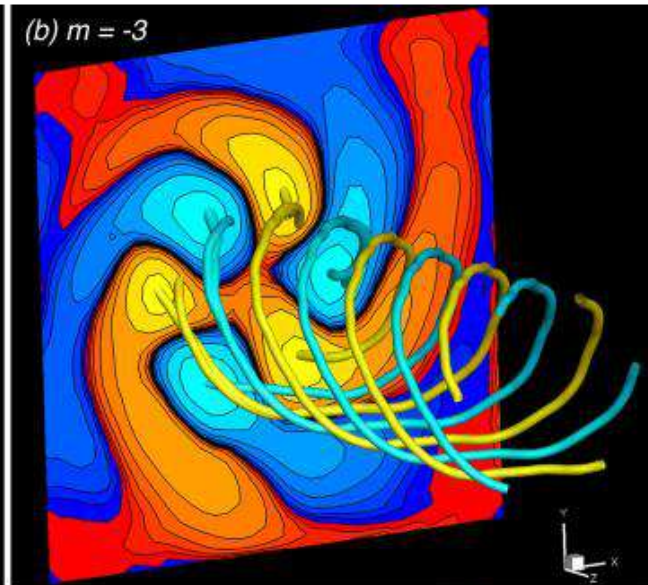
Francis Pagaud

Remerciements : Simon Dupouy

Plasma généré par une onde électromagnétique



Stenzel et Urrutia,
Phys. Plasmas (2015)



Stenzel et Urrutia,
Phys. Plasmas (2018)

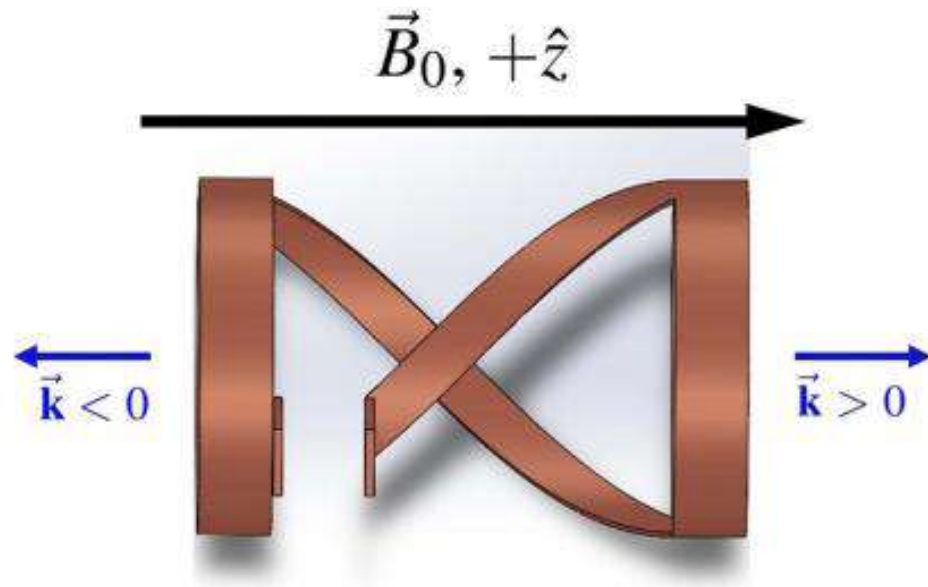
Fig. 1. Coil producing a transverse rf field across the plasma.

Boswell, Phys. Lett. (1970)

- Transfert d'énergie : onde hélicon → électrons
- Multitude d'ondes hélicons

Antenne et sélectivité des ondes

Antenne semi-hélicale



Stevenson *et al.*, Plasma Sources Sci. Technol. (2024)

Antenne birdcage

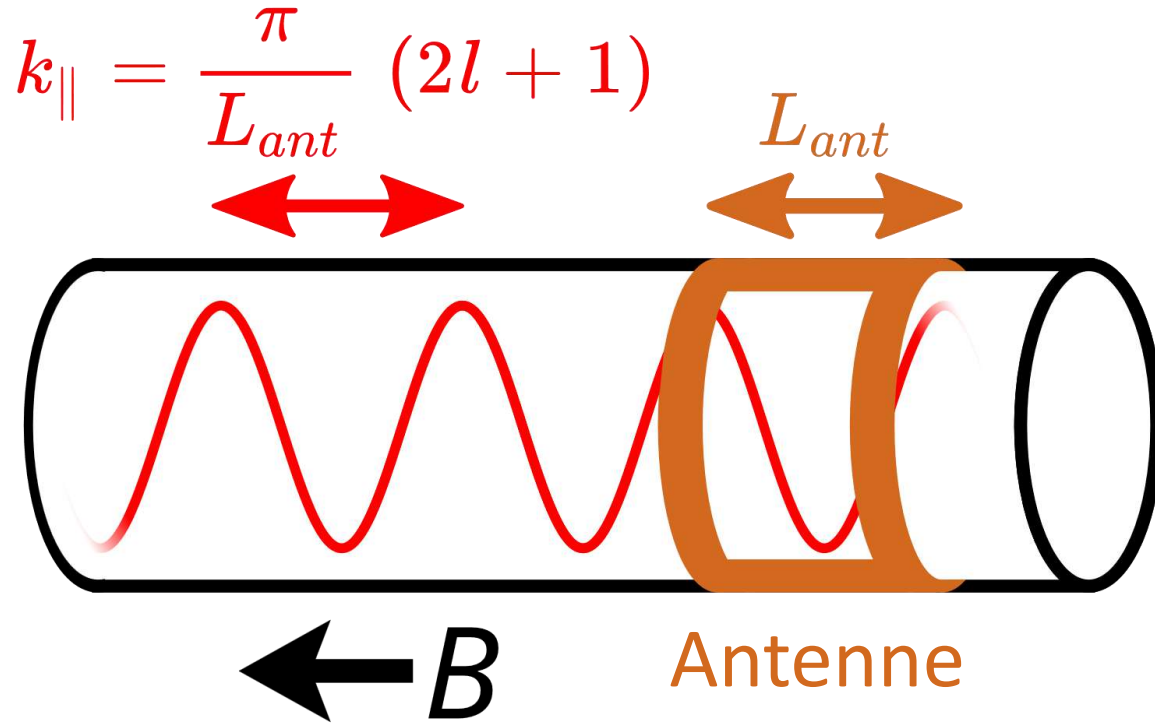


Guittienne *et al.*, Plasma Sources Sci. Technol. (2021)

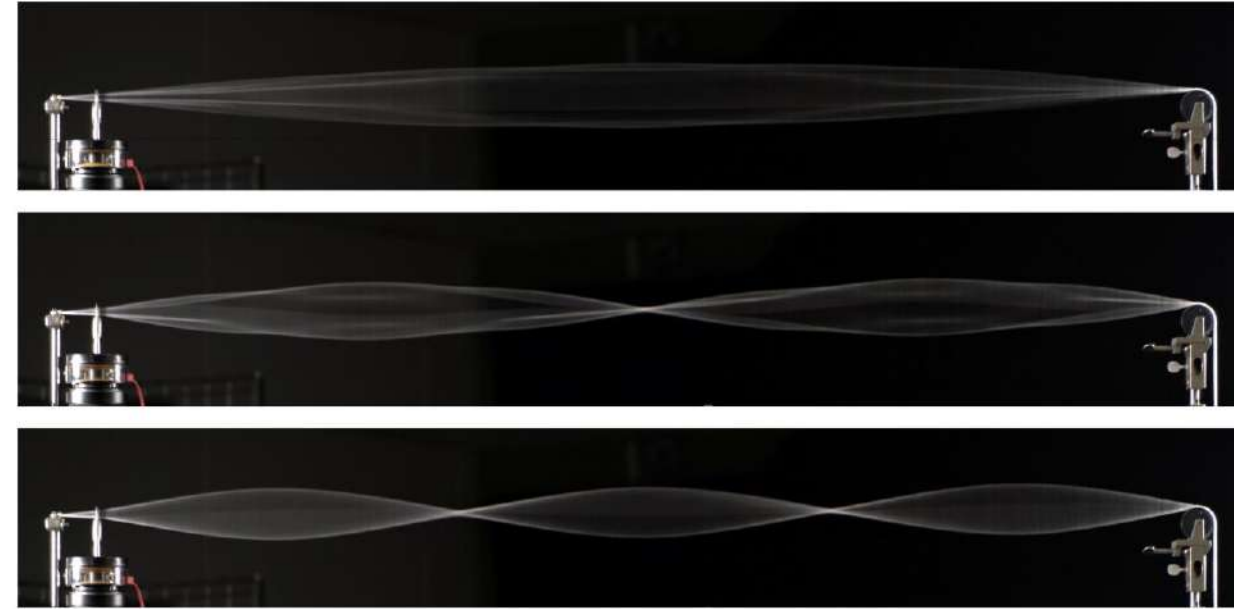
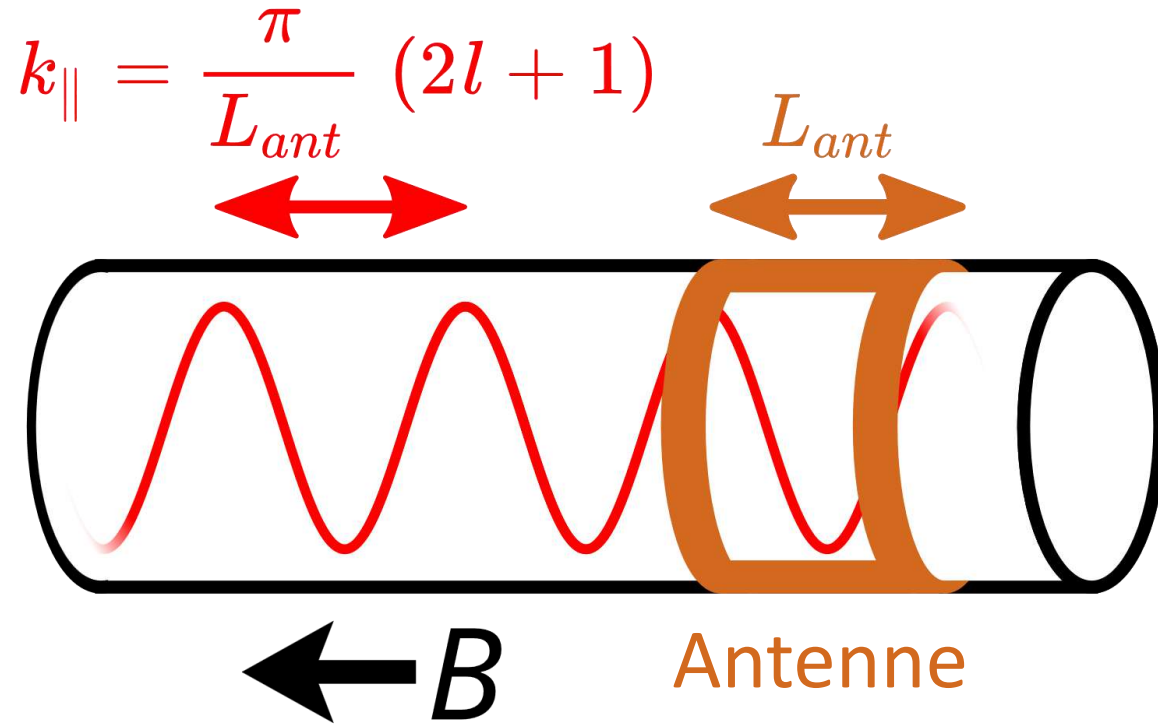
Vincent *et al.*, ArXiv (2024)

➤ L'antenne induit une sélectivité des ondes hélicons

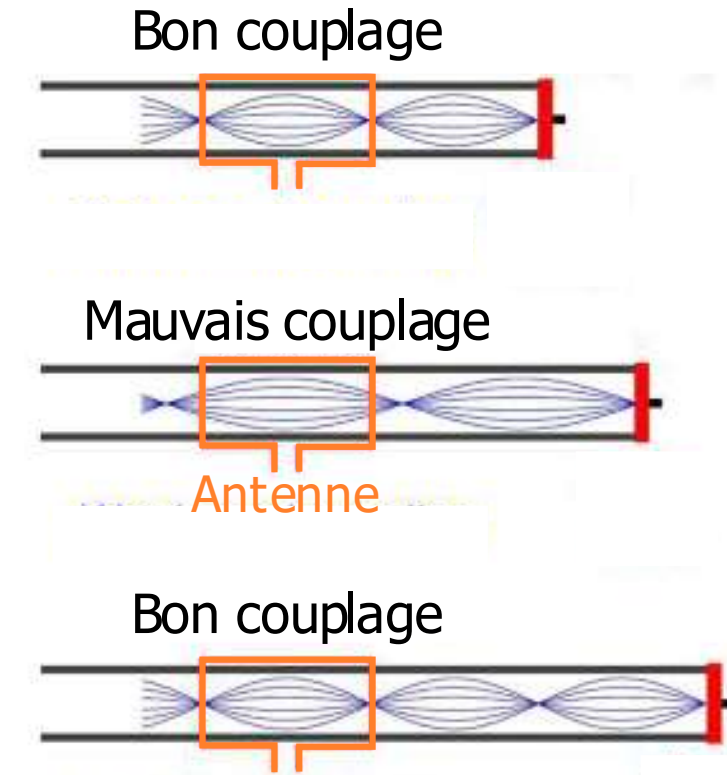
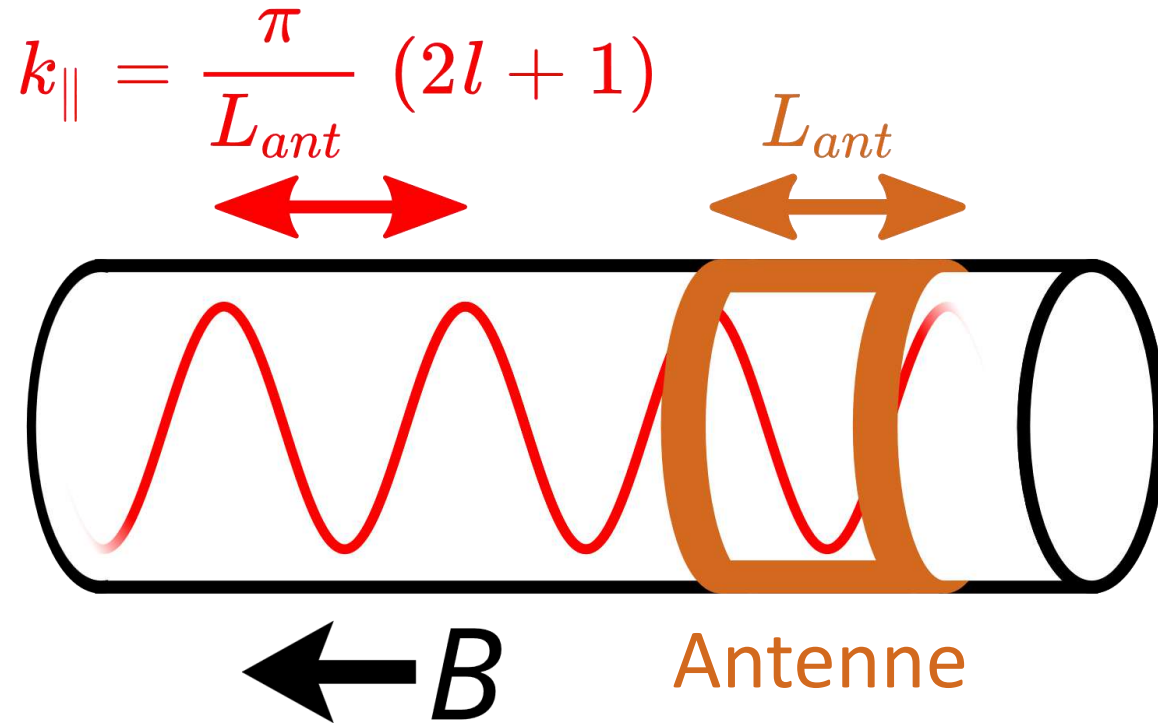
Onde hélicon dans une cavité



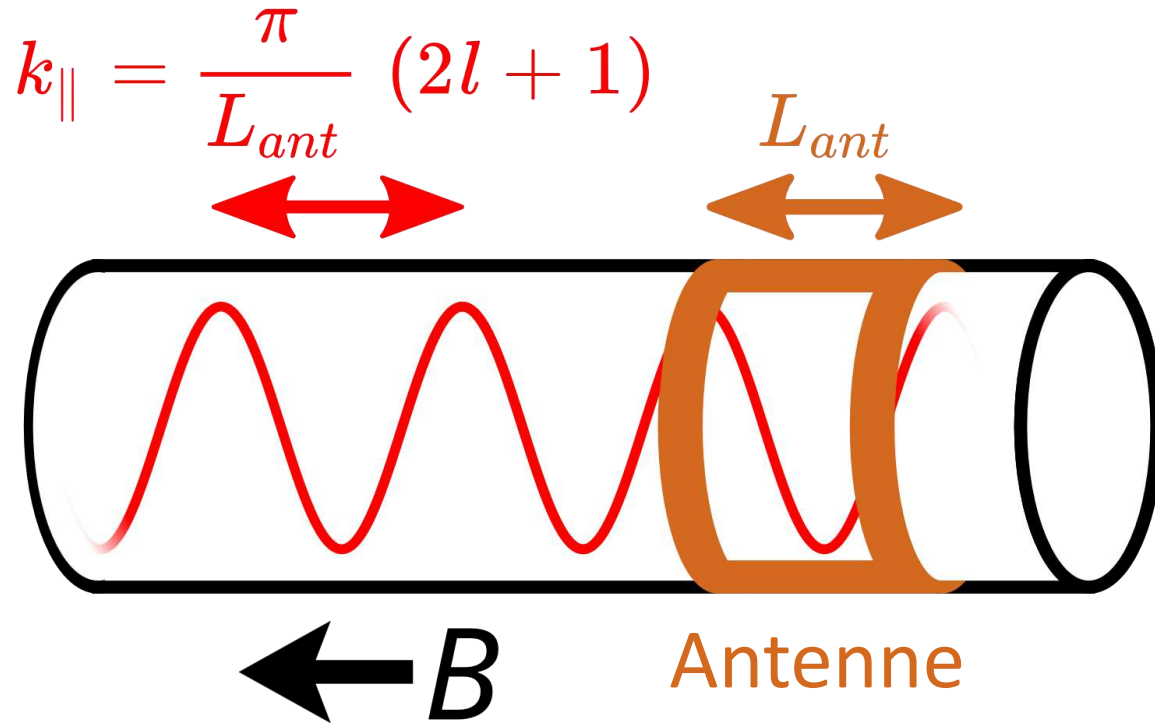
Onde hélicon dans une cavité



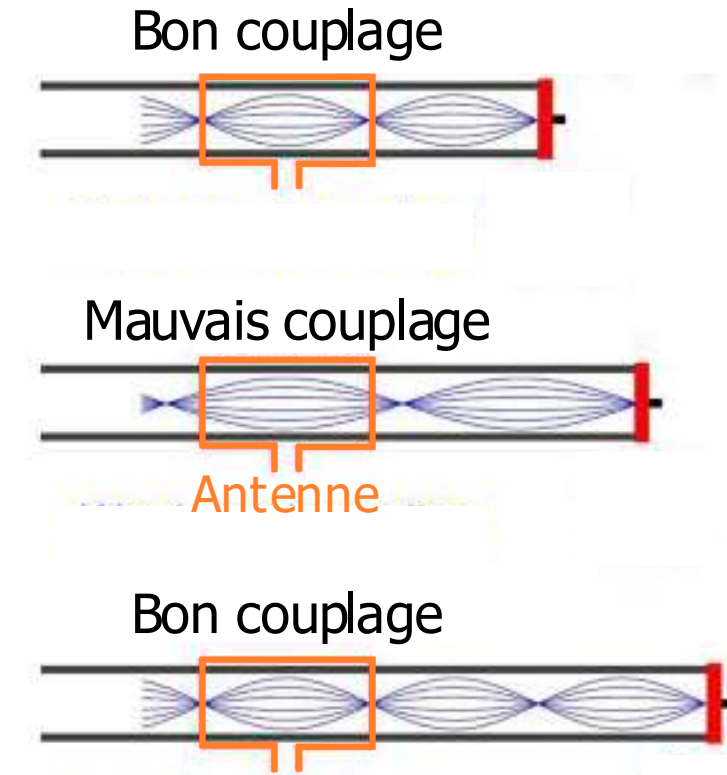
Onde hélicon dans une cavité



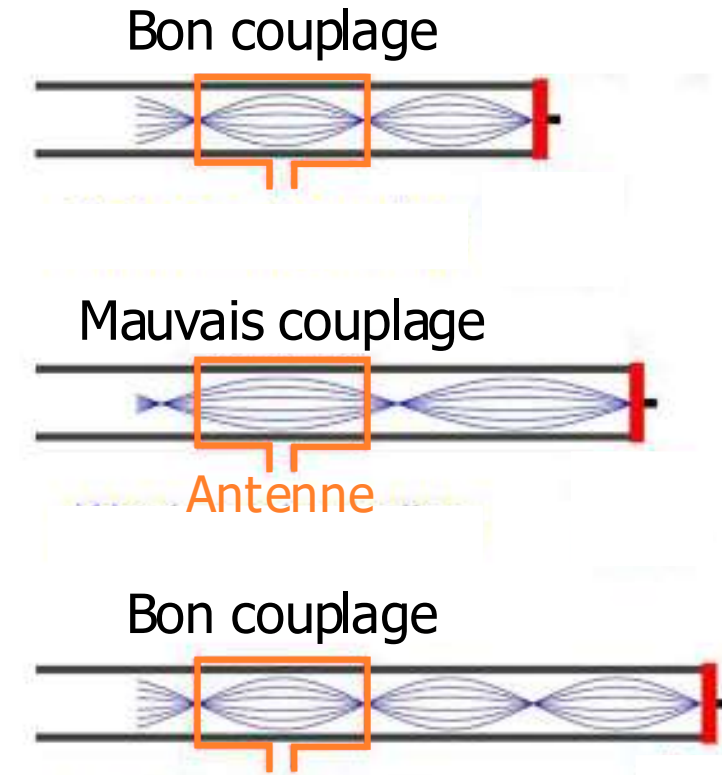
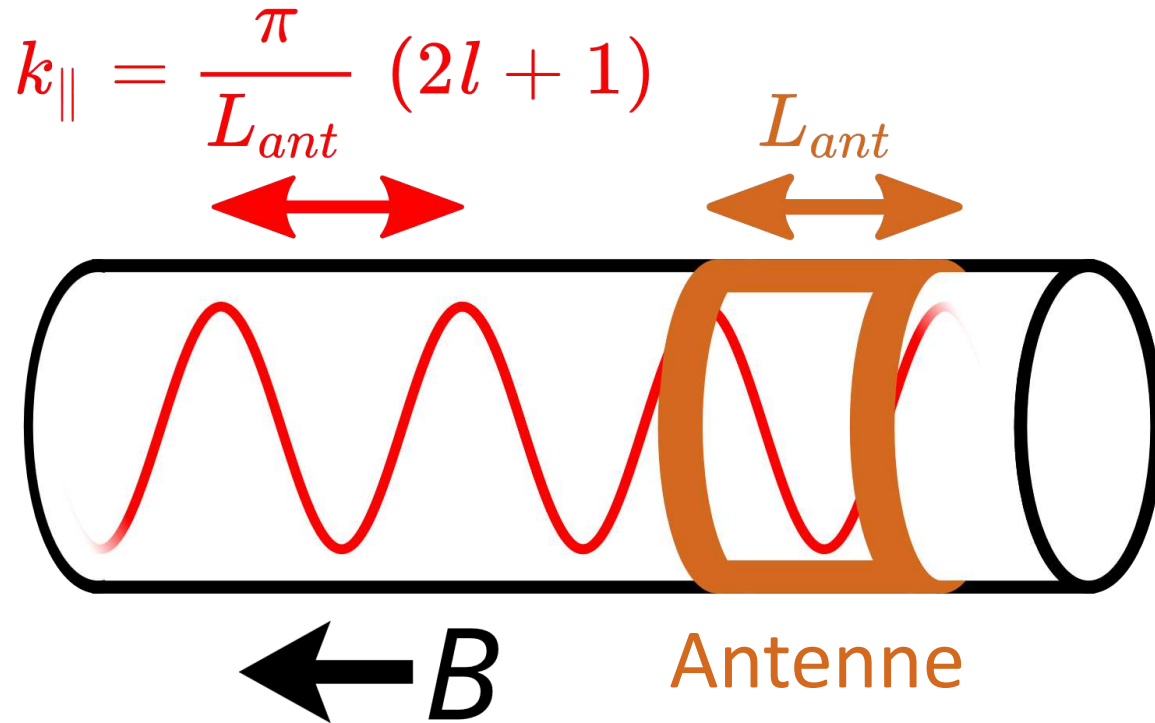
Onde hélicon dans une cavité



$$\frac{\omega}{k} = \frac{Bk_{||}}{ne\mu_0}$$

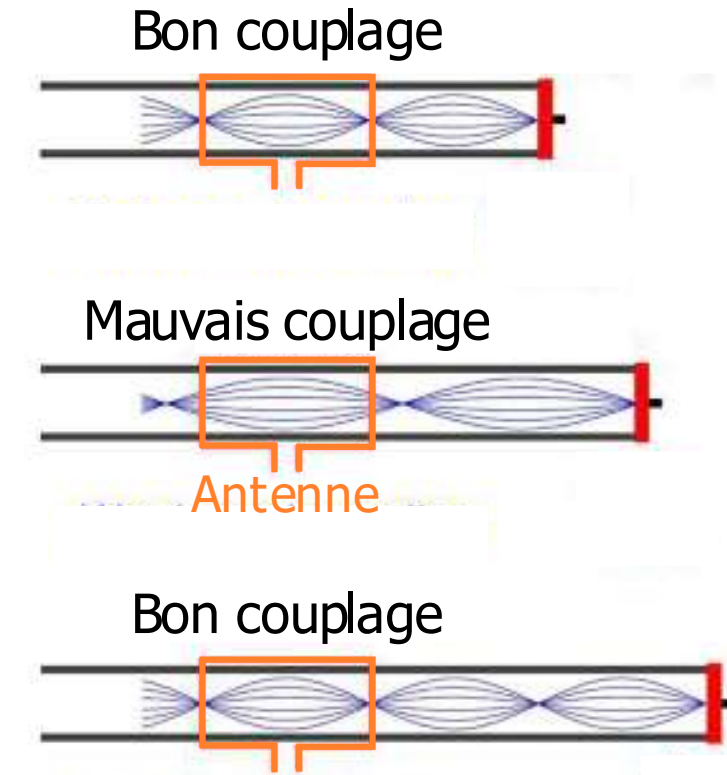
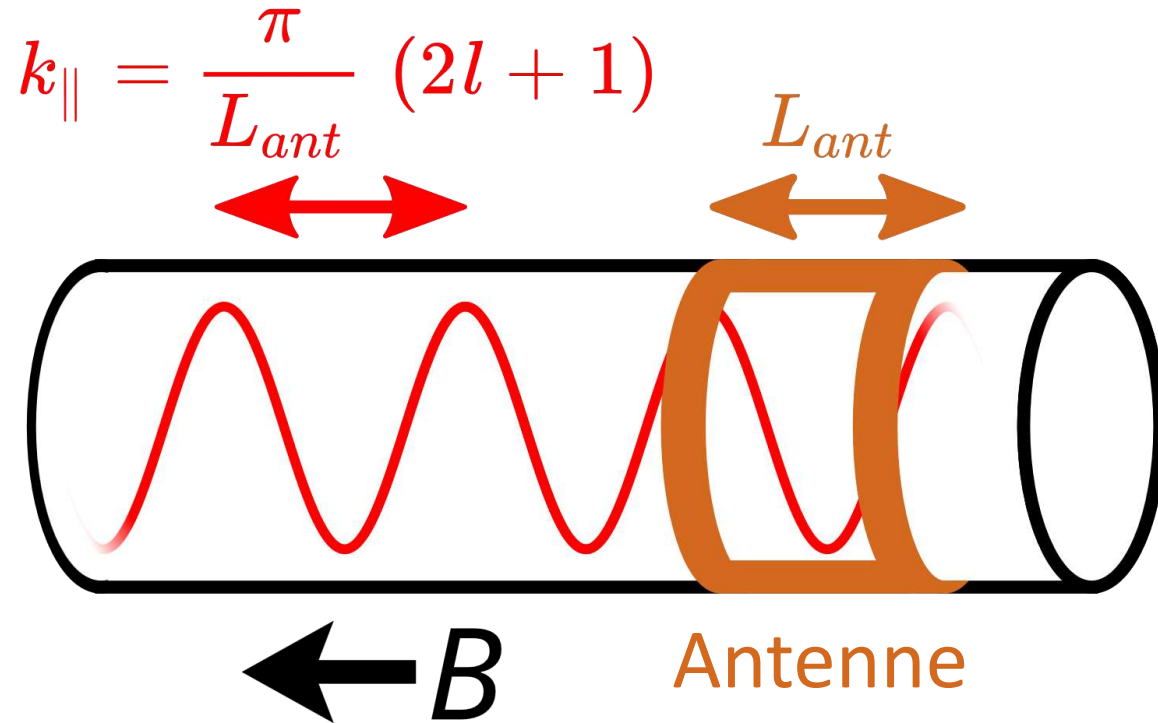


Onde hélicon dans une cavité



Radio-fréquence 13.56 MHz $\rightarrow \frac{\omega}{k} = \frac{Bk_{||}}{ne\mu_0} \leftarrow$ Constantes universelles

Onde hélicon dans une cavité



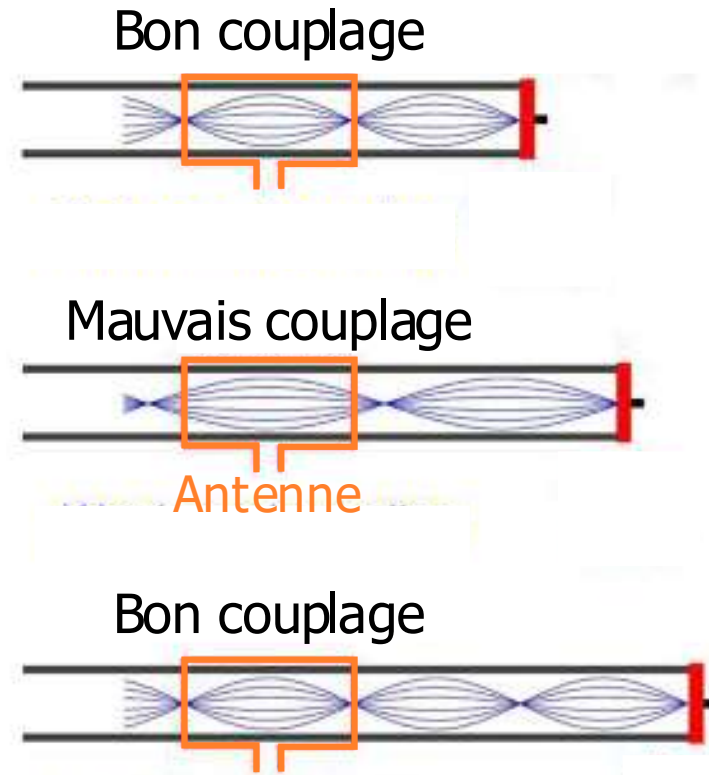
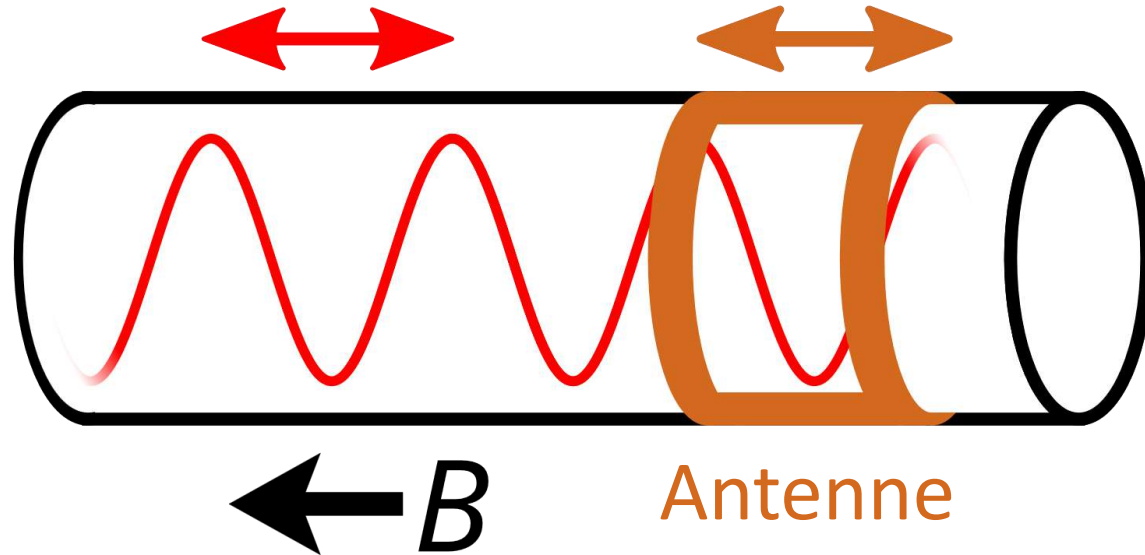
Radio-fréquence 13.56 MHz $\rightarrow \frac{\omega}{k} = \frac{B k_{||}}{n e \mu_0}$

$\leftarrow$ Dimensions du système

$\leftarrow$ Constantes universelles

Onde hélicon dans une cavité

$$k_{\parallel} = \frac{\pi}{L_{ant}} (2l + 1)$$



Paramètre de contrôle

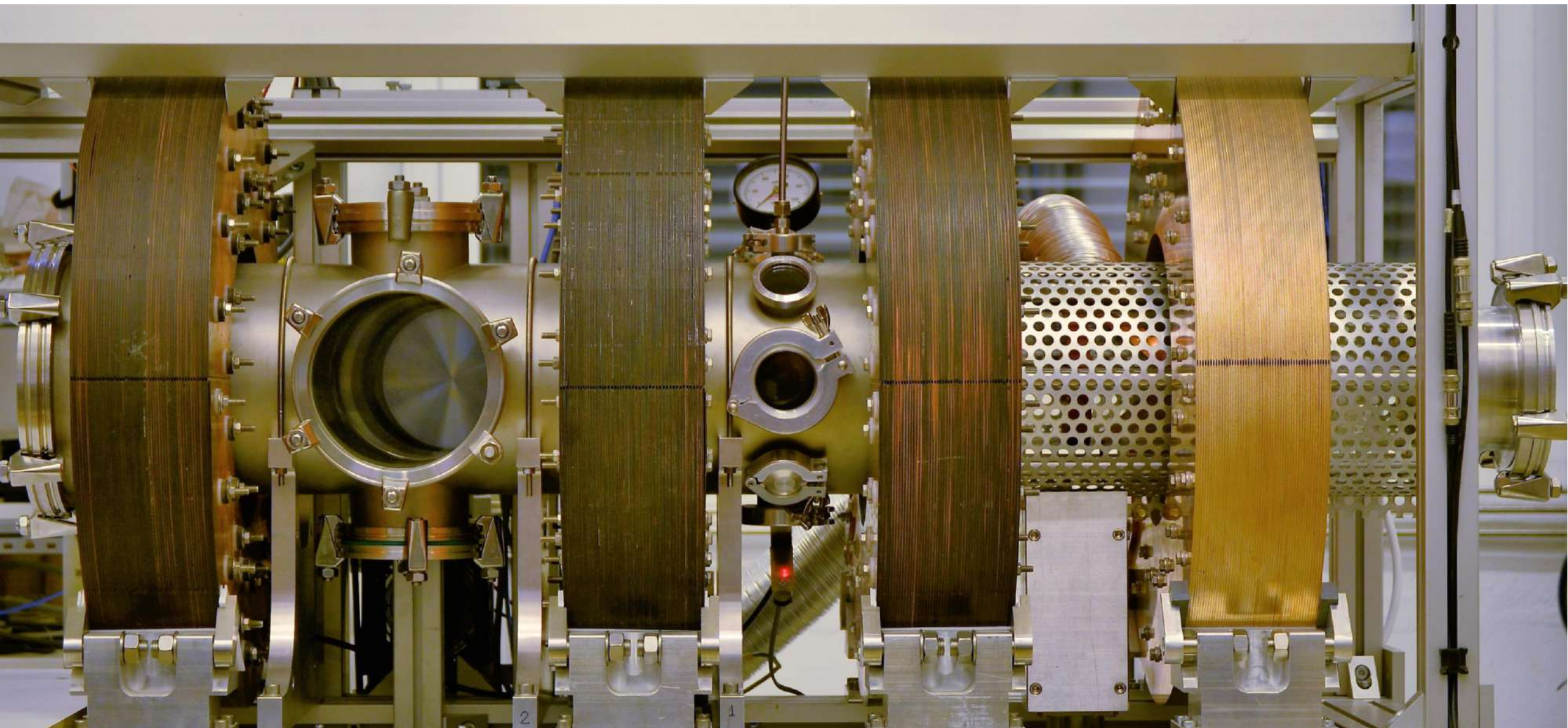
$$\frac{\omega}{k} = \frac{B k_{\parallel}}{n e \mu_0}$$

Radio-fréquence 13.56 MHz → ω

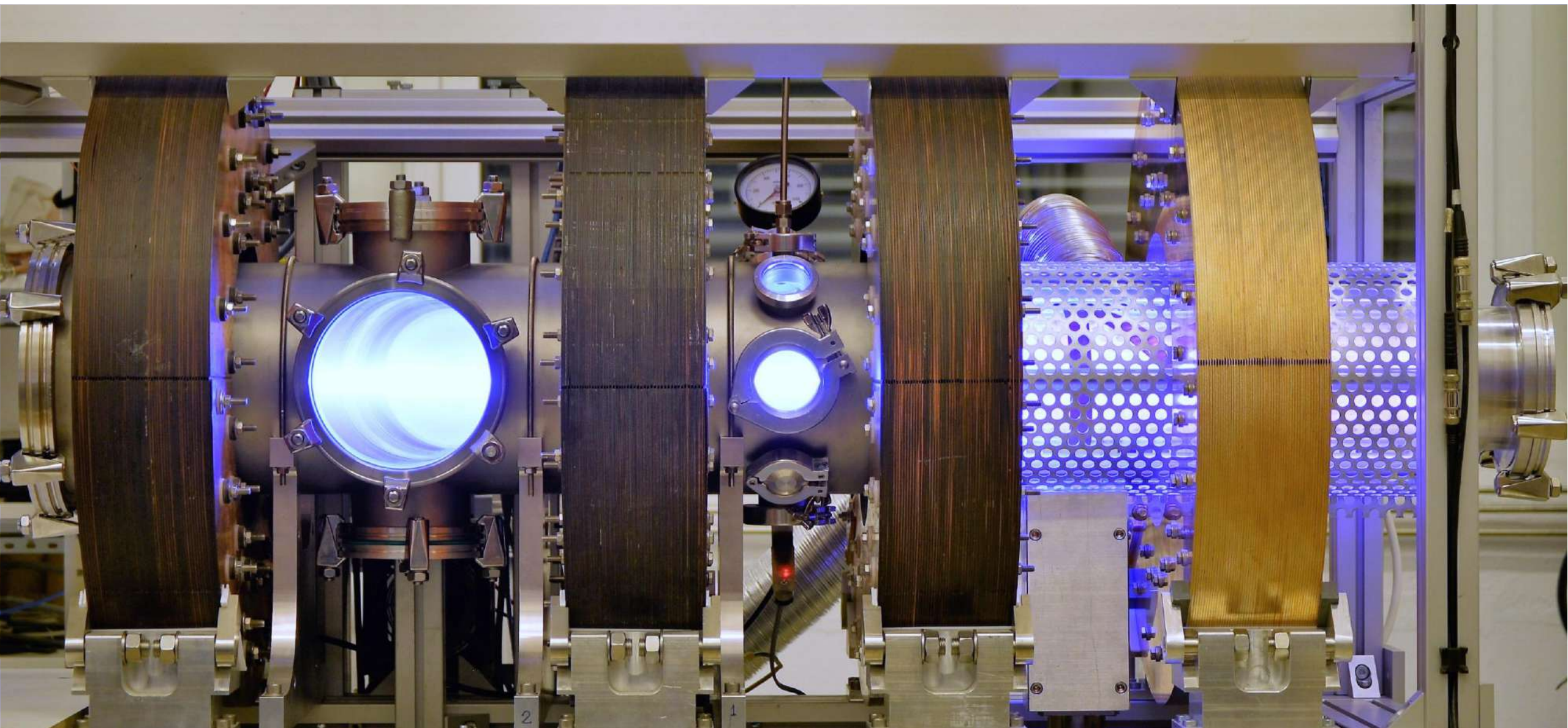
← $B k_{\parallel}$ Dimensions du système

← $n e \mu_0$ Constantes universelles

Installation d'une antenne hélicon



Installation d'une antenne hélicon



Installation d'une antenne hélicon

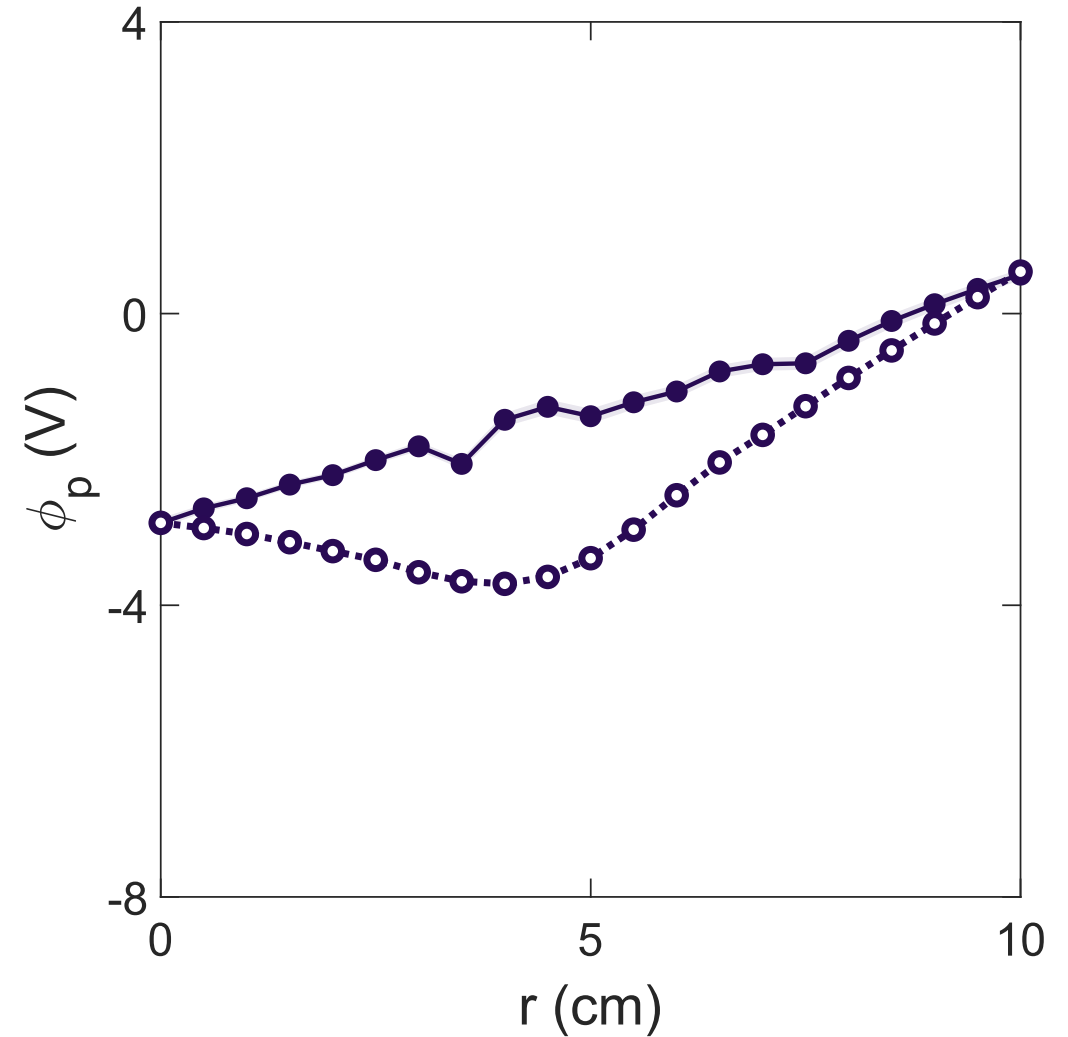
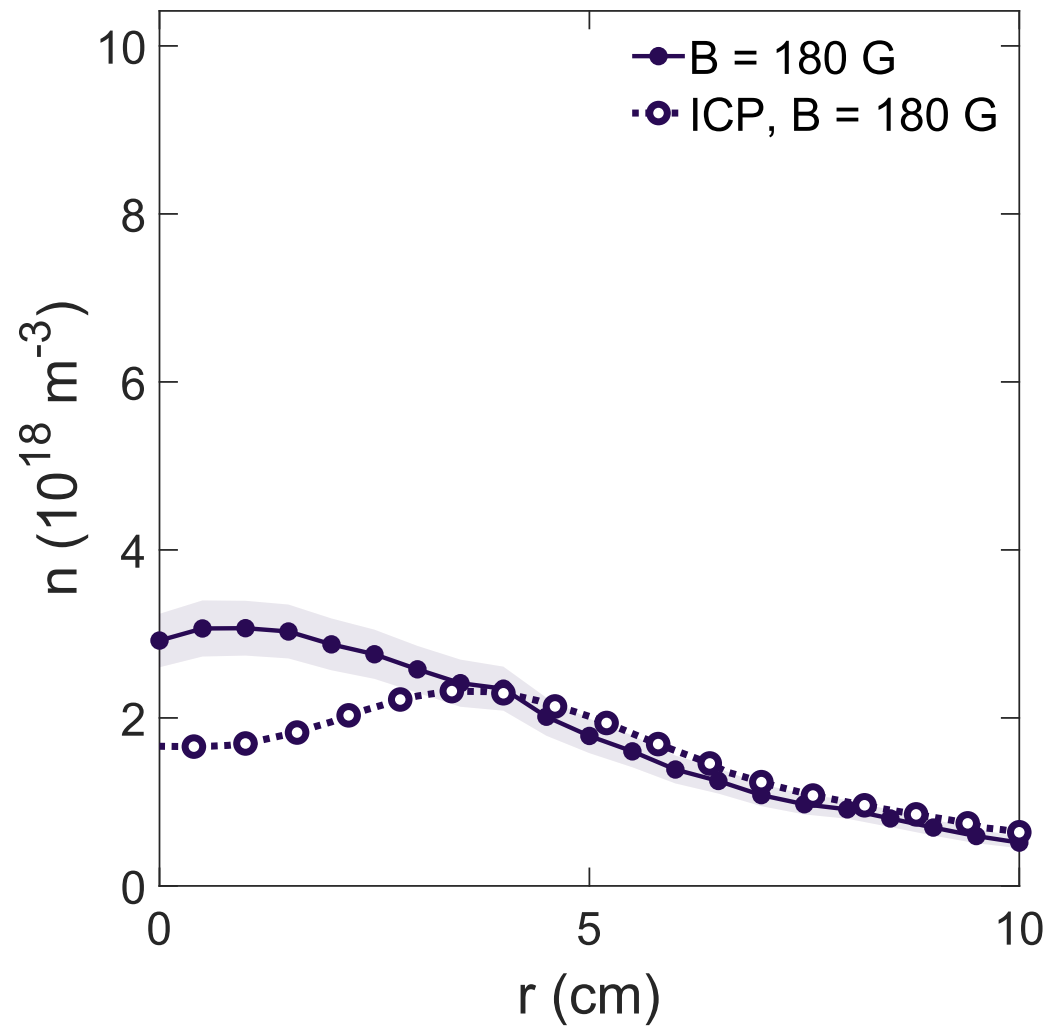
Motivation :

Caractérisation des régimes opératoires et de leur stabilité



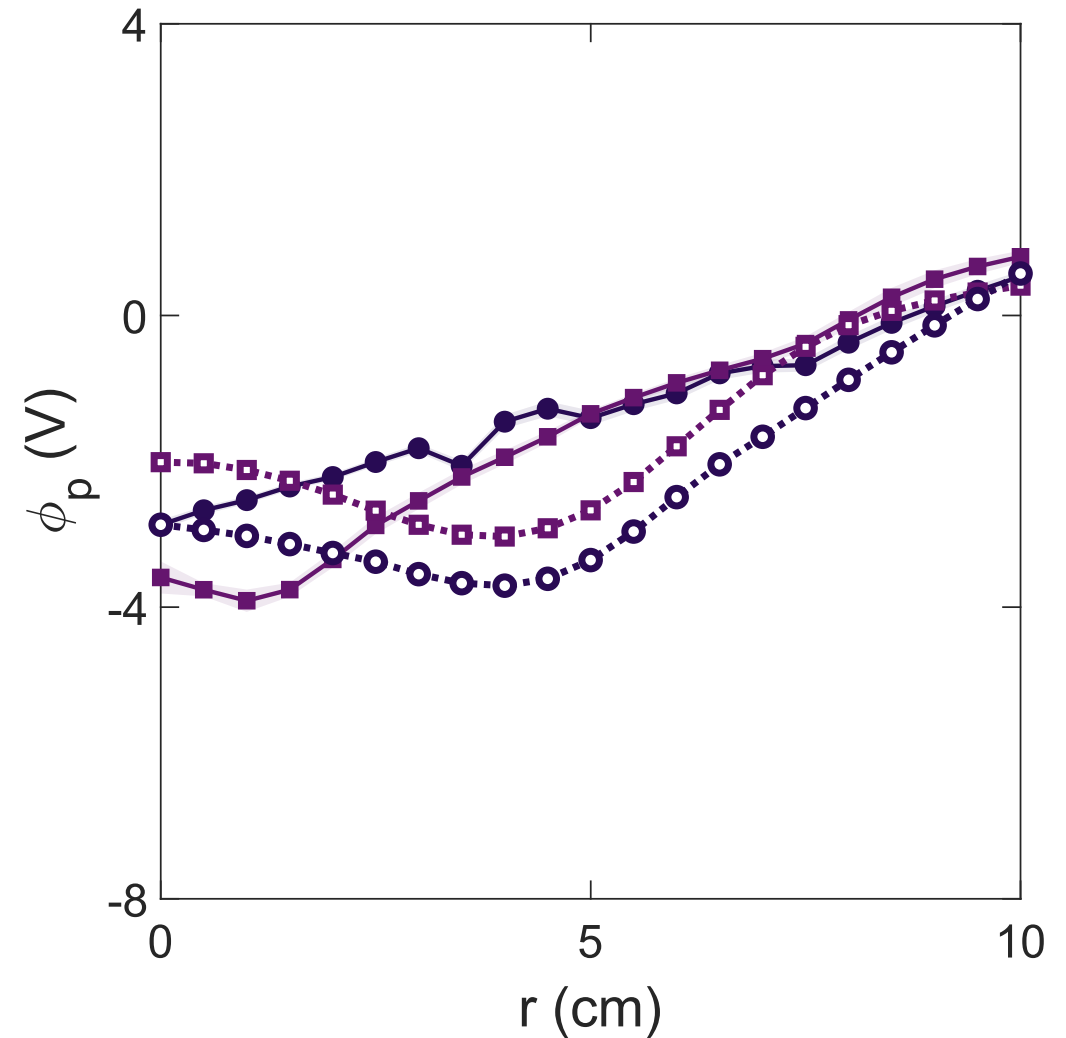
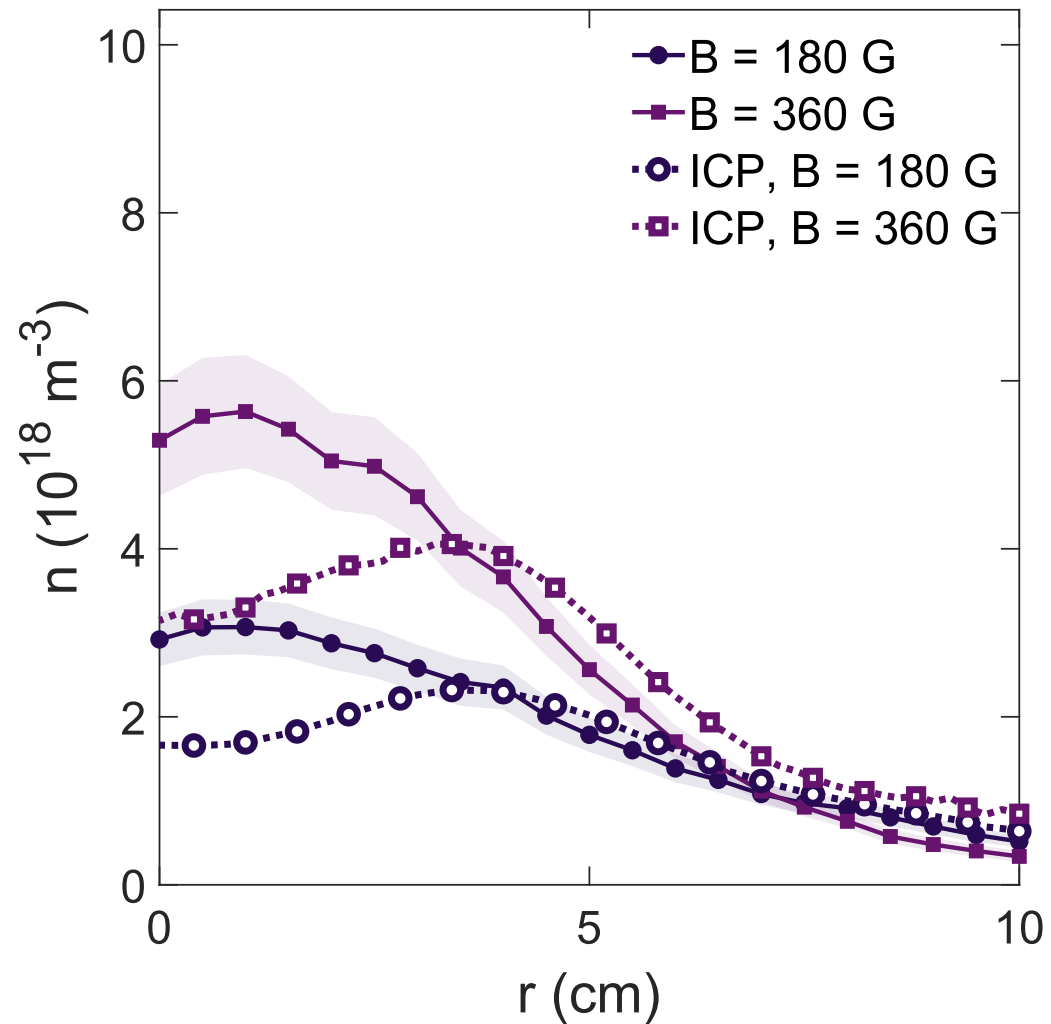
Caractérisation du plasma hélicon

Mode **hélicon**, 0.13 Pa, 1 kW



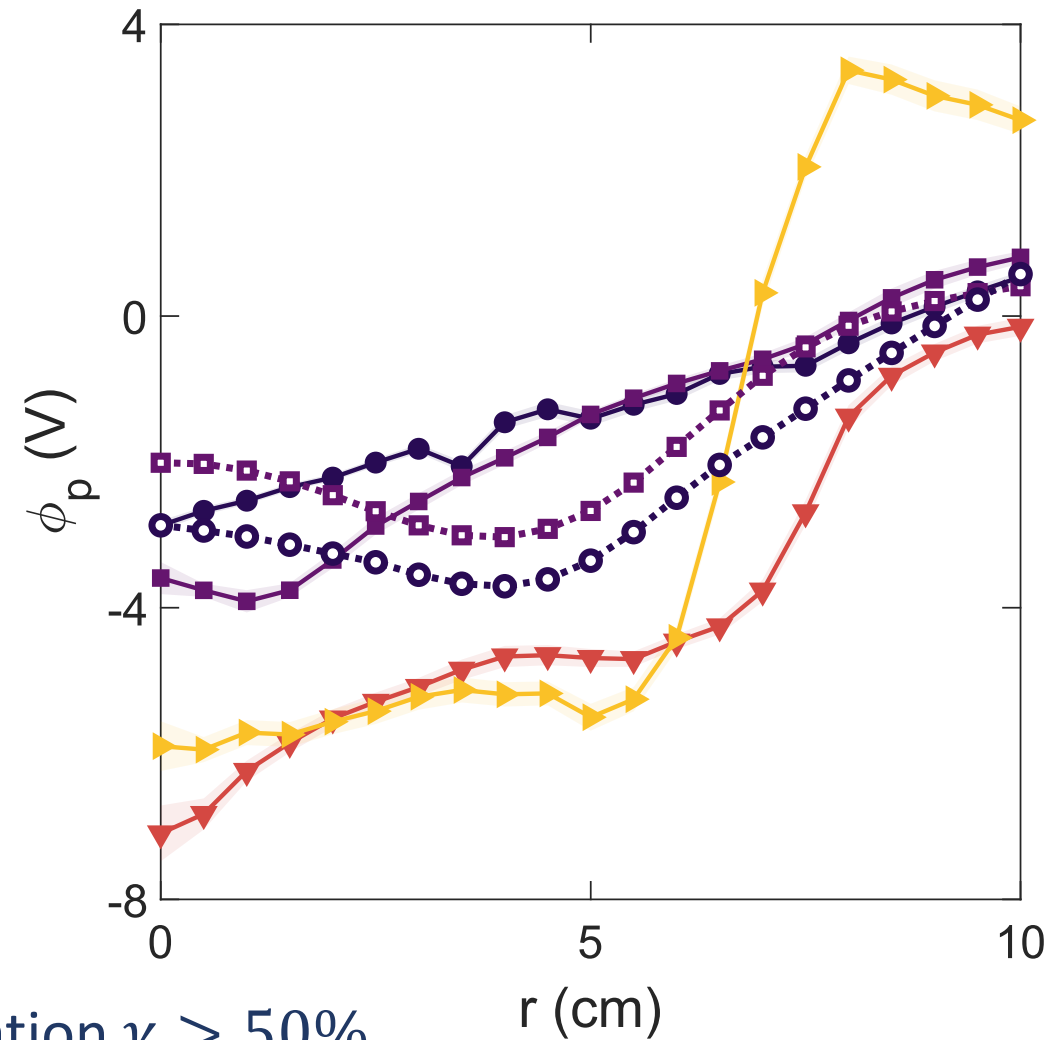
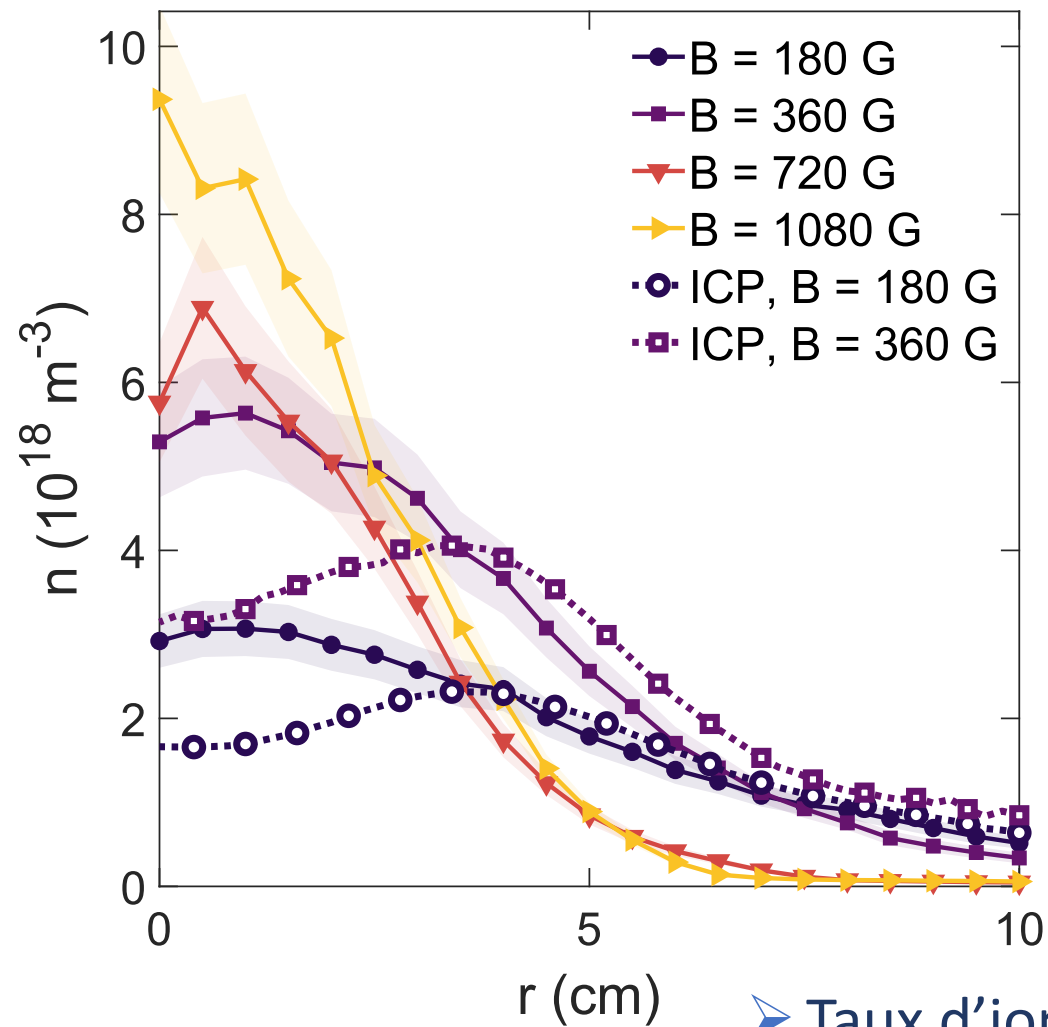
Caractérisation du plasma hélicon

Mode **hélicon**, 0.13 Pa, 1 kW



Caractérisation du plasma hélicon

Mode hélicon, 0.13 Pa, 1 kW



➤ Taux d'ionisation $\gamma > 50\%$

➤ Fort gradients radiaux lorsque $B \nearrow$

Diagramme d'opération

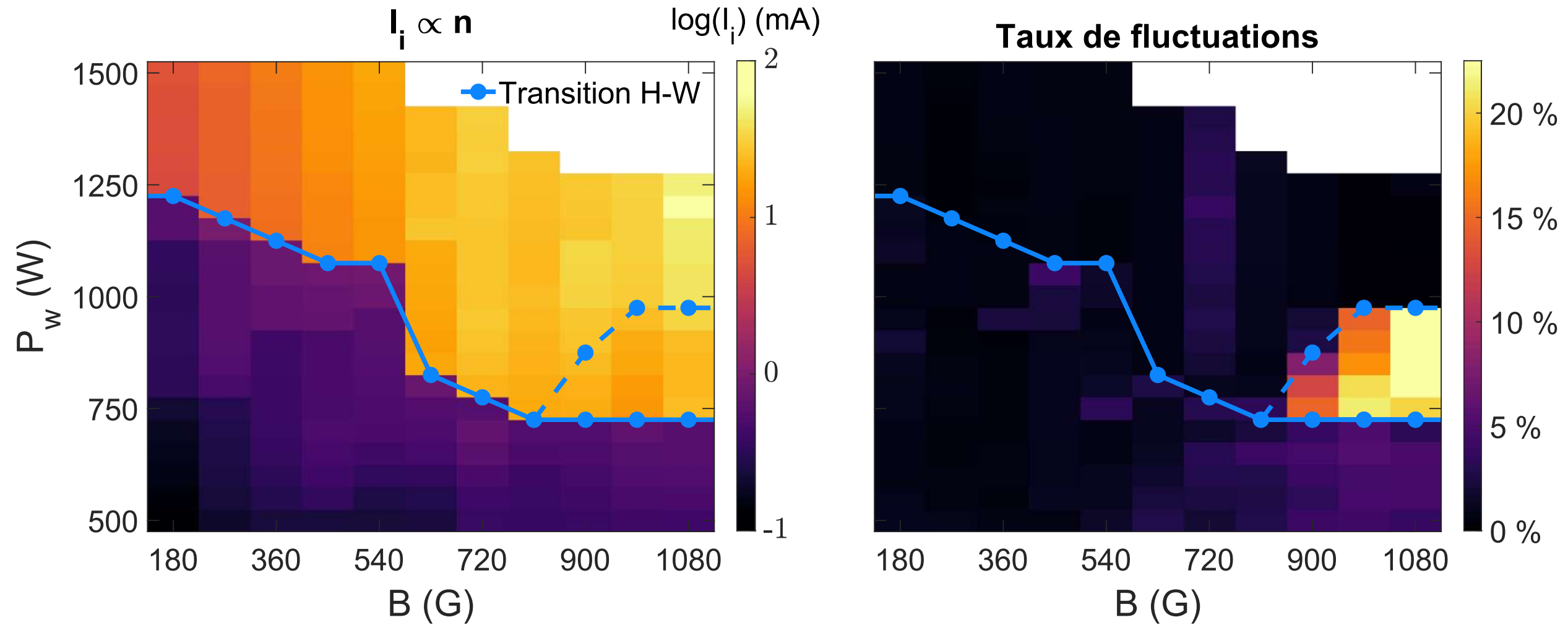


Diagramme d'opération

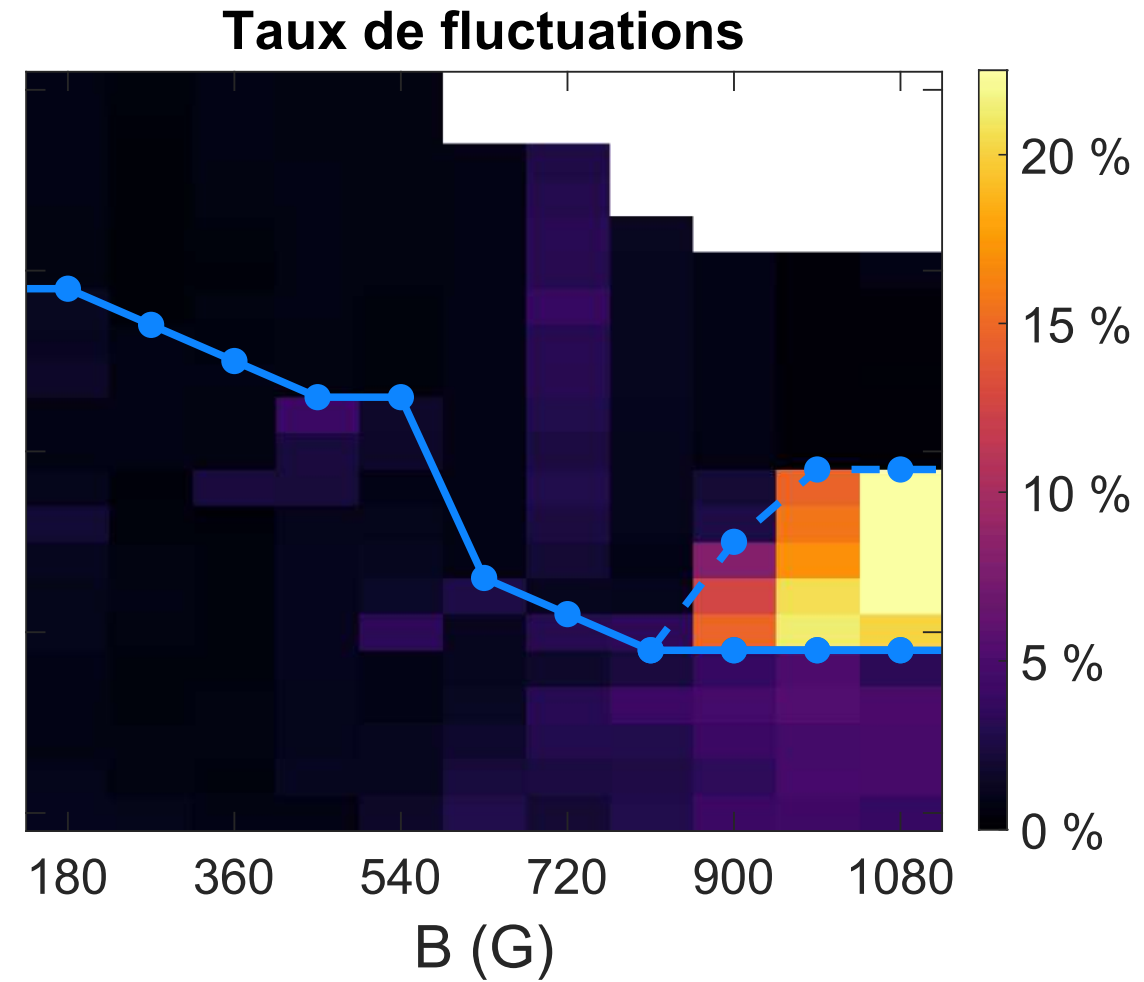
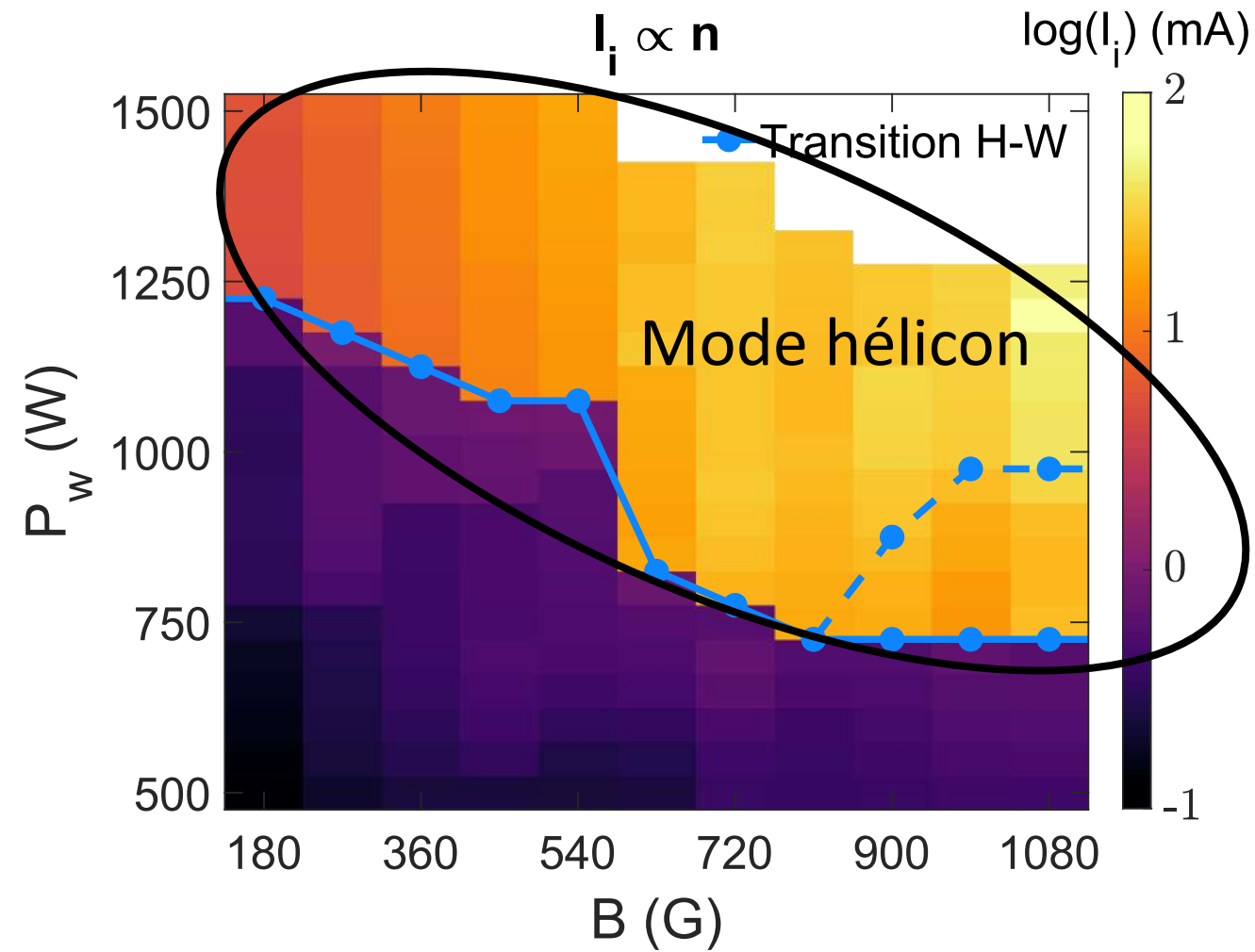


Diagramme d'opération

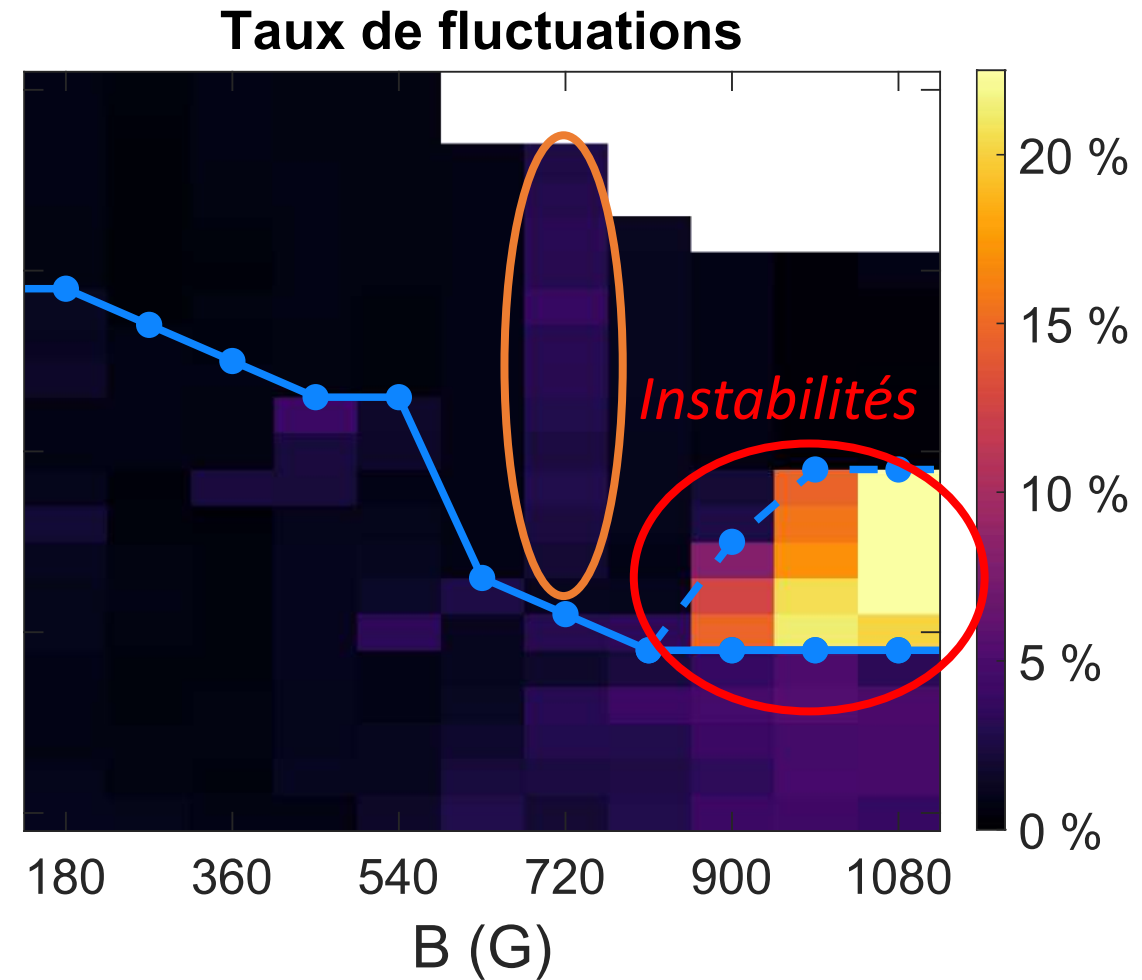
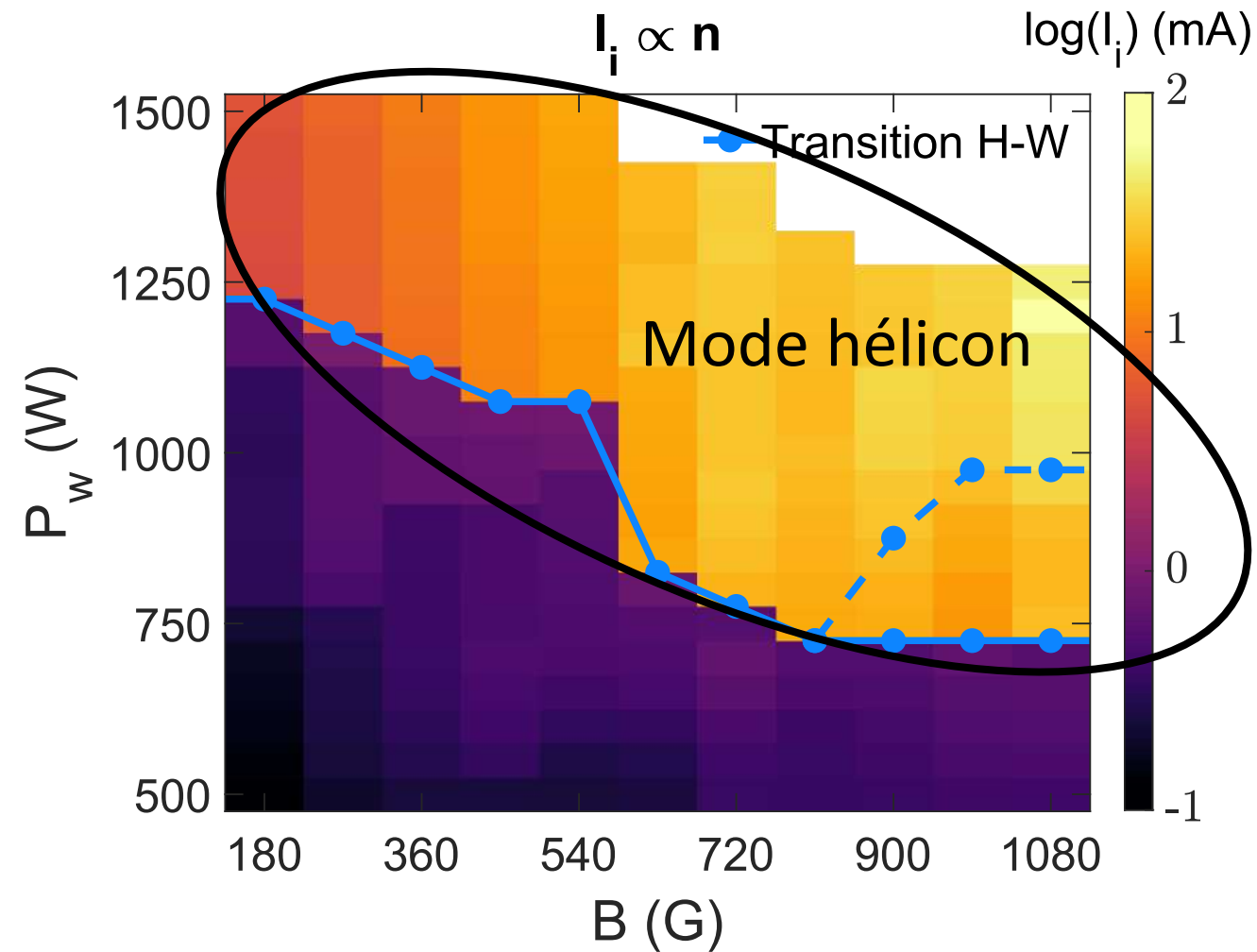
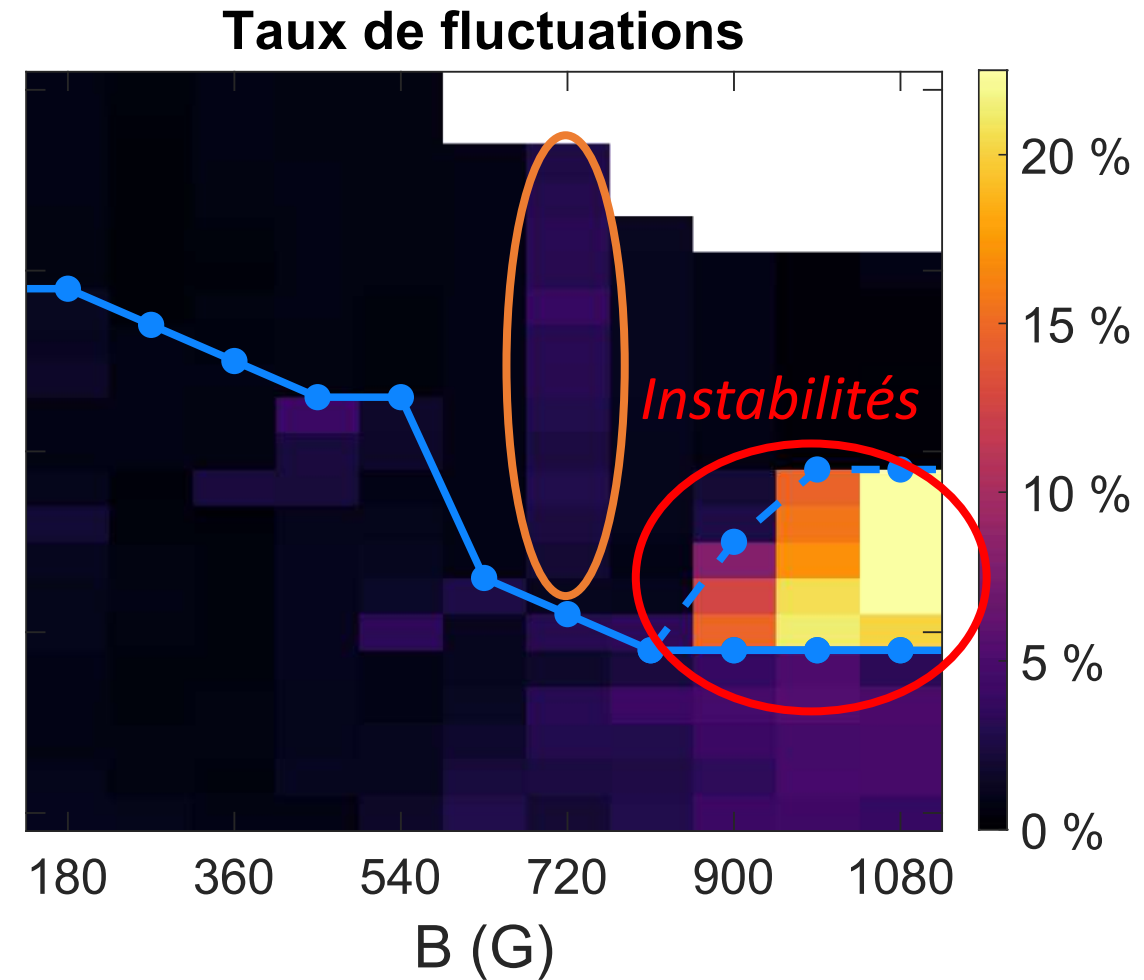
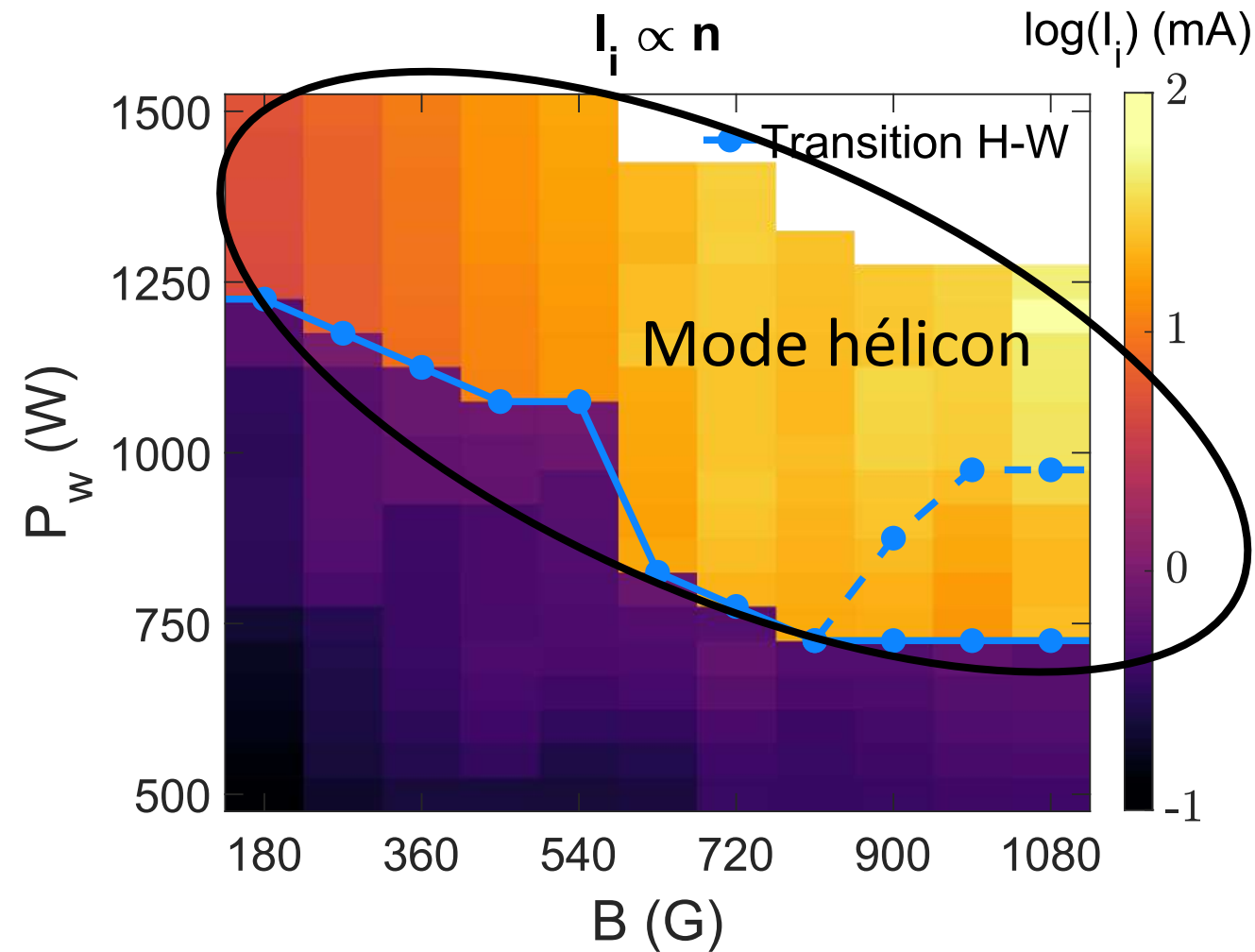
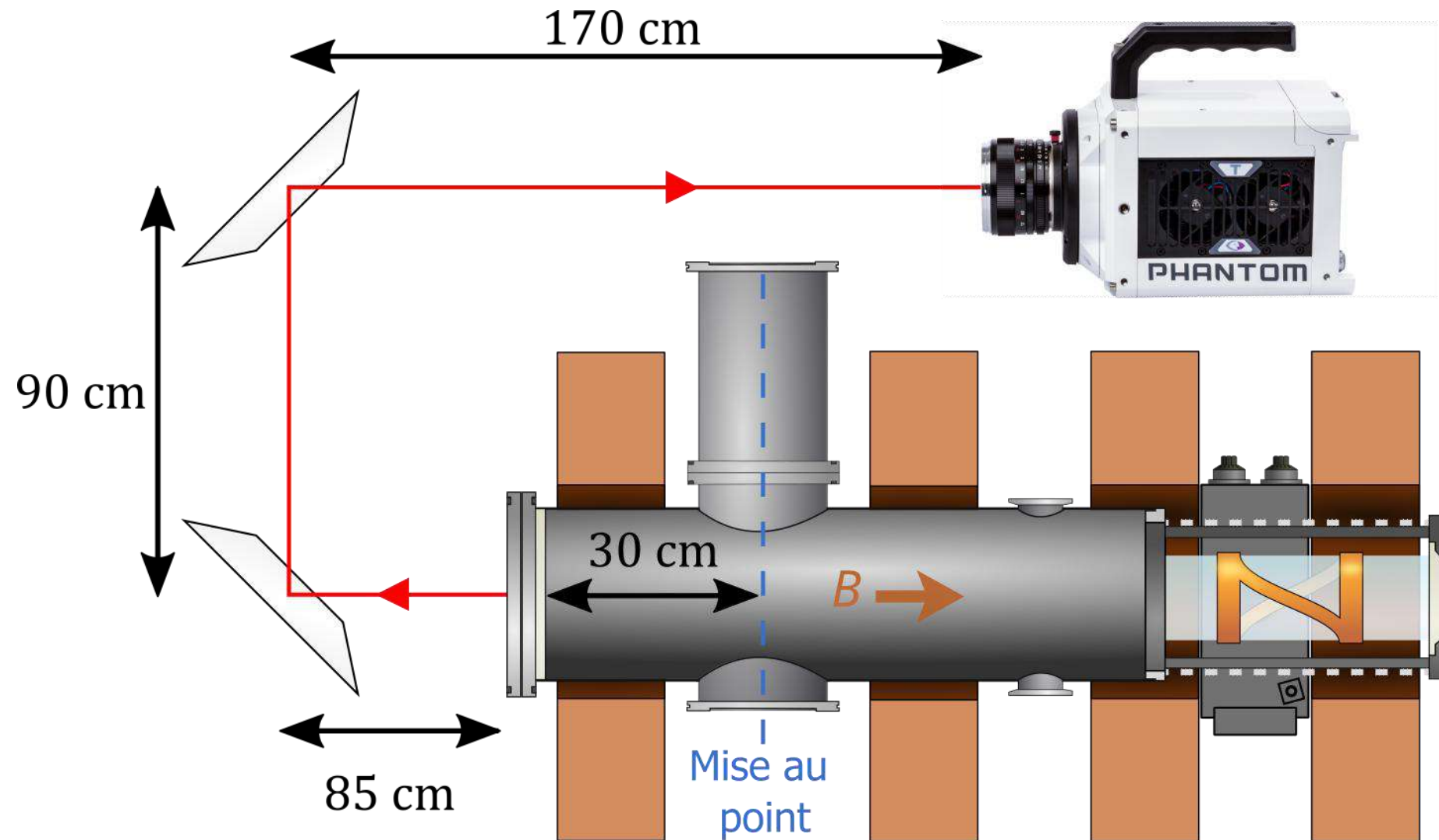


Diagramme d'opération



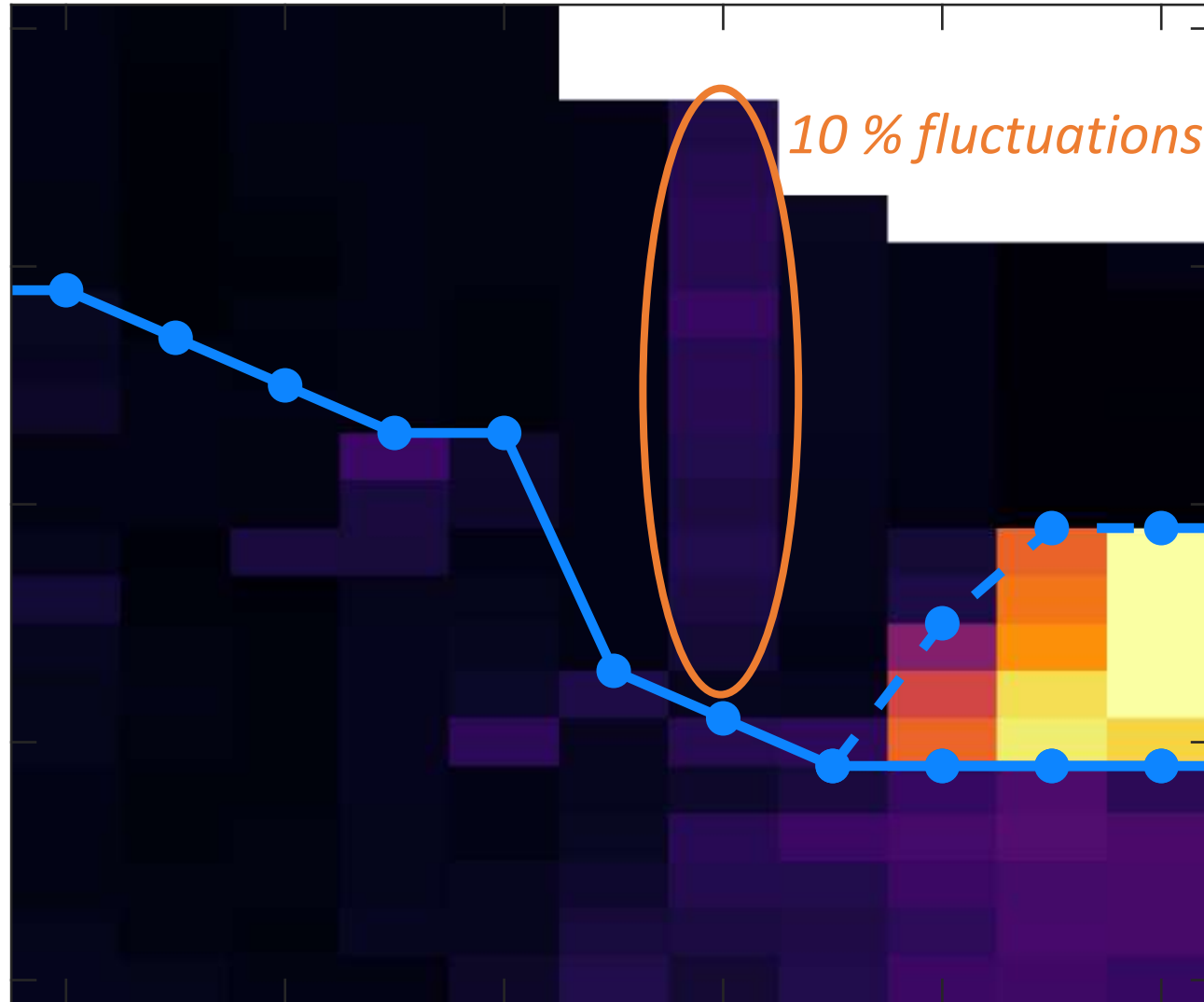
- Transition au mode hélicon
- Domaine de hautes fluctuations = instabilités

Imagerie rapide du plasma



- 200 000 FPS, filtre à 750 nm (Argon neutre)

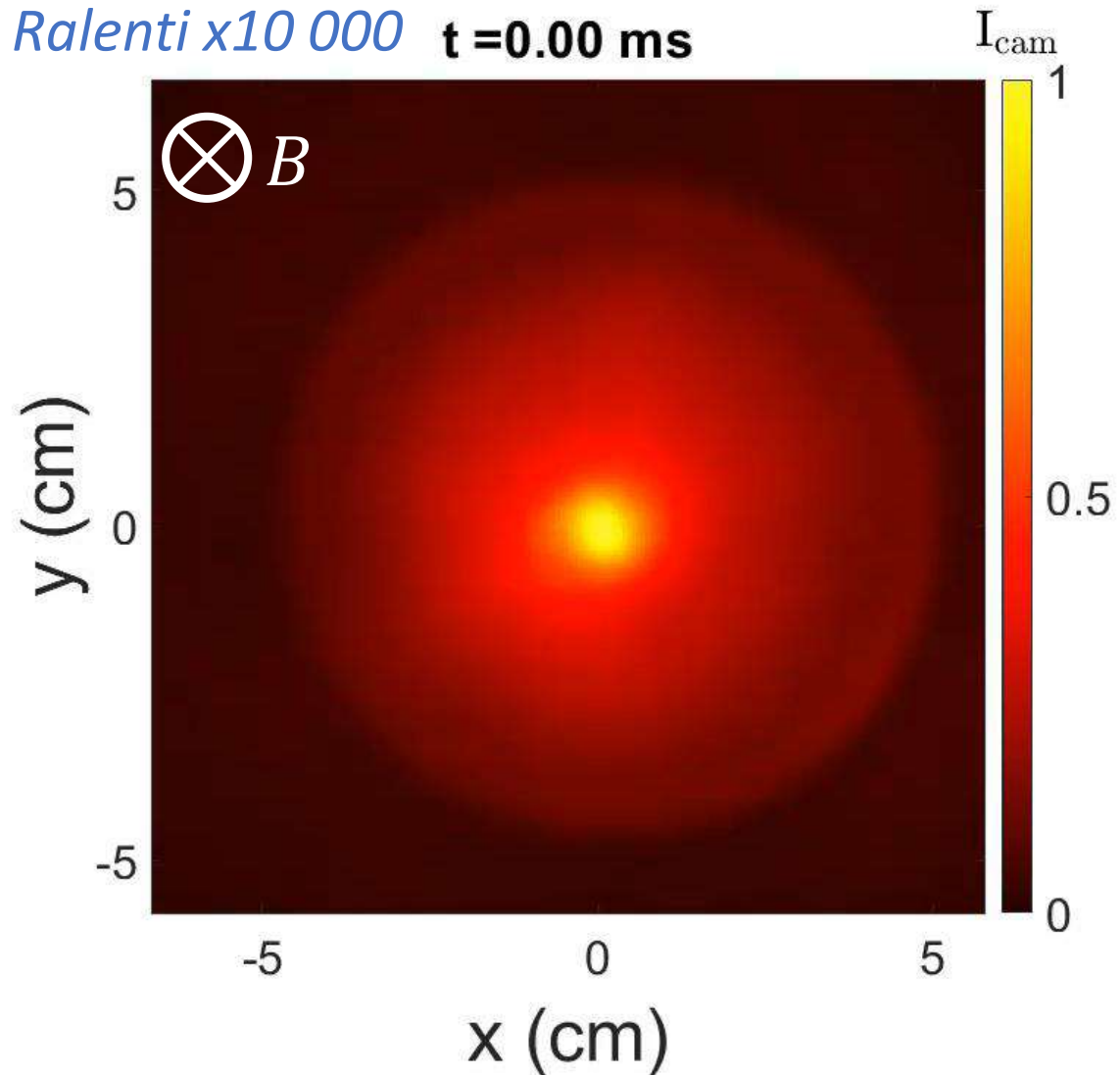
Imagerie rapide du plasma



Analyse des fluctuations de lumière

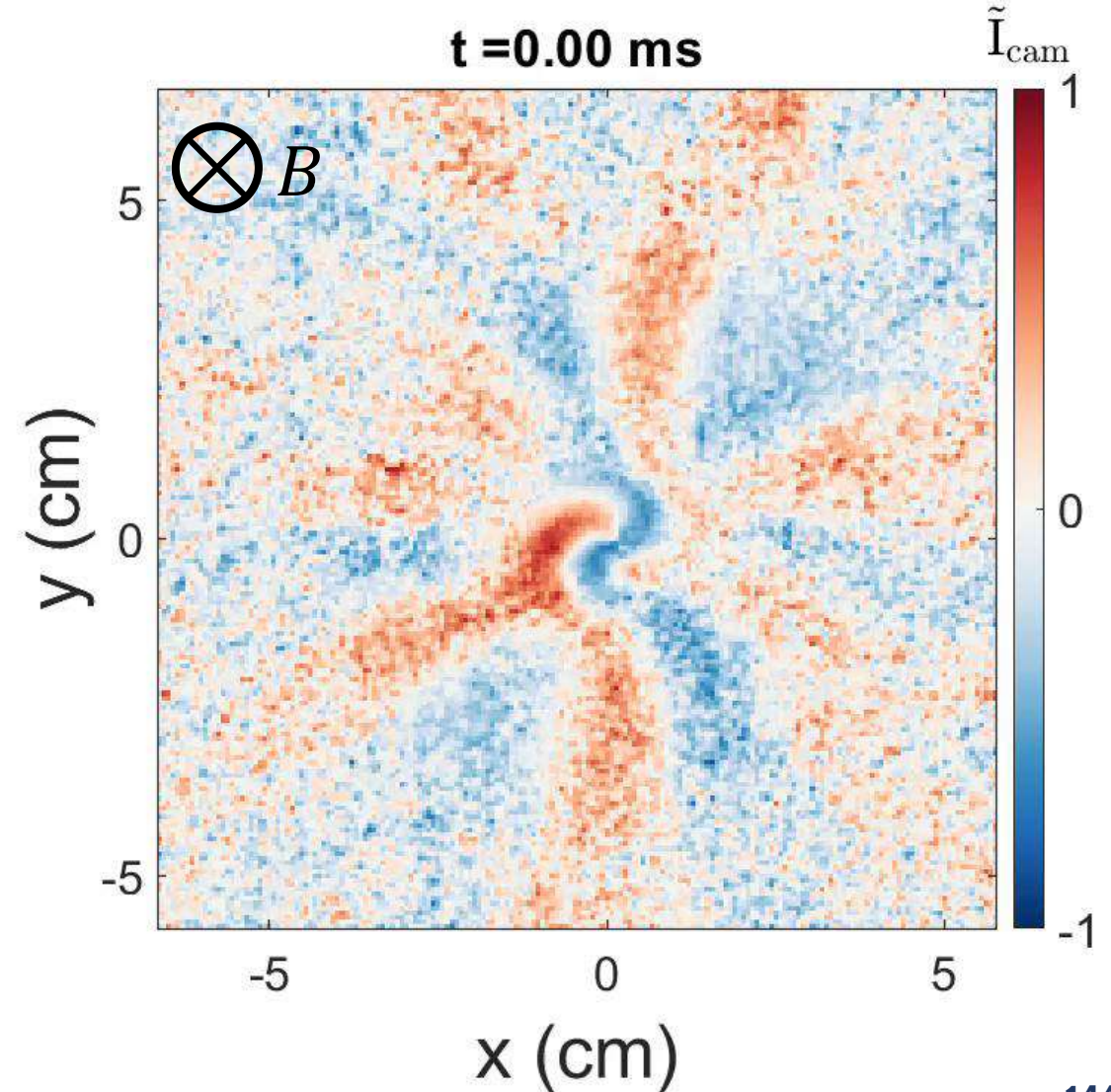
Images brutes

Ralenti x10 000 **t = 0.00 ms**



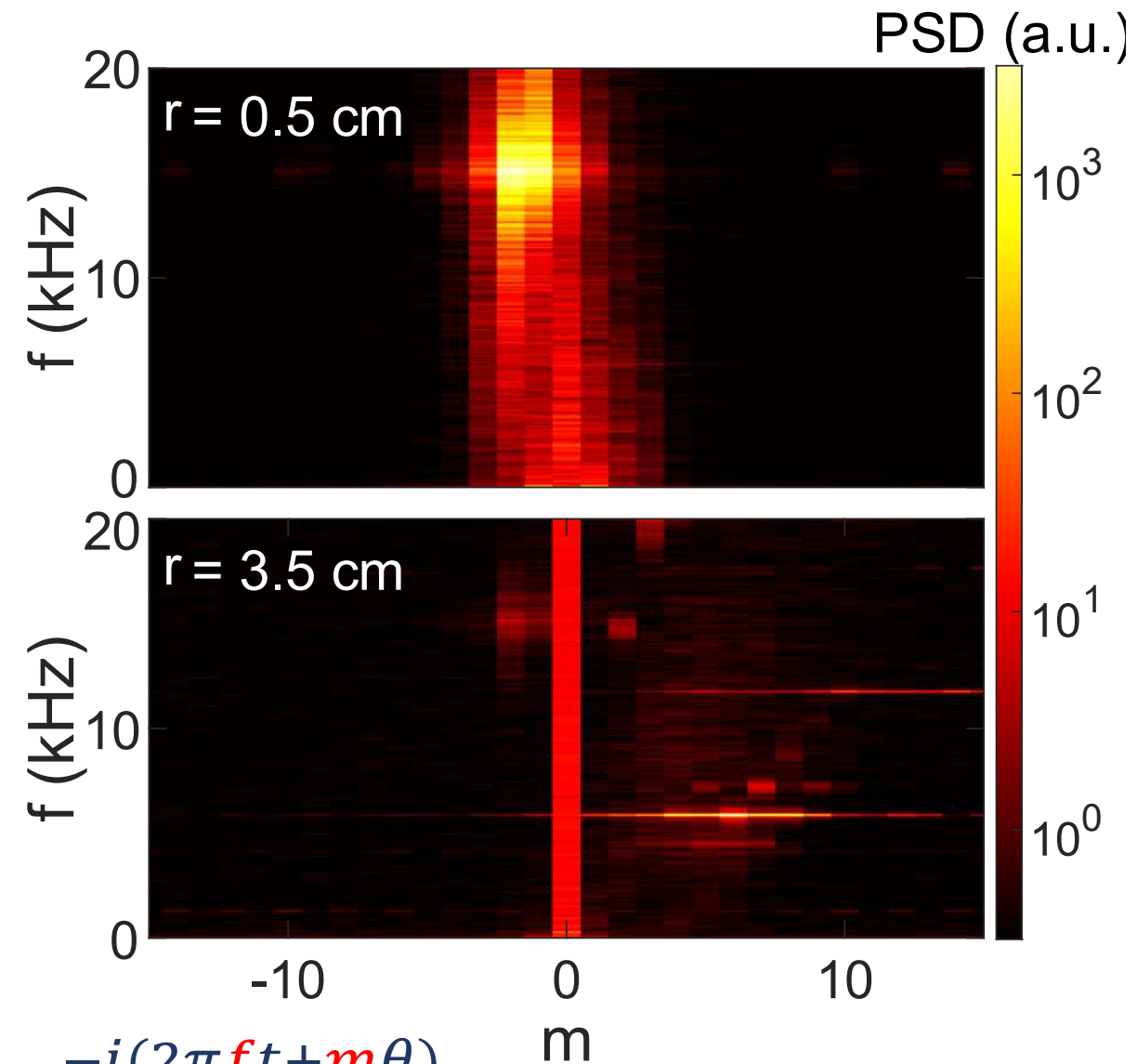
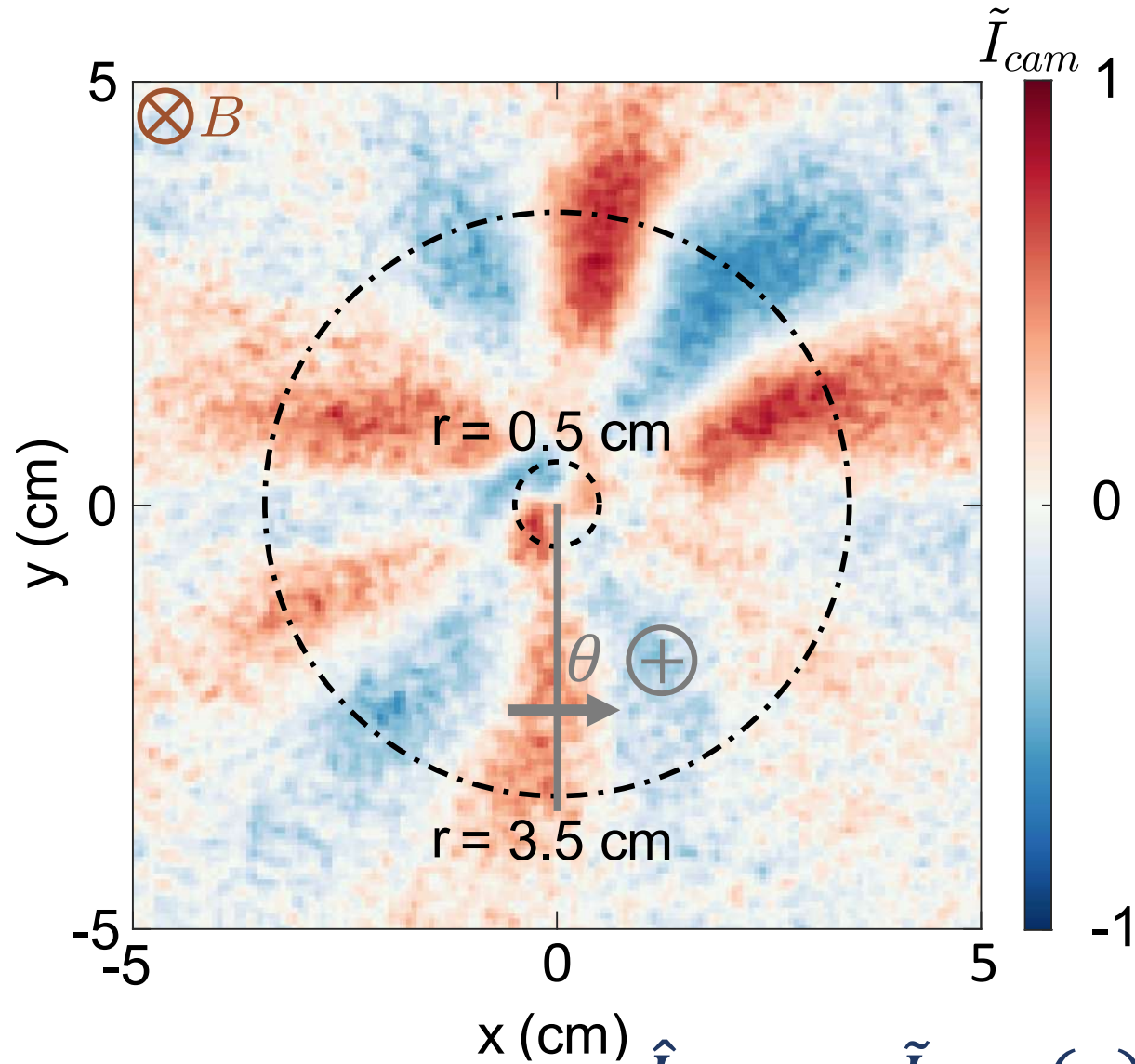
Fluctuations de luminosité

t = 0.00 ms



Paramètres expérimentaux : 900 G, 0.13 Pa, 1.2 kW

Transformée de Fourier 2D



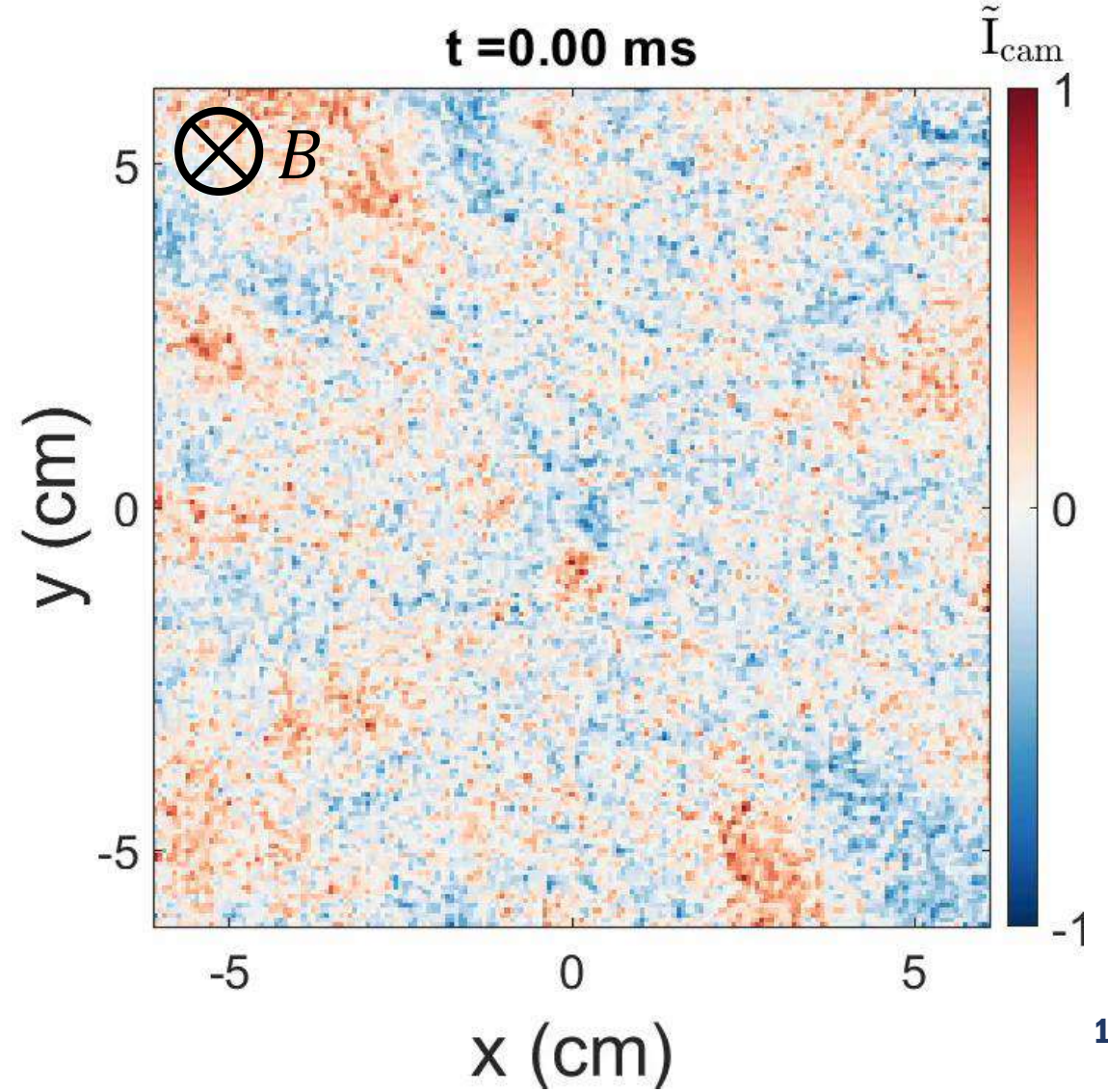
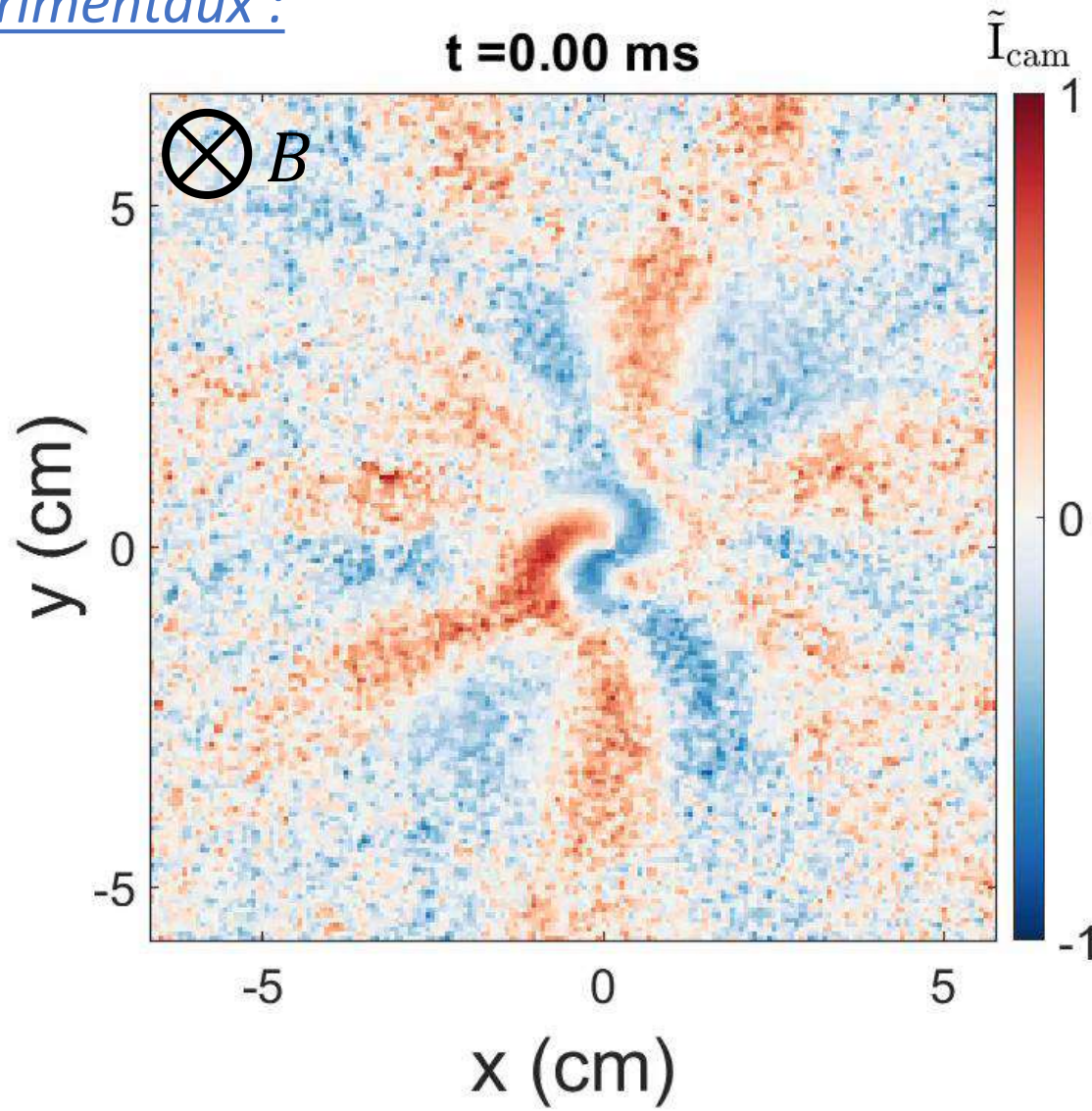
$$\hat{I}_{cam} = \tilde{I}_{cam}(r) \exp^{-i(2\pi f t + m \theta)}$$

Coexistence d'instabilités

Paramètres

900 G, 0.13 Pa, 1.2 kW

expérimentaux :



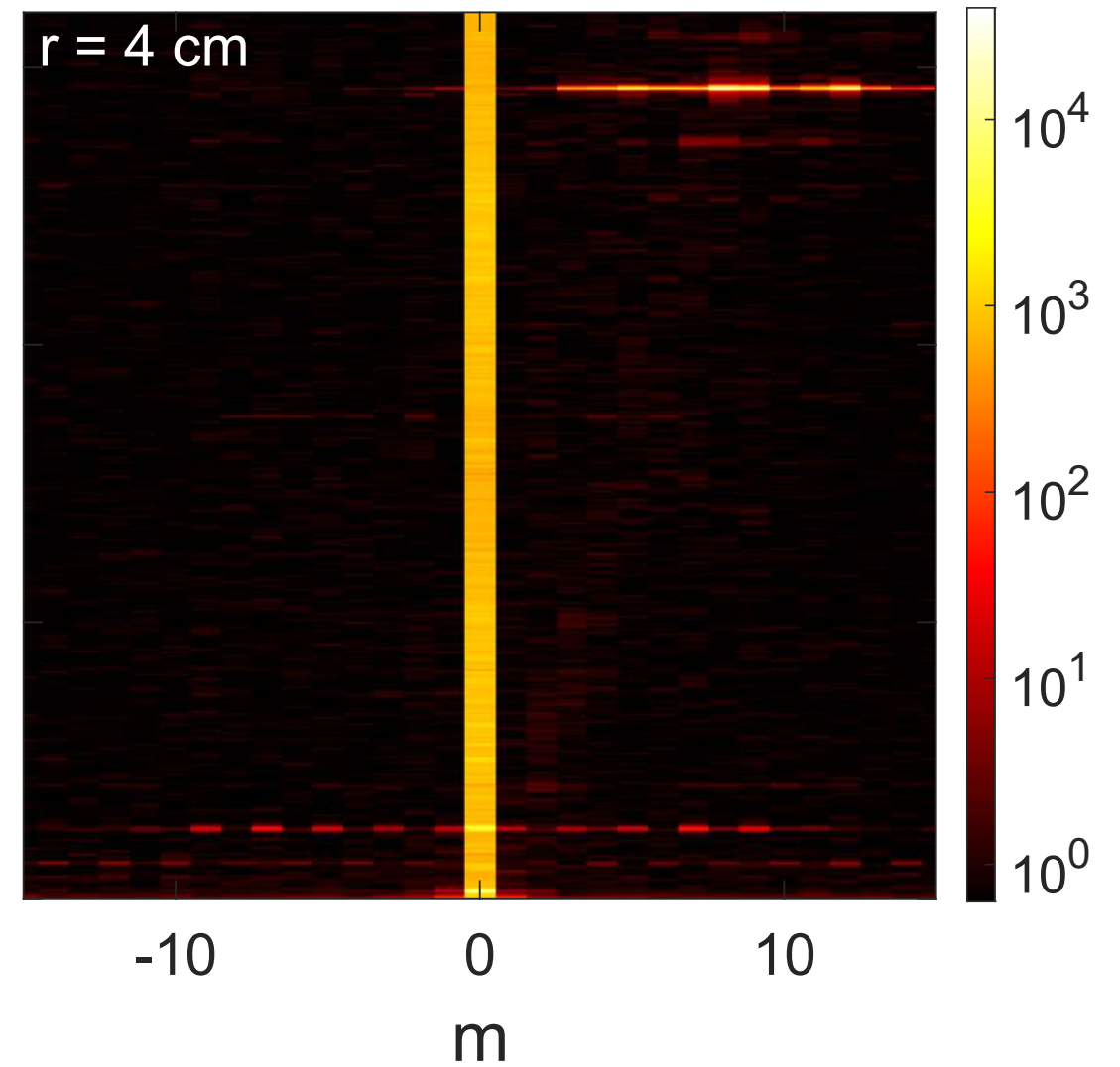
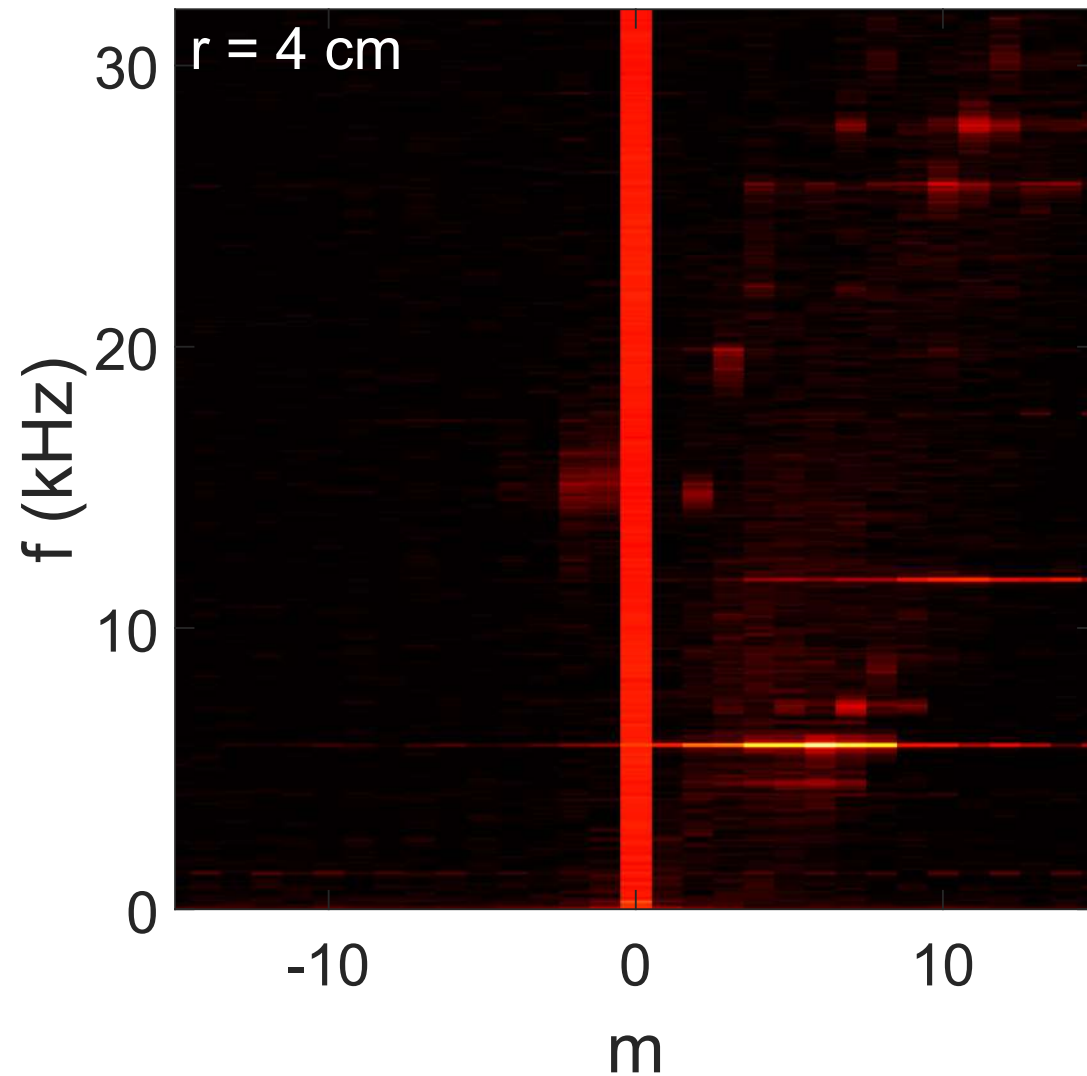
Coexistence d'instabilités

Paramètres

900 G, 0.13 Pa, 1.2 kW

expérimentaux :

1080 G, 0.13 Pa, 1.8 kW

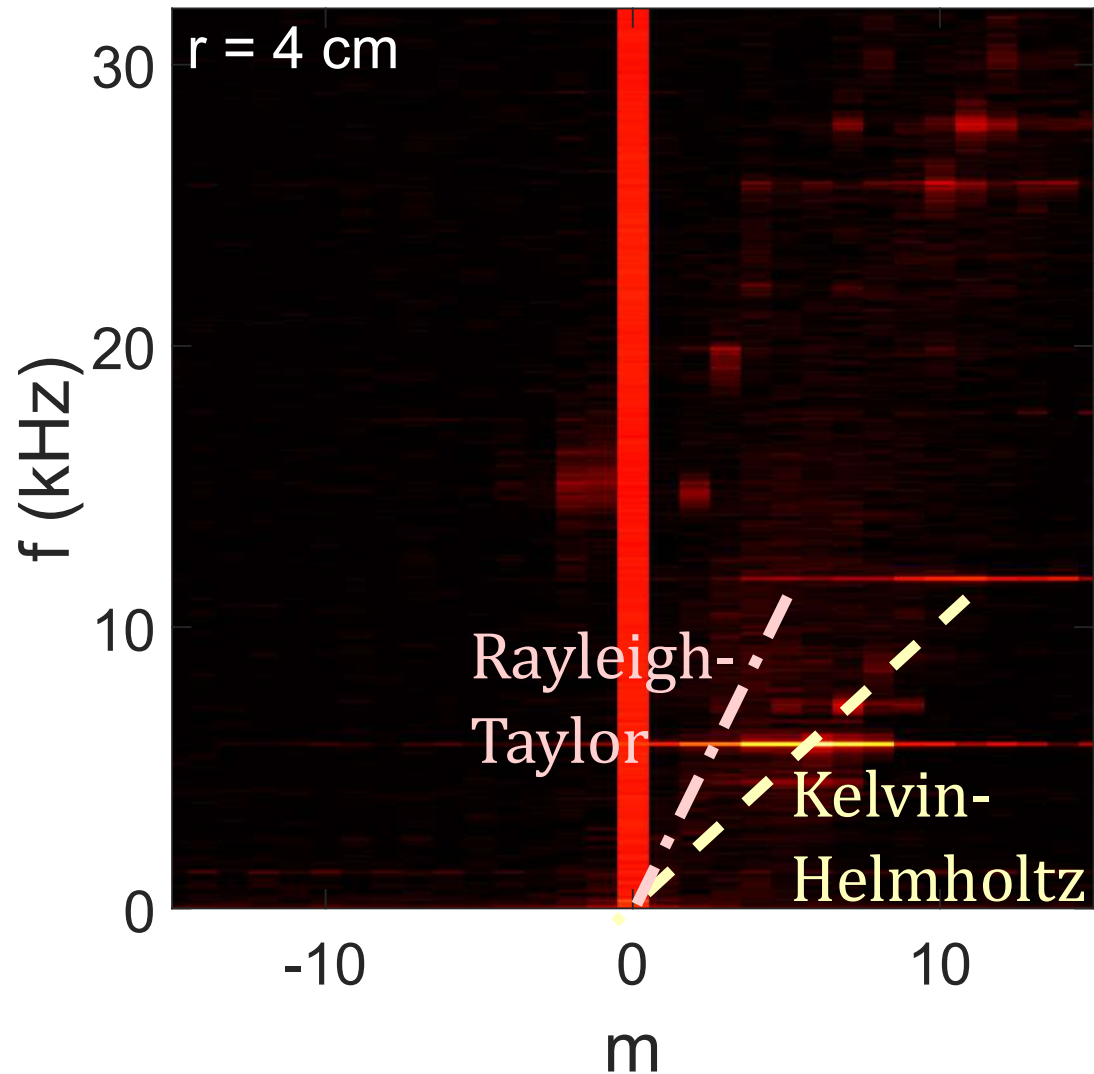


Coexistence d'instabilités

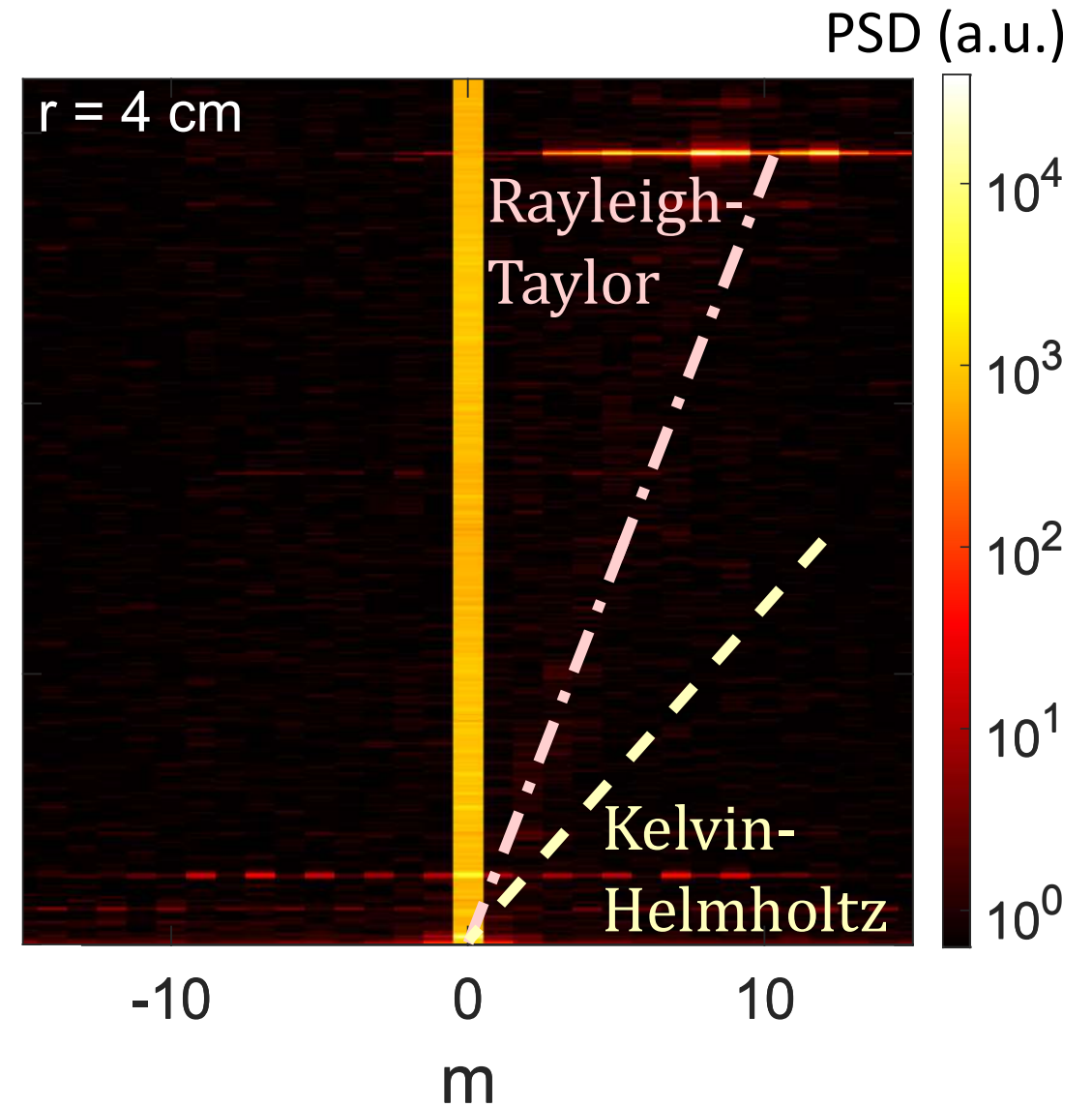
Paramètres

900 G, 0.13 Pa, 1.2 kW

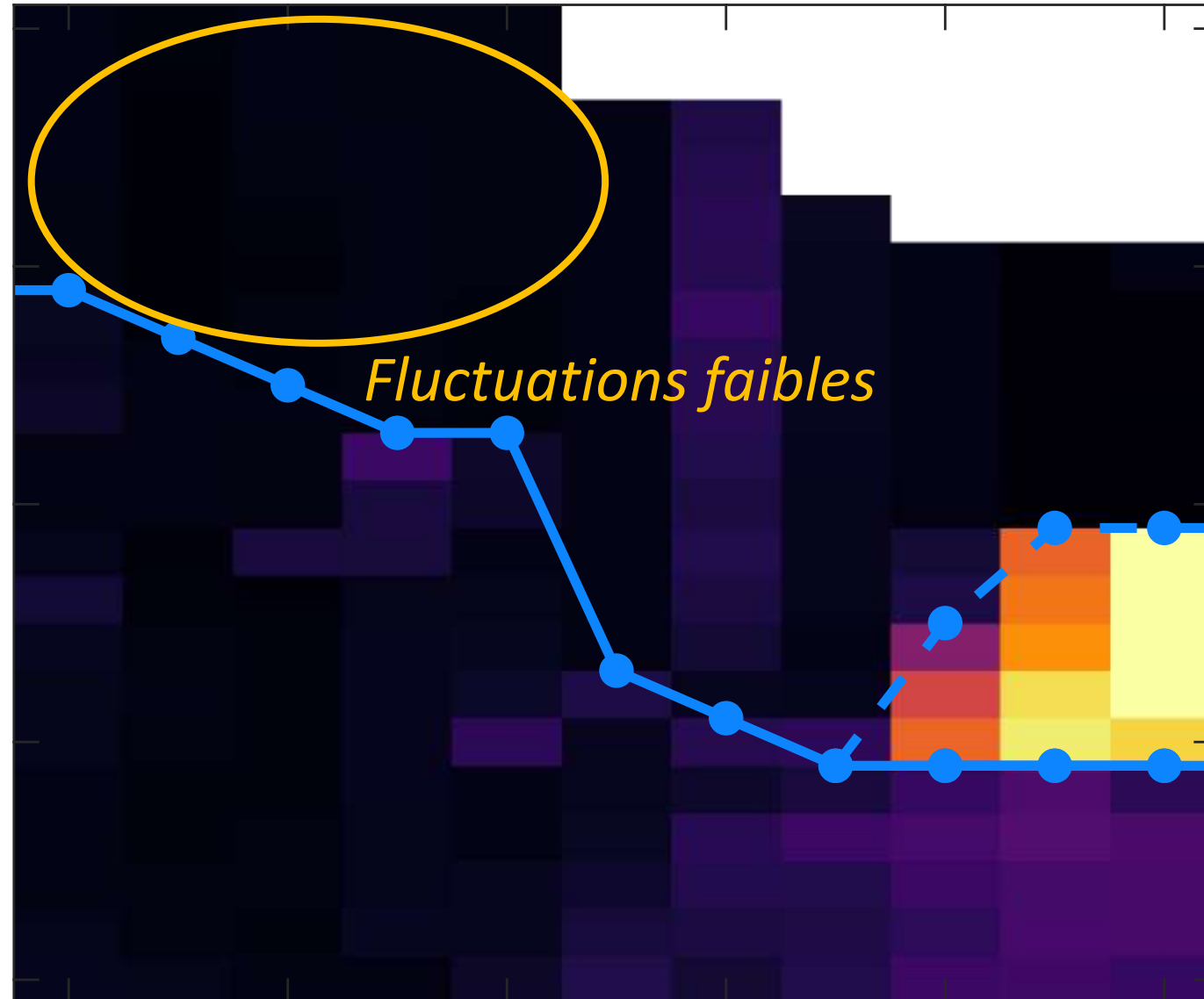
expérimentaux :



1080 G, 0.13 Pa, 1.8 kW

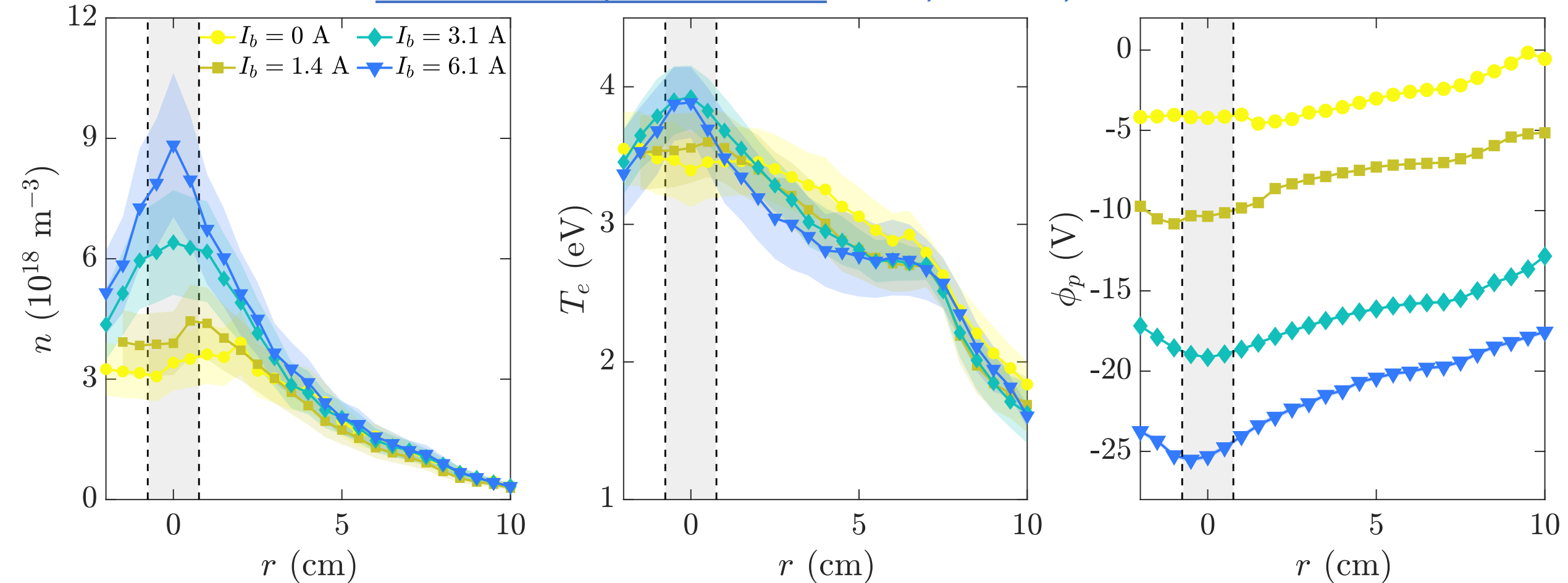


Coexistence d'instabilités



Contrôle via cathode émissive

Paramètres expérimentaux : 180 G, 0.13 Pa, 1 kW



➤ Contrôle de ϕ_p accru

➤ Impact sur la transition au mode hélicon

Contribution : Vassili Desages

CONCLUSION

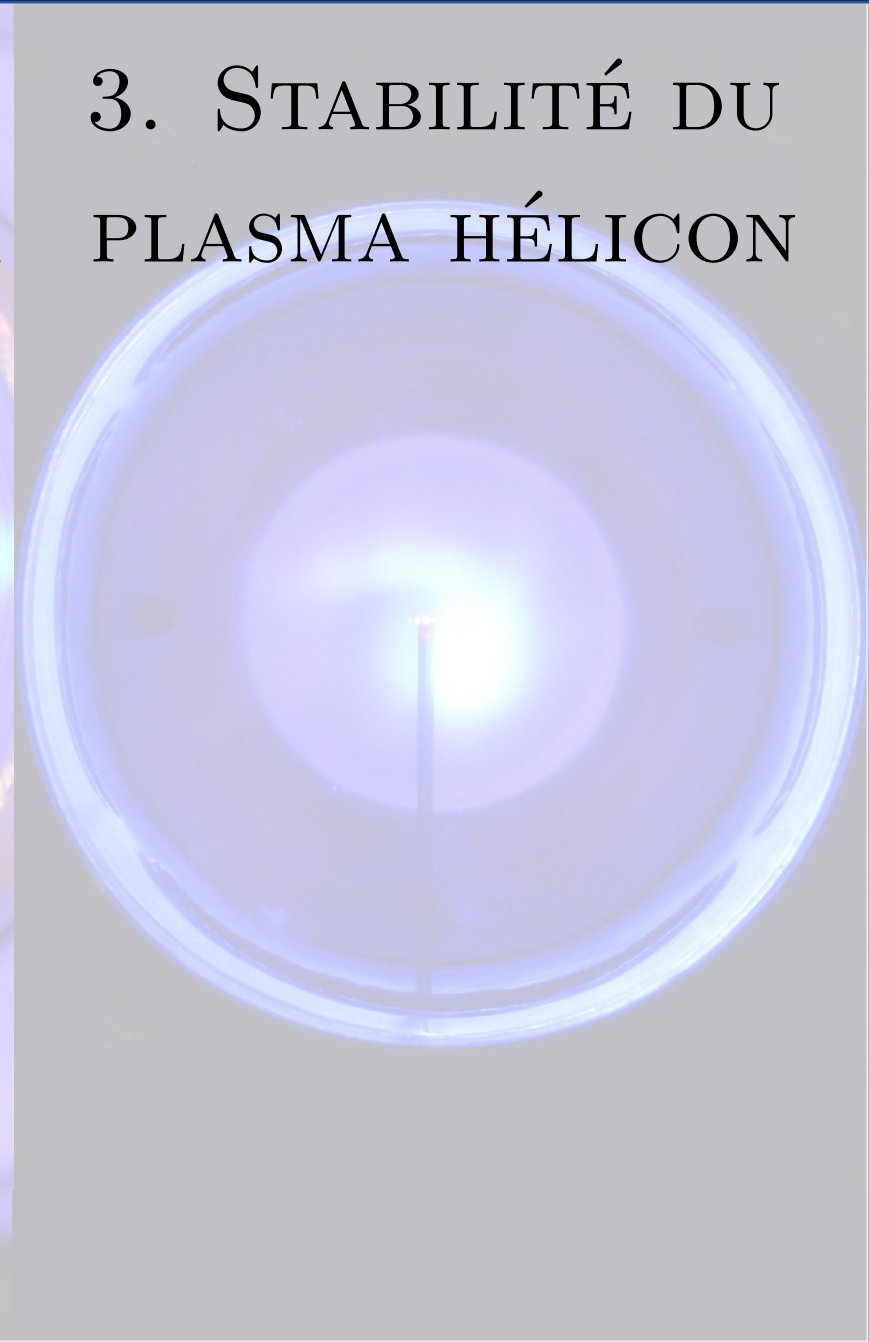
1. OPÉRATION D'UNE
CATHODE ÉMISSIVE



2. CONTRÔLE DU
POTENTIEL PLASMA



3. STABILITÉ DU
PLASMA HÉLICON

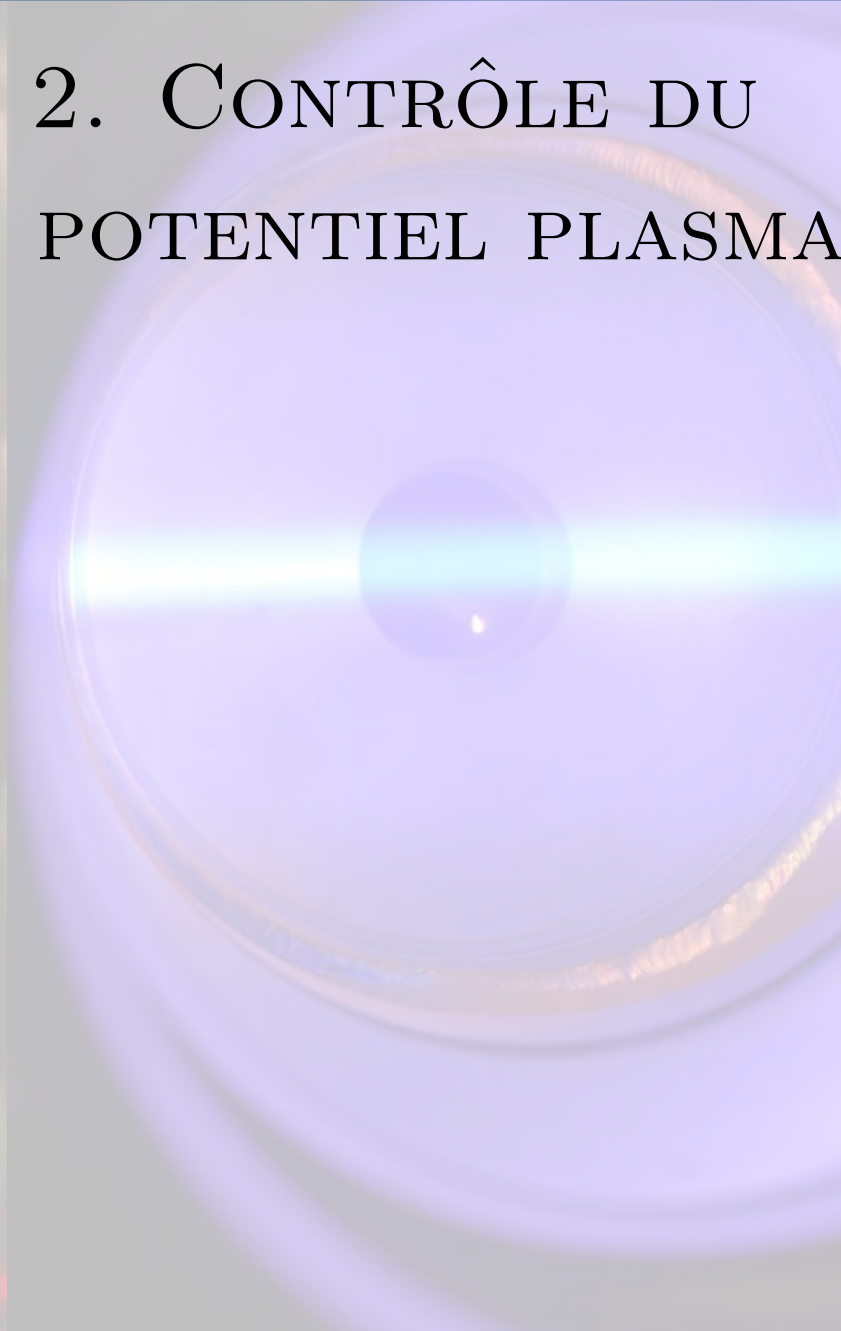


CONCLUSION

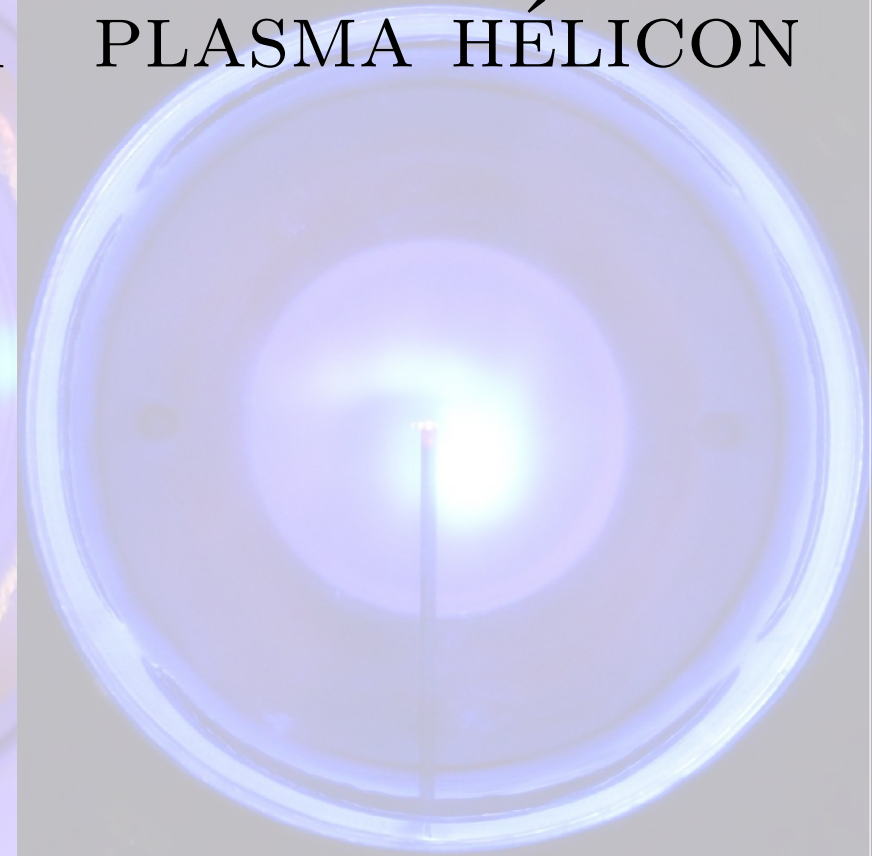
1. OPÉRATION D'UNE CATHODE ÉMISSIVE

- Nouveau modèle thermique complet
- Prédiction de I_b fiable
- Compréhension fine des interactions plasma-cathode

2. CONTRÔLE DU POTENTIEL PLASMA



3. STABILITÉ DU PLASMA HÉLICON



CONCLUSION

1. OPÉRATION D'UNE CATHODE ÉMISSIVE

- Nouveau modèle thermique complet
- **Prédiction de I_b fiable**
- Compréhension fine des interactions plasma-cathode

2. CONTRÔLE DU POTENTIEL PLASMA

- Cathode émissive comme **paramètre de contrôle**
- Confrontation au modèle
- Influence des processus collisionnels sur ϕ_p

3. STABILITÉ DU PLASMA HÉLICON

CONCLUSION

1. OPÉRATION D'UNE CATHODE ÉMISSIVE

- Nouveau modèle thermique complet
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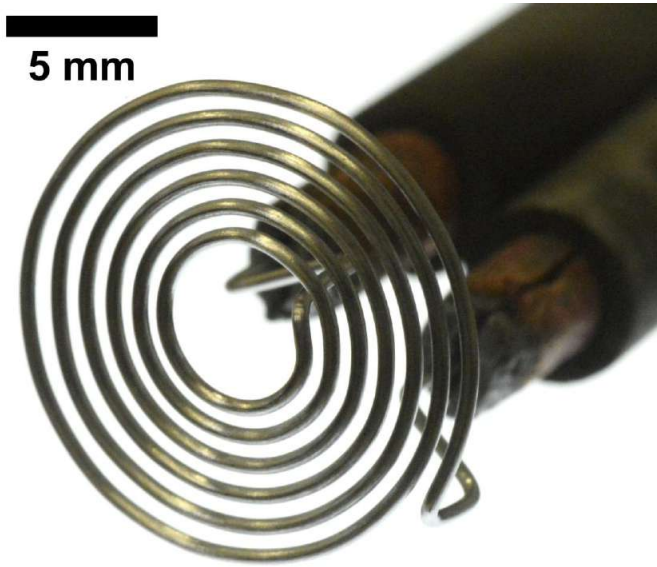
3. STABILITÉ DU PLASMA HÉLICON

- Implémentation réussie d'une source hélicon
- **Caractérisation** de l'espace des paramètres (p_0, B, P_w)
- Identification d'**instabilités**

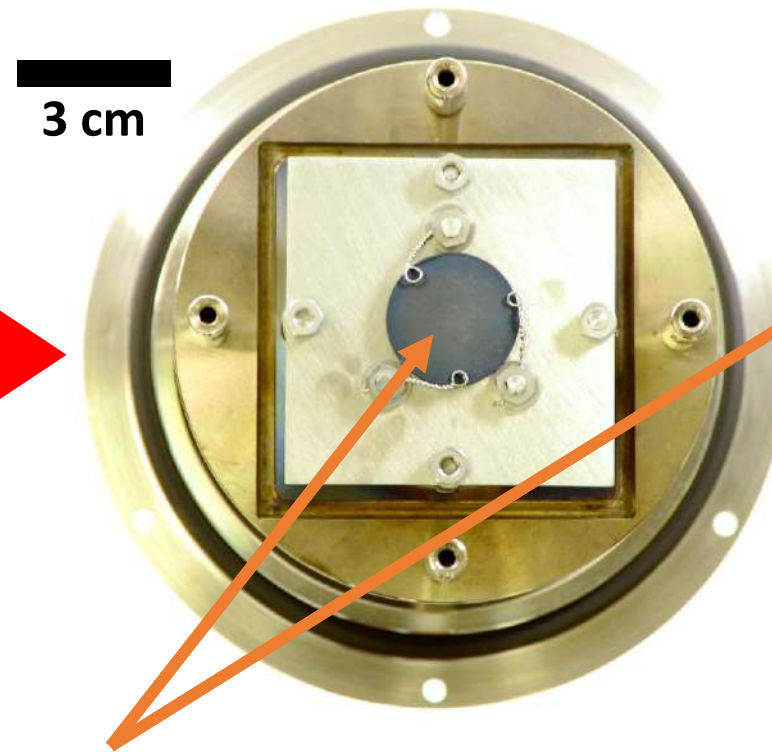
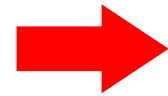
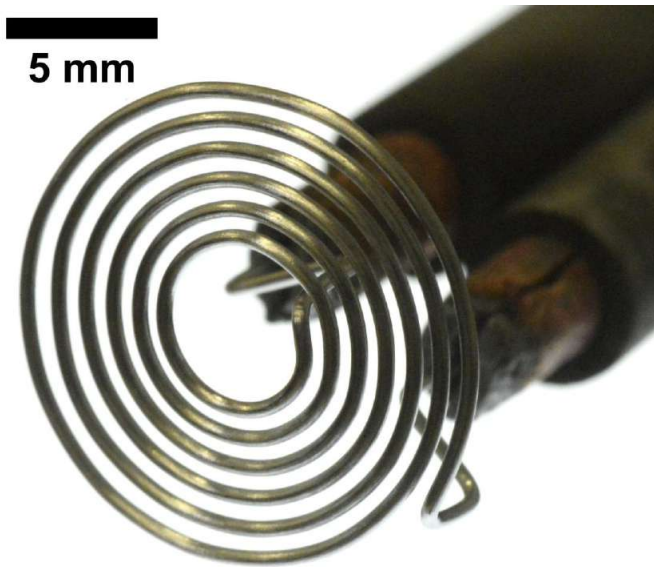
PERSPECTIVES



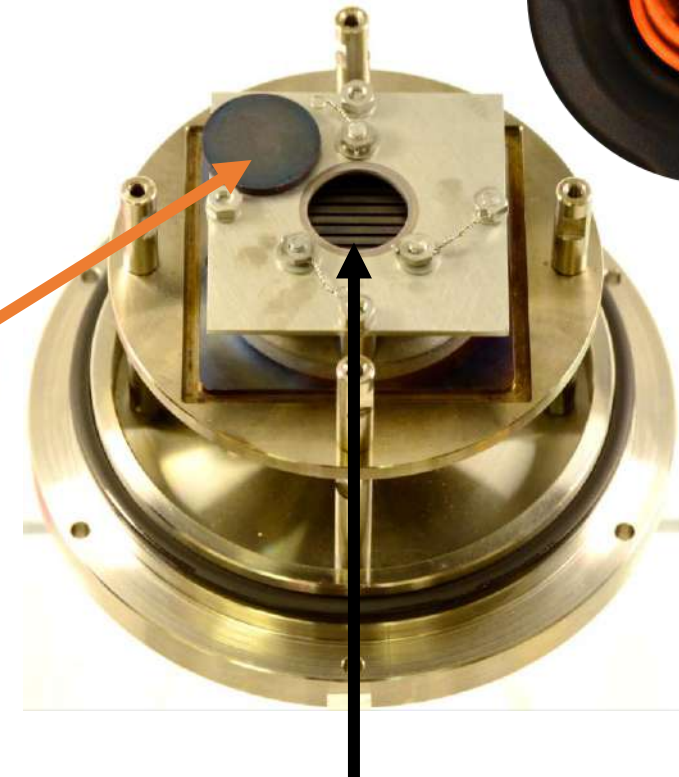
Nouvelle cathode LaB_6



Nouvelle cathode LaB_6



Disque LaB_6
émission électrons I_{em}

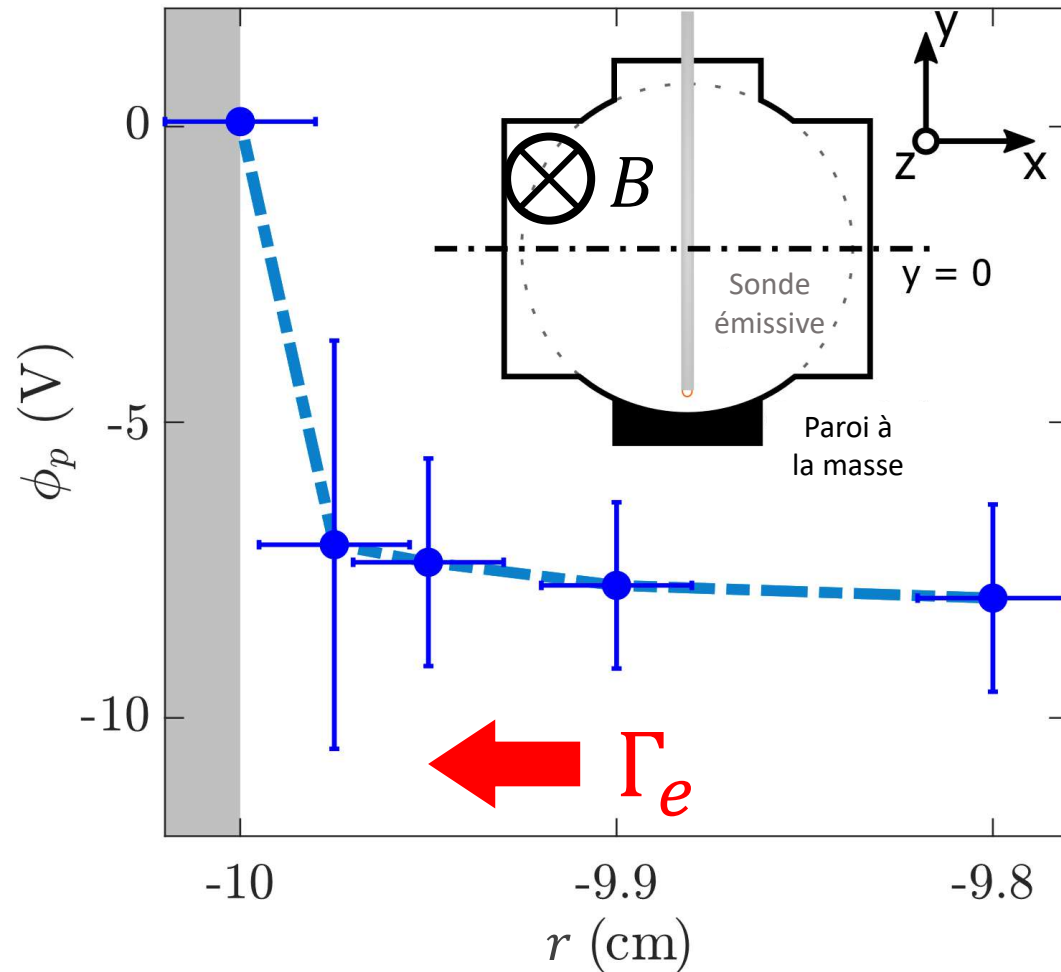
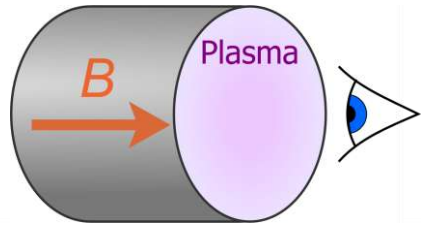
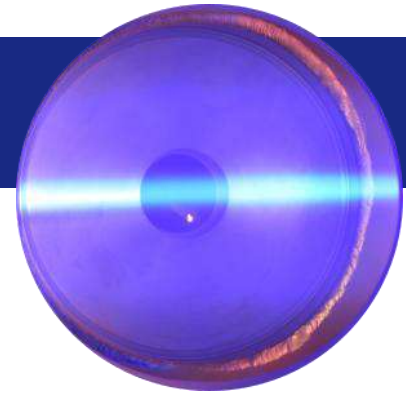


Filament carbone
Chauffage I_h



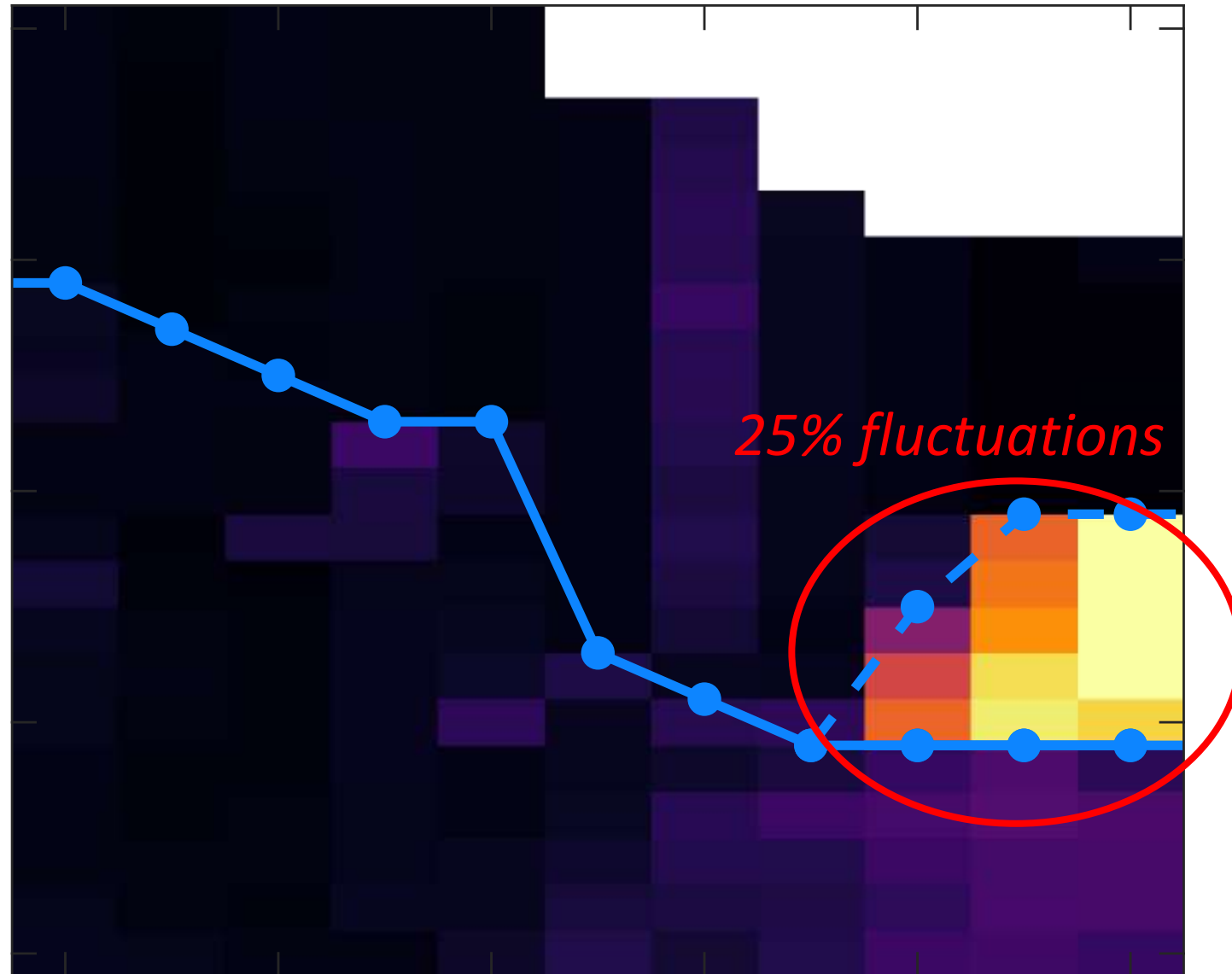
- Découplage de l'injection I_{em} et du chauffage I_h
 - Cathode large, travail de sortie W faible

Modèle de gaine anodique



- Effet du champ magnétique || anode
- Flux d'électrons incident Γ_e
- Modèle auto-consistant / Mesures exp
 - Contribution : Eitan Soussan

Fluctuations incohérentes multi-échelles

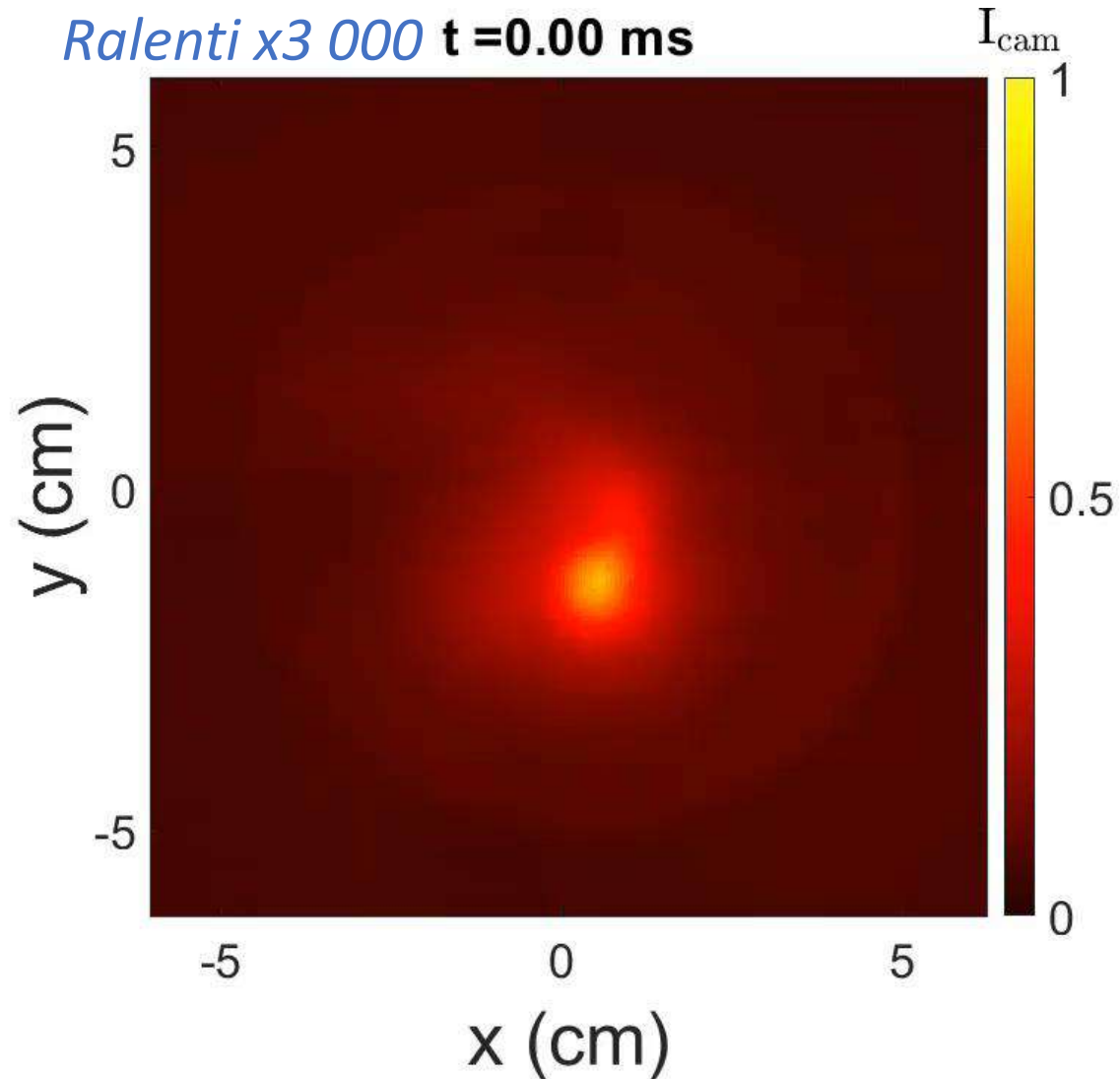


Fluctuations incohérentes multi-échelles

Paramètres expérimentaux : 1080 G, 0.07 Pa, 0.9 kW



Ralenti x3 000 **t = 0.00 ms**

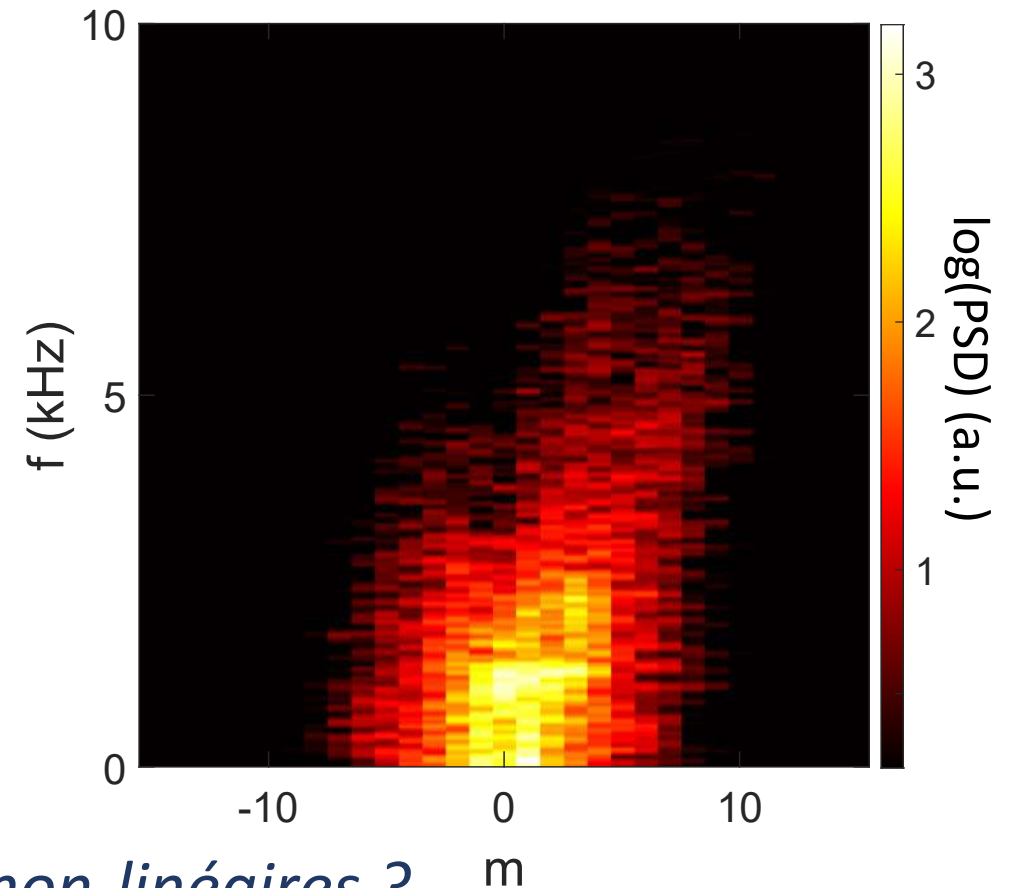
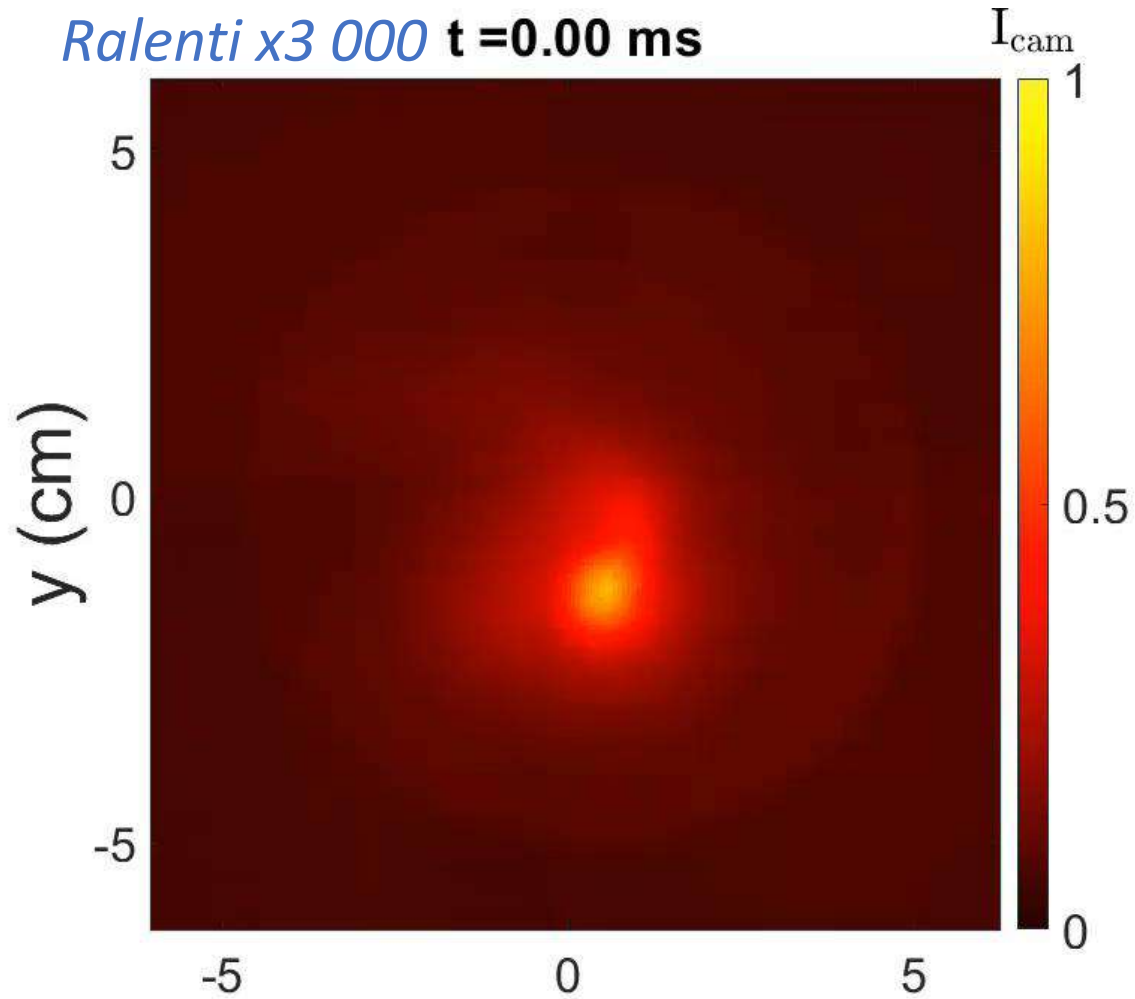


Fluctuations incohérentes multi-échelles

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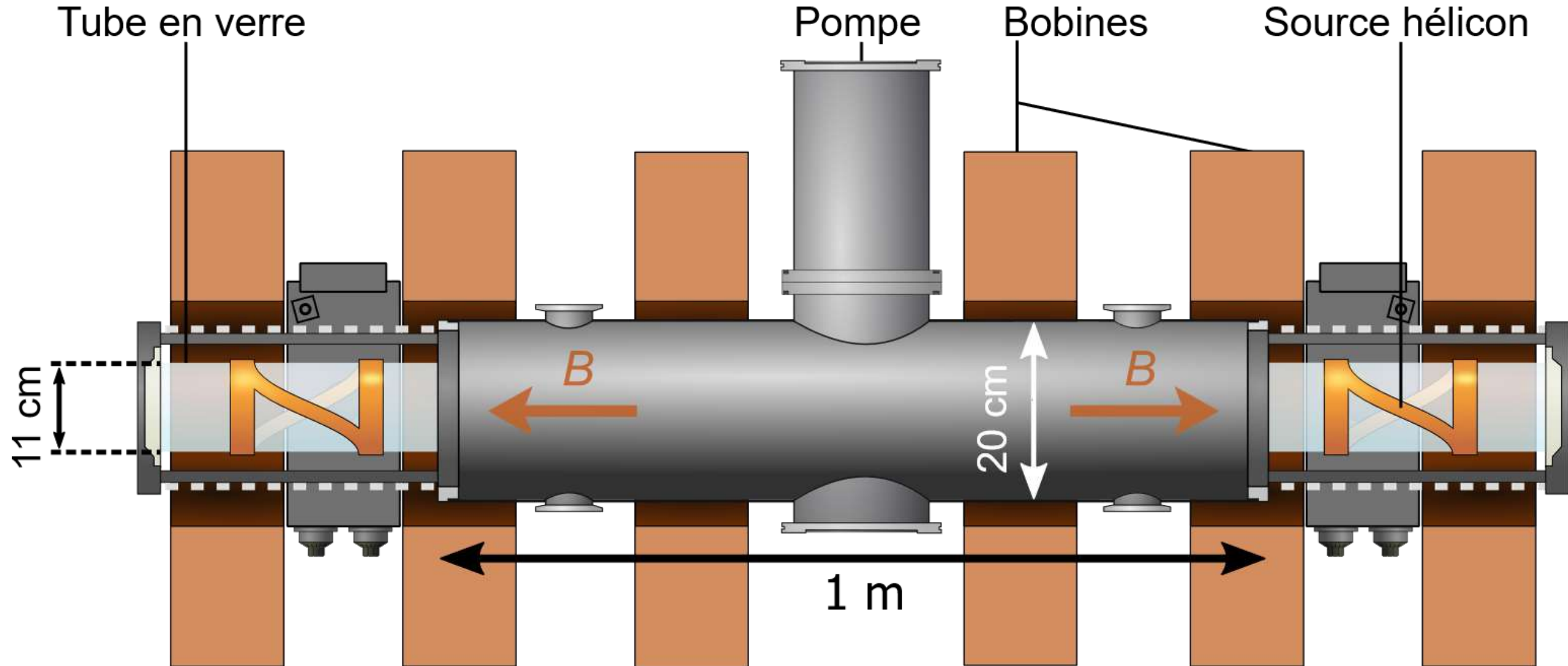
Ralenti x3 000 **t = 0.00 ms**



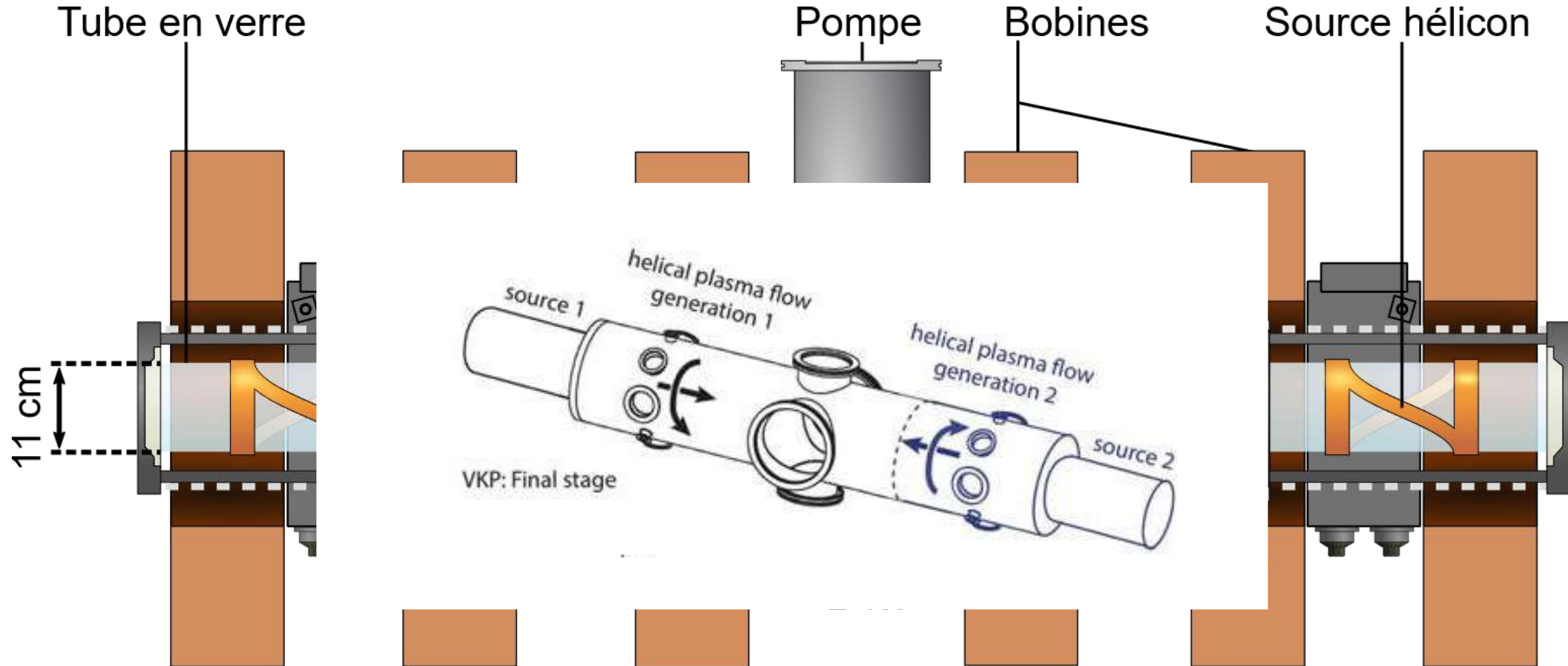
➤ *Interactions d'ondes non-linéaires ?*

➤ *Dépletion de neutres ?*

Confrontation de deux colonnes de plasma



Confrontation de deux colonnes de plasma

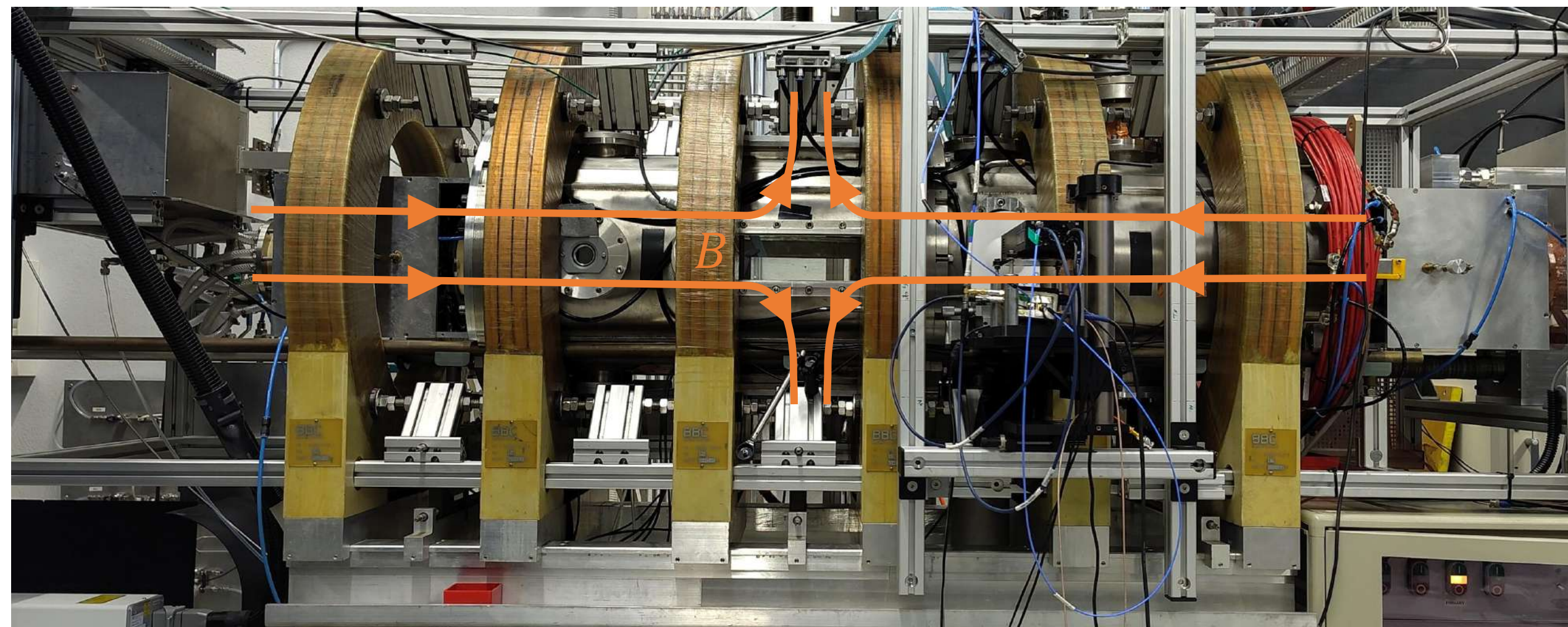


Collaboration internationale



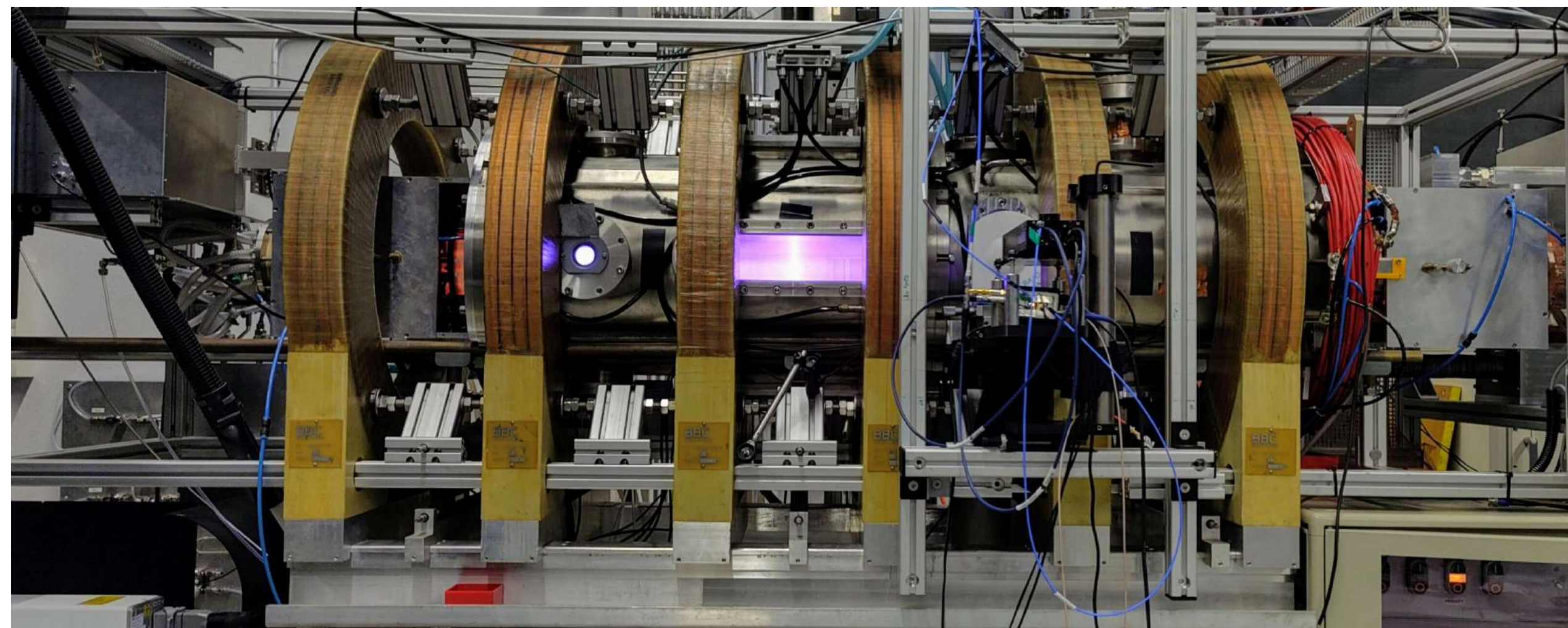
Dispositif RAID, Swiss Plasma Center, Lausanne

Collaboration internationale



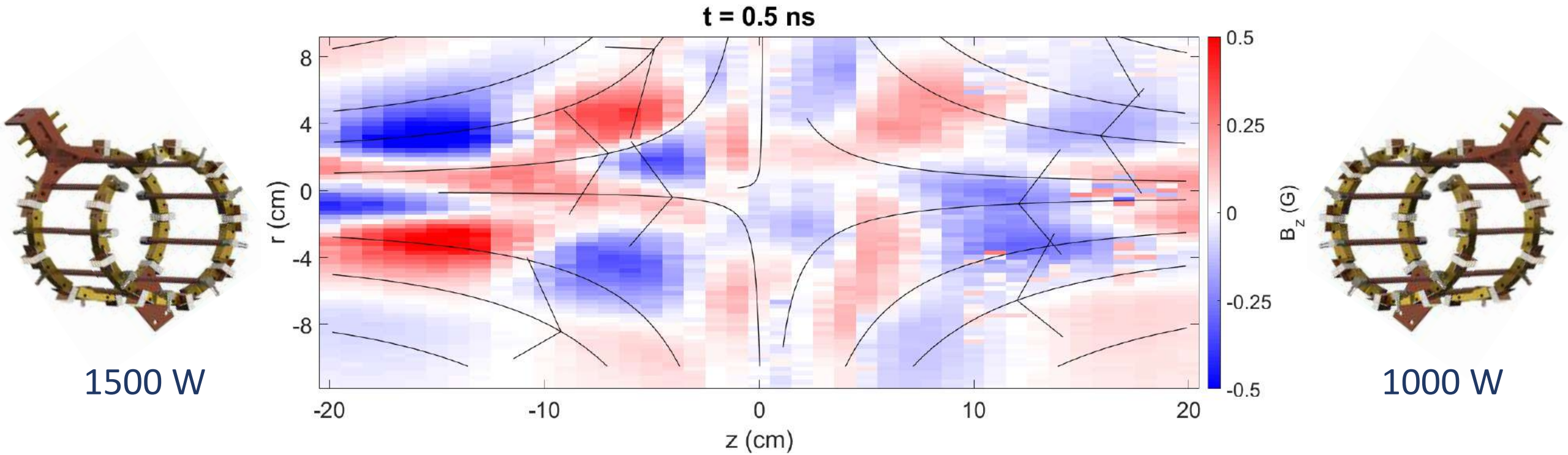
Dispositif RAID, Swiss Plasma Center, Lausanne

Collaboration internationale

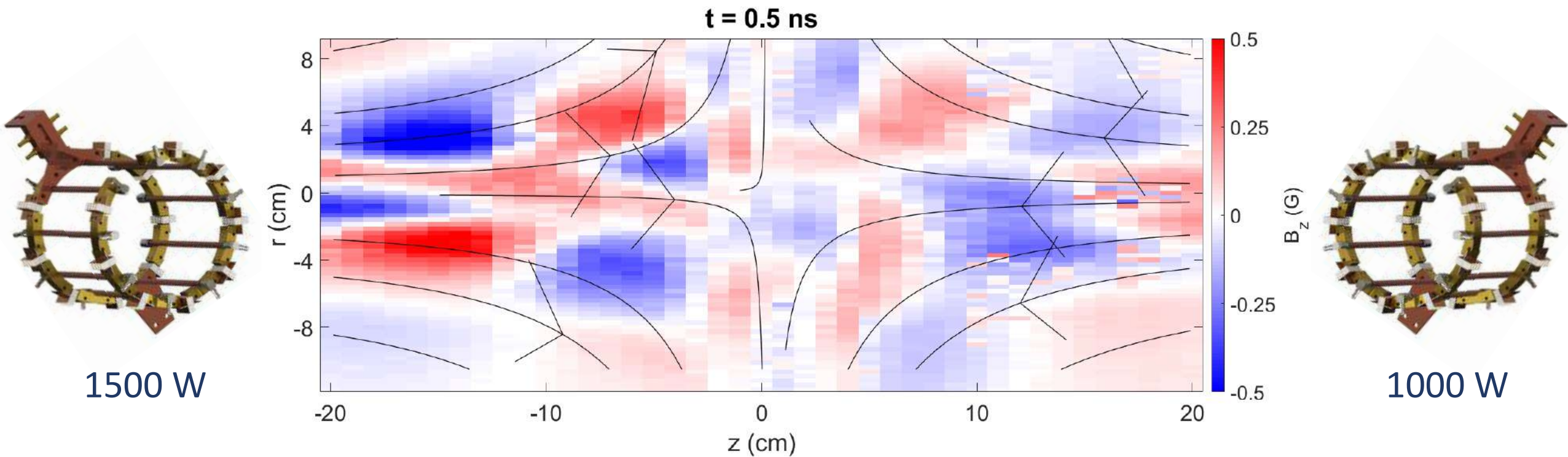


Dispositif RAID, Swiss Plasma Center, Lausanne

Interactions entre ondes hélicons



Interactions entre ondes hélicons



- Utilisation de deux sources hélicon « birdcage »
- Régimes expérimentaux complémentaires

Remerciements